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Friedman et al.

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(54) **COMPOSITIONS FOR TREATING CANCER**

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Related U.S. Application Data

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(51) **Int. Cl.**

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A61K 40/42 (2025.01)
A61K 48/00 (2006.01)
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C07K 14/725 (2006.01)
C07K 16/28 (2006.01)
C12N 15/86 (2006.01)

(52) **U.S. Cl.**

CPC **A61K 40/4215** (2025.01); **A61K 40/11** (2025.01); **A61K 40/15** (2025.01); **A61K 40/31** (2025.01); **A61K 48/005** (2013.01); **A61P 35/00** (2018.01); **C07K 14/7051** (2013.01); **C07K 16/2809** (2013.01); **C07K 16/2878** (2013.01); **C12N 15/86** (2013.01); **A61K 2239/13** (2023.05); **C07K 2317/622** (2013.01); **C12N 2740/15022** (2013.01); **C12N 2740/15023** (2013.01); **C12N 2740/15043** (2013.01); **C12N 2830/48** (2013.01); **C12N 2830/50** (2013.01)

(58) **Field of Classification Search**

None

See application file for complete search history.

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(57) **ABSTRACT**

The present disclosure provides compositions and methods comprising recombinant particles suitable for specifically delivering one or more chimeric antigen receptors to immune effector cells in vivo.

25 Claims, 20 Drawing Sheets

Specification includes a Sequence Listing.

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FIGURE 1

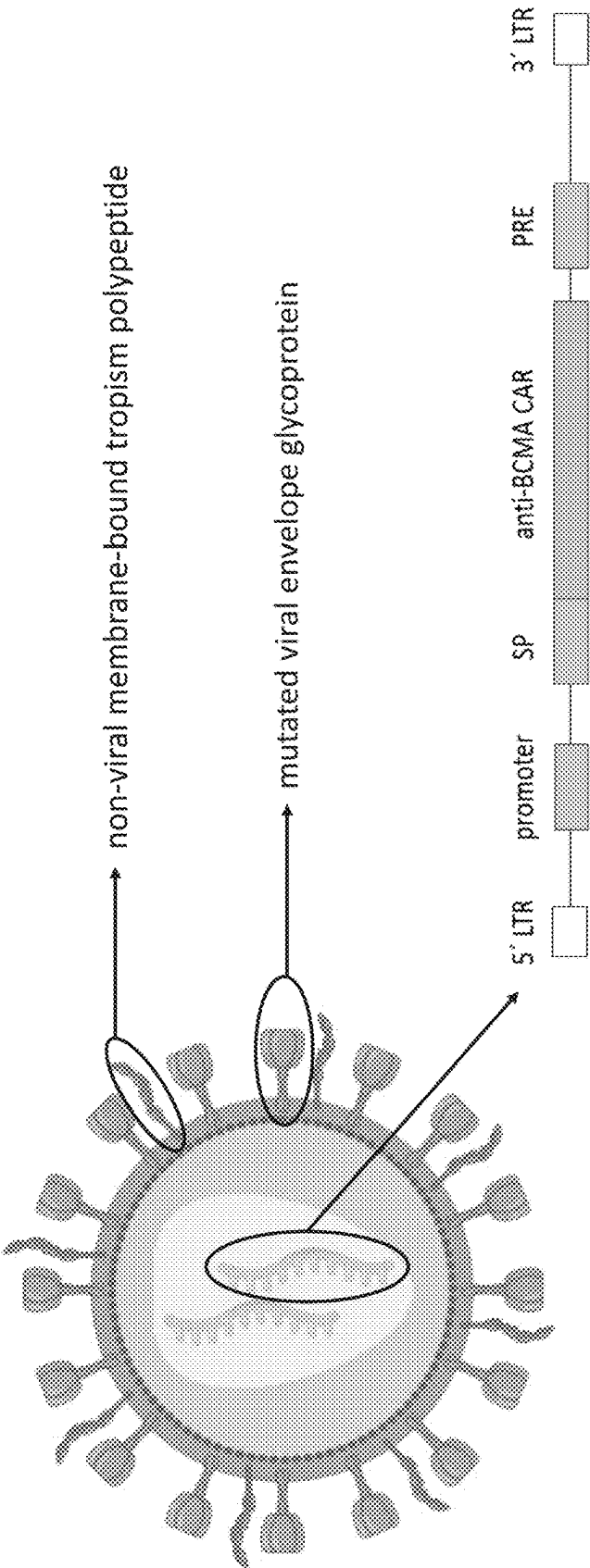


FIGURE 2A

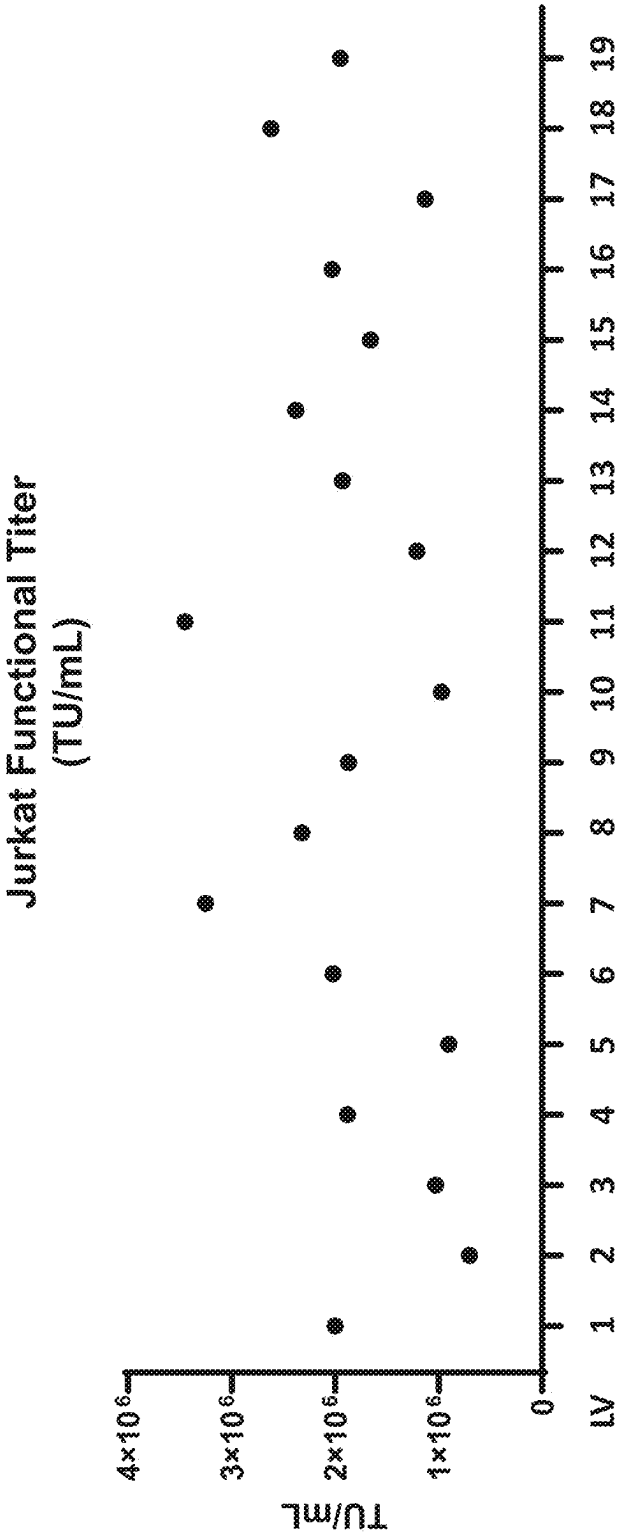


FIGURE 2B

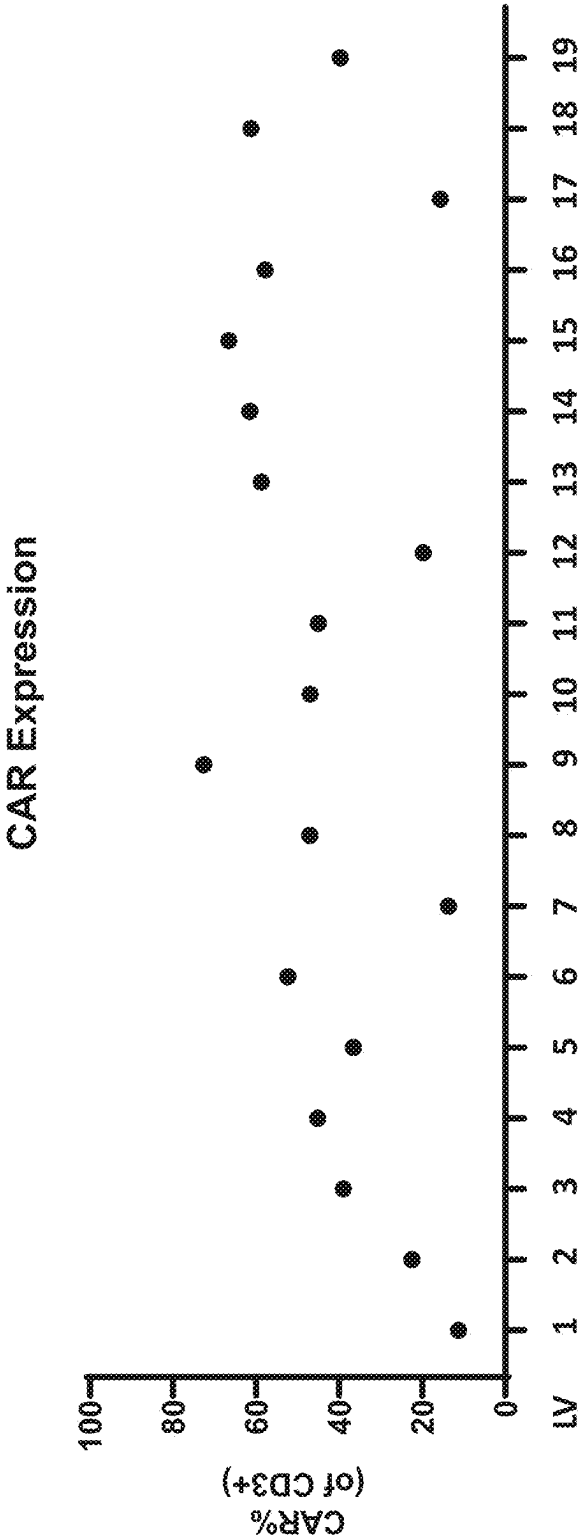


FIGURE 2C

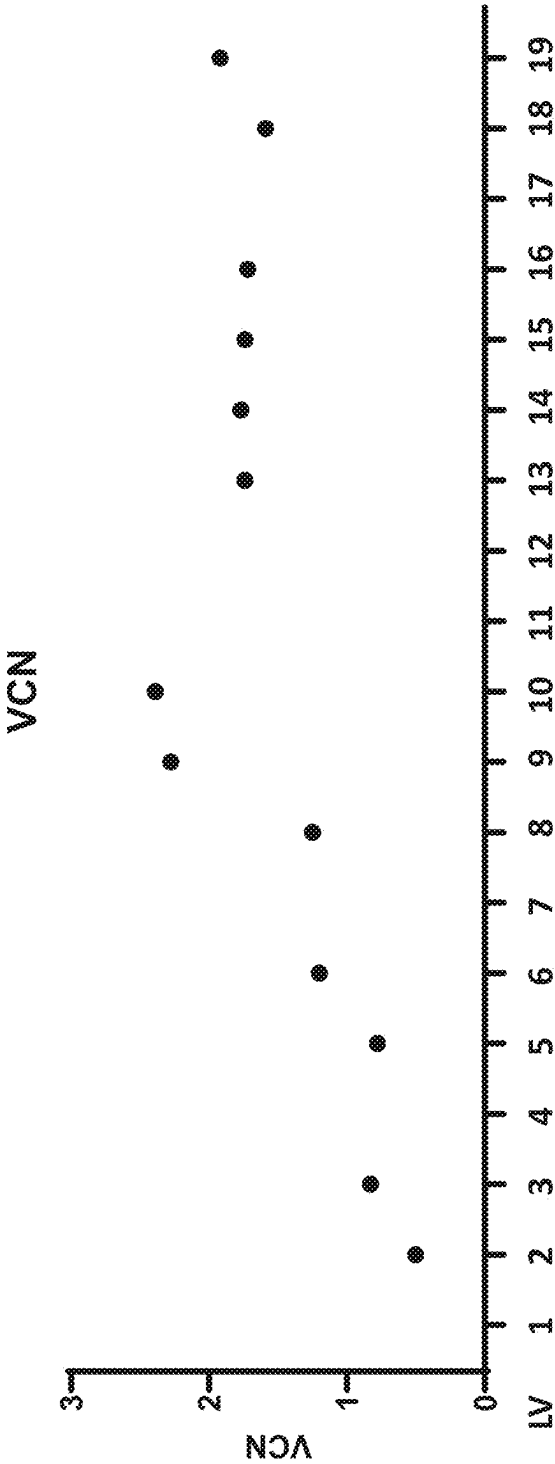


FIGURE 2D

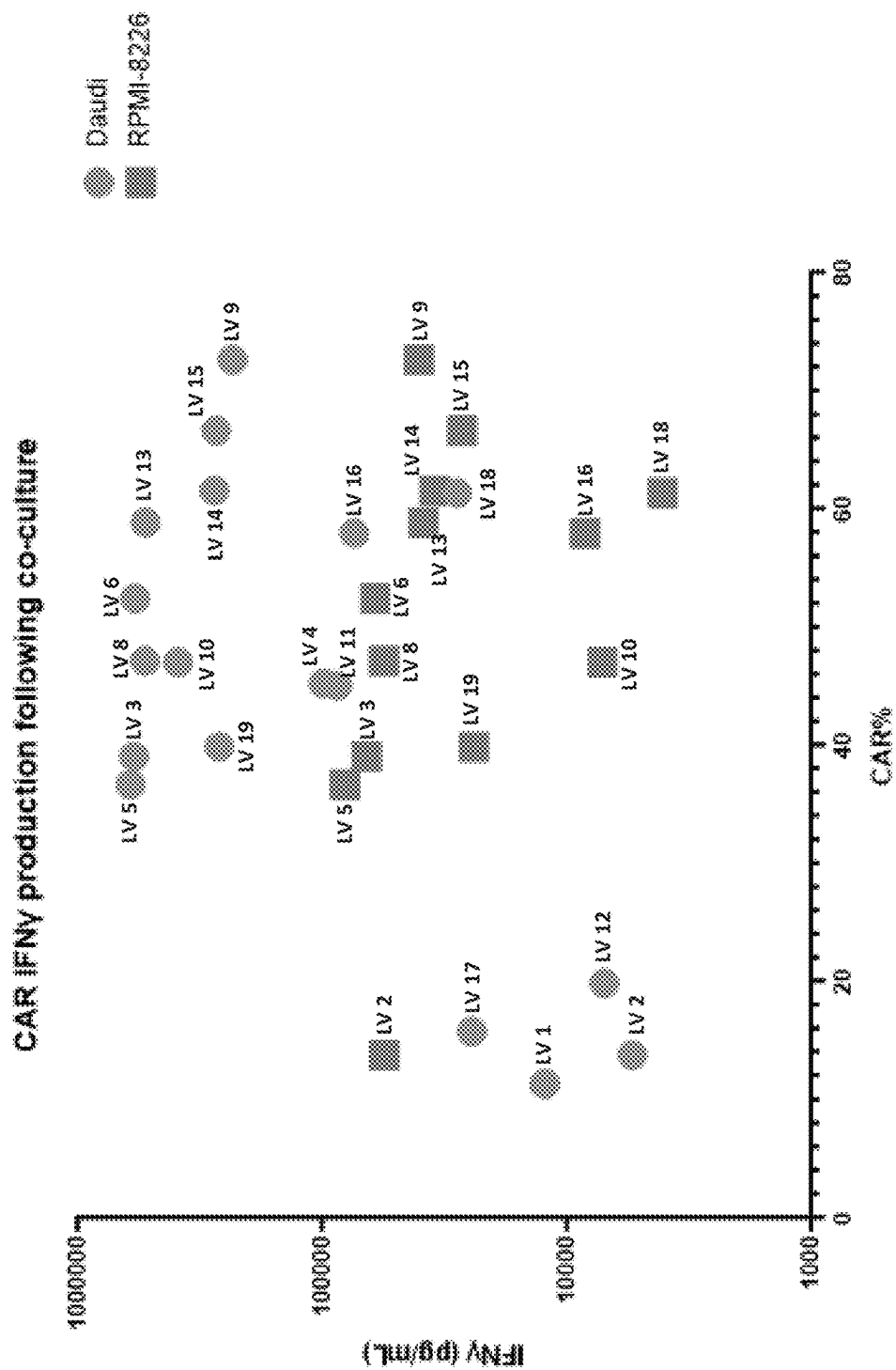


FIGURE 3A

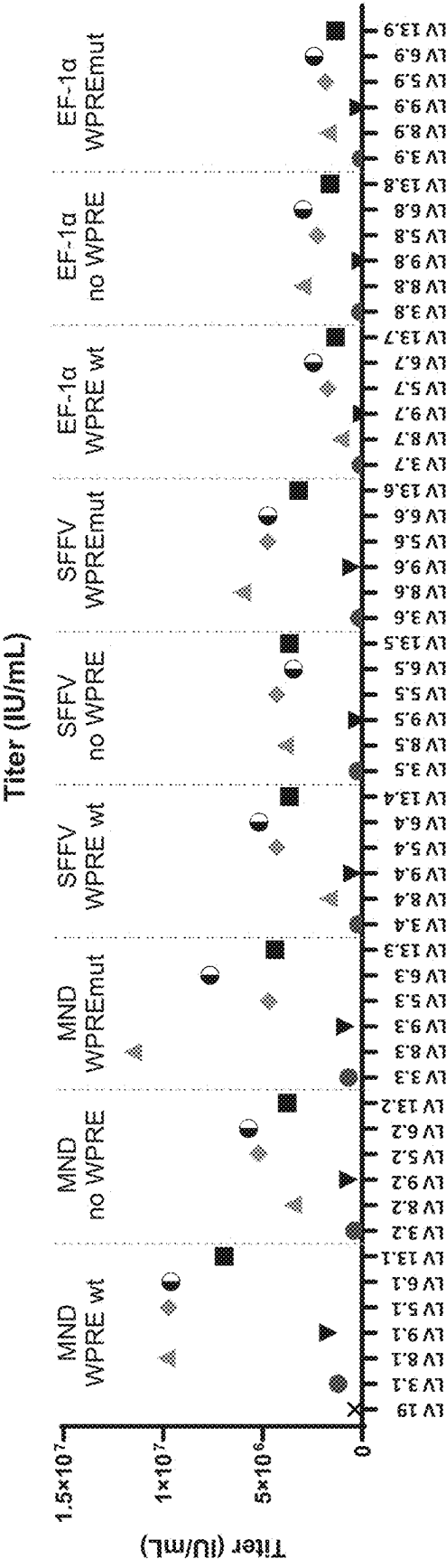


FIGURE 3B

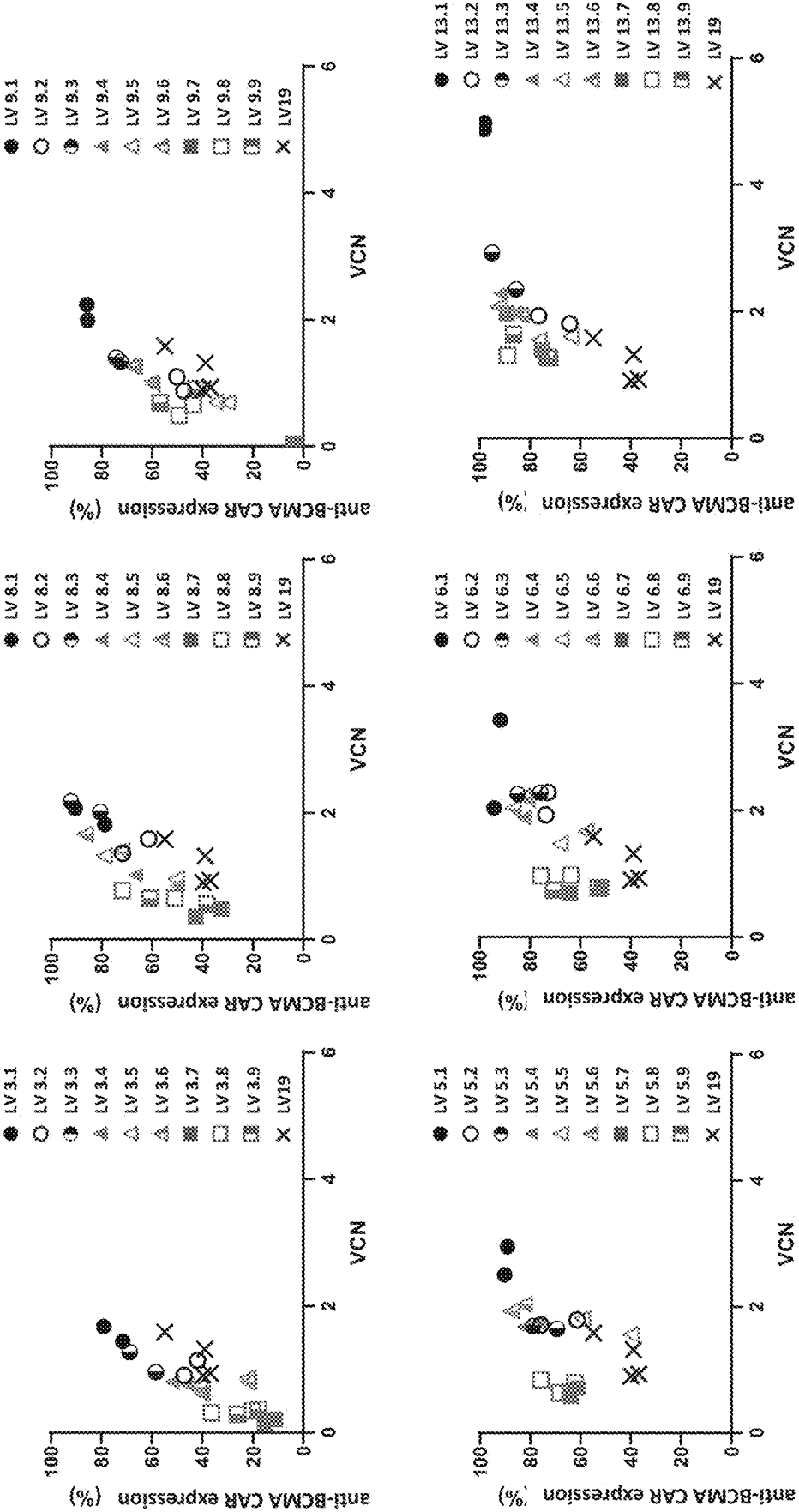


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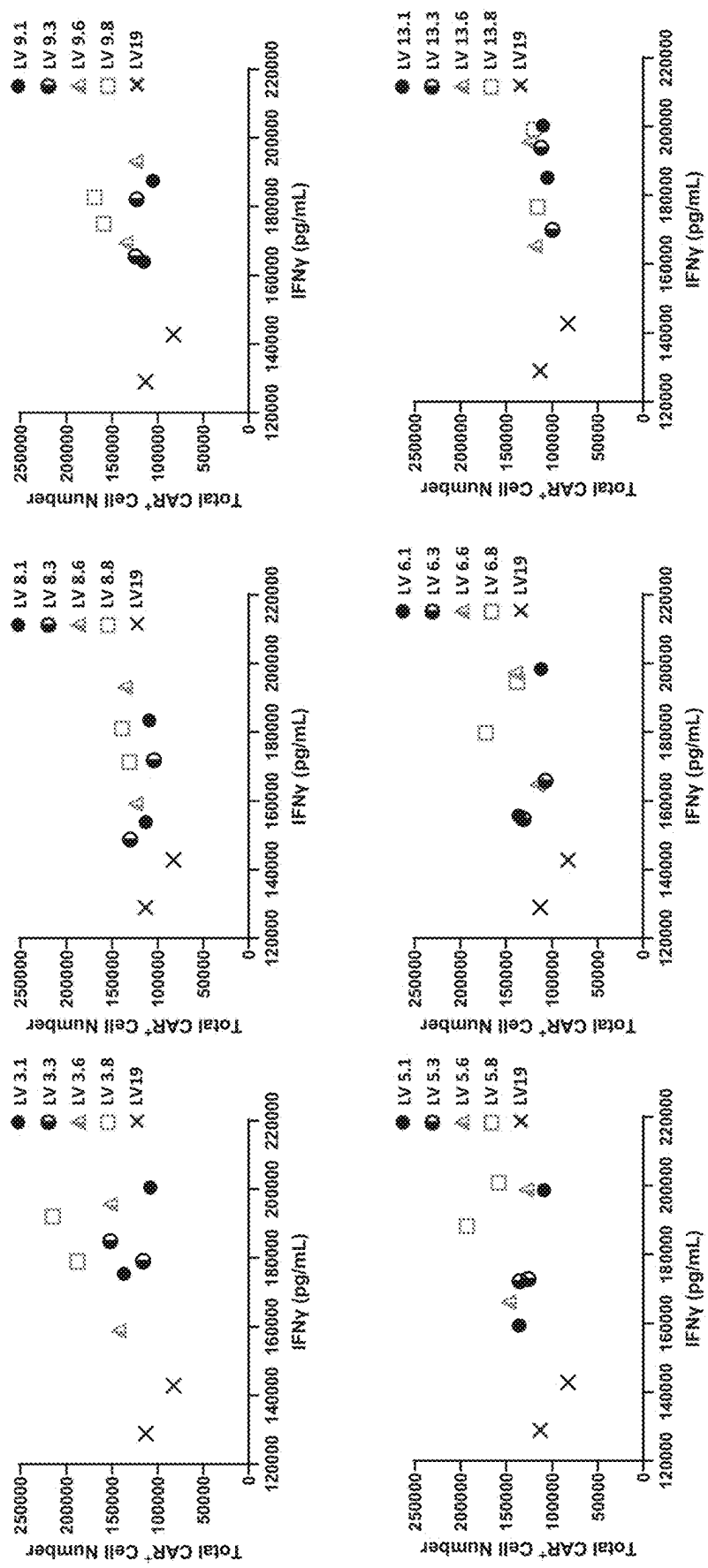


FIGURE 3D

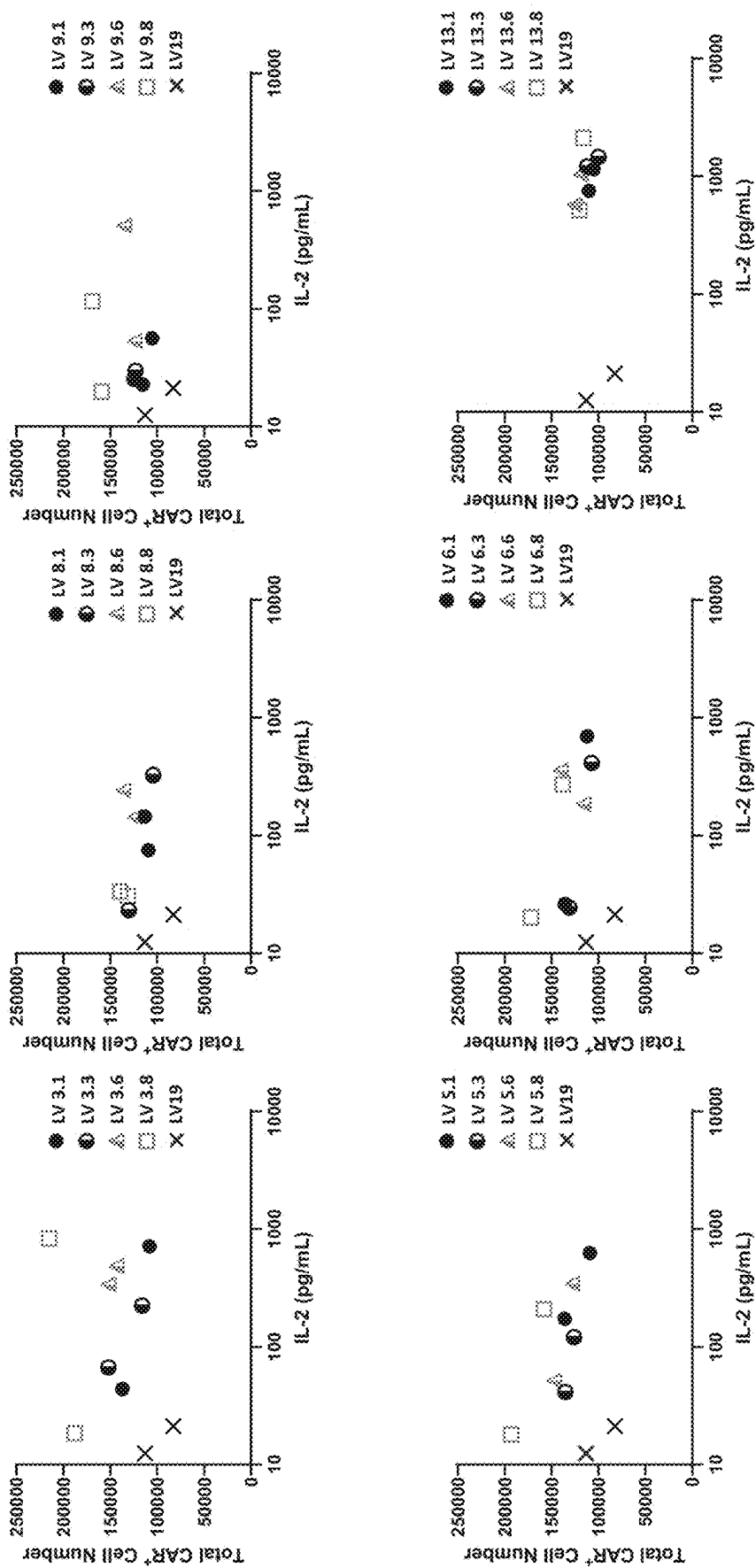


FIGURE 3E

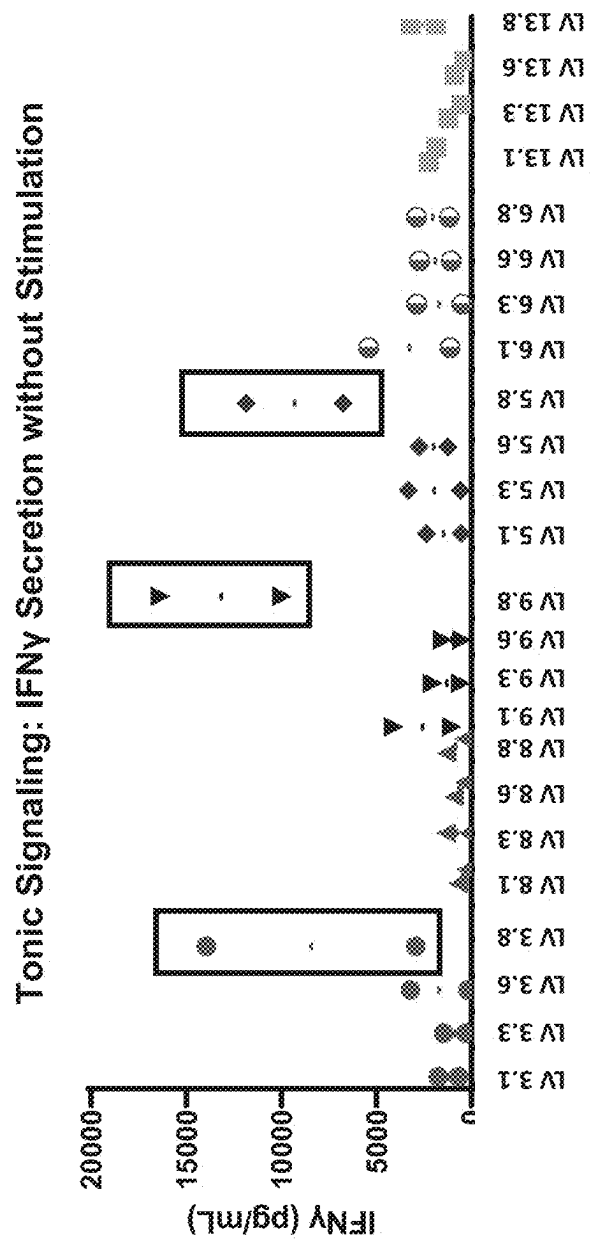


FIGURE 3F

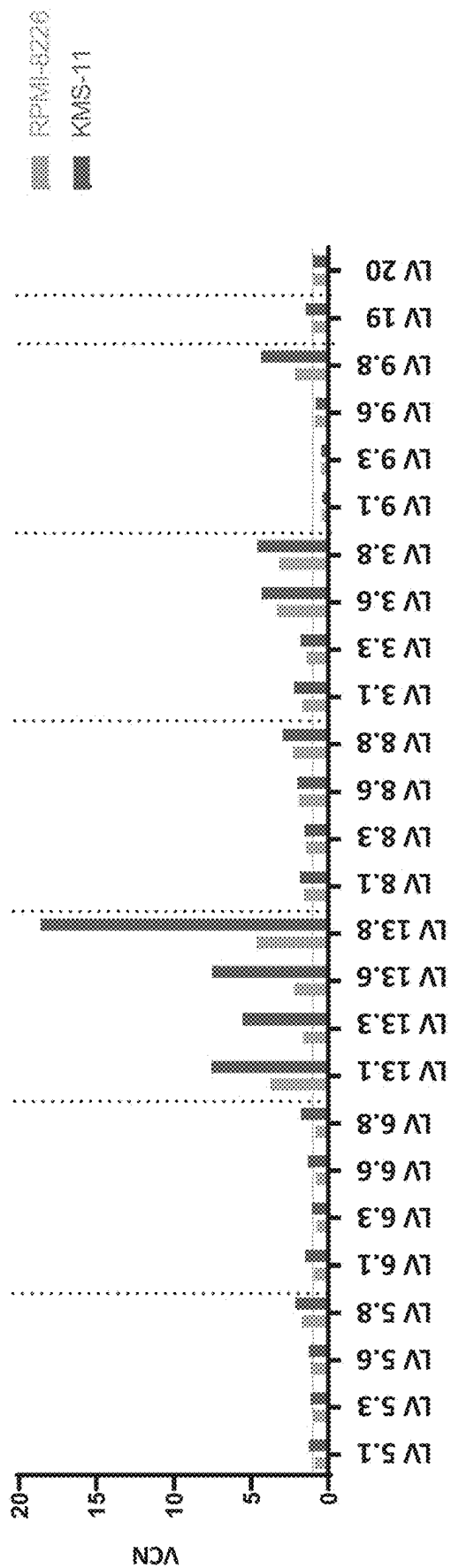
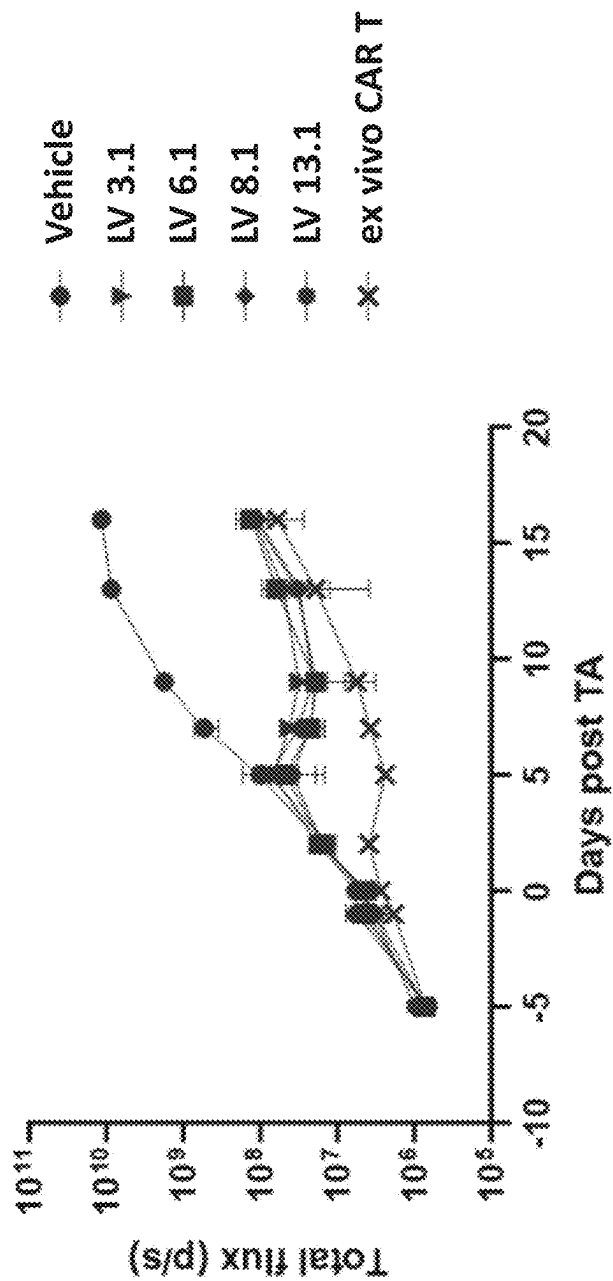


FIGURE 4A



ଅସମ୍ଭବ

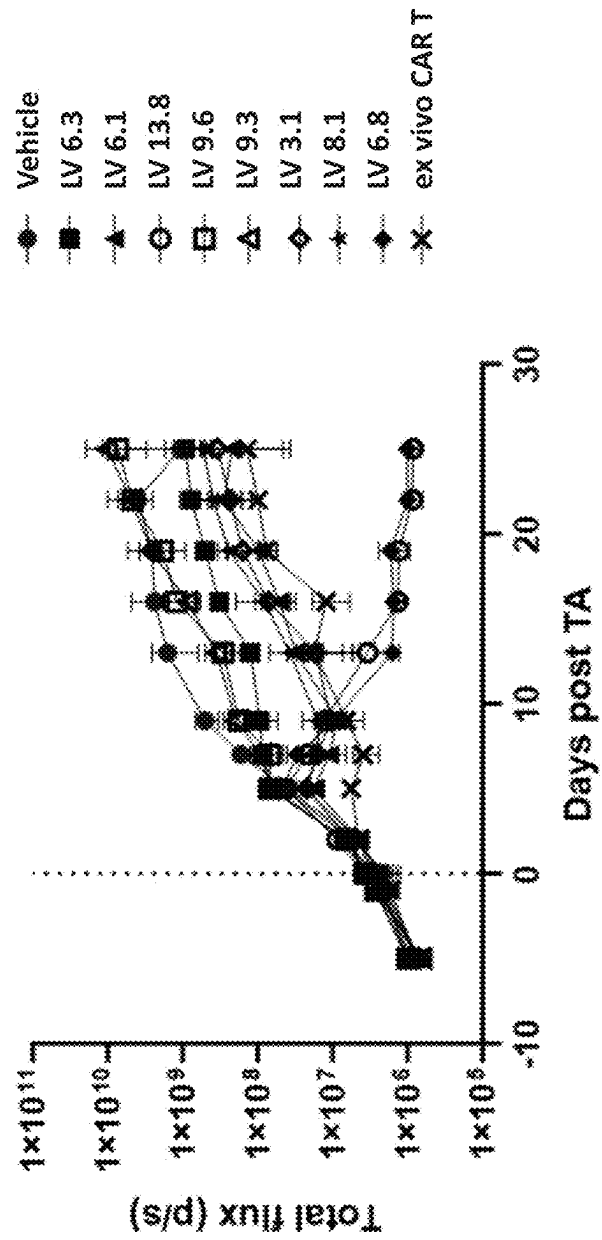


FIGURE 4C

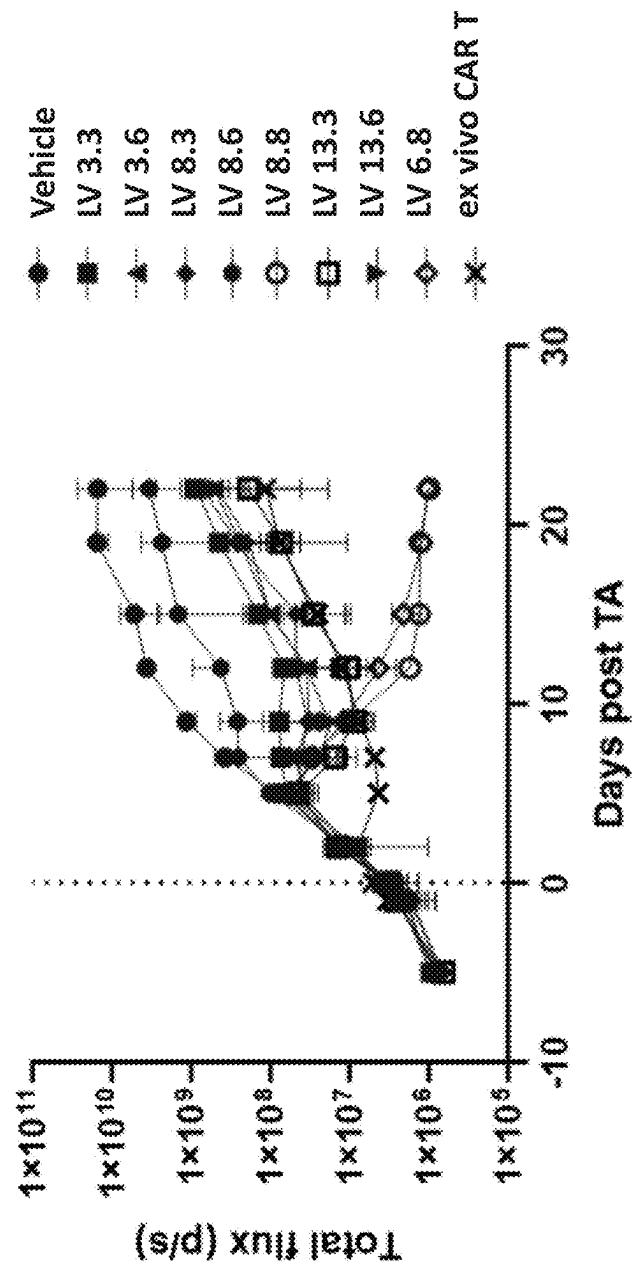


FIGURE 4D

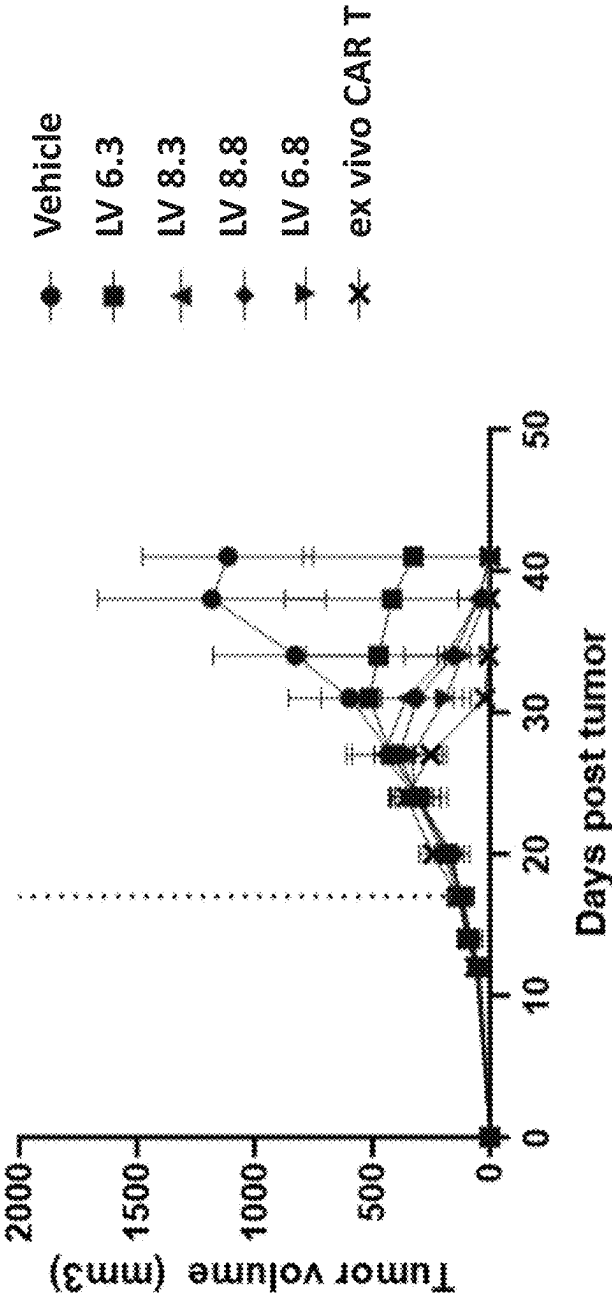


FIGURE 4E

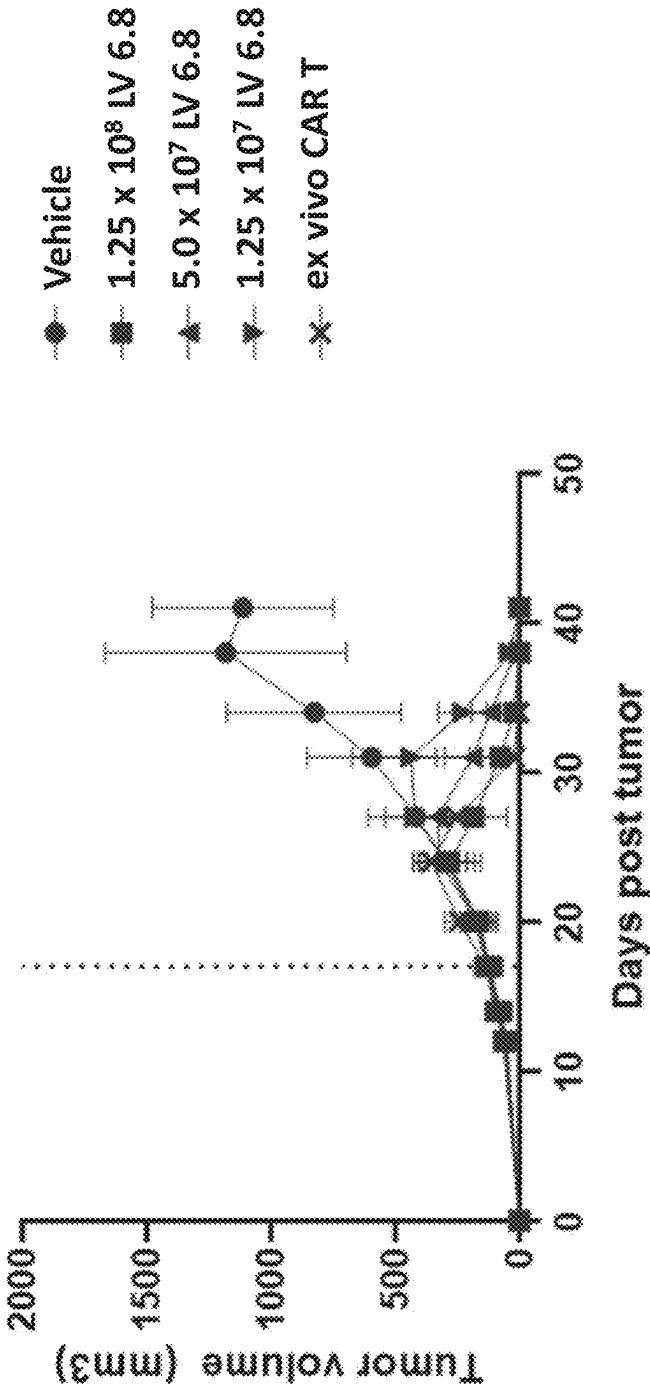


FIGURE 4F

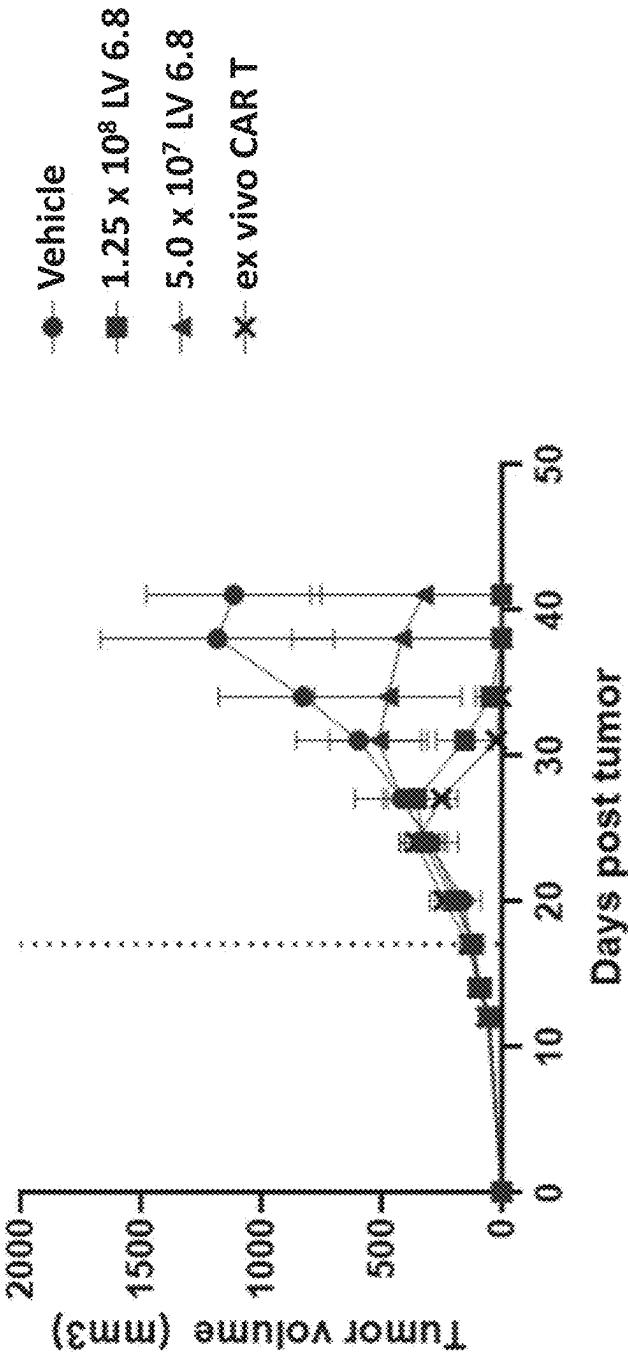


FIGURE 4G

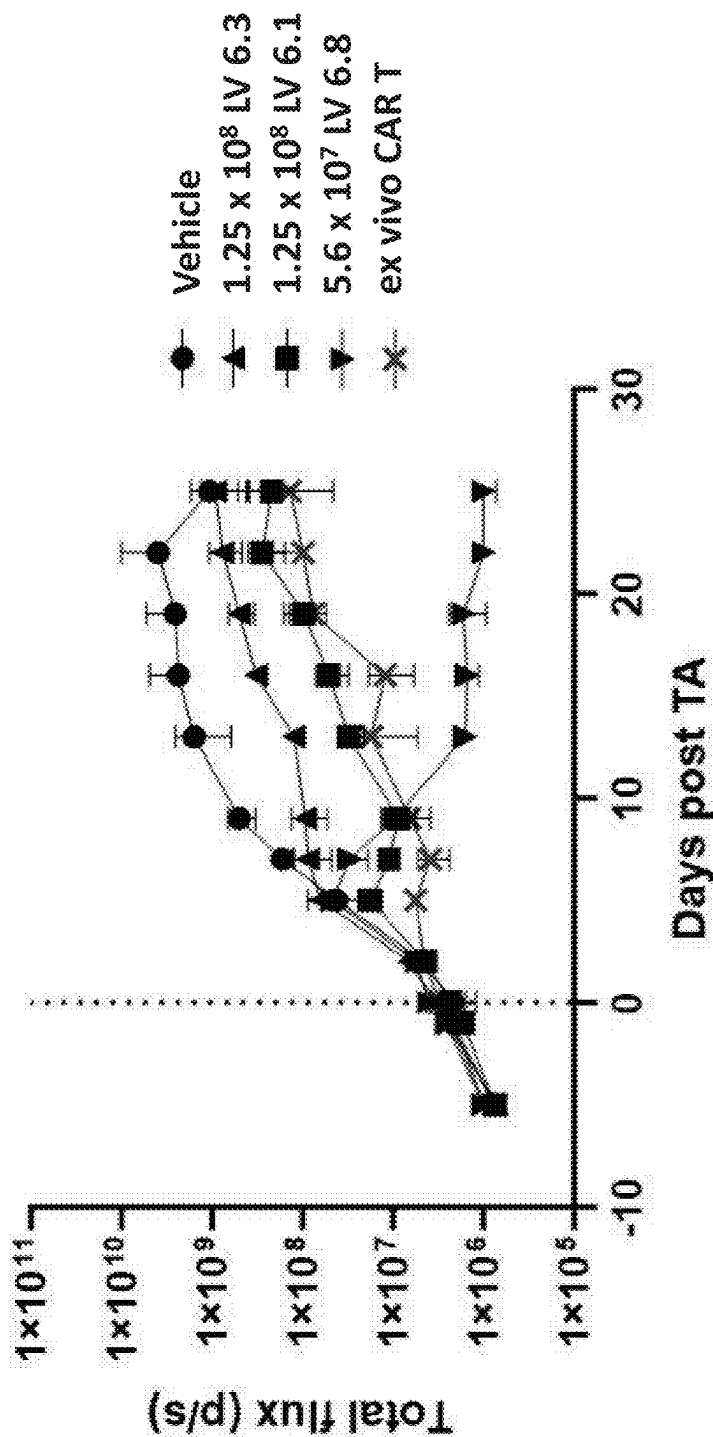


FIGURE 5A

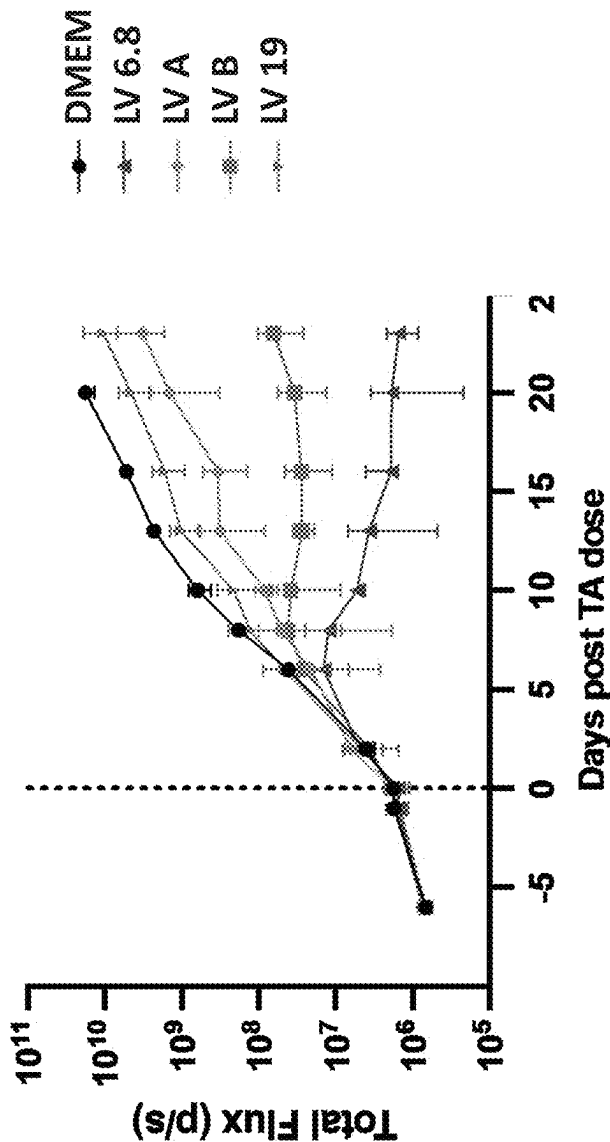
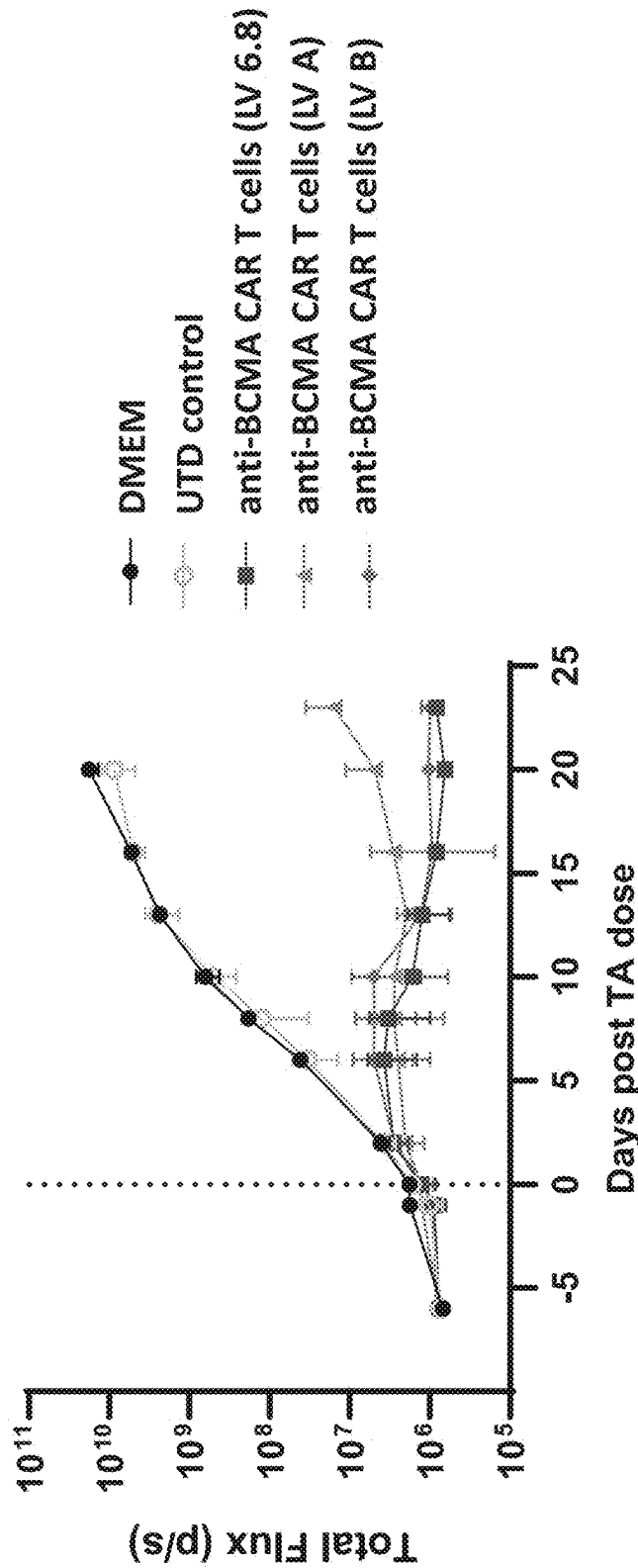


FIGURE 5B



COMPOSITIONS FOR TREATING CANCER**CROSS REFERENCE TO RELATED APPLICATIONS**

This application is a continuation application of International Application No. PCT/US2024/048301, filed Sep. 25, 2024, which claims the benefit of and priority to U.S. Provisional Application No. 63/540,336, filed Sep. 25, 2023, and U.S. Provisional Application No. 63/618,878, filed Jan. 8, 2024. The entire teachings of the applications are incorporated herein by reference.

STATEMENT REGARDING SEQUENCE LISTING

The Sequence Listing associated with this application is provided in XML format in lieu of a paper copy and is hereby incorporated by reference into the specification. The name of the XML file containing the Sequence Listing is KELO-012-102X_ST26.xml. The XML file is 385 KB, was created on Sep. 20, 2024, and is being submitted electronically via Patent Center, concurrent with the filing of the specification.

TECHNICAL FIELD

The present disclosure relates to recombinant particles engineered to deliver a chimeric antigen receptor to a cell. More particularly, the disclosure relates to recombinant particles engineered to deliver a chimeric antigen receptor to cells in vivo.

DESCRIPTION OF THE RELATED ART

B cell maturation antigen (BCMA) is a member of the tumor necrosis factor receptor superfamily and is also known as tumor necrosis factor receptor superfamily member 17 (TNFRSF17). BCMA is normally expressed in mature B lymphocytes and plasma cells. BCMA expression is also detected in various lymphomas and multiple myelomas. Multiple myeloma is an incurable plasma cell malignancy that originates in the bone marrow.

Multiple myeloma is the second most prevalent hematological malignancy after non-lymphoma. In 2020, an estimated 176,404 people world-wide were diagnosed with multiple myeloma and about 117,077 patients succumbed to the disease. In 2023, an estimated 35,730 people in the United States alone will be diagnosed with multiple myeloma and an estimated 12,590 multiple myeloma patients will pass from the disease or associated complications. The 5-year relative survival rate for multiple myeloma in the United States is only about 58%.

Multiple myeloma may initially be treated with an autologous stem cell transplantation (ASCT) and/or various drug combinations (e.g., proteasome inhibitors including bortezomib, carfilzomib, ixazomib; immunomodulatory drugs (IMiDs) including pomalidomide, lenalidomide, thalidomide; and corticosteroids like dexamethasone) but patients eventually relapse with the disease becoming refractory to treatment. Subsequent lines of treatment include monoclonal antibodies, bispecific antibodies, e.g., BiTEs, antibody-drug conjugates, and finally chimeric antigen receptor T cell therapy.

Ex vivo gene therapies are potentially one-time therapeutic modalities that generally involve harvesting cells from a subject, modifying the cells by culturing them with a gene

therapy vector, and delivering the modified cells back to the subject. Because ex vivo gene therapies are manufactured in a controlled environment, they do not generally require specialized targeting moieties and instead, targeting moieties with a broad tropism and that are highly efficient in delivering a gene therapy to most cell types are used.

In contrast, in vivo gene therapies are manufactured in the patient, in an uncontrolled environment. Accordingly, in vivo delivery of gene therapy vectors to specific cell types is orders of magnitude more complex than ex vivo delivery. In vivo gene therapy vectors encounter many non-target or off-target cells and may require a narrower or more specific tropism to deliver therapeutic payloads to a particular cell type. The potential of in vivo gene therapies has yet to be realized primarily due to inefficient delivery to desired cell types in combination with substantial off-target delivery. Use of specialized targeting moieties to deliver gene therapies in vivo has proven difficult in abrogating the off-target delivery to undesired cell types. In addition, on-target delivery of in vivo gene therapies using such specialized targeting moieties is often inefficient.

BRIEF SUMMARY

The present disclosure generally relates, in part, to a recombinant particle comprising a mutated vesiculovirus envelope glycoprotein, a tropism polypeptide that binds to an immune effector cell, and a lentiviral vector encoding or comprising a promoter operably linked to a polynucleotide encoding a chimeric antigen receptor that binds B cell maturation antigen (BCMA).

In various embodiments, the disclosure contemplates, in part, a recombinant lentiviral particle comprising: (a) a viral envelope comprising (i) a mutated coxal virus envelope glycoprotein (COCV-G) or a mutated vesicular stomatitis Indiana virus envelope glycoprotein (VSIV-G), wherein the mutated COCV-G or VSIV-G comprises amino acid substitutions at positions 47 and 354; and (ii) a non-viral membrane-bound tropism polypeptide comprising an anti-CD3ε scFv and a human CD8α hinge and transmembrane domain; and (b) a recombinant lentiviral vector comprising a polynucleotide encoding a myeloproliferative sarcoma virus enhancer, negative control region deleted, d1587rev primer-binding site substituted (MND) U3 promoter or an EF1α promoter operably linked to a polynucleotide encoding an anti-BCMA chimeric antigen receptor comprising an anti-BCMA scFv or anti-BCMA VHH, a CD8α hinge and transmembrane domain, a CD137 costimulatory domain, a CD3ζ primary signaling domain.

In particular embodiments, the mutated COCV-G or the mutated VSIV-G comprises amino acid substitutions selected from the group consisting of: K47A and R354A; K47A and R354Q; K47Q and R354A; and K47Q and R354Q.

In some embodiments, the mutated COCV-G or the mutated VSIV-G comprises the amino acid substitutions K47A and R354A.

In certain embodiments, the mutated COCV-G or the mutated VSIV-G comprises the amino acid substitutions K47A and R354Q.

In particular embodiments, the mutated COCV-G or the mutated VSIV-G comprises the amino acid substitutions K47Q and R354A.

In some embodiments, the mutated COCV-G or the mutated VSIV-G comprises the amino acid substitutions K47Q and R354Q.

In additional embodiments, the mutated COCV-G or the mutated VSIV-G comprises the amino acid sequence set forth in any one of SEQ ID NOs: 332, 333, 334, 335, 336, 337, 338, and 339.

In particular embodiments, the mutated VSIV-G comprises the amino acid sequence set forth in any one of SEQ ID NOs: 332, 333, 334, and 335.

In further embodiments, the mutated COCV-G comprises the amino acid sequence set forth in any one of SEQ ID NOs: 336, 337, 338, and 339.

In certain embodiments, the anti-CD3 ϵ scFv is isolated from an antibody selected from the group consisting of: OKT3, UCHT1, YTH12.5, TR66, and variants thereof.

In particular embodiments, the anti-CD3 ϵ scFv is isolated from OKT3.

In additional embodiments, the anti-CD3 ϵ scFv is isolated from UCHT1.

In some embodiments, the anti-CD3 ϵ scFv is isolated from YTH12.5.

In further embodiments, the anti-CD3 ϵ scFv is isolated from TR66.

In certain embodiments, the anti-CD3 ϵ scFv comprises an amino acid sequence set forth in any one of SEQ ID NOs: 153, 154, 163, 164, 173, 174, 183, 184, 193, 194, 203, 204, 213, 214, 223, and 224.

In particular embodiments, the non-viral membrane-bound tropism polypeptide comprises an amino acid sequence set forth in any one of SEQ ID NOs: 324, 325, 326, 327, 328, 329, 330, and 331.

In some embodiments, the MND U3 promoter comprises the polynucleotide sequence set forth in SEQ ID NO: 320.

In certain embodiments, the EF1 α promoter comprises the polynucleotide sequence set forth in SEQ ID NO: 319.

In particular embodiments, the anti-BCMA CAR comprises an anti-BCMA scFv comprising an amino acid sequence selected from the group consisting of: 19, 20, 29, 30, 39, 40, 49, 50, 59, 60, 69, 70, 79, 80, 89, 90, 99, and 100.

In additional embodiments, the anti-BCMA CAR comprises an anti-BCMA scFv comprising an amino acid sequence selected from the group consisting of: 20, 30, 39, 50, 59, 70, 80, 90, and 100.

In particular embodiments, the anti-BCMA CAR comprises an anti-BCMA scFv comprising an amino acid sequence selected from the group consisting of: 39, 59, 70, and 90.

In particular embodiments, the anti-BCMA CAR comprises an anti-BCMA VHH comprising an amino acid sequence selected from the group consisting of: 101, 105, 109, 113, 117, 121, 125, 129, 133, 137, and 141.

In some embodiments, the anti-BCMA CAR comprises an anti-BCMA VHH comprising an amino acid sequence selected from the group consisting of: 101 and 117.

In further embodiments, the anti-BCMA CAR comprises the amino acid sequence set forth in any one of SEQ ID NOs: 259, 263, 266, 270, 273 and 277, preferably SEQ ID NO: 266.

In certain embodiments, the polynucleotide encoding the anti-BCMA CAR comprises the polynucleotide sequence set forth in any one of SEQ ID NOs: 297, 299, 300, 302, 304, and 308.

In particular embodiments, the polynucleotide encoding the anti-BCMA CAR further comprises a polynucleotide sequence encoding a signal peptide.

In additional embodiments, the polynucleotide encoding the anti-BCMA CAR further comprises a polynucleotide sequence encoding a signal peptide isolated from a poly-

peptide selected from the group consisting of: CD8 α , mIgG κ , hIgG κ , CD33, tPA, SEAP, hGM-CSF, CSF2R, and B2M.

In certain embodiments, the polynucleotide encoding the anti-BCMA CAR further comprises a polynucleotide sequence encoding a signal peptide comprising an amino acid sequence set forth in any one of SEQ ID NOs: 245, 246, 247, 248, 249, 250, 251, 252, 253, and 254.

In particular embodiments, the polynucleotide encoding the signal peptide comprises the polynucleotide sequence set forth in SEQ ID NO: 294.

In further embodiments, the lentiviral vector further comprises a WPRE operably linked to the 3' end of the polynucleotide encoding the anti-BCMA CAR.

In some embodiments, the lentiviral vector further comprises a WPRE that comprises, consists essentially of, or consists of a polynucleotide sequence set forth in any one of SEQ ID NOs: 315, 316, and 317.

In various embodiments, the disclosure contemplates, in part, a recombinant lentiviral particle comprising: (a) a viral envelope comprising (i) a mutated viral envelope glycoprotein comprising an amino acid sequence set forth in any one of SEQ ID NOs: 332, 333, 334, and 335 and (ii) a non-viral membrane-bound tropism polypeptide comprising an amino acid sequence set forth in any one of SEQ ID NOs: 324, 325, 326, 327, 328, 329, 330, and 331; and (b) a recombinant lentiviral vector comprising a 5' long terminal repeat (LTR) comprising R and U5 regions; a Psi (Ψ) packaging signal, a cPPT/FLAP, a rev response element (RRE); a polynucleotide encoding an MND promoter or an EF1 α promoter operably linked to a polynucleotide encoding an anti-BCMA chimeric antigen receptor (CAR) comprising an amino acid sequence set forth in any one of SEQ ID NOs: 259, 263, 266, 270, 273, and 277 or an amino acid sequence at least 95%, at least 96%, at least 97%, at least 98%, at least 99% identical thereto; optionally a WPRE; a 3' LTR comprising U3 and R regions; a polyadenylation signal and a poly(A) tail.

In particular embodiments, the mutated viral envelope glycoprotein comprises an amino acid sequence set forth in SEQ ID NO: 332.

In some embodiments, the mutated viral envelope glycoprotein comprises an amino acid sequence set forth in SEQ ID NO: 333.

In certain embodiments, the mutated viral envelope glycoprotein comprises an amino acid sequence set forth in SEQ ID NO: 334.

In additional embodiments, the mutated viral envelope glycoprotein comprises an amino acid sequence set forth in SEQ ID NO: 335.

In various embodiments, the disclosure contemplates, in part, a recombinant lentiviral particle comprising: (a) a viral envelope comprising (i) a mutated viral envelope glycoprotein comprising an amino acid sequence set forth in any one of SEQ ID NOs: 336, 337, 338, and 339 and (ii) a non-viral membrane-bound tropism polypeptide comprising an amino acid sequence set forth in any one of SEQ ID NOs: 324, 325, 326, 327, 328, 329, 330, and 331; and (b) a recombinant lentiviral vector comprising a 5' long terminal repeat (LTR) comprising R and U5 regions; a Psi (Ψ) packaging signal, a cPPT/FLAP, a rev response element (RRE); a polynucleotide encoding an MND promoter or an EF1 α promoter operably linked to a polynucleotide encoding an anti-BCMA chimeric antigen receptor (CAR) comprising an amino acid sequence set forth in any one of SEQ ID NOs: 259, 263, 266, 270, 273, and 277 or an amino acid sequence at least 95%, at least 96%, at least 97%, at least 98%, at least 99%

5

identical thereto; optionally a WPRE; a 3' LTR comprising U3 and R regions; a polyadenylation signal and a poly(A) tail.

In some embodiments, the mutated viral envelope glycoprotein comprises an amino acid sequence set forth in SEQ ID NO: 336.

In further embodiments, the mutated viral envelope glycoprotein comprises an amino acid sequence set forth in SEQ ID NO: 337.

In certain embodiments, the mutated viral envelope glycoprotein comprises an amino acid sequence set forth in SEQ ID NO: 338.

In some embodiments, the mutated viral envelope glycoprotein comprises an amino acid sequence set forth in SEQ ID NO: 339.

In particular embodiments, the non-viral membrane-bound tropism polypeptide comprises an amino acid sequence set forth in SEQ ID NO: 324.

In additional embodiments, the non-viral membrane-bound tropism polypeptide comprises an amino acid sequence set forth in SEQ ID NO: 325.

In certain embodiments, the non-viral membrane-bound tropism polypeptide comprises an amino acid sequence set forth in SEQ ID NO: 326.

In particular embodiments, the non-viral membrane-bound tropism polypeptide comprises an amino acid sequence set forth in SEQ ID NO: 327.

In further embodiments, the non-viral membrane-bound tropism polypeptide comprises an amino acid sequence set forth in SEQ ID NO: 328.

In some embodiments, the non-viral membrane-bound tropism polypeptide comprises an amino acid sequence set forth in SEQ ID NO: 329.

In additional embodiments, the non-viral membrane-bound tropism polypeptide comprises an amino acid sequence set forth in SEQ ID NO: 330.

In particular embodiments, the non-viral membrane-bound tropism polypeptide comprises an amino acid sequence set forth in SEQ ID NO: 331.

In particular embodiments, the recombinant lentiviral vector is derived from HIV-1 or HIV-2.

In additional embodiments, the MND U3 promoter comprises the polynucleotide sequence set forth in SEQ ID NO: 320.

In some embodiments, the EF1 α promoter comprises the polynucleotide sequence set forth in SEQ ID NO: 319.

In particular embodiments, the polynucleotide encoding the anti-BCMA CAR comprises the polynucleotide sequence set forth in any one of SEQ ID NOs: 297, 299, 300, 302, 304, and 308.

In some embodiments, the polynucleotide encoding the anti-BCMA CAR further comprises a polynucleotide sequence encoding a signal peptide.

In further embodiments, the polynucleotide encoding the anti-BCMA CAR further comprises a polynucleotide sequence encoding a signal peptide isolated from a polypeptide selected from the group consisting of: CD8 α , mIgGk, hIgGk, CD33, tPA, SEAP, hGM-CSF, CSF2R, and B2M.

In particular embodiments, the polynucleotide encoding the anti-BCMA CAR further comprises a polynucleotide sequence encoding a signal peptide comprising an amino acid sequence set forth in any one of SEQ ID NOs: 245, 246, 247, 248, 249, 250, 251, 252, 253, and 254.

In certain embodiments, the lentiviral vector further comprises a WPRE operably linked to the 3' end of the polynucleotide encoding the anti-BCMA CAR.

6

In some embodiments, the lentiviral vector further comprises a WPRE operably linked to the 3' end of the polynucleotide encoding the anti-BCMA CAR, wherein the WPRE comprises, consists essentially of, or consists of the polynucleotide sequence set forth in any one of SEQ ID NOs: 315, 316, and 317.

In various embodiments, the disclosure contemplates, in part, a cell transduced with a particle contemplated herein.

In further embodiments, the cell is an immune effector cell.

In additional embodiments, the cell is a T cell or a natural killer T (NKT) cell.

In various embodiments, the disclosure contemplates, in part, a composition comprising a particle or a cell contemplated herein.

In various embodiments, the disclosure contemplates, in part, a pharmaceutical composition comprising a pharmaceutically acceptable carrier and a particle, a cell, or a composition contemplated herein.

In various embodiments, the disclosure contemplates, in part, a method of treating, preventing, or ameliorating at least one symptom of a disease, disorder or condition associated therewith in a subject, comprising administering to the subject an effective amount of a particle, a cell, a composition, or a pharmaceutical composition contemplated herein.

In some embodiments, the disease, disorder, or condition is a cancer.

In certain embodiments, the cancer is a multiple myeloma (MM).

In particular embodiments, the cancer is MM selected from the group consisting of: active multiple myeloma, smoldering multiple myeloma, light chain myeloma, non-secretory myeloma, IgD myeloma, IgE myeloma, osteosclerotic myeloma, solitary plasmacytoma of bone, and extramedullary plasmacytoma.

In further embodiments, the cancer is relapsed and/or refractory.

In various embodiments, the disclosure contemplates, in part, a method of treating a subject that has, or has been diagnosed with, a multiple myeloma, comprising administering the subject an effective amount of a particle, a cell, a composition, or a pharmaceutical composition contemplated herein.

In some embodiments, the administration is parenteral administration.

In particular embodiments, the administration is intravenous.

In various embodiments, the disclosure contemplates, in part, a method of transducing an immune effector cell in vivo, comprising administering to a subject a pharmaceutical composition comprising a pharmaceutically acceptable carrier and an effective amount of a particle, a cell, a composition, or a pharmaceutical composition contemplated herein.

In various embodiments, the disclosure contemplates, in part, a method of making a recombinant lentivirus comprising (a) transfecting a host cell with four polynucleotides: a first polynucleotide that encodes lentiviral gag-pol, a second polynucleotide that encodes lentiviral rev, a third polynucleotide that encodes a mutated viral envelope glycoprotein and a non-viral membrane-bound tropism polypeptide contemplated herein, and a fourth polynucleotide that is a transfer plasmid encoding the recombinant lentiviral vector contemplated herein; and b) culturing the transduced cell for about 1 to 3 days to produce the recombinant lentivirus.

In various embodiments, the disclosure contemplates, in part, a kit comprising the particle contemplated herein, a pharmaceutically acceptable carrier, and instructions for use.

BRIEF DESCRIPTION OF SEVERAL VIEWS OF THE DRAWINGS

FIG. 1 is a cartoon of a recombinant lentiviral particle comprising a viral envelope expressing a mutated viral envelope glycoprotein and non-viral membrane-bound tropism polypeptide and two copies of a lentiviral vector encoding a promoter operably linked to a polynucleotide encoding an anti-BCMA CAR and an optional post-transcriptional response element (PRE) operably linked to the 3' prime end of the polynucleotide encoding the anti-BCMA CAR.

FIG. 2A shows Jurkat cell titer of recombinant lentiviruses comprising a viral envelope expressing a mutated vesicular stomatitis Indiana virus envelope glycoprotein G (VSIV-G) and a non-viral membrane-bound tropism polypeptide that binds CD3 and a lentiviral vector encoding an MNDU3 promoter operably linked to a polynucleotide encoding a CD8 α signal peptide and an anti-BCMA CAR and a WPRE operably linked to the 3' prime end of the polynucleotide encoding the anti-BCMA CAR (18 anti-BCMA CARs were evaluated).

FIG. 2B shows anti-BCMA CAR expression on PBMCs transduced with recombinant lentiviral particles comprising a viral envelope expressing a mutated VSIV-G and a non-viral membrane-bound tropism polypeptide that binds CD3 and a lentiviral vector encoding an MNDU3 promoter operably linked to a polynucleotide encoding a CD8 α signal peptide and an anti-BCMA CAR and a WPRE operably linked to the 3' prime end of the polynucleotide encoding the anti-BCMA CAR (18 anti-BCMA CARs were evaluated).

FIG. 2C shows the vector copy number (VCN) in PBMCs transduced with recombinant lentiviral particles comprising a viral envelope expressing a mutated VSIV-G and a non-viral membrane-bound tropism polypeptide that binds CD3 and a lentiviral vector encoding an MNDU3 promoter operably linked to a polynucleotide encoding a CD8 α signal peptide and an anti-BCMA CAR and a WPRE operably linked to the 3' prime end of the polynucleotide encoding the anti-BCMA CAR (18 anti-BCMA CARs were evaluated).

FIG. 2D shows the anti-BCMA CAR activity measured as the amount of IFN γ produced in a co-culture assay. PBMCs transduced with recombinant lentiviral particles comprising a viral envelope expressing a mutated VSIV-G and a non-viral membrane-bound tropism polypeptide that binds CD3 and a lentiviral vector encoding an MNDU3 promoter operably linked to a polynucleotide encoding a CD8 α signal peptide and an anti-BCMA CAR and a WPRE operably linked to the 3' prime end of the polynucleotide encoding the anti-BCMA CAR (18 anti-BCMA CARs were evaluated) were co-cultured with Daudi cells (low BCMA expression) or RPMI-8226 cells (high BCMA expression) for 24 hours. After 24 hours, IFN γ levels in co-culture supernatant were measured and plotted as a function of % anti-BCMA CAR positive cells in the co-culture.

FIG. 3A shows Jurkat cell functional titer of recombinant lentivirus comprising a viral envelope expressing a mutated VSIV-G and a non-viral membrane-bound tropism polypeptide that binds CD3 and a lentiviral vector encoding either an MNDU3 promoter, an SFFV promoter, or an EF1 α promoter operably linked to a polynucleotide encoding a CD8 α signal peptide and an anti-BCMA CAR (6 anti-BCMA CARs were evaluated) and either no PRE or a wild-type or mutated

WPRE operably linked to the 3' prime end of the polynucleotide encoding the anti-BCMA CAR.

FIG. 3B shows the VCN in transduced PBMCs as a function of the percentage of PBMCs expressing an anti-BCMA CAR. Human PBMCs were transduced with a recombinant lentivirus comprising a viral envelope expressing a mutated VSIV-G and a non-viral membrane-bound tropism polypeptide that binds CD3 and a lentiviral vector encoding either an MNDU3 promoter, an SFFV promoter, or an EF1 α promoter operably linked to a polynucleotide encoding a CD8 α signal peptide and an anti-BCMA CAR (6 anti-BCMA CARs were evaluated) and either no PRE or a wild-type or mutated WPRE operably linked to the 3' prime end of the polynucleotide encoding the anti-BCMA CAR.

FIG. 3C shows the amount of IFN γ secreted from PBMCs expressing an anti-BCMA CAR co-cultured with RPMI-8226 cells (BCMA expressing cells) for 24 hours as a function of the CAR-expressing cells in the co-culture.

FIG. 3D shows the amount of IL-2 secreted from PBMCs expressing an anti-BCMA CAR co-cultured with RPMI-8226 cells (BCMA expressing cells) for 24 hours as a function of the CAR-expressing cells in the co-culture.

FIG. 3E shows the amount of IFN γ secreted from PBMCs expressing an anti-BCMA CAR in the absence of target cells. Human PBMCs were transduced with a recombinant lentivirus comprising a viral envelope expressing a mutated VSIV-G and a non-viral membrane-bound tropism polypeptide that binds CD3 and a lentiviral vector comprising one of the following lentiviral vector architectures, MNDU3 promoter and wild-type WPRE, MNDU3 promoter and a mutated WPRE, SFFV promoter and a mutated WPRE, and EF1 α promoter and no WPRE and encoding an anti-BCMA CAR (6 anti-BCMA CARs were evaluated).

FIG. 3F shows the levels of off-target transduction in BCMA expressing cells (RPMI-8226 and KMS-11) of recombinant lentiviruses comprising a viral envelope expressing a mutated VSIV-G and a non-viral membrane-bound tropism polypeptide that binds CD3 and a lentiviral vector comprising one of the following lentiviral vector architectures, MNDU3 promoter and wild-type WPRE, MNDU3 promoter and a mutated WPRE, SFFV promoter and a mutated WPRE, and EF1 α promoter and no WPRE and encoding an anti-BCMA CAR (6 anti-BCMA CARs were evaluated). Transduction was normalized to VCN in cells transduced with a recombinant lentivirus encoding GFP in place of an anti-BCMA CAR.

FIG. 4A shows the results from an in vivo Daudi mouse model. Mice were administered a recombinant lentivirus comprising a viral envelope expressing a mutated VSIV-G and a non-viral membrane-bound tropism polypeptide that binds CD3 and a lentiviral vector encoding either an MNDU3 promoter operably linked to a polynucleotide encoding a CD8 α signal peptide and an anti-BCMA CAR (4 anti-BCMA CARs were evaluated) and a wild-type WPRE operably linked to the 3' prime end of the polynucleotide encoding the anti-BCMA CAR; ex vivo CAR T cells, or vehicle.

FIG. 4B shows the results from an in vivo Daudi mouse model. Mice were administered a recombinant lentivirus comprising a viral envelope expressing a mutated VSIV-G and a non-viral membrane-bound tropism polypeptide that binds CD3 and a lentiviral vector comprising one of the following lentiviral vector architectures, MNDU3 promoter and wild-type WPRE, MNDU3 promoter and a mutated WPRE, SFFV promoter and a mutated WPRE, and EF1 α

promoter and no WPRE and encoding an anti-BCMA CAR (5 anti-BCMA CARs were evaluated); ex vivo CAR T cells, or vehicle.

FIG. 4C shows the results from an in vivo Daudi mouse model. Mice were administered a recombinant lentivirus comprising a viral envelope expressing a mutated VSIV-G and a non-viral membrane-bound tropism polypeptide that binds CD3 and a lentiviral vector comprising one of the following lentiviral vector architectures, MNDU3 promoter and a mutated WPRE, SFFV promoter and a mutated WPRE, and EF1 α promoter and no WPRE and encoding an anti-BCMA CAR (4 anti-BCMA CARs were evaluated); ex vivo CAR T cells, or vehicle.

FIG. 4D shows the results from an in vivo RPMI-8226 mouse model. Mice were administered a recombinant lentivirus comprising a viral envelope expressing a mutated VSIV-G and a non-viral membrane-bound tropism polypeptide that binds CD3 and a lentiviral vector comprising one of the following lentiviral vector architectures, MNDU3 promoter and a mutated WPRE, or EF1 α promoter and no WPRE and encoding an anti-BCMA CAR (2 anti-BCMA CARs were evaluated); ex vivo CAR T cells, or vehicle.

FIG. 4E shows the results from an in vivo RPMI-8226 mouse model. Mice were administered three doses (1.25×10^7 IU, 5.0×10^7 IU, or 1.25×10^8 IU) of a recombinant lentivirus comprising a viral envelope expressing a mutated VSIV-G and a non-viral membrane-bound tropism polypeptide that binds CD3 and a lentiviral vector comprising an EF1 α promoter operably linked to a polynucleotide encoding an anti-BCMA CAR without a PRE; ex vivo CAR T cells, or vehicle.

FIG. 4F shows the results from an in vivo RPMI-8226 mouse model. Mice were administered two doses (5.0×10^7 IU or 1.25×10^8 IU) of a recombinant lentivirus comprising a viral envelope expressing a mutated VSIV-G and a non-viral membrane-bound tropism polypeptide that binds CD3 and a lentiviral vector comprising an EF1 α promoter operably linked to a polynucleotide encoding an anti-BCMA CAR without a PRE; ex vivo CAR T cells, or vehicle.

FIG. 4G shows the results from an in vivo Daudi mouse model. Mice were administered a recombinant lentivirus comprising a viral envelope expressing a mutated VSIV-G and a non-viral membrane-bound tropism polypeptide that binds CD3 and a lentiviral vector and encoding an anti-BCMA CAR and comprising one of the following lentiviral vector architectures, MNDU3 promoter and a mutated WPRE or MNDU3 promoter and a mutated WPRE (at 1.25×10^8 IU) or an EF1 α promoter and no WPRE (5.6×10^7 IU); ex vivo CAR T cells, or vehicle.

FIG. 5A shows the results from a Daudi mouse model interrogated with in vivo lentivirus. Mice were administered vehicle control or recombinant lentivirus comprising a viral envelope expressing a mutated VSIV-G and a non-viral membrane-bound tropism polypeptide that binds CD3 and a lentiviral vector encoding one of three anti-BCMA CARs or a GFP control.

FIG. 5B shows the results from a Daudi mouse model interrogated with ex vivo manufactured CAR T cells. Mice were administered vehicle control, untransduced PBMCS, or PBMCs transduced with a recombinant lentivirus comprising a viral envelope expressing a mutated VSIV-G and a non-viral membrane-bound tropism polypeptide that binds CD3 and a lentiviral vector and encoding one of three anti-BCMA CARs.

BRIEF DESCRIPTION OF THE SEQUENCE IDENTIFIERS

SEQ ID NOs: 1-9 set forth amino acid sequences of fusogens.

SEQ ID NO: 10 sets forth an amino acid sequence of BCMA.

SEQ ID NOs: 11-224 set forth amino acid sequences of antibodies.

SEQ ID NOs: 225-230 set forth amino acid sequences of spacer/hinge domains.

SEQ ID NOs: 231-237 set forth amino acid sequences of transmembrane domains.

SEQ ID NOs: 238-244 set forth amino acid sequences of intracellular domains.

SEQ ID NOs: 245-254 set forth amino acid sequences of signal peptides.

SEQ ID NOs: 255-283 set forth amino acid sequences of chimeric antigen receptors.

SEQ ID NOs: 284-287 set forth nucleic acid sequences encoding spacer/hinge domains.

SEQ ID NOs: 288-289 set forth nucleic acid sequences encoding transmembrane domains.

SEQ ID NOs: 290-293 set forth nucleic acid sequences encoding intracellular signaling domains.

SEQ ID NO: 294 sets forth the nucleic acid sequence encoding a signal peptide.

SEQ ID NOs: 295-314 set forth nucleic acid sequences encoding chimeric antigen receptors without a signal peptide.

SEQ ID NOs: 315-317 set forth nucleic acid sequences of post-transcriptional response elements.

SEQ ID NOs: 318-323 set forth nucleic acid sequences of promoters.

SEQ ID NOs: 324-331 set forth amino acid sequences of non-viral membrane bound tropism polypeptides.

SEQ ID NOs: 332-339 set forth amino acid sequences of fusogens.

SEQ ID NOs: 340-341 set forth amino acid sequences of anti-BCMA CARs.

SEQ ID NOs: 342-354 set forth amino acid sequences of polypeptide linkers.

SEQ ID NOs: 355-374 set forth amino acid sequences of viral self-cleaving peptides.

In the foregoing sequences, X, if present, refers to any amino acid, a specified group of amino acids or the absence of an amino acid.

Throughout the disclosure, the amino acid positions of a fusogen are with reference to the fusogen lacking a signal sequence (i.e., the amino acid sequence after the signal peptide has been cleaved).

DETAILED DESCRIPTION

A. Overview

The field of ex vivo gene therapy is not new and has been evolving for decades. Despite huge potential, ex vivo gene therapy has been met with limited success. Moreover, substantial obstacles still plague the field of ex vivo gene therapy, obstacles including limited precision and lack of commercial viability are likely among the reasons that it has yet to see widespread adoption in a clinical setting.

Recently, ex vivo CAR T cell therapies that target B cell maturation antigen (BCMA) have been used to treat relapsed and refractory multiple myeloma. Although many multiple myeloma patients that have been treated with ex vivo anti-BCMA CAR T cell therapies experience partial or complete remissions, most relapse and succumb to the disease. There is a significant unmet need for a durable, one-time, and potentially curative treatment for multiple myeloma.

Recombinant lentiviral particles that enable delivery of vectors encoding chimeric antigen receptors to immune effector cells in vivo are new and offer the potential to deliver life-altering therapies on an unprecedented scale. In vivo CAR T cell therapy solves the commercial viability issues associated with the astronomical costs associated with ex vivo CAR T cell manufacturing. But in vivo CAR T cell therapy also comes with its own set of challenges related to the delivery mechanism, including potential off-target toxicity, low efficacy, and immunogenicity.

The present disclosure offers solutions to foregoing challenges and others that exist in the field of using recombinant lentiviral particles to efficiently and safely provide in vivo CAR T cell therapy.

The present disclosure generally relates to, in part, to an engineered cell-targeting particle (e.g., a fusosome; an extracellular vesicle, including a microvesicle, an apoptotic body, and an exosome; a lipid nanoparticle; a virus-like particle (VLPs); or a viral particle) that has a surface that expresses a non-viral tropism polypeptide engineered to bind an immune effector cell and a mutated viral glycoprotein that promotes fusion of the particle and the immune effector cell; and one or more copies of a vector that encodes or comprises a promoter operably linked to a polynucleotide encoding a chimeric antigen receptor that binds B cell maturation antigen (BCMA).

The disclosure contemplates, in part, recombinant lentiviral particles engineered to bind and transduce immune effector cells with a vector encoding an anti-BCMA CAR, in vivo. In various embodiments, a recombinant lentiviral particle comprises an envelope engineered to express a non-viral tropism polypeptide that bind an immune effector cell and a mutated vesiculovirus glycoprotein that does not bind its cognate receptor, e.g., low density lipoprotein receptor (LDLR), but that promotes fusion of the particle and the immune effector cell; and one or more copies of a lentiviral vector that encodes or comprises a promoter operably linked to a polynucleotide encoding an anti-BCMA CAR. The recombinant lentiviral particles may be used for ex vivo CAR T cell therapy but provide substantial advantages for use in in vivo CAR T cell therapy.

The disclosure further contemplates, in part, methods of making the recombinant lentiviral particles contemplated herein, along with methods of using the particles for treating a subject in need thereof.

In particular embodiments, the disclosure contemplates, methods of using a recombinant lentiviral particle contemplated herein to generate anti-BCMA immune effector cells, e.g., T cells, in vivo, to treat a disorder, disease, condition or symptoms associated therewith, preferably to treat cancer, and more preferably, to treat a multiple myeloma, e.g., relapsed refractory multiple myeloma.

Compositions, pharmaceutical compositions, and kits comprising one or more recombinant lentiviral particles contemplated herein and methods of making and using the same are also provided in particular embodiments.

Techniques for recombinant (i.e., engineered) DNA, peptide and oligonucleotide synthesis, immunoassays, tissue culture, transformation (e.g., electroporation, lipofection), enzymatic reactions, purification and related techniques and procedures may be generally performed as described in various general and more specific references in microbiology, molecular biology, biochemistry, molecular genetics, cell biology, virology and immunology as cited and discussed throughout the present specification. See, e.g., Sambrook et al., *Molecular Cloning: A Laboratory Manual*, 4th ed., Cold Spring Harbor Laboratory Press, Cold Spring

Harbor, N.Y.; *Current Protocols in Molecular Biology* (John Wiley and Sons, updated July 2008); *Short Protocols in Molecular Biology: A Compendium of Methods from Current Protocols in Molecular Biology*, Greene Pub. Associates and Wiley-Interscience (2002); Glover, *DNA Cloning: A Practical Approach*, vol. I & II (IRL Press, Oxford Univ. Press USA, 1985); *Current Protocols in Immunology* (Edited by: John E. Coligan, Ada M. Kruisbeek, David H. Margulies, Ethan M. Shevach, Warren Strober 2001 John Wiley & Sons, NY, NY); *Real-Time PCR: Current Technology and Applications*, Edited by Julie Logan, Kirstin Edwards and Nick Saunders, 2009, Caister Academic Press, Norfolk, UK; Anand, *Techniques for the Analysis of Complex Genomes*, (Academic Press, New York, 1992); Guthrie and Fink, *Guide to Yeast Genetics and Molecular Biology* (Academic Press, New York, 1991); *Oligonucleotide Synthesis* (N. Gait, Ed., 1984); *Nucleic Acid Hybridization* (B. Hames & S. Higgins, Eds., 1985); *Transcription and Translation* (B. Hames & S. Higgins, Eds., 1984); *Animal Cell Culture* (R. Freshney, Ed., 1986); Perbal, *A Practical Guide to Molecular Cloning* (1984); *Next-Generation Genome Sequencing* (Janitz, 2008 Wiley-VCH); *PCR Protocols* (Methods in Molecular Biology) (Park, Ed., 3rd Edition, 2010 Humana Press); *Immobilized Cells and Enzymes* (IRL Press, 1986); the treatise, *Methods in Enzymology* (Academic Press, Inc., N.Y.); *Gene Transfer Vectors for Mammalian Cells* (J. H. Miller and M. P. Calos eds., 1987, Cold Spring Harbor Laboratory); Harlow and Lane, *Antibodies*, (Cold Spring Harbor Laboratory Press, Cold Spring Harbor, N.Y., 1998); *Immunochemical Methods in Cell and Molecular Biology* (Mayer and Walker, eds., Academic Press, London, 1987); *Handbook of Experimental Immunology*, Volumes I-IV (D. M. Weir and CC Blackwell, eds., 1986); Roitt, *Essential Immunology*, 6th Edition, (Blackwell Scientific Publications, Oxford, 1988); *Current Protocols in Immunology* (Q. E. Coligan, A. M. Kruisbeek, D. H. Margulies, E. M. Shevach and W. Strober, eds., 1991); *Annual Review of Immunology*; as well as monographs in journals such as *Advances in Immunology*.

B. Definitions

Prior to setting forth this disclosure in more detail, it may be helpful to an understanding thereof to provide definitions of certain terms to be used herein.

Unless defined otherwise, all technical and scientific terms used herein have the same meaning as commonly understood by those of ordinary skill in the art to which the invention belongs. Although any methods and materials similar or equivalent to those described herein can be used in the practice or testing of particular embodiments, preferred embodiments of compositions, methods and materials are described herein. For the purposes of the present disclosure, the following terms are defined below.

The articles “a,” “an,” and “the” are used herein to refer to one or to more than one (i.e., to at least one, or to one or more) of the grammatical object of the article. By way of example, “an element” means one element or one or more elements.

The use of the alternative (e.g., “or”) should be understood to mean either one, both, or any combination of the recited alternatives.

The term “and/or” should be understood to mean either one of, or both of, the alternatives.

As used herein, the term “about” or “approximately” refers to a quantity, level, value, number, frequency, percentage, dimension, size, amount, weight or length that

varies by as much as 15%, 10%, 9%, 8%, 7%, 6%, 5%, 4%, 3%, 2% or 1% to a reference quantity, level, value, number, frequency, percentage, dimension, size, amount, weight or length. In one embodiment, the term "about" or "approximately" refers a range of quantity, level, value, number, frequency, percentage, dimension, size, amount, weight or length $\pm 15\%$, $\pm 10\%$, $\pm 9\%$, $\pm 8\%$, $\pm 7\%$, $\pm 6\%$, $\pm 5\%$, $\pm 4\%$, $\pm 3\%$, $\pm 2\%$, or $\pm 1\%$ of a reference quantity, level, value, number, frequency, percentage, dimension, size, amount, weight or length.

In one embodiment, a range, e.g., 1 to 5, about 1 to 5, or about 1 to about 5, refers to each numerical value encompassed by the range. For example, in one non-limiting and merely illustrative embodiment, the range "1 to 5" is equivalent to the expression 1, 2, 3, 4, 5; or 1.0, 1.5, 2.0, 2.5, 3.0, 3.5, 4.0, 4.5, or 5.0; or 1.0, 1.1, 1.2, 1.3, 1.4, 1.5, 1.6, 1.7, 1.8, 1.9, 2.0, 2.1, 2.2, 2.3, 2.4, 2.5, 2.6, 2.7, 2.8, 2.9, 3.0, 3.1, 3.2, 3.3, 3.4, 3.5, 3.6, 3.7, 3.8, 3.9, 4.0, 4.1, 4.2, 4.3, 4.4, 4.5, 4.6, 4.7, 4.8, 4.9, or 5.0.

As used herein, the term "substantially" refers to a quantity, level, value, number, frequency, percentage, dimension, size, amount, weight or length that is 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99% or higher compared to a reference quantity, level, value, number, frequency, percentage, dimension, size, amount, weight or length. In one embodiment, "substantially the same" refers to a quantity, level, value, number, frequency, percentage, dimension, size, amount, weight or length that produces an effect, e.g., a physiological effect, that is approximately the same as a reference quantity, level, value, number, frequency, percentage, dimension, size, amount, weight or length. In particular embodiments, "substantially lacks cell binding or cell attachment activity and/or cognate receptor binding activity" or "substantially abates cell binding or cell attachment activity and/or cognate receptor binding activity" refers to the negligible or undetectable or absent cell binding activity or cell attachment activity of a modified membrane-bound viral glycoprotein contemplated herein to bind or attach to its cognate receptor on the surface of a cell compared to the cell binding activity or cell attachment activity of the unmodified membrane-bound viral glycoprotein to bind or attach to its cognate receptor on the surface of the cell.

Throughout this specification, unless the context requires otherwise, the words "comprise", "comprises" and "comprising" will be understood to imply the inclusion of a stated step or element or group of steps or elements but not the exclusion of any other step or element or group of steps or elements. By "consisting of" is meant including, and limited to, whatever follows the phrase "consisting of." Thus, the phrase "consisting of" indicates that the listed elements are required or mandatory and that no other elements may be present. The phrase "consisting essentially of" means including any elements listed after the phrase and other elements that do not interfere with or contribute to the activity or action specified in the disclosure for the listed elements. Thus, the phrase "consisting essentially of" indicates that the listed elements are required or mandatory but that no other elements are present that materially affect the activity or action of the listed elements.

Reference throughout this specification to "one embodiment," "an embodiment," "a particular embodiment," "a related embodiment," "a certain embodiment," "an additional embodiment," or "a further embodiment" or combinations thereof means that a particular feature, structure or characteristic described in connection with the embodiment is included in at least one embodiment. Thus, the appear-

ances of the foregoing phrases in various places throughout this specification are not necessarily all referring to the same embodiment. Furthermore, the particular features, structures, or characteristics may be combined in any suitable manner in one or more embodiments. It is also understood that the positive recitation of a feature in one embodiment, serves as a basis for excluding the feature in a particular embodiment.

The terms "spacer," "spacer domain," or "spacer polypeptide" are used interchangeably and refer to a polypeptide domain or sequence of amino acids in a non-viral membrane-bound tropism polypeptide disposed between an extracellular antigen targeting domain and a transmembrane domain. A spacer positions the extracellular antigen targeting domain away from the particle surface to enable proper particle/target cell contact, attachment, or binding. A spacer may be derived either from a natural, synthetic, semi-synthetic, or recombinant source. Illustrative examples of spacer domains include but are not limited to hinge or stalk domains derived, obtained, or isolated from CD8 α , CD28, and CD45 isoforms, and polypeptide linkers of similar amino acid composition, rigidity, flexibility and/or length.

A "hinge domain," is a type of spacer domain that plays a role in positioning the antigen binding domain away from the effector cell surface to enable proper cell/cell contact, antigen binding and activation. A hinge domain is placed between a binding domain and a transmembrane domain (TM) of a non-viral membrane-bound tropism polypeptide or between an anti-BCMA antibody or antigen binding fragment thereof and a TM domain of a chimeric antigen receptor. A hinge domain may be derived either from a natural, synthetic, semi-synthetic, or recombinant source. A hinge domain may be altered by substituting one or more cysteine and/or proline residues in a naturally occurring immunoglobulin hinge domain with one or more other amino acid residues (e.g., one or more serine residues).

A "transmembrane domain" or "TM domain" refers to a hydrophobic portion of polypeptide that anchors the polypeptide to the plasma membrane of the cell. The TM domain may be derived either from a natural, synthetic, semi-synthetic, or recombinant source.

An "intracellular signaling domain" refers to a polypeptide domain that participates in transducing the message of effective binding of a target antigen by a chimeric antigen receptor expressed on an immune effector cell to the immune effector cell's interior to elicit one or more effector functions (an "effector function" refers to a specialized function of an immune effector cell), e.g., activation, cytokine production, proliferation and cytotoxic activity, including the release of cytotoxic factors, or other cellular responses elicited with antigen binding to the receptor expressed on the immune effector cell. "Intracellular signaling domains" include a polypeptide domain or functional fragment thereof, which transduces an effector function signal and that directs a cell to perform a specialized function. The term intracellular signaling domain is meant to include any truncated portion of an intracellular signaling domain sufficient to transduce effector function signal.

T cell activation can be said to be mediated by two distinct classes of intracellular signaling domains: primary signaling domains that initiate antigen-dependent primary activation through the TCR (e.g., a TCR/CD3 complex) and costimulatory signaling domains that act in an antigen-independent manner to provide a secondary or costimulatory signal.

A "primary signaling domain" refers to a signaling domain that regulates the primary activation of a TCR complex either in a stimulatory way, or in an inhibitory way.

Primary signaling domains that act in a stimulatory manner may contain one or more signaling motifs which are known as immunoreceptor tyrosine-based activation motifs or ITAMs.

A “costimulatory signaling domain” or “costimulatory domain” refers to an intracellular signaling domain of a costimulatory molecule. Costimulatory molecules are cell surface molecules other than antigen receptors or Fc receptors that provide a second signal required for efficient activation and function of T lymphocytes upon binding to antigen.

“Linker,” “peptide linker,” and “polypeptide linker” are used interchangeably and refer to a plurality of amino acid residues between various polypeptide domains added for appropriate spacing, conformation, and function. A polypeptide linker sequence may be employed to separate any two or more polypeptide components by a distance sufficient to ensure that each polypeptide folds into its appropriate secondary and tertiary structures so as to allow the polypeptide domains to exert their desired functions. Linkers include a “variable domain linking sequence,” an amino acid sequence that connects two or more domains of an antibody or antigen binding fragments thereof and provides a spacer function compatible with interaction of the two sub-binding domains so that the resulting polypeptide retains a specific binding affinity to the same target molecule as an antibody that comprises the same light and/or heavy chain variable domains. A linker may be 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, or 50 or more amino acids long. Illustrative examples of linkers include, but are not limited to the following amino acid sequences: TGEKP (SEQ ID NO: 342); (GGGGGS)_n wherein n=1, 2, 3, 4 or 5 (SEQ ID NOs: 343-347); EGKSSGSGSESKVD (SEQ ID NO: 348); KESGSVSSEQLAQFRSLD (SEQ ID NO: 349); LRQRDGERP (SEQ ID NO: 350); LRQKDDGGGSRP (SEQ ID NO: 351); LRQKD(GGGGS)₂ERP (SEQ ID NO: 352); GEGTSTGSGSGSGSGGAD (SEQ ID NO: 353), and GSTSGSGKPGSGEGSTKG (SEQ ID NO: 354).

The terms “individual” and “subject” are often used interchangeably and refer to any animal that exhibits a symptom of a disease, disorder, or condition, e.g., cancer, that can be treated with the recombinant particles, e.g., recombinant lentiviral particles contemplated elsewhere herein. Suitable subjects (e.g., patients) include laboratory animals (e.g., mouse, rat, rabbit, or guinea pig), farm animals, and domestic animals or pets (e.g., cat or dog). Non-human primates and, preferably, human patients, are preferred subjects.

A “patient” refers to a subject that has been diagnosed with a particular disease, disorder, or condition that can be treated with the recombinant particles disclosed elsewhere herein.

“Treatment” or “treating,” as used herein includes any beneficial or desirable effect on the symptoms or pathology of a disease or pathological condition and may include even minimal reductions in one or more measurable markers of the disease or condition being treated. Optionally, treatment can include reducing the disease burden or delaying disease progression. “Treatment” may, but does not necessarily, indicate complete eradication or cure of the disease or condition, or associated symptoms thereof.

“Prevent,” “prevention,” “preventing” and the like, as used herein, indicate an approach for preventing, inhibiting, or reducing the likelihood of occurrence or recurrence of a disease or condition. It also refers to delaying the onset or

recurrence of a disease or condition or delaying the occurrence or recurrence of the symptoms of a disease or condition. Prevention includes reducing the intensity, effect, symptoms and/or burden of a disease or condition prior to onset or recurrence of the disease or condition.

“Ameliorating at least one symptom of” as used herein, refers to decreasing one or more symptoms of the disease or condition for which a subject is being treated. In particular embodiments, the disease or condition being treated is a cancer, and the one or more symptoms ameliorated include, but are not limited to, weakness, fatigue, shortness of breath, easy bruising and bleeding, frequent infections, enlarged lymph nodes, distended or painful abdomen (due to enlarged abdominal organs), bone or joint pain, fractures, unplanned weight loss, poor appetite, night sweats, persistent mild fever, and decreased urination (due to impaired kidney function).

Additional definitions are set forth throughout this disclosure.

C. Recombinant Particles

Various particles may be used as gene delivery vehicles, e.g., a fusosome; an extracellular vesicle, including a microvesicle, an apoptotic body, and an exosome; a lipid nanoparticle; a virus-like particle (VLPs); or a recombinant viral particle. The present disclosure contemplates, in part, a recombinant lentiviral particle engineered to bind to an immune effector cell and transduce the cell with a vector encoding an anti-BCMA CAR.

Recombinant retroviral particles have been used as a gene delivery platform for treatments of severe genetic diseases and cancer. A “lentivirus” refers to a complex retrovirus. Among retroviruses, lentiviruses are the most efficient at transducing resting or growth-arrested cells. In preferred embodiments, a recombinant particle is a recombinant lentiviral particle or recombinant lentivirus. The terms “recombinant lentiviral particle” and “recombinant lentivirus” are used interchangeably.

Lentiviruses suitable for deriving or engineering recombinant lentiviruses contemplated in particular embodiments herein include but are not limited to human immunodeficiency virus (HIV), including HIV type 1 (HIV-1) and HIV type 2 (HIV-2); visna-maedi virus (VMV); caprine arthritis-encephalitis virus (CAEV); equine infectious anemia virus (EIAV); feline immunodeficiency virus (FIV); bovine immune deficiency virus (BIV); and simian immunodeficiency virus (SIV).

In particular embodiments, a recombinant lentiviral particle is derived or engineered from an HIV-1 or HIV-2 lentivirus.

In particular embodiments, a recombinant lentiviral particle comprises (i) a particle surface comprising (a) a mutated viral envelope glycoprotein and (b) a non-viral membrane-bound tropism polypeptide that binds an immune effector cell; and (ii) two copies of a lentiviral vector comprising or encoding a promoter operably linked to a polynucleotide encoding an anti-BCMA CAR.

D. Particle Surface

Recombinant particles are engineered to bind to an immune effector cell, e.g., a CD3+ cell and deliver one or more copies of a vector encoding an anti-BCMA chimeric antigen receptor to the cell.

In particular embodiments, a recombinant lentiviral particle comprises a viral envelope comprising a mutated

17

vesiculovirus envelope glycoprotein that does not bind its cognate receptor, e.g., LDLR, but that mediates virus-cell fusion and a non-viral membrane-bound tropism polypeptide that redirects the particle to immune effector cells that express CD3; and a lentiviral vector encoding or comprising a promoter operably linked to a polynucleotide encoding an anti-BCMA CAR. The foregoing engineering strategy enables highly efficient on-target delivery of vectors encoding CARs to immune effector cells while minimizing, reducing and/or eliminating delivery to undesired cell types.

18

1. Viral Envelope Glycoproteins

In particular embodiments, a recombinant lentiviral particle comprises a mutated vesiculovirus envelope glycoprotein in which the native or a heterologous signal peptide has been cleaved. Thus, amino acid positions identified in mutated vesiculovirus envelope glycoproteins are with reference to the mature polypeptide, in which the signal peptide has been cleaved.

In particular embodiments, a vesiculovirus is vesicular stomatitis Indiana virus (VSIV). In particular embodiments, a mutated viral envelope glycoprotein is derived from a VSIV envelope glycoprotein (VSIV-G) set forth in Table 1.

TABLE 1

VSIV-G polypeptides	
SEQ ID NO: AMINO ACID SEQUENCE	
1	KFTIVFPHNQKGNWKNVPSNYHYCPSSDDLNWHDNLIGTALQVKMPKSHKAIQADGWMCHASKWVTT CDFRWYGPKYI THSIRSF TSPSVEQCKESIEQTKQGTWLNPGFPPQSCGYATVIDAEAVIVQVTPHHV LVDEYTG EWVDSQFINGKCSNYICPTVHNSTTWHSDYKVKGLCDNLISMDITFFSEDGELSSLGKE GTGFRSNYFAYETGGKACKMQYCKHWGVRLPSGVWFEMADKDLFAAARFPECPEGSSISAPSQTSVD VSLIQDVERILDYSLCQETWSKIRAGLPISPDLSYLAPKNPGTGPAFTIINGTLKYFETRYIRVDI AAPILSRMVGMISGTTTERELWDDWAPYEDVEIGPNGVLR TSSGYKFP LYMIGHGMLSDHLHSSKA QVFEHPHIQDAASQLPDDSLFFGDTGLSKNPIELVEGWFSWKS SIASFFFIIGLIIGLFLVLRVG IHLCKLKHHTKKRQIYTDIEMNRLGK
2	KFTIVFPHNQKGNWKNVPSNYHYCPSSDDLNWHDNLIGTALQVKMPKSHKAIQADGWMCHASKWVTT CDFRWYGPKYI TQSIRSF TSPSVEQCKESIEQTKQGTWLNPGFPPQSCGYATVIDAEAVIVQVTPHHV LVDEYTG EWVDSQFINGKCSNYICPTVHNSTTWHSDYKVKGLCDNLISMDITFFSEDGELSSLGKE GTGFRSNYFAYETGGKACKMQYCKHWGVRLPSGVWFEMADKDLFAAARFPECPEGSSISAPSQTSVD VSLIQDVERILDYSLCQETWSKIRAGLPISPDLSYLAPKNPGTGPAFTIINGTLKYFETRYIRVDI AAPILSRMVGMISGTTTERELWDDWAPYEDVEIGPNGVLR TSSGYKFP LYMIGHGMLSDHLHSSKA QVFEHPHIQDAASQLPDDSLFFGDTGLSKNPIELVEGWFSWKS SIASFFFIIGLIIGLFLVLRVG IHLCKLKHHTKKRQIYTDIEMNRLGK
3	KFTIVFPHNQKGNWKNVPSNYHYCPSSDDLNWHDNLIGTALQVKMPKSHKAIQADGWMCHASKWVTT CDFRWYGPKYI THSIRSF TSPSVEQCKESIEQTKQGTWLNPGFPPQSCGYATVIDAEAVIVQVTPHHV LVDEYTG EWVDSQFINGKCSNYICPTVHNSTTWHSDYKVKGLCDNLISMDITFFSEDGELSSLGKE GTGFRSNYFAYETGGKACKMQYCKHWGVRLPSGVWFEMADKDLFAAARFPECPEGSSISAPSQTSVD VSLIQDVERILDYSLCQETWSKIRAGLPISPDLSYLAPKNPGTGPAFTIINGTLKYFETRYIRVDI AAPILSRMVGMISGTTTERELWDDWAPYEDVEIGPNGVLR TSSGYKFP LYMIGHGMLSDHLHSSKA QVFEHPHIQDAASQLPDDSLFFGDTGLSKNPIELVEGWFSWKS SIASFFFIIGLIIGLFLVLRVG IHLCKLKHHTKKRQIYTDIEMNRLGK
4	KFTIVFPHNQKGNWKNVPSNYHYCPSSDDLNWHDNLIGTALQVKMPKSHKAIQADGWMCHASKWVTT CDFRWYGPKYI TQSIRSF TSPSVEQCKESIEQTKQGTWLNPGFPPQSCGYATVIDAEAVIVQVTPHHV LVDEYTG EWVDSQFINGKCSNYICPTVHNSTTWHSDYKVKGLCDNLISMDITFFSEDGELSSLGKE GTGFRSNYFAYETGGKACKMQYCKHWGVRLPSGVWFEMADKDLFAAARFPECPEGSSISAPSQTSVD VSLIQDVERILDYSLCQETWSKIRAGLPISPDLSYLAPKNPGTGPAFTIINGTLKYFETRYIRVDI AAPILSRMVGMISGTTTERELWDDWAPYEDVEIGPNGVLR TSSGYKFP LYMIGHGMLSDHLHSSKA QVFEHPHIQDAASQLPDDSLFFGDTGLSKNPIELVEGWFSWKS SIASFFFIIGLIIGLFLVLRVG IHLCKLKHHTKKRQIYTDIEMNRLGK
5	KFTIVFPHNQKGNWKNVPSNYHYCPSSDDLNWHDNLIGTALQVKMPKSHKAIQADGWMCHASKWVTT CDFRWYGPKYI THSIRSF TSPSVEQCKESIEQTKQGTWLNPGFPPQSCGYATVIDAEAVIVQVTPHHV LVDEYTG EWVDSQFINGKCSNDICLTVHNSTTWHSDYKVKGLCDNLISMDITFFSEDGELSSLGKA GTGFRSNYFAYETGDKACKMQYCKHWGVRLPSGVWFEMADKDLFAAARFPECPEGSSISAPSQTSVD VSLIQDVERILDYSLCQETWSKIRAGLPISPDLSYLAPKNPGTGPAFTIINGTLKYFETRYIRVDI AAPILSRMVGMISGTTTERELWDDWAPYEDVEIGPNGVLR TSSGYKFP LYMIGHGMLSDHLHSSKV QVFEHPHIQDAASQLPDDSLFFGDTGLSKNPIELVEGWFSWKS SIASFFFIIGLIIGLFLVLRVG IYLCIKLKHTRKRKIYADIEMNRLGK
6	KFTTVFPHNKKGDWKNVPSNYHYCPSSDDLNWHDNLIGTALQVKMPKSHKAIQADGWMCHASKWVTT CDFRWYGPKYI THSIRSF TSPSVEQCKESIEQTKQGTWLNPGFPPQSCGYATVIDAEAVIVQVTPHHV LVDEYTG EWVDSQFINGKCSDDICPTVHNSTTWHSDYKVKGLCDNLISMDITFFSEDGELSSLGKE GTGFRSNYFAYETGDKACKMQYCKHWGVRLPSGVWFEMADKDLFAAARFPECPEGSSISAPSQTSVD VSLIQDVERILDYSLCQETWSKIRAGLPISPDLSYLAPKNPGTGPAFTIINGTLKYFETRYIRVDI AAPILSRMVGMISGTTTERELWDDWAPYEDVEIGPNGVLR TSSGYKFP LYMIGHGMLSDHLHSSKA QVFEHPHIPDATSQLPDDSLFFGDTGLSKDPIELVEGWFSWKS SIASFFFIIGLIIGLFFVLRIG VYLCIKLKHTRKRQIYTDIEMNRLGK
7	KFTIVFPHNQKGTWKNVPSNYHYCPSSDDLNWHDNLIGTALQVKMPKSHKAIQADGWMCHASKWVTT CDFRWYGPKYI THSIRSF TSPSVEQCKESIEQTKQGTWLNPGFPPQSCGYATVIDAEAVIVQVTPHHV LVDEYTG EWVDSQFINGKCSNDICPTVHNSTTWHSDYKVKGLCDNLISMDITFFSEDGELSSLGKE GTGFRSNHAYETGDKACKMQYCKHWGVRLPSGVWFEMADQDLFAAARFPECPEGSSISAPSQTSVD VSLIQDVERILDYSLCQETWSKIRAGLPISPDLSYLAPKNPGTGPAFTIINGTLKYFETRYIRVDI

TABLE 1-continued

VSIV-G polypeptides	
SEQ ID NO: AMINO ACID SEQUENCE	
	AAPILSRMVGMI SGTTTERELWDDWAPYEDVEIGPNGVLR TSSSGYKFPPLYMIGHGMLDSDLHLSSKA QVFEHPHIQDAASQLPDDDELFFGDTGLSKNPIELVEGWFSWKSSIASFFFIIGLIIGLFLVLRVG IYLCIKLKHTKKRQIYTDIEMNRLGK
8	KFTIVFPHNQKGNWKNVPSNYHYCPSSSDLNWNHNDLIGTGLQVKMPKSHKAIQADGWMCHASKWVTT CDFRWYGPKYITHSIRSFTPSVEQCKESIEQTKQGTWLNPGFPPQSCGYATVTDAAEAVIVQVTPHHV LVDEYTGWVDSQFINGKCSNDICPTVHNSTTWHSYKVKGLCDSNLISTDITFFSEDRELSLGKE GTGFRSNFYAYETGDKACKMQYCKHWGVR LPSGVWFEMADKDLFAAARFPECPEGSSISAPSQTSVD VSLIQDVERILDYSLCQETWSKIRAGLPISPVDLSYLAPKNPGTGPAFTIINGTLKYFETRYIRVDI AAPILSRMVGMI SGTTTERELWDDWAPYEDVEIGPNGVLR TSSSGYKFPPLYMIGHGMLDSDLHLSSKA QVFEHPHIQDAASQLPDDDELFFGDTGLSKNPIELVEGWFSWKSSIASFFFIIGLIIGLFLVLRVG IYLYIKLKHTKKRQIYTDIEMNRLGR

In particular embodiments, a mutated viral envelope glycoprotein is derived from a VSIV envelope glycoprotein (VSIV-G) comprising an amino acid sequence set forth in any one of SEQ ID NOs: 1-8 or an amino acid sequence at least 95%, 96%, 97%, 98%, or 99% identical thereto.

In particular embodiments, a mutated VSIV-G polypeptide comprises one or more amino acid substitutions at K47, I182, and/or R354 (substitution with any amino acid; a conservation substitution; a disruptive substitution; substitution with D, E, A, G, F, or Q; or substitution with A, G, F, or Q). In particular embodiments, a mutated VSIV-G polypeptide comprises amino acid substitutions at K47, I182, or R354; K47 and I182; K47 and R354; I182 and R354; or at K47, I182, and R354 of any one of SEQ ID NOs: 1-8.

In particular embodiments, a mutated VSIV-G polypeptide comprises one or more of the following amino acid substitutions: K47A, K47Q, I182E, I182D, R354A, and/or R354Q. In particular embodiments, a mutated VSIV-G polypeptide comprises the following amino acid substitutions: K47A, K47Q, I182E, I182D, R354A, or R354Q; K47A and

I182E; K47A and I182D; K47Q and I182E; K47Q and I182D; I182E and R354A; I182E and R354Q; I182D and R354A; I182D and R354Q; K47A and R354A; K47A and R354Q; K47Q and R354A; K47Q and R354Q; K47A, I182E, and R354A; K47A, I182D, and R354A; K47Q, I182E, and R354A; K47Q, I182D, and R354A; K47A, I182E, and R354Q; K47A, I182D, and R354Q; K47Q, I182E, and R354Q; or K47Q, I182D, and R354Q of any one of SEQ ID NOs: 1-8. In a preferred embodiment, a VSIV-G polypeptide comprises the amino acid substitutions K47Q and R354A of any one of SEQ ID NOs: 1, 2, 3, 4, 5, 6, 7, or 8.

In particular embodiments, a mutated VSIV-G polypeptide comprises an amino acid sequence set forth in Table 2 (SEQ ID NOs: 332, 333, 334, or 335) or an amino acid sequence at least 95%, 96%, 97%, 98%, or 99% identical thereto that mediates fusion of the lentiviral particle and an immune effector cell but that substantially ablates or ablates the polypeptide's ability to bind its cognate receptor expressed on a cell, e.g., LDL-R.

TABLE 2

SEQ ID NO: AMINO ACID SEQUENCE	
332	KFTIVFPHNQKGNWKNVPSNYHYCPSSSDLNWNHNDLIGTAIQVKMPQSHKAIQADGWMCHASKW VTTCDFRWYGPKYITQSIRSFTPSVEQCKESIEQTKQGTWLNPGFPPQSCGYATVTDAAEAVIVQ VTPHHVLVDEYTGWVDSQFINGKCSNYICPTVHNSTTWHSYKVKGLCDSNLISTDITFFSED GELSSLGKEGTGFRSNFYAYETGGKACKMQYCKHWGVR LPSGVWFEMADKDLFAAARFPECPEG SSISAPSQTSVDVSLIQDVERILDYSLCQETWSKIRAGLPISPVDLSYLAPKNPGTGPAFTIIN GTLKYFETRYIRVDIAAPILSRMVGMI SGTTTAEELWDDWAPYEDVEIGPNGVLR TSSSGYKFPPL YMIGHGMLDSDLHLSSKAQVFEHPHIQDAASQLPDDDELFFGDTGLSKNPIELVEGWFSWKSS IASFFFIIGLIIGLFLVLRVGIHLCLIKLKHTKKRQIYTDIEMNRLGK
333	KFTIVFPHNQKGNWKNVPSNYHYCPSSSDLNWNHNDLIGTALQVKMPQSHKAIQADGWMCHASKW VTTCDFRWYGPKYITQSIRSFTPSVEQCKESIEQTKQGTWLNPGFPPQSCGYATVTDAAEAVIVQ VTPHHVLVDEYTGWVDSQFINGKCSNYICPTVHNSTTWHSYKVKGLCDSNLISTDITFFSED GELSSLGKEGTGFRSNFYAYETGGKACKMQYCKHWGVR LPSGVWFEMADKDLFAAARFPECPEG SSISAPSQTSVDVSLIQDVERILDYSLCQETWSKIRAGLPISPVDLSYLAPKNPGTGPAFTIIN GTLKYFETRYIRVDIAAPILSRMVGMI SGTTTAEELWDDWAPYEDVEIGPNGVLR TSSSGYKFPPL YMIGHGMLDSDLHLSSKAQVFEHPHIQDAASQLPDDDELFFGDTGLSKNPIELVEGWFSWKSS IASFFFIIGLIIGLFLVLRVGIHLCLIKLKHTKKRQIYTDIEMNRLGK
334	KFTIVFPHNQKGNWKNVPSNYHYCPSSSDLNWNHNDLIGTAIQVKMPQSHKAIQADGWMCHASKW VTTCDFRWYGPKYITHSIRSFTPSVEQCKESIEQTKQGTWLNPGFPPQSCGYATVTDAAEAVIVQ VTPHHVLVDEYTGWVDSQFINGKCSNYICPTVHNSTTWHSYKVKGLCDSNLISTDITFFSED GELSSLGKEGTGFRSNFYAYETGGKACKMQYCKHWGVR LPSGVWFEMADKDLFAAARFPECPEG SSISAPSQTSVDVSLIQDVERILDYSLCQETWSKIRAGLPISPVDLSYLAPKNPGTGPAFTIIN GTLKYFETRYIRVDIAAPILSRMVGMI SGTTTAEELWDDWAPYEDVEIGPNGVLR TSSSGYKFPPL YMIGHGMLDSDLHLSSKAQVFEHPHIQDAASQLPDDDELFFGDTGLSKNPIELVEGWFSWKSS IASFFFIIGLIIGLFLVLRVGIHLCLIKLKHTKKRQIYTDIEMNRLGK

TABLE 2-continued

SEQ ID NO: AMINO ACID SEQUENCE
335 KFTIVFPFHQKGNWKNVPSNYHYCPSSSDLNWHNDLIGTALQVKMPQSHKAIQADGWMCHASKW VTTCDPRWYGPKYIITHSIRSFPTSPVEQCKESIEQTKQGTWLNPGFPPQSCGYATVTDAAEAVIVQ VTPHHVLVDEYTGWVDSQFINGKCSNYICPTVHNSTTWHSYKVKGLCDNLISMDITFFSED GELSSLGKEGTGFRSNFYAYETGGKACKMQYCKHWGVRPSPGVWFEMADKDLFAAARFPCEPEG SSISAPSQTSDVSLIQDVERILDYSLCQETWSKIRAGLPISPVDSLAPKNPMTGPAFTIIN GTLKYFETRYIRVDIAAPILSRMVGMI SGTTEAELWDDWAPYEDVEIGPNGVLRRTSSGYKFPFL YMIHGMLDSDLHLSSKAQVFEHPHIQDAASQLPDDSLFFGDTGLSKNPIELVEGWFSWKS IASFFFIIGLIIGLFLVLRVGIHLCKLKHKKRQIYTDIEMNRLGK

In particular embodiments, a vesiculovirus is coccal virus (COCV). In particular embodiments, a mutated viral envelope glycoprotein is derived from a COCV envelope glycoprotein (COCV-G) set forth in Table 3.

TABLE 3

SEQ ID NO AMINO ACID SEQUENCE
9 KFSIVFPQSQKGNWKNVPSNYHYCPSSSDQNWHDLLGITMKVKM PKTHKAIQADGWMCHAAKWITTCDFRWYGPKYIITHSIHQPTSE QCKESIKQTKQGTWMSPGFPPQNCGYATVTDVAVVVQATPHHVL VDEYTGWVDSQFNGKCEETECETVHNSTVWYSDYKVTGLCDAT LVDTEITFFSEDGKKEGKESIGKPTGYRSNYFAYEKGDKVCKMNYCK HAGVRLPSGVWFEPVDQDVYAAKLPECPVGATISAPTQTSVDVS LILDVERILDYSLCQETWSKIRSKQPVSPVDLSYLAPKNPMTGPA FTIINGTLKYFETRYIRIDIDNPIISKMGKISGSQTERELWTEW FPYEGVEIGPNGILKTPTYGKFPFLFMIGHGMLDSDLHKTSAEVEF EHPHLAEAPKQLPEEETLFFGDTGISKNPVELIEGWFSWKS TFFFAIGVFILLYVVARIVIAVRYRYQGSNNKRIYNDIEMSRFRK

In particular embodiments, a mutated viral envelope glycoprotein is derived from a COCV envelope glycoprotein (COCV-G) comprising an amino acid sequence set forth in SEQ ID NO: 9 or an amino acid sequence at least 95%, 96%, 97%, 98%, or 99% identical thereto.

In particular embodiments, a mutated COCV-G polypeptide comprises one or more amino acid substitutions at K47, V182, and/or R354 (substitution with any amino acid; a conservation substitution; a disruptive substitution; substi-

15 tution with D, E, A, G, F, or Q; or substitution with A, G, F, or Q). In particular embodiments, a mutated COCV-G polypeptide comprises amino acid substitutions at K47, V182, or R354; K47 and V182; K47 and R354; V182 and R354; or at K47, V182, and R354 of SEQ ID NO: 9.

20 In particular embodiments, a mutated COCV-G polypeptide comprises one or more of the following amino acid substitutions: K47A, K47Q, V182E, V182D, R354A, and/or R354Q. In particular embodiments, a mutated VSIV-G polypeptide comprises the following amino acid substitutions: 25 K47A, K47Q, V182E, V182D, R354A, or R354Q; K47A and V182E; K47A and V182D; K47Q and V182E; K47Q and V182D; V182E and R354A; V182E and R354Q; V182D and R354A; V182D and R354Q; K47A and R354A; K47A and R354Q; K47Q and R354A; K47Q and R354Q; 30 K47A, V182E, and R354A; K47A, V182D, and R354A; K47Q, V182E, and R354A; K47Q, V182D, and R354A; K47A, V182E, and R354Q; K47A, V182D, and R354Q; K47Q, V182E, and R354Q; or K47Q, V182D, and R354Q of SEQ ID NO: 9. In a preferred embodiment, a COCV-G polypeptide comprises the amino acid substitutions K47Q and R354A of SEQ ID NO: 9.

In particular embodiments, a mutated COCV-G polypeptide comprises an amino acid sequence set forth in Table 4 (SEQ ID NOs: 336, 337, 338, and 339) or an amino acid 40 sequence at least 95%, 96%, 97%, 98%, or 99% identical thereto that mediates fusion of the lentiviral particle and an immune effector cell but that substantially ablates or ablates the polypeptide's ability to bind its cognate receptor expressed on a cell, e.g., LDLR.

TABLE 4

SEQ ID NO: AMINO ACID SEQUENCE
336 KFSIVFPQSQKGNWKNVPSNYHYCPSSSDQNWHDLLGITMKVKMPATHKAIQADGWMCHAAKW ITTCDFRWYGPKYIITHSIHQPTSEQCKESIKQTKQGTWMSPGFPPQNCGYATVTDVAVVVQ ATPHHVLVDEYTGWVDSQFNGKCEETECETVHNSTVWYSDYKVTGLCDATLVDTEITFFSED GKKEGKESIGKPTGYRSNYFAYEKGDKVCKMNYCKHAGVRLPSGVWFEPVDQDVYAAKLPECPVG ATISAPTQTSVDVSLILDVERILDYSLCQETWSKIRSKQPVSPVDLSYLAPKNPMTGPAFTIIN GTLKYFETRYIRIDIDNPIISKMGKISGSQTEAELWTEWFPYEGVEIGPNGILKTPTYGKFPFL FMIGHGMLDSDLHKTSAEVEFHPHLAEAPKQLPEEETLFFGDTGISKNPVELIEGWFSWKS VTFFFAIGVFILLYVVARIVIAVRYRYQGSNNKRIYNDIEMSRFRK
337 KFSIVFPQSQKGNWKNVPSNYHYCPSSSDQNWHDLLGITMKVKMPATHKAIQADGWMCHAAKW ITTCDFRWYGPKYIITHSIHQPTSEQCKESIKQTKQGTWMSPGFPPQNCGYATVTDVAVVVQ ATPHHVLVDEYTGWVDSQFNGKCEETECETVHNSTVWYSDYKVTGLCDATLVDTEITFFSED GKKEGKESIGKPTGYRSNYFAYEKGDKVCKMNYCKHAGVRLPSGVWFEPVDQDVYAAKLPECPVG ATISAPTQTSVDVSLILDVERILDYSLCQETWSKIRSKQPVSPVDLSYLAPKNPMTGPAFTIIN GTLKYFETRYIRIDIDNPIISKMGKISGSQTEAELWTEWFPYEGVEIGPNGILKTPTYGKFPFL FMIGHGMLDSDLHKTSAEVEFHPHLAEAPKQLPEEETLFFGDTGISKNPVELIEGWFSWKS VTFFFAIGVFILLYVVARIVIAVRYRYQGSNNKRIYNDIEMSRFRK
338 KFSIVFPQSQKGNWKNVPSNYHYCPSSSDQNWHDLLGITMKVKMPQTHKAIQADGWMCHAAKW ITTCDFRWYGPKYIITHSIHQPTSEQCKESIKQTKQGTWMSPGFPPQNCGYATVTDVAVVVQ

TABLE 4-continued

SEQ ID NO: AMINO ACID SEQUENCE
ATPHHVLVDYETGEWIDSQFPNGKCETEECETVHNSTVWYSDYKVTGLCDATLVDTEITFFSED GKKEIGKPNTRYGSRNYFAYEKGDVKCMNYCKHAGVRLPSGVWFEFVDQDVYAAAKLPECVVG ATISAPTQTSVDVSLILDVERILDYSLCQETWSKIRSKQPVSPVDLSYLAPKNPGTGPAFTIIN GTLKYFETRYRIDIDNP IISKMGKISGSQTEAELWTEWFPYEGVEIGPNGLKPTGYKFPFL FMIGHGMLDSDLHKTSQAEVFEHPLAEAPKQLPEEETLFFPGDTGISKNPVELIEGWFSWSKST VVTFFFAIGVFILLYVVARIVIAVRYRYQGSNNKRIYNDIEMSRFRK
339 KFSIVFPQSQKGNWKNVPSSYHYCPSSSDQNWHNDLLGITMKVKMPQTHKAIQADGWMCHAAKW ITTCDFRWYGPKYITHSIHSIQPTSEQCKESIKQTKQGTWMSPGFPPQNGYATVTDVAVVVQ ATPHHVLVDYETGEWIDSQFPNGKCETEECETVHNSTVWYSDYKVTGLCDATLVDTEITFFSED GKKEIGKPNTRYGSRNYFAYEKGDVKCMNYCKHAGVRLPSGVWFEFVDQDVYAAAKLPECVVG ATISAPTQTSVDVSLILDVERILDYSLCQETWSKIRSKQPVSPVDLSYLAPKNPGTGPAFTIIN GTLKYFETRYRIDIDNP IISKMGKISGSQTEQELWTEWFPYEGVEIGPNGLKPTGYKFPFL FMIGHGMLDSDLHKTSQAEVFEHPLAEAPKQLPEEETLFFPGDTGISKNPVELIEGWFSWSKST VVTFFFAIGVFILLYVVARIVIAVRYRYQGSNNKRIYNDIEMSRFRK

2. Tropism Polypeptides

Recombinant particles contemplated herein comprise a tropism polypeptide that governs targeting the particle to an immune effector cell. A “tropism polypeptide” is a polypeptide that binds one or more antigens on a desired cell type. A “non-viral membrane bound tropism polypeptide” is a polypeptide that binds one or more antigens on a desired cell type; that is not native to, or derived from, either in whole or in part, a virus.

In particular embodiments, a recombinant lentiviral particle comprises an envelope comprising or expressing a mutated VSIV-G polypeptide comprising one or more amino acid substitutions at positions K47, I182, and/or R354 or a mutated COCV-G polypeptide comprising one or more amino acid substitutions at positions K47, V182, and/or R354; a non-viral membrane-bound tropism polypeptide that binds an antigen expressed on an immune effector cell, e.g., CD3; and one or more copies of a lentiviral vector encoding or comprising a promoter operably linked to a polynucleotide encoding an anti-BCMA CAR. In particular embodiments, a recombinant lentiviral particle comprises an envelope that further comprises a secondary non-viral membrane-bound tropism polypeptide that an antigen specifically expressed on an immune effector cell, e.g., CD80, CD86, OX40L, 4-1BBL, and ICOSL.

Non-viral membrane-bound tropism polypeptides contemplated herein comprise, consist essentially of, or consist of an extracellular antigen targeting domain, a spacer and a transmembrane domain. As used herein, an “extracellular antigen targeting domain” (also referred to as an “extracellular targeting domain,” “antigen targeting domain,” or “targeting domain”) refers to any naturally occurring, synthetic, semi-synthetic, or recombinantly produced binding partner that binds a biological molecule or target antigen expressed or displayed on the surface of a cell. In preferred embodiments, an extracellular antigen targeting domain binds an antigen, e.g., CD3ε, expressed on an immune effector cell.

In particular embodiments, a non-viral membrane-bound tropism polypeptide comprises an extracellular antigen targeting domain comprising an antibody or antigen binding fragment thereof that binds CD3R. In particular embodiments, an extracellular antigen targeting domain comprises an anti-CD3ε scFv isolated from an antibody selected from the group consisting of: OKT3, UCHT1, YTH12.5, TR66, and variants thereof.

In certain embodiments, a non-viral membrane-bound tropism polypeptide comprises an extracellular antigen targeting domain comprising an anti-CD3ε scFv comprising an amino acid sequence set forth in Table 5.

TABLE 5

anti-CD3 antibodies			
Ab ID	SEQ ID NO	ID	AMINO ACID SEQUENCE
CD3.9	145	VH	DVQLVQSGAEVKKPGASVKVCKASGYTFTRYTMHWVRQAPGQGL EWIGYINPSRGYTNADSVKGRFTITTDKSTSTAYMELSSLRSED TATYYCARYDDHYCLDYWGQGTTVTVSS
	146	CDRH1	RYTMH
	147	CDRH2	YINPSRGYTNADSVK
	148	CDRH3	YYDDHYCLDY
	149	VL	DIVLTQSPATLSLSPGERATLSCRASQSVSYMNWYQKPKGKAPKR WYDTSKVASGVPARFSGSGSGTDYSLTINSLEAEDAATYYCQQW SSNPLTFGGGTKVEIK
	150	CDRL1	RASQSVSYMN
	151	CDRL2	DTSKVAS
	152	CDRL3	QQWSSNPLT
	153	scFv	DVQLVQSGAEVKKPGASVKVCKASGYTFTRYTMHWVRQAPGQGL EWIGYINPSRGYTNADSVKGRFTITTDKSTSTAYMELSSLRSED TATYYCARYDDHYCLDYWGQGTTVTVSSSSGGGSGGGGSGGGGS DIVLTQSPATLSLSPGERATLSCRASQSVSYMNWYQKPKGKAPKR WYDTSKVASGVPARFSGSGSGTDYSLTINSLEAEDAATYYCQQW SSNPLTFGGGTKVEIK

TABLE 5-continued

anti-CD3 antibodies			
Ab ID	SEQ ID NO	ID	AMINO ACID SEQUENCE
CD3.10	154	scFv	DIVLTQSPATLSLSPGERATLSCRASQSVSYMNWYQQKPGKAPKR WIYDTSKVASGVPARFSGSGSGTDYSLTINSLEAEDAATYYCQQW SSNPLTFGGGTVKVEIKSGGGSGGGSGGGSDVQLVQSGAEVKK PGASVKVSKASGYTFTRYTMHWVRQAPGQGLEWIGYINPSRGYT NYADSVKGRFTITTDKSTSTAYMELSSLRSEDATATYYCARYYDDH YCLDYWGQGTITVTVSS
	155	VH	DVQLVQSGAEVKKPGASVKVSKASGYTFTRYTMHWVRQAPGQGL EWIGYINPSRGYTNYADSVKGRFTITTDKSTSTAYMELSSLRSED TATYYCARYYDDPYCLDYWGQGTITVTVSS
	156	CDRH1	RYTMH
	157	CDRH2	YINPSRGYTNYADSVKG
	158	CDRH3	YYDDPYCLDY
	159	VL	DIVLTQSPATLSLSPGERATLSCRASQSVSYMNWYQQKPGKAPKR WIYDTSKVASGVPARFSGSGSGTDYSLTINSLEAEDAATYYCQQW SSNPLTFGGGTVKVEIK
	160	CDRL1	RASQSVSYMN
	161	CDRL2	DTSKVAS
	162	CDRL3	QQWSSNPLT
	163	scFv	DVQLVQSGAEVKKPGASVKVSKASGYTFTRYTMHWVRQAPGQGL EWIGYINPSRGYTNYADSVKGRFTITTDKSTSTAYMELSSLRSED TATYYCARYYDDPYCLDYWGQGTITVTVSSSGGGSGGGSGGGGS DIVLTQSPATLSLSPGERATLSCRASQSVSYMNWYQQKPGKAPKR WIYDTSKVASGVPARFSGSGSGTDYSLTINSLEAEDAATYYCQQW SSNPLTFGGGTVKVEIK
	164	scFv	DIVLTQSPATLSLSPGERATLSCRASQSVSYMNWYQQKPGKAPKR WIYDTSKVASGVPARFSGSGSGTDYSLTINSLEAEDAATYYCQQW SSNPLTFGGGTVKVEIKSGGGSGGGSGGGSDVQLVQSGAEVKK PGASVKVSKASGYTFTRYTMHWVRQAPGQGLEWIGYINPSRGYT NYADSVKGRFTITTDKSTSTAYMELSSLRSEDATATYYCARYYDDP YCLDYWGQGTITVTVSS
CD3.11	165	VH	DVQLVQSGAEVKKPGASVKVSKASGYTFTRYTMHWVRQAPGQGL EWIGYINPSRGYTNYADSVKGRFTITTDKSTSTAYMELSSLRSED TATYYCARYYNDQYCLDYWGQGTITVTVSSS
	166	CDRH1	RYTMH
	167	CDRH2	YINPSRGYTNYADSVKG
	168	CDRH3	YYNDQYCLDY
	169	VL	DIVLTQSPATLSLSPGERATLSCRASQSVSYMNWYQQKPGKAPKR WIYDTSKVASGVPARFSGSGSGTDYSLTINSLEAEDAATYYCQQW SSNPLTFGGGTVKVEIK
	170	CDRL1	RASQSVSYMN
	171	CDRL2	DTSKVAS
	172	CDRL3	QQWSSNPLT
	173	scFv	DVQLVQSGAEVKKPGASVKVSKASGYTFTRYTMHWVRQAPGQGL EWIGYINPSRGYTNYADSVKGRFTITTDKSTSTAYMELSSLRSED TATYYCARYYNDQYCLDYWGQGTITVTVSSSSGGGSGGGSGGGG SDIVLTQSPATLSLSPGERATLSCRASQSVSYMNWYQQKPGKAPK RWIYDTSKVASGVPARFSGSGSGTDYSLTINSLEAEDAATYYCQQ WSSNPLTFGGGTVKVEIK
	174	scFv	DIVLTQSPATLSLSPGERATLSCRASQSVSYMNWYQQKPGKAPKR WIYDTSKVASGVPARFSGSGSGTDYSLTINSLEAEDAATYYCQQW SSNPLTFGGGTVKVEIKSGGGSGGGSGGGSDVQLVQSGAEVKK PGASVKVSKASGYTFTRYTMHWVRQAPGQGLEWIGYINPSRGYT NYADSVKGRFTITTDKSTSTAYMELSSLRSEDATATYYCARYYNDQ YCLDYWGQGTITVTVSSS
CD3.12	175	VH	DVQLVQSGAEVKKPGASVKVSKASGYTFTRYTMHWVRQAPGQGL EWIGYINPSRGYTNYADSVKGRFTITTDKSTSTAYMELSSLRSED TATYYCARYYDAHNCLDYWGQGTITVTVSS
	176	CDRH1	RYTMH
	177	CDRH2	YINPSRGYTNYADSVKG
	178	CDRH3	YYDAHNCLDY
	179	VL	DIVLTQSPATLSLSPGERATLSCRASQSVSYMNWYQQKPGKAPKR WIYDTSKVASGVPARFSGSGSGTDYSLTINSLEAEDAATYYCQQW SSNPLTFGGGTVKVEIK
	180	CDRL1	RASQSVSYMN
	181	CDRL2	DTSKVAS
	182	CDRL3	QQWSSNPLT
	183	scFv	DVQLVQSGAEVKKPGASVKVSKASGYTFTRYTMHWVRQAPGQGL EWIGYINPSRGYTNYADSVKGRFTITTDKSTSTAYMELSSLRSED TATYYCARYYDAHNCLDYWGQGTITVTVSSSGGGSGGGSGGGGS DIVLTQSPATLSLSPGERATLSCRASQSVSYMNWYQQKPGKAPKR WIYDTSKVASGVPARFSGSGSGTDYSLTINSLEAEDAATYYCQQW SSNPLTFGGGTVKVEIK

TABLE 5-continued

anti-CD3 antibodies			
Ab ID	SEQ ID NO	ID	AMINO ACID SEQUENCE
CD3.13	184	scFv	DIVLTQSPATLSLSPGERATLSCRASQSVSYMNWYQQKPKGKAPKR WIYDTSKVASGVPARFSGSGSGTDYSLTINSLEAEDAATYYCQQW SSNPLTFGGGTVKVEIKSGGGSGGGSGGGSDVQLVQSGAEVKK PGASVKVSKASGYTFTRYTMHWVRQAPGQGLEWIGYINPSRGYT NYADSVKGRFTITTDKSTSTAYMELSSLRSEDATATYYCARYYDAH NCLDYWGQTTTVTVSS
	185	VH	DVQLVQSGAEVKKPGASVKVSKASGYTFTRYTMHWVRQAPGQGL EWIGYINPSRGYTNYADSVKGRFTITTDKSTSTAYMELSSLRSED TATYYCARYSDHHCYCLDYWGQTTTVTVSS
	186	CDRH1	RYTMH
	187	CDRH2	YINPSRGYTNYADSVKG
	188	CDRH3	YSDHHCYCLDY
	189	VL	DIVLTQSPATLSLSPGERATLSCRASQSVSYMNWYQQKPKGKAPKR WIYDTSKVASGVPARFSGSGSGTDYSLTINSLEAEDAATYYCQQW SSNPLTFGGGTVKVEIK
	190	CDRL1	RASQSVSYMN
	191	CDRL2	DTSKVAS
	192	CDRL3	QQWSSNPLT
	193	scFv	DVQLVQSGAEVKKPGASVKVSKASGYTFTRYTMHWVRQAPGQGL EWIGYINPSRGYTNYADSVKGRFTITTDKSTSTAYMELSSLRSED TATYYCARYSDHHCYCLDYWGQTTTVTVSSSGGGSGGGSGGGGS DIVLTQSPATLSLSPGERATLSCRASQSVSYMNWYQQKPKGKAPKR WIYDTSKVASGVPARFSGSGSGTDYSLTINSLEAEDAATYYCQQW SSNPLTFGGGTVKVEIK
	194	scFv	DIVLTQSPATLSLSPGERATLSCRASQSVSYMNWYQQKPKGKAPKR WIYDTSKVASGVPARFSGSGSGTDYSLTINSLEAEDAATYYCQQW SSNPLTFGGGTVKVEIKSGGGSGGGSGGGSDVQLVQSGAEVKK PGASVKVSKASGYTFTRYTMHWVRQAPGQGLEWIGYINPSRGYT NYADSVKGRFTITTDKSTSTAYMELSSLRSEDATATYYCARYSDH YCLDYWGQTTTVTVSS
CD3.14	195	VH	DVQLVQSGAEVKKPGASVKVSKASGYTFTRYTMHWVRQAPGQGL EWIGYINPSRGYTNYADSVKGRFTITTDKSTSTAYMELSSLRSED TATYYCARYSDHHCYCLDYWGQTTTVTVSS
	196	CDRH1	RYTMH
	197	CDRH2	YINPSRGYTNYADSVKG
	198	CDRH3	YSDHHCYCLDY
	199	VL	DIVLTQSPATLSLSPGERATLSCRASQSVSYMNWYQQKPKGKAPKR WIYDTSKVASGVPARFSGSGSGTDYSLTINSLEAEDAATYYCQQW SSNPLTFGGGTVKVEIK
	200	CDRL1	RASQSVSYMN
	201	CDRL2	DTSKVAS
	202	CDRL3	QQWSSNPLT
	203	scFv	DVQLVQSGAEVKKPGASVKVSKASGYTFTRYTMHWVRQAPGQGL EWIGYINPSRGYTNYADSVKGRFTITTDKSTSTAYMELSSLRSED TATYYCARYSDHHCYCLDYWGQTTTVTVSSSGGGSGGGSGGGGS DIVLTQSPATLSLSPGERATLSCRASQSVSYMNWYQQKPKGKAPKR WIYDTSKVASGVPARFSGSGSGTDYSLTINSLEAEDAATYYCQQW SSNPLTFGGGTVKVEIK
	204	scFv	DIVLTQSPATLSLSPGERATLSCRASQSVSYMNWYQQKPKGKAPKR WIYDTSKVASGVPARFSGSGSGTDYSLTINSLEAEDAATYYCQQW SSNPLTFGGGTVKVEIKSGGGSGGGSGGGSDVQLVQSGAEVKK PGASVKVSKASGYTFTRYTMHWVRQAPGQGLEWIGYINPSRGYT NYADSVKGRFTITTDKSTSTAYMELSSLRSEDATATYYCARYSDH YCLDYWGQTTTVTVSS
	205	VH	DVQLVQSGAEVKKPGASVKVSKASGYTFTRYTMHWVRQAPGQGL EWIGYINPSRGYTNYADSVKGRFTITTDKSTSTAYMELSSLRSED TATYYCARYYDDNYCLDYWGQTTTVTVSS
	206	CDRH1	RYTMH
	207	CDRH2	YINPSRGYTNYADSVKG
	208	CDRH3	YDDNYCLDY
CD3.15	209	VL	DIVLTQSPATLSLSPGERATLSCRASQSVSYMNWYQQKPKGKAPKR WIYDTSKVASGVPARFSGSGSGTDYSLTINSLEAEDAATYYCQQW SSNPLTFGGGTVKVEIK
	210	CDRL1	RASQSVSYMN
	211	CDRL2	DTSKVAS
	212	CDRL3	QQWSSNPLT
	213	scFv	DVQLVQSGAEVKKPGASVKVSKASGYTFTRYTMHWVRQAPGQGL EWIGYINPSRGYTNYADSVKGRFTITTDKSTSTAYMELSSLRSED TATYYCARYYDDNYCLDYWGQTTTVTVSSSGGGSGGGSGGGGS SDIVLTQSPATLSLSPGERATLSCRASQSVSYMNWYQQKPKGKAPK RWIYDTSKVASGVPARFSGSGSGTDYSLTINSLEAEDAATYYCQQ WSSNPLTFGGGTVKVEIK

TABLE 5-continued

anti-CD3 antibodies			
Ab ID	SEQ ID NO	ID	AMINO ACID SEQUENCE
CD3.16	214	scFv	DIVLTQSPATLSLSPGERATLSCRASQSVSYMNWYQQKPKGKAPKR WIYDTSKVASGVPARFSGSGSGTDYSLTINSLEAEDAATYYCQW SSNPLTFGGGTKEIKSGGGSGGGSGGGSDVQLVQSGAEVKK PGASVKVSKASGYTFTRYTMHWRQAPGQGLEWIGYINPSRGYT NYADSVKGRFTITTDKSTSTAYMELSSLRSEDATYYCARYYDDN YCLDYWGQGTITVTVSS
	215	VH	DVQLVQSGAEVKKPGASVKVSKASGYTFTRYTMHWRQAPGQGL EWIGYINPSRGYTNYADSVKGRFTITTDKSTSTAYMELSSLRSED TATYYCARYYDDQYCLDYWGQGTITVTVSS
	216	CDRH1	RYTMH
	217	CDRH2	YINPSRGYTNYADSVKG
	218	CDRH3	YYDDQYCLDY
	219	VL	DIVLTQSPATLSLSPGERATLSCRASQSVSYMNWYQQKPKGKAPKR WIYDTSKVASGVPARFSGSGSGTDYSLTINSLEAEDAATYYCQW SSNPLTFGGGTKEIK
	220	CDRL1	RASQSVSYMN
	221	CDRL2	DTSKVAS
	222	CDRL3	QQWSSNPLT
	223	scFv	DVQLVQSGAEVKKPGASVKVSKASGYTFTRYTMHWRQAPGQGL EWIGYINPSRGYTNYADSVKGRFTITTDKSTSTAYMELSSLRSED TATYYCARYYDDQYCLDYWGQGTITVTVSSSGGGSGGGSGGGGS DIVLTQSPATLSLSPGERATLSCRASQSVSYMNWYQQKPKGKAPKR WIYDTSKVASGVPARFSGSGSGTDYSLTINSLEAEDAATYYCQW SSNPLTFGGGTKEIK
	224	scFv	DIVLTQSPATLSLSPGERATLSCRASQSVSYMNWYQQKPKGKAPKR WIYDTSKVASGVPARFSGSGSGTDYSLTINSLEAEDAATYYCQW SSNPLTFGGGTKEIKSGGGSGGGSGGGSDVQLVQSGAEVKK PGASVKVSKASGYTFTRYTMHWRQAPGQGLEWIGYINPSRGYT NYADSVKGRFTITTDKSTSTAYMELSSLRSEDATYYCARYYDDQ YCLDYWGQGTITVTVSS

In particular embodiments, a non-viral membrane-bound tropism polypeptide comprises an extracellular antigen targeting domain comprising an anti-CD3ε scFv that comprises an amino acid sequence set forth in any one of SEQ ID NOs: 153, 154, 163, 164, 173, 174, 183, 184, 193, 194, 203, 204, 213, 214, 223, and 224. In particular preferred embodiments, the anti-CD3ε scFv comprises an amino acid sequence set forth in SEQ ID NO: 153 or 154. In particular embodiments, the anti-CD3ε scFv comprises an amino acid sequence set forth in SEQ ID NO: 163 or 164. In particular embodiments, the anti-CD3ε scFv comprises an amino acid sequence set forth in SEQ ID NO: 173 or 174. In particular embodiments, the anti-CD3ε scFv comprises an amino acid sequence set forth in SEQ ID NO: 183 or 184. In particular embodiments, the anti-CD3ε scFv comprises an amino acid sequence set forth in SEQ ID NO: 193 or 194. In particular embodiments, the anti-CD3ε scFv comprises an amino acid sequence set forth in SEQ ID NO: 203 or 204. In particular embodiments, the anti-CD3ε scFv comprises an amino acid sequence set forth in SEQ ID NO: 213 or 214. In particular embodiments, the anti-CD3ε scFv comprises an amino acid sequence set forth in SEQ ID NO: 223 or 224.

Non-viral membrane-bound tropism polypeptides contemplated herein comprise, consist essentially of, or consist of an anti-CD3ε scFv comprising an amino acid sequence set forth in any one of SEQ ID NOs: 153, 154, 163, 164, 173, 174, 183, 184, 193, 194, 203, 204, 213, 214, 223, and 224, a spacer domain comprising a hinge domain isolated from a polypeptide selected from the group consisting of CD4, CD8α, CD28, IgG4, or IgG1 and a transmembrane domain.

Non-viral membrane-bound tropism polypeptides contemplated herein comprise, consist essentially of, or consist of an anti-CD3ε scFv comprising an amino acid sequence set forth in any one of SEQ ID NOs: 153, 154, 163, 164, 173,

174, 183, 184, 193, 194, 203, 204, 213, 214, 223, and 224, a spacer domain comprising an amino acid sequence set forth in Table 6 and a transmembrane domain.

TABLE 6

Spacer Domains	
SEQ ID NO:	AMINO ACID SEQUENCE
225	TTTPAPRPPTPAPTIASQPLSLRPEACRPAAGAVHTRGLDFACD
226	IEVMYPPPYLDNEKSNGTIIHVKGKHLCPSPFPGPSKP
227	SGQVLLESNIKVLPTWSTPVQP
228	ESKYGPCCPPCP
229	ESKYGPCCPPCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVNV DVSQEDPEVQFNWYVDGVEVHNAKTPREEQFQSTYRVVSVLTVL HQDWLNGKEYKCKVSNKGLPSSIEKTI SKAKGQPREPQVYITLPPS QEEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTTTPVLD SDGSFPLYSLRTVDKSRWQEGNVFSCSVMEALHNHYTQKSLSLS LGK
230	LEPKSCDKTHTCPPCP

Non-viral membrane-bound tropism polypeptides contemplated herein comprise, consist essentially of, or consist of an anti-CD3ε scFv comprising an amino acid sequence set forth in any one of SEQ ID NOs: 153, 154, 163, 164, 173, 174, 183, 184, 193, 194, 203, 204, 213, 214, 223, and 224, a spacer domain comprising an amino acid sequence set forth in any one of SEQ ID NOs: 225, 226, 227, 228, 229, and 230 and a transmembrane domain.

31

Non-viral membrane-bound tropism polypeptides contemplated herein comprise, consist essentially of, or consist of an anti-CD3 ϵ scFv comprising an amino acid sequence set forth in any one of SEQ ID NOs: 153, 154, 163, 164, 173, 174, 183, 184, 193, 194, 203, 204, 213, 214, 223, and 224, a spacer domain comprising an amino acid sequence set forth in any one of SEQ ID NOs: 225, 226, 227, 228, 229, and 230 and a transmembrane domain isolated from a polypeptide selected from the group consisting of CD3c, CD4, CD8 α , CD28, CD134, CD137, and CD278.

Non-viral membrane-bound tropism polypeptides contemplated herein comprise, consist essentially of, or consist of an anti-CD3 ϵ scFv comprising an amino acid sequence set forth in any one of SEQ ID NOs: 153, 154, 163, 164, 173, 174, 183, 184, 193, 194, 203, 204, 213, 214, 223, and 224, a spacer domain comprising an amino acid sequence set forth in any one of SEQ ID NOs: 225, 226, 227, 228, 229, and 230 and a transmembrane domain comprising an amino acid sequence set forth in Table 7.

TABLE 7

Transmembrane Domains	
SEQ ID NO:	AMINO ACID SEQUENCE
231	IYIWAPLAGTCGVLLSLVITLYC

32

TABLE 7-continued

Transmembrane Domains	
SEQ ID NO:	AMINO ACID SEQUENCE
232	IISFFLALTSTALLFLFFLTFRFSVV
233	FWVLVVVGVLACYSLLVTVAFIIFWV
234	VAAILGLGLVGLGLPLAILL
235	WLPIGCAAFVVVCILGCILICWL
236	VMSVATIVIVDICITGGLLLLVYYWS
237	MALIVLGGVAGLLFIGLGIF

Non-viral membrane-bound tropism polypeptides contemplated herein comprise, consist essentially of, or consist of an anti-CD3 ϵ scFv comprising an amino acid sequence set forth in any one of SEQ ID NOs: 153, 154, 163, 164, 173, 174, 183, 184, 193, 194, 203, 204, 213, 214, 223, and 224, a spacer domain comprising an amino acid sequence set forth in any one of SEQ ID NOs: 225, 226, 227, 228, 229, and 230 and a transmembrane domain comprising an amino acid sequence set forth in any one of SEQ ID NOs: 231, 232, 233, 234, 235, 236, and 237.

In particular embodiments, a non-viral membrane bound tropism polypeptide comprises, consists essentially of, or consists of an amino acid sequence set forth in Table 8.

TABLE 8

Non-viral membrane bound tropism polypeptides	
SEQ ID NO:	AMINO ACID SEQUENCE
324	DVQLVQSGAEVKKPGASVKVSKASGYTFTRYTMHWVRQAPGGGLEWIGYINPSRGYTNADSVKGRFTITTDKSTSTAYMELSSLRSEDTATYYCARYYDDHYCLDYWGQGTITVTVSSGGGGSGGGSGGGGSDIVLTQSPATLSLSPGERATLSCRASQSVSYMNWYQQKPKGKAPKRWIYDTSKVASGVPARFSGSGSGTDYSLTINSLEAEDAATYYCQQWSSNPLTFGGGTQKVEIKTTTPAPRPPTPAPTIAAQPLSLRPEACRPAAGGAVHTRGLDFACDIYIWAPLAGTCGVLLSLVITLYC
325	DVQLVQSGAEVKKPGASVKVSKASGYTFTRYTMHWVRQAPGGGLEWIGYINPSRGYTNADSVKGRFTITTDKSTSTAYMELSSLRSEDTATYYCARYYDDPYCLDYWGQGTITVTVSSGGGGSGGGSGGGGSDIVLTQSPATLSLSPGERATLSCRASQSVSYMNWYQQKPKGKAPKRWIYDTSKVASGVPARFSGSGSGTDYSLTINSLEAEDAATYYCQQWSSNPLTFGGGTQKVEIKTTTPAPRPPTPAPTIAAQPLSLRPEACRPAAGGAVHTRGLDFACDIYIWAPLAGTCGVLLSLVITLYC
326	DVQLVQSGAEVKKPGASVKVSKASGYTFTRYTMHWVRQAPGGGLEWIGYINPSRGYTNADSVKGRFTITTDKSTSTAYMELSSLRSEDTATYYCARYYNDQYCLDYWGQGTITVTVSSGGGGSGGGSGGGGSDIVLTQSPATLSLSPGERATLSCRASQSVSYMNWYQQKPKGKAPKRWIYDTSKVASGVPARFSGSGSGTDYSLTINSLEAEDAATYYCQQWSSNPLTFGGGTQKVEIKTTTPAPRPPTPAPTIAAQPLSLRPEACRPAAGGAVHTRGLDFACDIYIWAPLAGTCGVLLSLVITLYC
327	DVQLVQSGAEVKKPGASVKVSKASGYTFTRYTMHWVRQAPGGGLEWIGYINPSRGYTNADSVKGRFTITTDKSTSTAYMELSSLRSEDTATYYCARYYDAHNCCLDYWGQGTITVTVSSGGGGSGGGSGGGGSDIVLTQSPATLSLSPGERATLSCRASQSVSYMNWYQQKPKGKAPKRWIYDTSKVASGVPARFSGSGSGTDYSLTINSLEAEDAATYYCQQWSSNPLTFGGGTQKVEIKTTTPAPRPPTPAPTIAAQPLSLRPEACRPAAGGAVHTRGLDFACDIYIWAPLAGTCGVLLSLVITLYC
328	DVQLVQSGAEVKKPGASVKVSKASGYTFTRYTMHWVRQAPGGGLEWIGYINPSRGYTNADSVKGRFTITTDKSTSTAYMELSSLRSEDTATYYCARYSDDHYCLDYWGQGTITVTVSSGGGGSGGGSGGGGSDIVLTQSPATLSLSPGERATLSCRASQSVSYMNWYQQKPKGKAPKRWIYDTSKVASGVPARFSGSGSGTDYSLTINSLEAEDAATYYCQQWSSNPLTFGGGTQKVEIKTTTPAPRPPTPAPTIAAQPLSLRPEACRPAAGGAVHTRGLDFACDIYIWAPLAGTCGVLLSLVITLYC
329	DVQLVQSGAEVKKPGASVKVSKASGYTFTRYTMHWVRQAPGGGLEWIGYINPSRGYTNADSVKGRFTITTDKSTSTAYMELSSLRSEDTATYYCARYSDDRYCLDYWGQGTITVTVSSGGGGSGGGSGGGGSDIVLTQSPATLSLSPGERATLSCRASQSVSYMNWYQQKPKGKAPKRWIYDTSKVASGVPARFSGSGSGTDYSLTINSLEAEDAATYYCQQWSSNPLTFGGGTQKVEIKTTTPAPRPPTPAPTIAAQPLSLRPEACRPAAGGAVHTRGLDFACDIYIWAPLAGTCGVLLSLVITLYC

TABLE 8-continued

Non-viral membrane bound tropism polypeptides	
SEQ ID NO: AMINO ACID SEQUENCE	
330	DVQLVQSGAEVKKPGASVKVSCKASGYTFTRYTMHWVRQAPGGLEWIGYINPSRGYTNYADS VKGRFTITTDKSTSTAYMELSSLRSEDATYYCARYYDDNYCLDYWGQGTITVTVSSSSGGGGS GGGGSGGGSDIVLTQSPATLSLSPGERATLSCRASQSVSYMNWYQQKPKAPKRWIYDTSKV ASGVPARFSGSGSDYSLTINSLEAEDAATYYCQWSSNPLTFGGGKVEIKTTTPAPRPPT PAPTIASQPLSLRPEACRPAAGGAVHTRGLDFACDIYIWAPLAGTCGVLLLSLVITLYC
331	DVQLVQSGAEVKKPGASVKVSCKASGYTFTRYTMHWVRQAPGGLEWIGYINPSRGYTNYADS VKGRFTITTDKSTSTAYMELSSLRSEDATYYCARYYDDNYCLDYWGQGTITVTVSSSSGGGGS GGGGSGGGSDIVLTQSPATLSLSPGERATLSCRASQSVSYMNWYQQKPKAPKRWIYDTSKVA SGVPARFSGSGSDYSLTINSLEAEDAATYYCQWSSNPLTFGGGKVEIKTTTPAPRPPT PAPTIASQPLSLRPEACRPAAGGAVHTRGLDFACDIYIWAPLAGTCGVLLLSLVITLYC

In particular embodiments, a non-viral membrane bound tropism polypeptide comprises an amino acid sequence set forth in any one of SEQ ID NOs: 324, 325, 326, 327, 328, 329, 330, and 331, preferably SEQ ID NO: 324.

In particular embodiments, a non-viral membrane bound tropism polypeptide consists essentially of an amino acid sequence set forth in any one of SEQ ID NOs: 324, 325, 326, 327, 328, 329, 330, and 331.

In particular embodiments, a non-viral membrane bound tropism polypeptide consists of an amino acid sequence set forth in any one of SEQ ID NOs: 324, 325, 326, 327, 328, 329, 330, and 331.

E. Chimeric Antigen Receptors

Recombinant particles contemplated herein are engineered to efficiently deliver one or more copies of a vector encoding or comprising a promotor operably linked to a polynucleotide encoding an anti-BCMA chimeric antigen receptor. Chimeric antigen receptors (CARs) contemplated herein are fusion polypeptides that exploit antibody-based specificity for BCMA to redirect immune effector cell specificity thereby triggering proliferation, cytokine production, phagocytosis or production of molecules that can mediate cell death of BCMA-expressing cells in a major histocompatibility (MHC) independent manner. As used herein, the term "chimeric" refers to a molecule that is composed of two or more polypeptides, or polynucleotides, of different origins.

The present disclosure contemplates improved anti-BCMA CARs that are suitable for in vivo modification, or ex vivo manufacture, of immune effector cells to redirect cytotoxicity toward BCMA-expressing cells (e.g., B cells, plasma cells).

In various embodiments, a CAR comprises an anti-BCMA scFv or VHH; a hinge domain; a transmembrane domain; one or more costimulatory signaling domains; and a primary signaling domain.

In particular embodiments, a CAR comprises an extracellular antigen binding domain that comprises an antibody or antigen binding fragment thereof that specifically binds to a human BCMA polypeptide. The term "binding domain" or "extracellular antigen binding domain" are used interchangeably and refer to one or more antibodies or antigen binding fragments thereof that bind BCMA. The binding domain may be derived either from a natural, synthetic, semi-synthetic, or recombinant source.

B cell maturation antigen (BCMA) is a member of the tumor necrosis factor receptor superfamily 17 (TNFRSF17) and is highly expressed on the plasma cells of multiple myeloma (MM) patients. The restricted expression of BCMA makes it a suitable therapeutic target for treating multiple myeloma. The present disclosure contemplates antibodies and antigen binding fragments thereof that bind BCMA. An "antibody" refers to a polypeptide or antigen binding fragment thereof that comprises at least a light chain immunoglobulin variable region and/or a heavy chain immunoglobulin variable region, which specifically recognizes and binds one or more epitopes of a BCMA polypeptide, e.g., SEQ ID NO: 10

(MLQMAGQCSQNEYFDSLHACIPCQLRCSNTPLTCQRYCNASVTNSV
KGTNAILWTCLGLSLIISLAVFVLMFLLRKINSEPLKDEFKNTSGLLGM
ANIDLEKSRTGDEIILPRGLETVVEECTCEDCIKSPKVDSDHCFPLPAM
EEGATILVTTKTNDYCKSLPAALSATEIEKSISAR).

In various embodiments, a CAR comprises an anti-BCMA antibody or antigen binding fragment thereof comprising an amino acid sequence set forth in Table 9; a hinge domain; a transmembrane domain; one or more costimulatory signaling domains; and a primary signaling domain.

TABLE 9

anti-BCMA antibodies				
AB ID	SEQ ID NO	ID	AMINO ACID SEQUENCE	
BCMA.1	11	VH	QITLKESGPTLVKPTQTLTLCTFSGFSLSTSGVGVGWIRQPPG KALEWLALIYWDEKRYSPSLKSRLTITKDTSKNQVVLTMTNMD PVDTAVYYCARDEYGGFDIWGQGTMTVTSS	
	12	CDRH1	TSGVGVG	
	13	CDRH2	LIYWDEKRYSPSLKS	

TABLE 9-continued

anti-BCMA antibodies			
AB ID	SEQ ID NO	ID	AMINO ACID SEQUENCE
	14	CDRH3	DEYGGFDI
	15	VL	EIVLTQSPATLSLSPGERATLSCRASQSVSSYLAWYQQKPGQAP RLLIYDASNRTGIPARFSGSGSGTDFTLTISSELEPEDFAVYYC QQRVVYPITFGGGTKVEIK
	16	CDRL1	RASQSVSSYLA
	17	CDRL2	DASNRTAT
	18	CDRL3	QQRVVYPIT
	19	scFv	QITLKESGPTLVKPTQTTLTCTFSGFSLSTSGVGVGWIRQPPG KALEWLALIIYWNDEKRYSPSLKSLRTITKDTSKNQVVLMTNMD PVDTAVYYCARDEYGGFDIWGGTMTVTSSSGGGSGGGSGGGG SGGGGSEIVLTQSPATLSLSPGERATLSCRASQSVSSYLAWYQQ KPGQAPRLLIYDASNRTGIPARFSGSGSGTDFTLTISSELEPED FAVYYCQQRVVYPITFGGGTKVEIK
	20	scFv	EIVLTQSPATLSLSPGERATLSCRASQSVSSYLAWYQQKPGQAP RLLIYDASNRTGIPARFSGSGSGTDFTLTISSELEPEDFAVYYC QQRVVYPITFGGGTKVEIKGGGGSGGGSGGGSGGGGSGGGG ESGPTLVKPTQTTLTCTFSGFSLSTSGVGVGWIRQPPGKALEW LALIIYWNDEKRYSPSLKSLRTITKDTSKNQVVLMTNMDPVDTA VYYCARDEYGGFDIWGGTMTVTSS
BCMA.2	21	VH	QITLKESGPTLVKPTQTTLTCTFSGFSLSTSGVGVGWIRQPPG KALEWLALIIYWNDDKRYSPSLKSLRTITKDTSKNQVVLMTNMD PVDTAVYYCARDEYGGFDIWGGTMTVTSS
	22	CDRH1	TSGVGVG
	23	CDRH2	LIYWNDDKRYSPSLKS
	24	CDRH3	DEYGGFDI
	25	VL	EIVLTQSPATLSLSPGERATLSCRASQSVSSYLAWYQQKPGQAP RLLIYDASNRTGIPARFSGSGSGTDFTLTISSELEPEDFAVYYC QQRFDYPITFGGGTKVEIK
	26	CDRL1	RASQSVSSYLA
	27	CDRL2	DASNRTAT
	28	CDRL3	QQRFDYPIT
	29	scFv	QITLKESGPTLVKPTQTTLTCTFSGFSLSTSGVGVGWIRQPPG KALEWLALIIYWNDDKRYSPSLKSLRTITKDTSKNQVVLMTNMD PVDTAVYYCARDEYGGFDIWGGTMTVTSSSGGGSGGGSGGGG SGGGGSEIVLTQSPATLSLSPGERATLSCRASQSVSSYLAWYQQ KPGQAPRLLIYDASNRTGIPARFSGSGSGTDFTLTISSELEPED FAVYYCQQRFDYPITFGGGTKVEIK
	30	scFv	EIVLTQSPATLSLSPGERATLSCRASQSVSSYLAWYQQKPGQAP RLLIYDASNRTGIPARFSGSGSGTDFTLTISSELEPEDFAVYYC QQRFDYPITFGGGTKVEIKGGGGSGGGSGGGSGGGGSGGGG ESGPTLVKPTQTTLTCTFSGFSLSTSGVGVGWIRQPPGKALEW LALIIYWNDDKRYSPSLKSLRTITKDTSKNQVVLMTNMDPVDTA VYYCARDEYGGFDIWGGTMTVTSS
BCMA.3	31	VH	QVQLVESGGGVVQPGRLRLSCAASGFTFSSYGMHWVRQAPGKG LEWVAVISYEGSNKYADSVKGRFTISRDNKNTLYLQMNSLRA EDTAVYYCARELGDMGVWGQTTVTSS
	32	CDRH1	SYGMH
	33	CDRH2	VISYEGSNKYADSVKG
	34	CDRH3	ELGDGMDV
	35	VL	EIVLTQSPATLSLSPGERATLSCRASQSVSSYLAWFQQKPGQAP RLLIYDASNRTGIPARFSGSGSGTDFTLTISSELEPEDFAVYYC QQRVDLWTFGGGTVEIK
	36	CDRL1	RASQSVSSYLA
	37	CDRL2	DASNRTAT
	38	CDRL3	QQRVDLWT
	39	scFv	QVQLVESGGGVVQPGRLRLSCAASGFTFSSYGMHWVRQAPGKG LEWVAVISYEGSNKYADSVKGRFTISRDNKNTLYLQMNSLRA EDTAVYYCARELGDMGVWGQTTVTSSSGGGSGGGSGGGG GGGGSEIVLTQSPATLSLSPGERATLSCRASQSVSSYLAWFQQK PGQAPRLLIYDASNRTGIPARFSGSGSGTDFTLTISSELEPEDF AVYYCQQRVDLWTFGGGTVEIK
	40	scFv	EIVLTQSPATLSLSPGERATLSCRASQSVSSYLAWFQQKPGQAP RLLIYDASNRTGIPARFSGSGSGTDFTLTISSELEPEDFAVYYC QQRVDLWTFGGGTVEIKGGGGSGGGSGGGSGGGGSGGGG SGGGVVQPGRLRLSCAASGFTFSSYGMHWVRQAPGKLEWVAV ISYEGSNKYADSVKGRFTISRDNKNTLYLQMNSLRAEDTAVY YCARELGDMGVWGQTTVTSS
BCMA.4	41	VH	EVQLVESGGGLVKPGGSLRLSCAASGFTFSDYYMSWIRQAPGKG LEWVSYISSSGSTIYADSVKGRFTISRDNKNTLYLQMNSLRA EDTAVYYCARDQNGYGVWVGQTTVTSS

TABLE 9-continued

anti-BCMA antibodies				
AB ID	SEQ ID NO	ID	AMINO ACID SEQUENCE	
	42	CDRH1	DYMS	
	43	CDRH2	YISSSGSTIYYADSVKG	
	44	CDRH3	DQGNYGVDV	
	45	VL	DIQMTQSPSTLSASVGDRTITCRASQSISSWLAWYQQKPGKAP KLLIYDASSLESQVPSRFSGSGSGTEFTLTISLQPDFFATYYC QQVSSLPTFTGGGKVEIK	
	46	CDRL1	RASQSISSWLA	
	47	CDRL2	DASSLES	
	48	CDRL3	QQVSSLPT	
	49	scFv	EVQLVESGGGLVHPGGSLRLSCAASGFTFSDIYMSWIRQAPGKG LEWVSYISSSGSTIYYADSVKGRFTISRDNKNSLYLQMNSLRA EDTAVYYCARDQGNYGVDVWGQGTITVTVSS SGGGGSDIQMTQSPSTLSASVGDRTITCRASQSISSWLAWYQQ KPGKAPKLLIYDASSLESQVPSRFSGSGSGTEFTLTISLQPDFFATYYCQQVSSLPTFTGGGKVEIK DIQMTQSPSTLSASVGDRTITCRASQSISSWLAWYQQKPGKAP KLLIYDASSLESQVPSRFSGSGSGTEFTLTISLQPDFFATYYC QQVSSLPTFTGGGKVEIKGGGGSGGGSGGGSGGGSEVQLV ESGGGLVHPGGSLRLSCAASGFTFSDIYMSWIRQAPGKLEWVS YISSSGSTIYYADSVKGRFTISRDNKNSLYLQMNSLRAEDTAV YYCARDQGNYGVDVWGQGTITVTVSS	
BCMA.5	51	VH	QVQLVESGGGLVHPGGSLRLSCAASGFTFSDIYMSWIRQAPGKG LEWVSYISSSGSTIYYADSVKGRFTISRDNKNSLYLQMNSLRA EDTAVYYCARDQGNYGVDVWGQGTITVTVSS	
	52	CDRH1	DYMS	
	53	CDRH2	YISSSGSTIYYADSVKG	
	54	CDRH3	DQGNYGVDV	
	55	VL	DIQMTQSPSTLSASVGDRTITCRASQSISSWLAWYQQKPGKAP KLLIYEASSLESQVPSRFSGSGSGTEFTLTISLQPDFFATYYC QQSDSHPIFTGGGKVEIK	
	56	CDRL1	RASQSISSWLA	
	57	CDRL2	EASSLES	
	58	CDRL3	QQSDSHPI	
	59	scFv	QVQLVESGGGLVHPGGSLRLSCAASGFTFSDIYMSWIRQAPGKG LEWVSYISSSGSTIYYADSVKGRFTISRDNKNSLYLQMNSLRA EDTAVYYCARDQGNYGVDVWGQGTITVTVSS SGGGGSDIQMTQSPSTLSASVGDRTITCRASQSISSWLAWYQQ KPGKAPKLLIYEASSLESQVPSRFSGSGSGTEFTLTISLQPDFFATYYCQQSDSHPIFTGGGKVEIK DIQMTQSPSTLSASVGDRTITCRASQSISSWLAWYQQKPGKAP KLLIYEASSLESQVPSRFSGSGSGTEFTLTISLQPDFFATYYC QQSDSHPIFTGGGKVEIKGGGGSGGGSGGGSGGGSEVQLV ESGGGLVHPGGSLRLSCAASGFTFSDIYMSWIRQAPGKLEWVS YISSSGSTIYYADSVKGRFTISRDNKNSLYLQMNSLRAEDTAV YYCARDQGNYGVDVWGQGTITVTVSS	
	60	scFv	DIQMTQSPSTLSASVGDRTITCRASQSISSWLAWYQQKPGKAP KLLIYEASSLESQVPSRFSGSGSGTEFTLTISLQPDFFATYYC QQSDSHPIFTGGGKVEIKGGGGSGGGSGGGSGGGSEVQLV ESGGGLVHPGGSLRLSCAASGFTFSDIYMSWIRQAPGKLEWVS YISSSGSTIYYADSVKGRFTISRDNKNSLYLQMNSLRAEDTAV YYCARDQGNYGVDVWGQGTITVTVSS	
BCMA.6	61	VH	EVQLVESGGGLVHPGGSLRLSCAASGFTFSDIYMSWIRQAPGKG LEWVSYISSSGSTIYYADSVKGRFTISRDNKNSLYLQMNSLRA EDTAVYYCARDQGNYGVDVWGQGTITVTVSS	
	62	CDRH1	DYMS	
	63	CDRH2	YISSSGSTIYYADSVKG	
	64	CDRH3	DQGNYGVDV	
	65	VL	DIQMTQSPSTLSASVGDRTITCRASQSISSWLAWYQQKPGKAP KLLIYEASSLESQVPSRFSGSGSGTEFTLTISLQPDFFATYYC QQANSHPIFTGGGKVEIK	
	66	CDRL1	RASQSISSWLA	
	67	CDRL2	EASSLES	
	68	CDRL3	QQANSHPI	
	69	scFv	EVQLVESGGGLVHPGGSLRLSCAASGFTFSDIYMSWIRQAPGKG LEWVSYISSSGSTIYYADSVKGRFTISRDNKNSLYLQMNSLRA EDTAVYYCARDQGNYGVDVWGQGTITVTVSS SGGGGSDIQMTQSPSTLSASVGDRTITCRASQSISSWLAWYQQ KPGKAPKLLIYEASSLESQVPSRFSGSGSGTEFTLTISLQPDFFATYYCQQANSHPIFTGGGKVEIK DIQMTQSPSTLSASVGDRTITCRASQSISSWLAWYQQKPGKAP KLLIYEASSLESQVPSRFSGSGSGTEFTLTISLQPDFFATYYC QQANSHPIFTGGGKVEIKGGGGSGGGSGGGSGGGSEVQLV ESGGGLVHPGGSLRLSCAASGFTFSDIYMSWIRQAPGKLEWVS YISSSGSTIYYADSVKGRFTISRDNKNSLYLQMNSLRAEDTAV YYCARDQGNYGVDVWGQGTITVTVSS	
	70	scFv	DIQMTQSPSTLSASVGDRTITCRASQSISSWLAWYQQKPGKAP KLLIYEASSLESQVPSRFSGSGSGTEFTLTISLQPDFFATYYC QQANSHPIFTGGGKVEIKGGGGSGGGSGGGSGGGSEVQLV ESGGGLVHPGGSLRLSCAASGFTFSDIYMSWIRQAPGKLEWVS YISSSGSTIYYADSVKGRFTISRDNKNSLYLQMNSLRAEDTAV YYCARDQGNYGVDVWGQGTITVTVSS	

TABLE 9-continued

anti-BCMA antibodies				
AB ID	SEQ ID NO	ID	AMINO ACID SEQUENCE	
BCMA.7	71	VH	EVQLLES GGLVQP GGSRLRLSCAASGFTFSNYAMSWVRQAPGKG LEWVSAISGSGGSTYYADSVKGRFTISRDN SKNTLYLQMNSLRA EDTAVYYCARPGDGYEGVYFDYWGQGLVTVSS	
	72	CDRH1	NYAMS	
	73	CDRH2	AISGSGGSTYYADSVKG	
	74	CDRH3	PGDGYEGVYFDY	
	75	VL	DIQMTQSPSSLASVGDRTITCRASQSISSYLNWYQQKPKGKAP KLLIYAASSLQSGVPSRFSGSGSGTDFTLTISSLQPEDFATYYC QQAHS SPITFGGGTKVEIK	
	76	CDRL1	RASQSISSYLN	
	77	CDRL2	AASSLQS	
	78	CDRL3	QQAHS SPIT	
	79	scFv	EVQLLES GGLVQP GGSRLRLSCAASGFTFSNYAMSWVRQAPGKG LEWVSAISGSGGSTYYADSVKGRFTISRDN SKNTLYLQMNSLRA EDTAVYYCARPGDGYEGVYFDYWGQGLVTVSSGGGGSGGGGS GGGGSGGGSDIQMTQSPSSLASVGDRTITCRASQSISSYLN WYQQKPKGKAPKLLIYAASSLQSGVPSRFSGSGSGTDFTLTISSL QPEDFATYYCQQAHS SPITFGGGTKVEIK	
	80	scFv	DIQMTQSPSSLASVGDRTITCRASQSISSYLNWYQQKPKGKAP KLLIYAASSLQSGVPSRFSGSGSGTDFTLTISSLQPEDFATYYC QQAHS SPITFGGGTKVEIKGGGGSGGGSGGGSGGGSEVQLL ESGGLVQP GGSRLRLSCAASGFTFSNYAMSWVRQAPGKLEWVS AISGSGGSTYYADSVKGRFTISRDN SKNTLYLQMNSLRAEDTAV YYCARPGDGYEGVYFDYWGQGLVTVSS	
BCMA.8	81	VH	QITLKESGPTLVKPTQTTLTCTFSGFSLSTSGVGVGWIRQPPG KALEWLALIYWDEKRYSPSLKSLRTITKDT SKNQVVLMTNMND PVD TAVYYCAREGSHDYKSSNWEDPWGQGLVTVSS	
	82	CDRH1	TSGVGVG	
	83	CDRH2	LIYWDEKRYSPSLKS	
	84	CDRH3	EGSHDYKSSNWEDP	
	85	VL	DIQMTQSPSSLASVGDRTITCQASQDIANYLNWYQQKPKGKAP KLLIYDASNLETGVP SRFSGSGSGTDFTFTISSLQPEDIATYYC QQHFNLP LTFGGGTKEIK	
	86	CDRL1	QASQDIANYLN	
	87	CDRL2	DASNLET	
	88	CDRL3	QQHFNLP LTF	
	89	scFv	QITLKESGPTLVKPTQTTLTCTFSGFSLSTSGVGVGWIRQPPG KALEWLALIYWDEKRYSPSLKSLRTITKDT SKNQVVLMTNMND PVD TAVYYCAREGSHDYKSSNWEDPWGQGLVTVSSGGGGSGGG GSGGGSGGGSDIQMTQSPSSLASVGDRTITCQASQDIANY LNWYQQKPKGKAPKLLIYDASNLETGVP SRFSGSGSGTDFTFTIS SLQPEDIATYYCQQHFNLP LTFGGGTKEIK	
	90	scFv	DIQMTQSPSSLASVGDRTITCQASQDIANYLNWYQQKPKGKAP KLLIYDASNLETGVP SRFSGSGSGTDFTFTISSLQPEDIATYYC QQHFNLP LTFGGGTKEIKGGGGSGGGSGGGSGGGSQITLK ESGPTLVKPTQTTLTCTFSGFSLSTSGVGVGWIRQPPGKALEW LALIYWDEKRYSPSLKSLRTITKDT SKNQVVLMTNMNDPVD TA VYYCAREGSHDYKSSNWEDPWGQGLVTVSS	
BCMA.9	91	VH	EVQLVESGGGLVKPGGSLRLSCAASGFTFSYS MNWVRQAPGKG LEWVSSISSSSSYIYYADSVKGRFTISRDN AKNSLYLQMNSLRA EDTAVYYCARAGDTYSAADYYMDVWGKGT TTVTVSS	
	92	CDRH1	SYS MN	
	93	CDRH2	SISSSSSYIYYADSVKG	
	94	CDRH3	AGDTYSAADYYMDV	
	95	VL	DIVMTQSPSLPVP TPGEPASISCRSSQSL LHSNGYNYLDWYLQK PGQSPQLLIYLG SNRASGV PDRFSGSGSGTDFTLKI SRVEADV GVYYCMQALGLITFGGGTKVEIK	
	96	CDRL1	RSSQSL LHSNGYNYLD	
	97	CDRL2	LGSN RAS	
	98	CDRL3	MQALGLIT	
	99	scFv	EVQLVESGGGLVKPGGSLRLSCAASGFTFSYS MNWVRQAPGKG LEWVSSISSSSSYIYYADSVKGRFTISRDN AKNSLYLQMNSLRA EDTAVYYCARAGDTYSAADYYMDVWGKGT TTVTVSSGGGGSGGG GSGGGSGGGSDIVMTQSPSLPVP TPGEPASISCRSSQSL LHS NGYNYLDWYLQKPGQSPQLLIYLG SNRASGV PDRFSGSGSGTDF TLKISRVEADVGVYYCMQALGLITFGGGTKVEIK	
	100	scFv	DIVMTQSPSLPVP TPGEPASISCRSSQSL LHSNGYNYLDWYLQK PGQSPQLLIYLG SNRASGV PDRFSGSGSGTDFTLKI SRVEADV GVYYCMQALGLITFGGGTKVEIKGGGGSGGGSGGGSGGGSE	

TABLE 9-continued

anti-BCMA antibodies			
AB ID	SEQ ID NO	ID	AMINO ACID SEQUENCE
			VQLVESGGGLVQPGGSLRLSCAASGFTFSSYSMNWVRQAPGKGL EWVSSISSSSYIYYADSVKGRFTISRDNKNTLYLQMNSLRAE DTAVYYCARAGDTYSAADYYMDVWGKGTQTVTVSS
BCMA.10	101	VHH	EVQLLESGGGLVQPGGSLRLSCAASGFTFGSEAMSWVRQAPGKE RELVSIAISGSGEVITYYADSVKGRFTISRDNKNTLYLQMNSLRA EDTAVYYCQRLVEAKRHWGQGTQTVTVSS
	102	CDRH1	SEAMS
	103	CDRH2	AISGSGEVITYYADSVKG
	104	CDRH3	LVEAKRH
BCMA.11	105	VHH	EVQLLESGGGLVQPGGSLRLSCAASGFTFESEAMSWYRQAPGKE RELVSVITSEGSTYYADSVKGRFTISRDNKNTLYLQMNSLRAE DTAVYYCAHIEWETRLNWGQGTQTVTVSS
	106	CDRH1	SEAMS
	107	CDRH2	VITSEGSTYYADSVKG
	108	CDRH3	IEWETRLN
BCMA.12	109	VHH	EVQLLESGGGLVQPGGSLRLSCAASGFTFDEYTMHWFRQAPGKE REFVSAISGGGSETYYADSVKGRFTISRDNKNTLYLQMNSLRA EDTAVYYCAAGGEEAGVGWGGTQTVTVSS
	110	CDRH1	EYTMH
	111	CDRH2	AISGGGSETYYADSVKG
	112	CDRH3	GGEEAGVG
BCMA.13	113	VHH	EVQLLESGGGLVQPGGSLRLSCAASGFTFEDYAMSWFRQAPGKE REGVSAISGKGGSTYYADSVKGRFTISRDNKNTLYLQMNSLRA EDTAVYYCAVLDEEAGAEGGYWGQGTQTVTVSS
	114	CDRH1	DYAMS
	115	CDRH2	AISGKGGSTYYADSVKG
	116	CDRH3	LDEEAGAEGGY
BCMA.14	117	VHH	EVQLLESGGGLVQPGGSLRLSCAASGFTFDYAMSWFRQAPGKE REGVSAISTSGDSTYYADSVKGRFTISRDNKNTLYLQMNSLRA EDTAVYYCAVLDEEAGAEGGYWGQGTQTVTVSS
	118	CDRH1	RYAMS
	119	CDRH2	AISTSGDSTYYADSVKG
	120	CDRH3	LDEEAGAEGGY
BCMA.15	121	VHH	EVQLLESGGGLVQPGGSLRLSCAASGFTFASDAMSWYRQAPGKE RELVSIAISGSGGSTYYADSVKGRFTISRDNKNTLYLQMNSLRA EDTAVYYCAAHDSGEAYLAFDYWGQGTQTVTVSS
	122	CDRH1	SDAMS
	123	CDRH2	AISGSGGSTYYADSVKG
	124	CDRH3	HDSGEAYLAFDY
BCMA.16	125	VHH	EVQLLESGGGLVQPGGSLRLSCAASGFTFDSYTMWYRQAPGKE RELVSIAISGHGDSTYYADSVKGRFTISRDNKNTLYLQMNSLRA EDTAVYYCTRISITTEWLAGDYWGQGTQTVTVSS
	126	CDRH1	SYTMS
	127	CDRH2	AISGHGDSTYYADSVKG
	128	CDRH3	ISITTEWLAGDY
BCMA.17	129	VHH	EVQLLESGGGLVQPGGSLRLSCAASGFTFSSYAMSWFRQAPGKE REFVSFISGSGDSTYYADSVKGRFTISRDNKNTLYLQMNSLRA EDTAVYYCTRWPYDFEEPSEPGVYWGQGTQTVTVSS
	130	CDRH1	SYAMS
	131	CDRH2	FISGSGDSTYYADSVKG
	132	CDRH3	WPYDFEEPSEPGVY
BCMA.18	133	VHH	EVQLLESGGGLVQPGGSLRLSCAASGFTFTDYDMSWYRQAPGKE RELVSVIHSGGSTYYADSVKGRFTISRDNKNTLYLQMNSLRAE DTAVYYCAPGYSDLSFDYINFDYWGQGTQTVTVSS
	134	CDRH1	DYDMS
	135	CDRH2	VIHSGGSTYYADSVKG
	136	CDRH3	GYYSDSLFDYINFDY
BCMA.19	137	VHH	EVQLLESGGGLVQPGGSLRLSCAASGFTFSDYAMHWFRQAPGKE RVLVSSIDSGGSTYYADSVKGRFTISRDNKNTLYLQMNSLRAE DTAVYYCNAGFKGDHPHKDAFDIWGQGTQTVTVSS
	138	CDRH1	DYAMH
	139	CDRH2	SIDSGGSTYYADSVKG
	140	CDRH3	GFKGDHPHKDAFDI

TABLE 9-continued

anti-BCMA antibodies			
AB ID	SEQ ID NO	ID	AMINO ACID SEQUENCE
BCMA.20	141	VHH	EVQLLESGGGLVQPGGSLRLSCAASGFTFSSEGMSWVRQAPGKE RELVSVAISGSGDHTYYADSVRGRTISRDNKNTLYLQMNLSRA EDTAVYYCNALEGGPTTAIQPGGPDYWGQGTQVTVSS
	142	CDRH1	SEGMS
	143	CDRH2	AISGSGDHTYYADSVRG
	144	CDRH3	LEGGPTTAIQPGGPDY

In particular embodiments, a CAR comprises a binding domain that comprises an anti-BCMA scFv comprising an amino acid sequence set forth in any one of SEQ ID NOs: 19, 20, 29, 30, 39, 40, 49, 50, 59, 60, 69, 70, 79, 80, 89, 90, 99, and 100; a hinge domain; a transmembrane domain; one or more costimulatory signaling domains; and a primary signaling domain. In particular embodiments, a CAR comprises a binding domain that comprises an anti-BCMA scFv comprising an amino acid sequence set forth in any one of SEQ ID NOs: 20, 30, 39, 50, 59, 70, 80, 90, and 100; a hinge domain; a transmembrane domain; one or more costimulatory signaling domains; and a primary signaling domain. In particular embodiments, a CAR comprises a binding domain that comprises an anti-BCMA scFv comprising an amino acid sequence set forth in any one of SEQ ID NOs: 39, 59, 70, and 90; a hinge domain; a transmembrane domain; one or more costimulatory signaling domains; and a primary signaling domain.

In particular embodiments, a CAR comprises a binding domain that comprises an anti-BCMA VHH comprising an amino acid sequence set forth in any one of SEQ ID NOs: 101, 105, 109, 113, 117, 121, 125, 129, 133, 137, and 141; a hinge domain; a transmembrane domain; one or more costimulatory signaling domains; and a primary signaling domain. In particular embodiments, a CAR comprises a binding domain that comprises an anti-BCMA VHH comprising an amino acid sequence set forth in SEQ ID NO: 101 or 117; a hinge domain; a transmembrane domain; one or more costimulatory signaling domains; and a primary signaling domain.

Chimeric antigen receptors contemplated herein comprise a hinge domain disposed between the extracellular antigen binding domain and the transmembrane domain of a CAR. A hinge domain plays a role in positioning the extracellular antigen binding domain away from the effector cell surface to enable proper cell/cell contact, antigen binding and activation. A hinge domain may be derived from a naturally occurring polypeptide or from a synthetic, semi-synthetic, or recombinant source.

In particular embodiments, a CAR comprises a binding domain that comprises an anti-BCMA scFv comprising an amino acid sequence set forth in any one of SEQ ID NOs: 19, 20, 29, 30, 39, 40, 49, 50, 59, 60, 69, 70, 79, 80, 89, 90, 99, and 100; a hinge domain isolated from CD4, CD8 α , CD28, IgG1, and IgG4; a transmembrane domain; one or more costimulatory signaling domains; and a primary signaling domain. In particular embodiments, a CAR comprises a binding domain that comprises an anti-BCMA scFv comprising an amino acid sequence set forth in any one of SEQ ID NOs: 39, 59, 70, and 90; a hinge domain comprising an amino acid sequence set forth in any one of SEQ ID NOs:

225, 226, 227, 228, 229 and 230; a transmembrane domain; one or more costimulatory signaling domains; and a primary signaling domain.

In particular embodiments, a CAR comprises a binding domain that comprises an anti-BCMA VHH comprising an amino acid sequence set forth in any one of SEQ ID NOs: 101, 105, 109, 113, 117, 121, 125, 129, 133, 137, and 141; a hinge domain isolated from CD4, CD8 α , CD28, IgG1, and IgG4; a transmembrane domain; one or more costimulatory signaling domains; and a primary signaling domain. In particular embodiments, a CAR comprises a binding domain that comprises an anti-BCMA VHH comprising an amino acid sequence set forth in SEQ ID NO: 101 or 117; a hinge domain comprising an amino acid sequence set forth in any one of SEQ ID NOs: 225, 226, 227, 228, 229 and 230; a transmembrane domain; one or more costimulatory signaling domains; and a primary signaling domain.

Chimeric antigen receptors contemplated herein comprise a transmembrane domain. The transmembrane domain is a hydrophobic domain that fuses the extracellular and intracellular portions of the CAR and anchors the CAR to the plasma membrane of the immune effector cell. The transmembrane domain may be derived either from a natural, synthetic, semi-synthetic, or recombinant source. In particular embodiments, the CAR further comprises a short oligo- or polypeptide linker, preferably between 1, 2, 3, 4, 5, 6, 7, 8, 9, or 10 amino acids in length disposed between the transmembrane domain and the intracellular domains of the CAR.

In particular embodiments, a CAR comprises a binding domain that comprises an anti-BCMA scFv comprising an amino acid sequence set forth in any one of SEQ ID NOs: 19, 20, 29, 30, 39, 40, 49, 50, 59, 60, 69, 70, 79, 80, 89, 90, 99, and 100; a hinge domain isolated from CD4, CD8 α , CD28, IgG1, and IgG4; a transmembrane domain isolated or derived from a polypeptide selected from the group consisting of CD3 ϵ , CD4, CD8 α , CD28, CD33, CD134, CD137, and CD278; one or more costimulatory signaling domains; and a primary signaling domain. In particular embodiments, a CAR comprises a binding domain that comprises an anti-BCMA scFv comprising an amino acid sequence set forth in any one of SEQ ID NOs: 39, 59, 70, and 90; a hinge domain comprising an amino acid sequence set forth in any one of SEQ ID NOs: 225, 227, 228, 229, and 230; a transmembrane domain comprising an amino acid sequence set forth in any one of SEQ ID NOs: 231, 323, 233, 234, 235, 236, and 237; one or more costimulatory signaling domains; and a primary signaling domain.

In particular embodiments, a CAR comprises a binding domain that comprises an anti-BCMA VHH comprising an amino acid sequence set forth in any one of SEQ ID NOs: 101, 105, 109, 113, 117, 121, 125, 129, 133, 137, and 141; a hinge domain isolated from CD4, CD8 α , CD28, IgG1, and

IgG4; a transmembrane domain isolated or derived from a polypeptide selected from the group consisting of CD3ε, CD4, CD8α, CD28, CD33, CD134, CD137, and CD278; one or more costimulatory signaling domains; and a primary signaling domain. In particular embodiments, a CAR comprises a binding domain that comprises an anti-BCMA VHH comprising an amino acid sequence set forth in SEQ ID NO: 101 or 117; a hinge domain comprising an amino acid sequence set forth in any one of SEQ ID NOs: 225, 227, 228, 229, and 230; a transmembrane domain comprising an amino acid sequence set forth in any one of SEQ ID NOs: 231, 323, 233, 234, 235, 236, and 237; one or more costimulatory signaling domains; and a primary signaling domain.

Chimeric antigen receptors contemplated herein comprise on or more intracellular signaling domains that function to transduce a signal of extracellular antigen recognition to the interior of the immune effector cell and elicit one or more effector cell functions including but not limited to activation, cytokine production, proliferation and cytotoxic activity. T cell activation is mediated by two distinct classes of intracellular signaling domains: primary signaling domains that initiate antigen-dependent primary activation through the TCR (e.g., a TCR/CD3 complex) and costimulatory signaling domains that act in an antigen-independent manner to provide a secondary or costimulatory signal. The intracellular primary signaling and costimulatory signaling domains may be linked in any order in tandem to the carboxyl terminus of the transmembrane domain.

In particular embodiments, a CAR comprises a binding domain that comprises an anti-BCMA scFv comprising an

set forth in any one of SEQ ID NOs: 225, 227, 228, 229, and 230; a transmembrane domain comprising an amino acid sequence set forth in any one of SEQ ID NOs: 231, 323, 233, 234, 235, 236, and 237; a costimulatory signaling domain comprising an amino acid sequence set forth in any one of SEQ ID NOs: 239, 240, 241, 242, 243, and 244; and a primary signaling domain comprising an amino acid sequence set forth in SEQ ID NO: 238.

In particular embodiments, a CAR comprises a binding domain that comprises an anti-BCMA VHH comprising an amino acid sequence set forth in any one of SEQ ID NOs: 101, 105, 109, 113, 117, 121, 125, 129, 133, 137, and 141; a hinge domain isolated from CD4, CD8α, CD28, IgG1, and IgG4; a transmembrane domain isolated or derived from a polypeptide selected from the group consisting of CD3ε, CD4, CD8α, CD28, CD33, CD134, CD137, and CD278; a costimulatory signaling domain isolated or derived from a polypeptide selected from the group consisting of CD27, CD28, CD134, CD137, CD278, and TNRF2; and a CD3ζ primary signaling domain. In particular embodiments, a CAR comprises a binding domain that comprises an anti-BCMA VHH comprising an amino acid sequence set forth in SEQ ID NO: 101 or 117; a hinge domain comprising an amino acid sequence set forth in any one of SEQ ID NOs: 225, 227, 228, 229, and 230; a transmembrane domain comprising an amino acid sequence set forth in any one of SEQ ID NOs: 231, 323, 233, 234, 235, 236, and 237; a costimulatory signaling domain comprising an amino acid sequence set forth in any one of SEQ ID NOs: 239, 240, 241, 242, 243, and 244; and a primary signaling domain comprising an amino acid sequence set forth in SEQ ID NO: 238.

TABLE 10

SEQ ID NO:	DOMAIN	AMINO ACID SEQUENCE
238	PRIMARY	RVKFSRSADAPAYQQGQNLYNELNLGRREYDVLDRKRRDPEMGGKPRKKNPQEGLYNELQKDKMAEAYSEIGMKGERRRGKGHDGLYQGLSTATKDTYDALHMQALPPR
239	COSTIM	KRGRKKLLYIFKQPFMRPVQTTQEEDGCSCRFPEEEEGGCEL
240	COSTIM	RSKRSRLHSDYMNMTPRRPGPTRKHYPYAPPRDFAAYRS
241	COSTIM	ALYLLRRDQRLPPDAHKKPPGGSFRTPIQEEQADAHSLAKI
242	COSTIM	TKKKYSSSVHDPNGEYMFMRVNTAKKSRLTDVTL
243	COSTIM	QRRKYRSNKGESPVPEAPCHYSCPREEGSTIPIQEDYRKPEPACSP
244	COSTIM	KKKPLCLQREAKVPHLPADKARGTQGPQQHLLITAPSSSSSSLESSASALDRRAPTRNQAPGVEASGAGEARASTGSSDSSPGHGHTQVNVTCIVNVCSSSDHSSQCSSQASSTMGDTDSSPSESPPKDEQVPFSGKEECAFRSQLETPETLLGSTEEKPLPLGVPDAGMKPS

amino acid sequence set forth in any one of SEQ ID NOs: 19, 20, 29, 30, 39, 40, 49, 50, 59, 60, 69, 70, 79, 80, 89, 90, 99, and 100; a hinge domain isolated from CD4, CD8α, CD28, IgG1, and IgG4; a transmembrane domain isolated or derived from a polypeptide selected from the group consisting of CD3ε, CD4, CD8α, CD28, CD33, CD134, CD137, and CD278; a costimulatory signaling domain isolated or derived from a polypeptide selected from the group consisting of CD27, CD28, CD134, CD137, CD278, and TNRF2; and a CD3 ζ primary signaling domain. In particular embodiments, a CAR comprises a binding domain that comprises an anti-BCMA scFv comprising an amino acid sequence set forth in any one of SEQ ID NOs: 39, 59, 70, and 90; a hinge domain comprising an amino acid sequence

In particular embodiments, a CAR comprises an scFv or VHH comprising an amino acid sequence set forth in Table 9; a CD8α hinge, a CD8α transmembrane domain; a CD137 costimulatory signaling domain; and a CD3ζ primary signaling domain.

In particular embodiments, a CAR comprises a binding domain that comprises an anti-BCMA scFv comprising an amino acid sequence set forth in any one of SEQ ID NOs: 19, 20, 29, 30, 39, 40, 49, 50, 59, 60, 69, 70, 79, 80, 89, 90, 99, and 100; a CD8α hinge, a CD8α transmembrane domain; a CD137 costimulatory signaling domain; and a CD3 primary signaling domain. In particular embodiments, a CAR comprises a binding domain that comprises an anti-BCMA scFv comprising an amino acid sequence set

47

forth in any one of SEQ ID NOs: 39, 59, 70, and 90; a CD8 α hinge, a CD8 α transmembrane domain; a CD137 costimulatory signaling domain; and a CD3 ζ primary signaling domain.

In particular embodiments, a CAR comprises a binding domain that comprises an anti-BCMA VHH comprising an amino acid sequence set forth in any one of SEQ ID NOs: 101, 105, 109, 113, 117, 121, 125, 129, 133, 137, and 141; a CD8 α hinge, a CD8 α transmembrane domain; a CD137

48

costimulatory signaling domain; and a CD3 ζ primary signaling domain. In particular embodiments, a CAR comprises a binding domain that comprises an anti-BCMA VHH comprising an amino acid sequence set forth in SEQ ID NO: 101 or 117; a CD8 α hinge, a CD8 α transmembrane domain; a CD137 costimulatory signaling domain; and a CD3 ζ primary signaling domain.

In some embodiments, a CAR comprises an amino acid sequence set forth in Table 11.

TABLE 11

Chimeric antigen receptors	
SEQ ID NO.	AMINO ACID SEQUENCE
255	QITLKESGPTLVKPTQTTLTLCTFSGFSLSTSGVGVGWIRQPPGKALEWLALIIYNDEKRYSPS LKSRLTITKDTSKNQVVLMTNMDPVDTA VYYCARDEYGGFDIWGGTMTVTVSSGGGSGGGGS GGGSGGGGSEIVLTQSPATLSLSPGERATLSCRASQSVSSYLAWYQQKPGQAPRLLIYDASNR ATGIPARFSGSGSDFTLTITSSLEPEDFAVYYCQQRVYPIITFGGGTKVEIKTTTPAPRPPTP APTIASQPLSLRPEACRPAAGGAVHTRGLDFACDIYIWAPLAGTCGVLLSLVITLYCKRGRKK LLYIFKQPFMRPVQTTQEEDGCS CRFP EEEEGGCELRVKFSRSADAPAYQQGQNLINELNLGR REEYDVLDKRRGRDPEMGGKPRRKNPQEGLYNELQKDKMAEAYSEIGMKGERRRGKGHDGLYQG LSTATKDTYDALHMQALPPR
256	EIVLTQSPATLSLSPGERATLSCRASQSVSSYLAWYQQKPGQAPRLLIYDASNRATGIPARFSG SGSGDFTLTITSSLEPEDFAVYYCQQRVYPIITFGGGTKVEIKGGGSGGGGSGGGGSGGGGSQ ITLKESGPTLVKPTQTTLTLCTFSGFSLSTSGVGVGWIRQPPGKALEWLALIIYNDEKRYSPSL KSRLTITKDTSKNQVVLMTNMDPVDTA VYYCARDEYGGFDIWGGTMTVTVSSTTPAPRPPTP APTIASQPLSLRPEACRPAAGGAVHTRGLDFACDIYIWAPLAGTCGVLLSLVITLYCKRGRKK LLYIFKQPFMRPVQTTQEEDGCS CRFP EEEEGGCELRVKFSRSADAPAYQQGQNLINELNLGR REEYDVLDKRRGRDPEMGGKPRRKNPQEGLYNELQKDKMAEAYSEIGMKGERRRGKGHDGLYQG LSTATKDTYDALHMQALPPR
257	QITLKESGPTLVKPTQTTLTLCTFSGFSLSTSGVGVGWIRQPPGKALEWLALIIYNDDKRYSPS LKSRLTITKDTSKNQVVLMTNMDPVDTA VYYCARDEYGGFDIWGGTMTVTVSSGGGSGGGGS GGGSGGGGSEIVLTQSPATLSLSPGERATLSCRASQSVSSYLAWYQQKPGQAPRLLIYDASNR ATGIPARFSGSGSDFTLTITSSLEPEDFAVYYCQQRFDYPIITFGGGTKVEIKTTTPAPRPPTP APTIASQPLSLRPEACRPAAGGAVHTRGLDFACDIYIWAPLAGTCGVLLSLVITLYCKRGRKK LLYIFKQPFMRPVQTTQEEDGCS CRFP EEEEGGCELRVKFSRSADAPAYQQGQNLINELNLGR REEYDVLDKRRGRDPEMGGKPRRKNPQEGLYNELQKDKMAEAYSEIGMKGERRRGKGHDGLYQG LSTATKDTYDALHMQALPPR
258	EIVLTQSPATLSLSPGERATLSCRASQSVSSYLAWYQQKPGQAPRLLIYDASNRATGIPARFSG SGSGDFTLTITSSLEPEDFAVYYCQQRFDYPIITFGGGTKVEIKGGGSGGGGSGGGGSGGGGSQ ITLKESGPTLVKPTQTTLTLCTFSGFSLSTSGVGVGWIRQPPGKALEWLALIIYNDDKRYSPSL KSRLTITKDTSKNQVVLMTNMDPVDTA VYYCARDEYGGFDIWGGTMTVTVSSTTPAPRPPTP APTIASQPLSLRPEACRPAAGGAVHTRGLDFACDIYIWAPLAGTCGVLLSLVITLYCKRGRKK LLYIFKQPFMRPVQTTQEEDGCS CRFP EEEEGGCELRVKFSRSADAPAYQQGQNLINELNLGR REEYDVLDKRRGRDPEMGGKPRRKNPQEGLYNELQKDKMAEAYSEIGMKGERRRGKGHDGLYQG LSTATKDTYDALHMQALPPR
259	QVQLVESGGGVVQPGSRSLRSCAASGFTFSYGMHWVRQAPGKGLEWVAVISYEGSNKYADSV KGRFTISRDNKNTLYQMNSLRAEDTA VYYCARELGDMVWGQGTITVTVSSGGGSGGGGS GGGSGGGGSEIVLTQSPATLSLSPGERATLSCRASQSVSSYLAWYQQKPGQAPRLLIYDASNR ATGIPARFSGSGSDFTLTITSSLEPEDFAVYYCQQRVDLWTFGGGTKEIKTTTPAPRPPTP APTIASQPLSLRPEACRPAAGGAVHTRGLDFACDIYIWAPLAGTCGVLLSLVITLYCKRGRKKLL YIFKQPFMRPVQTTQEEDGCS CRFP EEEEGGCELRVKFSRSADAPAYQQGQNLINELNLGR REEYDVLDKRRGRDPEMGGKPRRKNPQEGLYNELQKDKMAEAYSEIGMKGERRRGKGHDGLYQGLS TATKDTYDALHMQALPPR
260	EIVLTQSPATLSLSPGERATLSCRASQSVSSYLAWYQQKPGQAPRLLIYDASNRATGIPARFSG SGSGDFTLTITSSLEPEDFAVYYCQQRVDLWTFGGGTKEIKGGGSGGGGSGGGGSGGGGSQV QLVESGGGVVQPGSRSLRSCAASGFTFSYGMHWVRQAPGKGLEWVAVISYEGSNKYADSVK RFTISRDNKNTLYQMNSLRAEDTA VYYCARELGDMVWGQGTITVTVSSTTPAPRPPTP APTIASQPLSLRPEACRPAAGGAVHTRGLDFACDIYIWAPLAGTCGVLLSLVITLYCKRGRKKLL YIFKQPFMRPVQTTQEEDGCS CRFP EEEEGGCELRVKFSRSADAPAYQQGQNLINELNLGR REEYDVLDKRRGRDPEMGGKPRRKNPQEGLYNELQKDKMAEAYSEIGMKGERRRGKGHDGLYQGLS TATKDTYDALHMQALPPR
261	EVQLVESGGGLVKPGGSLRLSCAASGFTFSYGMHWVRQAPGKGLEWVSYISSSGSTIYYADSV KGRFTISRDNKNTLYQMNSLRAEDTA VYYCARDQGNVGVWVGQGTITVTVSSGGGSGGGGS GGGSGGGGDIQMTQSPSTLSASVGRVITCRASQSISSWLAWYQQKPKAPKLLIYDASSL ESGVPSPRSFGSGSGTEFTLTITSSLPDDFATYYCQVSSLPPTFGGGTKVEIKTTTPAPRPPTP APTIASQPLSLRPEACRPAAGGAVHTRGLDFACDIYIWAPLAGTCGVLLSLVITLYCKRGRKK LLYIFKQPFMRPVQTTQEEDGCS CRFP EEEEGGCELRVKFSRSADAPAYQQGQNLINELNLGR REEYDVLDKRRGRDPEMGGKPRRKNPQEGLYNELQKDKMAEAYSEIGMKGERRRGKGHDGLYQG LSTATKDTYDALHMQALPPR

TABLE 11-continued

Chimeric antigen receptors	
SEQ ID NO.	AMINO ACID SEQUENCE
262	DIQMTQSPSTLSASVGDRTTITCRASQSISSWLAWYQQKPKGAPKLLIYDASSLESQVPSRFSG SGSGTEFTLTITSSLPDDFATYYCQQVSSLPTFGGGTKVEIKGGGSGGGSGGGSGGGSE VQLVESGGGLVPGGSLRLSCAASGFTFSYYMSWIRQAPGKGLEWVSYISSSGSTIYYADSVK GRFTISRDNANKNSLYLQMNSLRAEDTAVYYCARDQGNYGVDVWGQGTITVTVSSTTTPAPRPPTP APTIASQPLSLRPEACRPAAGGAVHTRGLDFACDIYIWAPLAGTCGVLLLSLVITLYCKRGRKK LLYIFKQPFMRPVQTTQEEDGCSRFPEEEEGGCELRVKFSRSADAPAYQQGQNQLYNELNLGR REEYDVLDKRRGRDPEMGGKPRRKNPQEGLYNELQDKMAEAYSIEIGMKGERRRGKGHDLGYQG LSTATKDTYDALHMQALPPR
263	QVQLVESGGGLVPGGSLRLSCAASGFTFSYYMSWIRQAPGKGLEWVSYISSSGSTIYYADSV KGRFTISRDNANKNSLYLQMNSLRAEDTAVYYCARDQGNYGVDVWGQGTITVTVSSTTTPAPRPPTP APTIASQPLSLRPEACRPAAGGAVHTRGLDFACDIYIWAPLAGTCGVLLLSLVITLYCKRGRKK LLYIFKQPFMRPVQTTQEEDGCSRFPEEEEGGCELRVKFSRSADAPAYQQGQNQLYNELNLGR REEYDVLDKRRGRDPEMGGKPRRKNPQEGLYNELQDKMAEAYSIEIGMKGERRRGKGHDLGYQG LSTATKDTYDALHMQALPPR
264	DIQMTQSPSTLSASVGDRTTITCRASQSISSWLAWYQQKPKGAPKLLIYEASSLESQVPSRFSG SGSGTEFTLTITSSLPDDFATYYCQQSDSHPIFGGGTKVEIKGGGSGGGSGGGSGGGGSG VQLVESGGGLVPGGSLRLSCAASGFTFSYYMSWIRQAPGKGLEWVSYISSSGSTIYYADSVK GRFTISRDNANKNSLYLQMNSLRAEDTAVYYCARDQGNYGVDVWGQGTITVTVSSTTTPAPRPPTP APTIASQPLSLRPEACRPAAGGAVHTRGLDFACDIYIWAPLAGTCGVLLLSLVITLYCKRGRKK LLYIFKQPFMRPVQTTQEEDGCSRFPEEEEGGCELRVKFSRSADAPAYQQGQNQLYNELNLGR REEYDVLDKRRGRDPEMGGKPRRKNPQEGLYNELQDKMAEAYSIEIGMKGERRRGKGHDLGYQG LSTATKDTYDALHMQALPPR
265	EVQLVESGGGLVPGGSLRLSCAASGFTFSYYMSWIRQAPGKGLEWVSYISSSGSTIYYADSV KGRFTISRDNANKNSLYLQMNSLRAEDTAVYYCARDQGNYGVDVWGQGTITVTVSSTTTPAPRPPTP APTIASQPLSLRPEACRPAAGGAVHTRGLDFACDIYIWAPLAGTCGVLLLSLVITLYCKRGRKK LLYIFKQPFMRPVQTTQEEDGCSRFPEEEEGGCELRVKFSRSADAPAYQQGQNQLYNELNLGR REEYDVLDKRRGRDPEMGGKPRRKNPQEGLYNELQDKMAEAYSIEIGMKGERRRGKGHDLGYQG LSTATKDTYDALHMQALPPR
266	DIQMTQSPSTLSASVGDRTTITCRASQSISSWLAWYQQKPKGAPKLLIYEASSLESQVPSRFSG SGSGTEFTLTITSSLPDDFATYYCQQANSHPIFGGGTKVEIKGGGSGGGSGGGSGGGGSG VQLVESGGGLVPGGSLRLSCAASGFTFSYYMSWIRQAPGKGLEWVSYISSSGSTIYYADSVK GRFTISRDNANKNSLYLQMNSLRAEDTAVYYCARDQGNYGVDVWGQGTITVTVSSTTTPAPRPPTP APTIASQPLSLRPEACRPAAGGAVHTRGLDFACDIYIWAPLAGTCGVLLLSLVITLYCKRGRKK LLYIFKQPFMRPVQTTQEEDGCSRFPEEEEGGCELRVKFSRSADAPAYQQGQNQLYNELNLGR REEYDVLDKRRGRDPEMGGKPRRKNPQEGLYNELQDKMAEAYSIEIGMKGERRRGKGHDLGYQG LSTATKDTYDALHMQALPPR
267	EVQLLESGGGLVQPGGSLRLSCAASGFTFSNYAMSWVRQAPGKGLEWVSAISGGSGSTYYADSV KGRFTISRDNANKNSLYLQMNSLRAEDTAVYYCARPGDGYEGVYFDYWGQGTITVTVSSTTTPAPRPPTP APTIASQPLSLRPEACRPAAGGAVHTRGLDFACDIYIWAPLAGTCGVLLLSLVITLYCKRGRKK LLYIFKQPFMRPVQTTQEEDGCSRFPEEEEGGCELRVKFSRSADAPAYQQGQNQLYNELNLGR REEYDVLDKRRGRDPEMGGKPRRKNPQEGLYNELQDKMAEAYSIEIGMKGERRRGKGHDLGYQG LSTATKDTYDALHMQALPPR
268	DIQMTQSPSSLSASVGDRTTITCRASQSISSYLNWYQQKPKGAPKLLIYAASSLQSGVPSRESG SGSGTDFTLTITSSLPDDFATYYCQQAHSSPIFGGGTKVEIKGGGSGGGSGGGSGGGGSG VQLLESGGGLVQPGGSLRLSCAASGFTFSNYAMSWVRQAPGKGLEWVSAISGGSGSTYYADSVK GRFTISRDNANKNSLYLQMNSLRAEDTAVYYCARPGDGYEGVYFDYWGQGTITVTVSSTTTPAPRPPTP APTIASQPLSLRPEACRPAAGGAVHTRGLDFACDIYIWAPLAGTCGVLLLSLVITLYCKRGRKK LLYIFKQPFMRPVQTTQEEDGCSRFPEEEEGGCELRVKFSRSADAPAYQQGQNQLYNELNLGR REEYDVLDKRRGRDPEMGGKPRRKNPQEGLYNELQDKMAEAYSIEIGMKGERRRGKGHDLGYQG LSTATKDTYDALHMQALPPR
269	QITLKESGPTLVKPTQLTLCTFSGSLSTSGVGWIRQPPGKALEWLALIIWNDEKRYSPS LKSRLTITKDTSKNQVLTMTNMDPVDVAVYYCAREGSHDYKSSNWFDPWGQGTITVTVSSTTTPAPRPPTP APTIASQPLSLRPEACRPAAGGAVHTRGLDFACDIYIWAPLAGTCGVLLLSLVITLYCKRGRKK LLYIFKQPFMRPVQTTQEEDGCSRFPEEEEGGCELRVKFSRSADAPAYQQGQNQLYNELNLGR REEYDVLDKRRGRDPEMGGKPRRKNPQEGLYNELQDKMAEAYSIEIGMKGERRRGKGHDLGYQG LSTATKDTYDALHMQALPPR

TABLE 11-continued

Chimeric antigen receptors	
SEQ ID NO.	AMINO ACID SEQUENCE
270	DIQMTQSPSSLSASVGDRTTITCQASQDIANYLNWYQQKPGKAPKLLIYDASNLETGVPSPRFSG SGSGTDFTTTIISSLPEDIATYYCQQHFNLPITFGGGTKVEIKGGGGSGGGSGGGSGGGSG ITLKESGPTLVKQTTLTCTFSGFSLSTSGVGVGWIRQPPGKALEWLALIYWNDEKRYSPSL KSRLTITKDTSKNQVVLMTNMDPVDTAVYYCAREGSHDYKSSNWFDPWGGQTLVTVSSTTTPA PRPPTPAPTIASQPLSLRPEACRPAAGGAVHTRGLDFACDIYIWAPLAGTCGVLLLSLVITLYC KRGRKKLLYIFKQPFMRPVQTTQEEDGCS CRFP EEEEGGCEL RVKFSRSADAPAYQQGQNQLYN ELNLGRREEYDVLDKRRGRDPEMGGKPRRKNPQEGLYNELQKDKMAEAYSEIGMKGERRRGKH DGLYQGLSTATKDTYDALHMQALPPR
271	EVQLVESGGGLVQPGGSLRLSCAASGFTFSSYSMNWVRQAPGKLEWVSSISSSSYIYADSV KGRFTISRDNKNSLYLQMNSLRAEDTAVYYCARAGDTYSADYYMDVWGKGTITVTVSSGGG SGGGSGGGSGGGSDIVMTQSPSLSPVTPGEPASISCRSSQSLHNSNGYNLDWYLQKPGQS PQLLIYLGSRASGVDPDRFSGSGGTDFTLKISRVEADVGVYCMQALGLITFGGGTKVEIKT TTPAPRPPTPAPTIASQPLSLRPEACRPAAGGAVHTRGLDFACDIYIWAPLAGTCGVLLLSLV ITLYCKRGRKKLLYIFKQPFMRPVQTTQEEDGCS CRFP EEEEGGCEL RVKFSRSADAPAYQQGQN QLYNELNLGRREEYDVLDKRRGRDPEMGGKPRRKNPQEGLYNELQKDKMAEAYSEIGMKGERRR GKGHDGLYQGLSTATKDTYDALHMQALPPR
272	DIVMTQSPSLSPVTPGEPASISCRSSQSLHNSNGYNLDWYLQKPGQSPQLLIYLGSRASGVP DRFSGSGSGTDFTLKISRVEADVGVYCMQALGLITFGGGTKVEIKGGGGSGGGSGGGSGG GGSEVQLVESGGGLVQPGGSLRLSCAASGFTFSSYSMNWVRQAPGKLEWVSSISSSSYIY DSVKGGRFTISRDNKNSLYLQMNSLRAEDTAVYYCARAGDTYSADYYMDVWGKGTITVTVSS TTPAPRPPTPAPTIASQPLSLRPEACRPAAGGAVHTRGLDFACDIYIWAPLAGTCGVLLLSLV ITLYCKRGRKKLLYIFKQPFMRPVQTTQEEDGCS CRFP EEEEGGCEL RVKFSRSADAPAYQQGQN QLYNELNLGRREEYDVLDKRRGRDPEMGGKPRRKNPQEGLYNELQKDKMAEAYSEIGMKGERRR GKGHDGLYQGLSTATKDTYDALHMQALPPR
273	EVQLLESGGGLVQPGGSLRLSCAASGFTFGSEAMSWVRQAPGKERELVSAISGSGEVITYADSV KGRFTISRDNKNSLYLQMNSLRAEDTAVYYCQRLVEAKRHWGQGTQVTVSSTTTPAPRPPTPA PTIASQPLSLRPEACRPAAGGAVHTRGLDFACDIYIWAPLAGTCGVLLLSLVITLYCKRGRKK LLYIFKQPFMRPVQTTQEEDGCS CRFP EEEEGGCEL RVKFSRSADAPAYQQGQNQLYNELNLGR REEYDVLDKRRGRDPEMGGKPRRKNPQEGLYNELQKDKMAEAYSEIGMKGERRRGKGHDGLYQGL STATKDTYDALHMQALPPR
274	EVQLLESGGGLVQPGGSLRLSCAASGFTFESEAMSWVRQAPGKERELVSVITSEGSTYYADSVK GRFTISRDNKNSLYLQMNSLRAEDTAVYYCAHIEWETRLNWGQGTQVTVSSTTTPAPRPPTPA PTIASQPLSLRPEACRPAAGGAVHTRGLDFACDIYIWAPLAGTCGVLLLSLVITLYCKRGRKK LLYIFKQPFMRPVQTTQEEDGCS CRFP EEEEGGCEL RVKFSRSADAPAYQQGQNQLYNELNLGR REEYDVLDKRRGRDPEMGGKPRRKNPQEGLYNELQKDKMAEAYSEIGMKGERRRGKGHDGLYQGL STATKDTYDALHMQALPPR
275	EVQLLESGGGLVQPGGSLRLSCAASGFTFDEYTMHWFRQAPGKERELVSAISGGGSETYYADSV KGRFTISRDNKNSLYLQMNSLRAEDTAVYYCAAGGEEAGVGYWGQGTQVTVSSTTTPAPRPPT PAPTIASQPLSLRPEACRPAAGGAVHTRGLDFACDIYIWAPLAGTCGVLLLSLVITLYCKRGRK KLLYIFKQPFMRPVQTTQEEDGCS CRFP EEEEGGCEL RVKFSRSADAPAYQQGQNQLYNELNLGR REEYDVLDKRRGRDPEMGGKPRRKNPQEGLYNELQKDKMAEAYSEIGMKGERRRGKGHDGLYQGL GLSTATKDTYDALHMQALPPR
276	EVQLLESGGGLVQPGGSLRLSCAASGFTFEDYAMSWFRQAPGKERELVSAISGKGGSTYYADSV KGRFTISRDNKNSLYLQMNSLRAEDTAVYYCAVLDEEAGGEGYWGQGTQVTVSSTTTPAPRP PTPAPTIASQPLSLRPEACRPAAGGAVHTRGLDFACDIYIWAPLAGTCGVLLLSLVITLYCKR RKKLLYIFKQPFMRPVQTTQEEDGCS CRFP EEEEGGCEL RVKFSRSADAPAYQQGQNQLYNELN LGRREEYDVLDKRRGRDPEMGGKPRRKNPQEGLYNELQKDKMAEAYSEIGMKGERRRGKGHDGL YQGLSTATKDTYDALHMQALPPR
277	EVQLLESGGGLVQPGGSLRLSCAASGFTFDRYAMSWFRQAPGKERELVSAISTSGDSTYYADSV KGRFTISRDNKNSLYLQMNSLRAEDTAVYYCAVLDEEAGGEGYWGQGTQVTVSSTTTPAPRP PTPAPTIASQPLSLRPEACRPAAGGAVHTRGLDFACDIYIWAPLAGTCGVLLLSLVITLYCKR RKKLLYIFKQPFMRPVQTTQEEDGCS CRFP EEEEGGCEL RVKFSRSADAPAYQQGQNQLYNELN LGRREEYDVLDKRRGRDPEMGGKPRRKNPQEGLYNELQKDKMAEAYSEIGMKGERRRGKGHDGL YQGLSTATKDTYDALHMQALPPR
278	EVQLLESGGGLVQPGGSLRLSCAASGFTFASDAMSWVRQAPGKERELVSAISGSGGSTYYADSV KGRFTISRDNKNSLYLQMNSLRAEDTAVYYCAHDSGEAYLAFDYWGQGTQVTVSSTTTPAPR PPTPAPTIASQPLSLRPEACRPAAGGAVHTRGLDFACDIYIWAPLAGTCGVLLLSLVITLYCKR GRKKLLYIFKQPFMRPVQTTQEEDGCS CRFP EEEEGGCEL RVKFSRSADAPAYQQGQNQLYNEL NLGRREEYDVLDKRRGRDPEMGGKPRRKNPQEGLYNELQKDKMAEAYSEIGMKGERRRGKGHDG LYQGLSTATKDTYDALHMQALPPR
279	EVQLLESGGGLVQPGGSLRLSCAASGFTFDSYTMWVRQAPGKERELVSAISGHGDSYYADSV KGRFTISRDNKNSLYLQMNSLRAEDTAVYYCTRISITTEWLAGDYWGQGTQVTVSSTTTPAPR PPTPAPTIASQPLSLRPEACRPAAGGAVHTRGLDFACDIYIWAPLAGTCGVLLLSLVITLYCKR GRKKLLYIFKQPFMRPVQTTQEEDGCS CRFP EEEEGGCEL RVKFSRSADAPAYQQGQNQLYNEL

TABLE 11-continued

Chimeric antigen receptors	
SEQ ID NO.	AMINO ACID SEQUENCE
	NLGRREEYDVLDKRRGRDPEMGGKPRRKNPQEGLYNELQKDKMAEAYSEIGMKGERRRGKGGHDG LYQGLSTATKDTYDALHMQALPPR
280	EVQLESGLGGLVQPGGSLRLSCAASGFTFSSYAMSWFRQAPGKEREFVSFISGGSDSTYYADSV KGRFTISRDNKNTLYLQMNLSRAEDTAVYYCTRWPYDFEPESEPGVYWGQGTQVTVSSTTPA PRPPTPAPTIASQPLSLRPEACRPAAGGAVHTRGLDFACDIYIWAPLAGTCGVLLLSLVITLYC KRGRKKLLYIFKQPFMRPVQTTQEEDGCSCRFPEEEGGGCELRVKFSRSADAPAYQQGQNQLYN ELNLGRREEYDVLDKRRGRDPEMGGKPRRKNPQEGLYNELQKDKMAEAYSEIGMKGERRRGKGGH DGLYQGLSTATKDTYDALHMQALPPR
281	EVQLESGLGGLVQPGGSLRLSCAASGFTFTDYMSWYRQAPGKERELVSVIHSGGSTYYADSVK GRFTISRDNKNTLYLQMNLSRAEDTAVYYCAPGYSDLSFDYINFDYWGQGTQVTVSSTTPA PRPPTPAPTIASQPLSLRPEACRPAAGGAVHTRGLDFACDIYIWAPLAGTCGVLLLSLVITLYC KRGRKKLLYIFKQPFMRPVQTTQEEDGCSCRFPEEEGGGCELRVKFSRSADAPAYQQGQNQLYN ELNLGRREEYDVLDKRRGRDPEMGGKPRRKNPQEGLYNELQKDKMAEAYSEIGMKGERRRGKGGH DGLYQGLSTATKDTYDALHMQALPPR
282	EVQLESGLGGLVQPGGSLRLSCAASGFTFSDYAMHWFRQAPGKERVLSIDSGGSTYYADSVK GRFTISRDNKNTLYLQMNLSRAEDTAVYYCNAAGFGKGDHPKDAFDIWGGGTQVTVSSTTPA PRPPTPAPTIASQPLSLRPEACRPAAGGAVHTRGLDFACDIYIWAPLAGTCGVLLLSLVITLYC KRGRKKLLYIFKQPFMRPVQTTQEEDGCSCRFPEEEGGGCELRVKFSRSADAPAYQQGQNQLYN ELNLGRREEYDVLDKRRGRDPEMGGKPRRKNPQEGLYNELQKDKMAEAYSEIGMKGERRRGKGGH DGLYQGLSTATKDTYDALHMQALPPR
283	EVQLESGLGGLVQPGGSLRLSCAASGFTFSSEGMSWVRQAPGKERELVSAISGGSDHTYYADSV RGRFTISRDNKNTLYLQMNLSRAEDTAVYYCNALEGGPTTAIQPGGPDYWGQGTQVTVSSTTT PAPRPPTPAPTIASQPLSLRPEACRPAAGGAVHTRGLDFACDIYIWAPLAGTCGVLLLSLVITL YCKRGRKKLLYIFKQPFMRPVQTTQEEDGCSCRFPEEEGGGCELRVKFSRSADAPAYQQGQNQL YNELNLGRREEYDVLDKRRGRDPEMGGKPRRKNPQEGLYNELQKDKMAEAYSEIGMKGERRRGK GHDGLYQGLSTATKDTYDALHMQALPPR

In particular embodiments, a CAR comprises an amino acid sequence set forth in any one of SEQ ID NOs: 255, 256, 257, 258, 259, 260, 261, 262, 263, 264, 265, 266, 267, 268, 269, 270, 271, 272, 273, 274, 275, 276, 277, 278, 279, 280, 281, 282, and 283.

In particular embodiments, a CAR comprises an amino acid sequence set forth in any one of SEQ ID NOs: 256, 258, 259, 262, 263, 266, 268, 270, and 272.

In particular embodiments, a CAR comprises an amino acid sequence set forth in any one of SEQ ID NOs: 259, 263, 266, and 270.

In particular embodiments, a CAR comprises an amino acid sequence set forth in SEQ ID NO: 273 or 277.

F. Polypeptides

Polypeptides, fusion polypeptides, and polypeptide variants are contemplated in particular embodiments. Exemplary polypeptides contemplated herein include but not limited to fusion polypeptides, fusogens, tropism polypeptides, chimeric antigen receptors (CARs) and components thereof, and variants and/or fragments thereof, e.g., SEQ ID NOs: 1-283 and 324-374. Polypeptides contemplated herein also include those encoded by polynucleotide sequences set forth in any one of SEQ ID NOs: 284-323.

Polypeptide," "polypeptide," "peptide," and "protein" are used interchangeably, unless specified to the contrary, and according to conventional meaning, i.e., as a sequence of amino acids. In particular embodiments, a "polypeptide" is a fusion polypeptide or polypeptide variant. Polypeptides can be prepared using any of a variety of well-known recombinant and/or synthetic techniques. Polypeptides are not limited to a specific length, e.g., they may comprise a full-length protein sequence, a fragment of a full-length

protein, or a fusion protein, and may include post-translational modifications, e.g., glycosylations, acetylations, phosphorylations and the like, as well as other modifications known in the art, both naturally occurring and non-naturally occurring.

An "isolated peptide," "isolated protein" or an "isolated polypeptide" as used herein, refers to isolation, separation, and/or purification of a polypeptide molecule from a cellular environment, and from association with other components of the cell, i.e., it is not significantly associated with in vivo substances.

Polypeptides include "polypeptide variants." In particular embodiments, a polypeptide variant is referred to as a "modified polypeptide." Polypeptide variants may differ from a naturally occurring polypeptide in one or more amino acid substitutions, deletions, additions and/or insertions. For example, in particular embodiments, it may be desirable to modulate one or more biological activities of a chimeric antigen receptor by introducing one or more amino acid substitutions, deletions, additions and/or insertions into the polypeptide. Such variants may be naturally occurring or may be synthetically generated. In particular embodiments, polypeptides include polypeptide variants having at least about 65%, 66%, 67%, 68%, 69%, 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 86%, 97%, 98%, or 99% amino acid identity to any reference sequence contemplated herein, typically where the variant maintains at least one biological activity of the reference sequence.

Polypeptides variants include "polypeptide fragments." Illustrative examples of polypeptide fragments include but are not limited to anti-BCMA antibodies or antigen binding fragments thereof, anti-CD3 antibodies or antigen binding

fragments thereof, spacer domains, hinges, transmembrane domains, intracellular signaling domains, and the like. In particular embodiments, a polypeptide fragment is a biologically active polypeptide fragment. As used herein, the term "biologically active polypeptide fragment" refers to a polypeptide fragment that retains at least 100%, at least 95%, at least 90%, at least 85%, at least 80%, at least 75%, at least 70%, at least 65%, at least 60%, at least 55%, or at least 50% of the naturally occurring polypeptide activity. In certain embodiments, a polypeptide fragment comprises an amino acid sequence at least 5 to about 500 amino acids long. It will be appreciated that in certain embodiments, fragments are at least 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 55, 60, 65, 70, 75, 80, 85, 90, 95, 100, 110, 150, 200, 250, 300, 350, 400, 450, or 500 or more amino acids long.

In particular embodiments, polypeptides contemplated herein may comprise one or more amino acids denoted as "X" or "X_n" wherein n is an integer that denotes the particular X amino acid. "X" if present in an amino acid SEQ ID NO, refers to any one or more amino acids or particular amino acids if disclosed.

In particular embodiments, a polypeptide comprises one or more amino acid substitutions, deletions, truncations, or insertions using methods that are well known in the art. See, for example, Kunkel (Proc. Natl. Acad. Sci. USA. 82: 488-492. (1985)), Kunkel et al., (*Methods in Enzymol.* 154: 367-382. (1987)), U.S. Pat. No. 4,873,192, Watson, J. D. et al., (*Molecular Biology of the Gene*, Fourth Edition, Benjamin/Cummings, Menlo Park, Calif. (1987)) and the references cited therein. Guidance as to appropriate amino acid substitutions that do not affect biological activity of the protein of interest may be found in the model of Dayhoff et al., *Atlas of Protein Sequence and Structure* (Natl. Biomed. Res. Found., Washington, D.C. (1978)).

In certain embodiments, a polypeptide variant comprises one or more conservative substitutions or disruptive substitutions. A "conservative substitution" is one in which an amino acid is substituted for another amino acid that has similar properties. In particular embodiments, polypeptide variants contemplated herein comprise one or more conservative amino acid changes compared to a reference polypeptide. In particular embodiments, a conservative amino acid substitution involves substituting an amino acid with an amino acid having a related side chain. A "disruptive substitution" is one in which an amino acid is substituted for another amino acid that has different properties, e.g., polar vs. non-polar, bulky vs. non-bulky, charged vs. uncharged, acidic vs. basic. In particular embodiments, polypeptide variants contemplated herein comprise one or more disruptive amino acid changes compared to a reference polypeptide. In particular embodiments, a disruptive amino acid substitution involves substituting an amino acid with an amino acid having an unrelated side chain or side change with a different chemical property. Guidance in determining which amino acid residues can be substituted, inserted, or deleted can be found using computer programs well known in the art, such as DNASTAR, DNA Strider, Geneious, Mac Vector, or Vector NTI software.

Naturally occurring amino acids are generally divided into four families: acidic (aspartate, glutamate), basic (lysine, arginine, histidine), non-polar (alanine, valine, leucine, isoleucine, proline, phenylalanine, methionine, tryptophan), and uncharged polar (glycine, asparagine, glutamine, cysteine, serine, threonine, tyrosine) amino acids. Phenylalanine, tryptophan, and tyrosine are sometimes classified jointly as

aromatic amino acids. In particular embodiments, a conservative amino acid substitution refers to substituting amino acids within the same group or family.

Those of skill in this art recognize that, in general, conservative single amino acid substitutions in non-essential regions of a polypeptide do not substantially alter biological activity (see, e.g., Watson et al. *Molecular Biology of the Gene*, 4th Edition, 1987, The Benjamin/Cummings Pub. Co., p. 224), whereas disruptive single amino acid substitutions may.

In particular embodiments, a conservative amino acid substitution refers to substituting amino acids having a similar hydropathic index or score. The importance of the hydropathic amino acid index in conferring interactive biological function on a protein is generally understood in the art (Kyte and Doolittle, 1982, incorporated herein by reference). Each amino acid has been assigned a hydropathic index on the basis of its hydrophobicity and charge characteristics (Kyte and Doolittle, 1982). These values are: isoleucine (+4.5); valine (+4.2); leucine (+3.8); phenylalanine (+2.8); cysteine/cysteine (+2.5); methionine (+1.9); alanine (+1.8); glycine (-0.4); threonine (-0.7); serine (-0.8); tryptophan (-0.9); tyrosine (-1.3); proline (-1.6); histidine (-3.2); glutamate (-3.5); glutamine (-3.5); aspartate (-3.5); asparagine (-3.5); lysine (-3.9); and arginine (-4.5). In particular embodiments, a conservative amino acid substitution refers to substituting amino acids having a similar hydropathic index or score. In particular embodiments, substitution of amino acids whose hydropathic indices are within ± 2 is preferred, those within ± 1 are particularly preferred, and those within ± 0.5 are even more particularly preferred. It is also understood in the art that the substitution of like amino acids can be made effectively on the basis of hydrophilicity.

In particular embodiments, a conservative amino acid substitution refers to substituting amino acids having a similar hydrophilic index or score. As detailed in U.S. Pat. No. 4,554,101, the following hydrophilicity values have been assigned to amino acid residues: arginine (+3.0); lysine (+3.0); aspartate (+3.0 $\pm$ 1); glutamate (+3.0 $\pm$ 1); serine (+0.3); asparagine (+0.2); glutamine (+0.2); glycine (0); threonine (-0.4); proline (-0.5 $\pm$ 1); alanine (-0.5); histidine (-0.5); cysteine (-1.0); methionine (-1.3); valine (-1.5); leucine (-1.8); isoleucine (-1.8); tyrosine (-2.3); phenylalanine (-2.5); tryptophan (-3.4). In particular embodiments, a conservative amino acid substitution refers to substituting amino acids having a similar hydrophilic index or score. In particular embodiments, substitution of amino acids whose hydrophilic indices are substitution of amino acids whose hydrophilicity values are within ± 2 is preferred, those within ± 1 are particularly preferred, and those within ± 0.5 are even more particularly preferred.

In particular embodiments, a conservative amino acid substitution may be based on the relative similarity of the amino acid side-chain substituents, for example, their hydrophobicity, hydrophilicity, charge, size, and the like. In particular embodiments, a disruptive amino acid substitution may be based on the relative dissimilarity of the amino acid side-chain substituents, for example, their hydrophobicity, hydrophilicity, charge, size, and the like.

In particular embodiments, a polypeptide comprises a vesiculovirus envelope glycoprotein G comprising one or more amino acid substitutions at positions 47 and/or 354 of the mature polypeptide lacking a signal peptide. In particular embodiments, a polypeptide comprises a mutated coccal virus envelope glycoprotein (COCV-G) or a mutated vesicular stomatitis Indiana virus envelope glycoprotein (VSIV-

G), wherein the COCV-G or VSIV-G comprises amino acid substitutions at positions 47 and 354.

Polypeptides contemplated in particular embodiments include fusion polypeptides. In particular embodiments, fusion polypeptides and polynucleotides encoding fusion polypeptides are provided. Fusion polypeptides can include one or more polypeptide domains or segments including but not limited to signal peptides, antibodies or antigen binding fragments thereof, polypeptide linkers, spacer domains, transmembrane domains, intracellular signaling domains, and polypeptide cleavage signals. Fusion proteins and polypeptides are typically linked C-terminus to N-terminus, although they can also be linked C-terminus to C-terminus, N-terminus to N-terminus, or N-terminus to C-terminus. Fusion polypeptides and fusion proteins refer to a polypeptide having at least two, three, four, five, six, seven, eight, nine, or ten polypeptide segments.

Fusion polypeptides may optionally comprise a polypeptide linker contemplated elsewhere herein that can be used to link one or more polypeptides or domains within a polypeptide.

In particular embodiments, a polypeptide or fusion polypeptide comprises a non-viral membrane bound tropism polypeptide that binds an antigen expressed on an immune effector cell. In particular embodiments, a polypeptide comprises a non-viral membrane-bound tropism polypeptide comprising an anti-CD3 ϵ scFv and a human CD8 α hinge and transmembrane domain.

In particular embodiments, a polypeptide or fusion polypeptide comprises an anti-BCMA chimeric antigen receptor comprising an anti-BCMA antibody or antigen binding fragment thereof comprising an amino acid sequence set forth in any one of SEQ ID NOs: 39, 59, 70, 90, 101 and 117; a CD8 α hinge, a CD8 α transmembrane domain; a CD137 costimulatory signaling domain; and a CD3 ζ primary signaling domain.

In particular embodiments, a polypeptide comprises signal peptide set forth in any one of SEQ ID NOs: 245, 246, 247, 248, 249, 250, 251, 252, 253, and 254 that is subsequently cleaved from the polypeptide. Signal peptides are short 16 to 30 amino acid N-terminal sequences of nascently synthesized polypeptide chains that mediate protein targeting to the membrane of the endoplasmic reticulum (ER). Typically, signal peptides are cleaved cotranslationally by signal peptidase, a heterooligomeric polypeptide complex. In particular embodiments, a polypeptide comprises a signal peptide. In preferred embodiments, a polynucleotide encoding a polypeptide comprises a polynucleotide encoding a signal polypeptide; and the translated polypeptide does not comprise a signal peptide. Exemplary signal peptides are set forth in Table 12.

TABLE 12

Exemplary Signal Peptides	
SEQ ID NO	AMINO ACID SEQUENCE
245	MALPVTALLLPLALLHAARP
246	METDTLLLWVLLWVPGSTG
247	MDMRVPAQLLGLLLWLRGARC
248	MPLLLLLPLWAGALA
249	MDAMKRGCLCCVLLLCGAVFVSPS
250	MLLLLLLLGLRLQLSLG

TABLE 12-continued

Exemplary Signal Peptides	
SEQ ID NO	AMINO ACID SEQUENCE
251	MWLQSLLLLGTVACSIS
252	MLLLVTSLLLCELPHAPALLIP
253	MSRSVALAVLALLSLSGLEA
254	MLLLLLLLLLLALALA

In particular embodiments, a polypeptide comprises a signal peptide set forth in any one of SEQ ID NOs: 245, 246, 247, 248, 249, 250, 251, 252, 253, and 254 and a chimeric antigen receptor comprising an amino acid sequence set forth in any one of SEQ ID NOs: 255, 256, 257, 258, 259, 260, 261, 262, 263, 264, 265, 266, 267, 268, 269, 270, 271, 272, 273, 274, 275, 276, 277, 278, 279, 280, 281, 282, and 283.

In particular embodiments, a polypeptide comprises a signal peptide set forth in any one of SEQ ID NOs: 245, 246, 247, 248, 249, 250, 251, 252, 253, and 254 and a chimeric antigen receptor encoded by a polynucleotide sequence set forth in any one of SEQ ID NOs: 295, 296, 297, 298, 299, 300, 301, 302, 303, 304, 305, 306, 307, 308, 309, 310, 311, 312, 313 and 314.

In particular embodiments, two or more polypeptides can be expressed as a fusion polypeptide that comprises one or more polypeptide cleavage signals disposed between the two or more polypeptides.

Exemplary polypeptide cleavage signals include, but are not limited to, protease cleavage sites, nuclease cleavage sites and ribosomal skipping polypeptide or self-cleaving viral polypeptides (see, e.g., in Ryan et al., 1997. *J. Gen. Virol.* 78, 699-722; deFelipe and Ryan, 2004. *Traffic*, 5(8); 616-26; and Scymczak et al. (2004) *Nature Biotech.* 5, 589-594).

Exemplary protease cleavage sites include, but are not limited to the cleavage sites of potyvirus NIa proteases (e.g., tobacco etch virus protease), poty virus HC proteases, potyvirus PI (P35) proteases, byovirus NIa proteases, byovirus RNA-2-encoded proteases, aphthovirus L proteases, enterovirus 2A proteases, rhinovirus 2A proteases, picorna 3C proteases, comovirus 24K proteases, nepovirus 24K proteases, RTSV (rice tungro spherical virus) 3C-like protease, PYVF (parsnip yellow fleck virus) 3C-like protease, heparin, thrombin, factor Xa and enterokinase.

Illustrative examples of ribosomal skipping polypeptides include but are not limited to: a viral 2A peptide or sequence (Donnelly et al, 2001. *J. Gen. Virol.* 82: 1027-1041). In a particular embodiment, the viral 2A peptide is an aphthovirus 2A peptide, a potyvirus 2A peptide, or a cardiovirus 2A peptide.

In one embodiment, the viral 2A peptide is selected from the group consisting of: a foot-and-mouth disease virus (FMDV) 2A peptide, an equine rhinitis A virus (ERAV) 2A peptide, a Thosaea asigna virus (TaV) 2A peptide, a porcine teschovirus-1 (PTV-1) 2A peptide, a Theilovirus 2A peptide, and an encephalomyocarditis virus 2A peptide.

Illustrative examples of viral 2A sequences include, but are not limited to:

59

(SEQ ID NO: 355)
 GSGATNFSLLKQAGDVEENPGP;
 (SEQ ID NO: 356)
 ATNFSLLKQAGDVEENPGP;
 (SEQ ID NO: 357)
 LLKQAGDVEENPGP;
 (SEQ ID NO: 358)
 GSGEGRGSLLTCGDVEENPGP;
 (SEQ ID NO: 359)
 EGRGSLLTCGDVEENPGP;
 (SEQ ID NO: 360)
 LLTCGDVEENPGP;
 (SEQ ID NO: 361)
 GSGQCTNYALLKLAGDVESNPGP;
 (SEQ ID NO: 362)
 QCTNYALLKLAGDVESNPGP;
 (SEQ ID NO: 363)
 LLKLAGDVESNPGP;
 (SEQ ID NO: 364)
 GSGVKQTLNFDLLKLAGDVESNPGP;
 (SEQ ID NO: 365)
 VKQTLNFDLLKLAGDVESNPGP;
 (SEQ ID NO: 366)
 LLNFDLLKLAGDVESNPGP;
 (SEQ ID NO: 367)
 TLNFDLLKLAGDVESNPGP;
 (SEQ ID NO: 368)
 NFDLLKLAGDVESNPGP;
 (SEQ ID NO: 369)
 QLLNFDLLKLAGDVESNPGP;
 (SEQ ID NO: 370)
 APVKQTLNFDLLKLAGDVESNPGP;
 (SEQ ID NO: 371)
 VTELLYRMKRAETYCPRLAIHPTEARHKQKIVAPVKQT;
 (SEQ ID NO: 372)
 LNFDLLKLAGDVESNPGP;
 (SEQ ID NO: 373)
 LLAIHPTEARHKQKIVAPVKQTLNFDLLKLAGDVESNPGP;
 and
 (SEQ ID NO: 374)
 EARHKQKIVAPVKQTLNFDLLKLAGDVESNPGP.

G. Polynucleotides

Polynucleotides comprising or encoding fusogens, tropism polypeptides, chimeric antigen receptors (CARs) and components thereof, and variants and/or fragments thereof, vectors, promoters, enhancers, Kozak sequences, polyadenylation signals, untranslated regions, and posttranscriptional response elements as well as other polynucleotides are contemplated in various embodiments.

As used herein, the terms “polynucleotide” or “nucleic acid” refer to deoxyribonucleic acid (DNA), ribonucleic acid (RNA) and DNA/RNA hybrids. Polynucleotides may be single-stranded or double-stranded and either recombinant, synthetic, or isolated. Polynucleotides include but are not limited to: pre-messenger RNA (pre-mRNA), messenger RNA (mRNA), RNA, circular RNA (circRNA), synthetic

60

RNA, short interfering RNA (siRNA), short hairpin RNA (shRNA), microRNA (miRNA), ribozymes, genomic RNA (gRNA), viral genomic RNA, plus strand RNA (RNA(+)), minus strand RNA (RNA(-)), tracrRNA, crRNA, single guide RNA (sgRNA), Doggybone DNA (dbDNA), linear DNA, circular DNA, PCR amplified DNA, complementary DNA (cDNA), synthetic DNA, or recombinant DNA. Polynucleotides refer to a polymeric form of nucleotides of at least 5, at least 10, at least 15, at least 20, at least 25, at least 30, at least 40, at least 50, at least 100, at least 200, at least 300, at least 400, at least 500, at least 1000, at least 5000, at least 10000, or at least 15000 or more nucleotides in length, either ribonucleotides or deoxyribonucleotides or a modified form of either type of nucleotide, as well as all intermediate lengths. It will be readily understood that “intermediate lengths,” in this context, means any length between the quoted values, such as 6, 7, 8, 9, etc., 101, 102, 103, etc., 151, 152, 153, etc., 201, 202, 203, etc.

As used herein, “isolated polynucleotide” refers to a polynucleotide that has been isolated from or purified from the sequences which flank it in a naturally-occurring state. In particular embodiments, an isolated polynucleotide is a synthetic polynucleotide, a semi-synthetic polynucleotide, or a polynucleotide obtained or derived from a recombinant source, or other polynucleotide that does not exist in nature and that has been made by the hand of man.

In particular embodiments, polynucleotides contemplated herein are polynucleotide variants. As used herein, the terms “polynucleotide variant” and “variant” and the like refer to polynucleotides displaying substantial sequence identity with a reference polynucleotide sequence or polynucleotides that hybridize with a reference sequence under stringent conditions that are defined hereinafter. These terms also encompass polynucleotides that are distinguished from a reference polynucleotide by the addition, deletion, substitution, or modification of one or more nucleotides. Accordingly, the terms “polynucleotide variant” and “variant” include polynucleotides in which one or more nucleotides have been added or deleted, or modified, or replaced with different nucleotides. In this regard, it is well understood in the art that certain alterations inclusive of mutations, additions, deletions and substitutions can be made to a reference polynucleotide whereby the altered polynucleotide retains the biological function or activity of the reference polynucleotide or wherein the function or activity of the altered polynucleotide is modulated. In particular embodiments, polynucleotides or polynucleotide variants have at least or about 50%, 55%, 60%, 65%, 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99% or 100% sequence identity to a reference sequence.

In particular embodiments, a polynucleotide variant includes a polynucleotide fragment that encodes biologically active polypeptide fragments or variants. As used herein, the term “polynucleotide fragment” refers to a polynucleotide fragment at least 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 55, 60, 65, 70, 75, 80, 85, 90, 95, 100, 110, 150, 200, 250, 300, 350, 400, 450, 500, 550, 600, 650, 700, 750, 800, 850, 900, 950, 1000, 1100, 1200, 1300, 1400, 1500, 1600, 1700 or more nucleotides in length that encodes a polypeptide variant that retains at least 100%, at least 90%, at least 80%, at least 70%, at least 60%, at least 50%, at least 40%, at least 30%, at least 20%, at least 10%, or at least 5% of the naturally occurring polypeptide activity. Polynucle-

61

otide fragments refer to a polynucleotide that encodes a polypeptide that has an amino-terminal deletion, a carboxyl-terminal deletion, and/or an internal deletion or substitution of one or more amino acids of a naturally occurring or recombinantly-produced polypeptide.

Terms that describe the orientation of polynucleotides include: 5' (normally the end of the polynucleotide having a free phosphate group) and 3' (normally the end of the polynucleotide having a free hydroxyl (OH) group). Polynucleotide sequences can be annotated in the 5' to 3' orientation or the 3' to 5' orientation. For DNA and mRNA, the 5' to 3' strand is designated the "sense," "plus," or "coding" strand because its sequence is identical to the sequence of the RNA [except for uracil (U) in RNA, instead of thymine (T) in DNA]. For DNA and mRNA, the complementary 3' to 5' strand which is the strand transcribed by the RNA polymerase is designated as "template," "antisense," "minus," or "non-coding" strand. As used herein, the term "reverse orientation" refers to a 5' to 3' sequence written in the 3' to 5' orientation or a 3' to 5' sequence written in the 5' to 3' orientation.

As used herein, the phrases "sequence identity" or, for example, comprising a "sequence 50% identical to," refer to the extent that sequences are identical on a nucleotide-by-nucleotide basis or an amino acid-by-amino acid basis over a window of comparison. A "percentage of sequence identity" may be calculated by comparing two optimally aligned sequences over the window of comparison, determining the number of positions at which the identical nucleic acid base (e.g., A, T, C, G, I) or the identical amino acid residue (e.g., Ala, Pro, Ser, Thr, Gly, Val, Leu, Ile, Phe, Tyr, Trp, Lys, Arg, His, Asp, Glu, Asn, Gln, Cys and Met) occurs in both sequences to yield the number of matched positions, dividing the number of matched positions by the total number of positions in the window of comparison (i.e., the window size), and multiplying the result by 100 to yield the percentage of sequence identity. In particular embodiments, poly-

62

nucleotides and polypeptides comprise at least about 50%, 55%, 60%, 65%, 66%, 67%, 68%, 69%, 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 86%, 97%, 98%, or 99% sequence identity to any of the reference sequences described herein, e.g., SEQ ID NOs: 1-339.

Illustrative examples of polynucleotides include, but are not limited to, polynucleotide sequences set forth in any one of SEQ ID NOs: 284-323 and polynucleotides encoding polypeptides set forth in SEQ ID NOs: 1-283 and 324-374.

In various embodiments, a polynucleotide encodes a polypeptide comprising an amino acid sequence set forth in any one of SEQ ID NOs: 1-283 and 324-374.

In particular embodiments, a polynucleotide encodes a chimeric antigen receptor comprising an amino acid sequence set forth in any one of SEQ ID NOs: 255, 256, 257, 258, 259, 260, 261, 262, 263, 264, 265, 266, 267, 268, 269, 270, 271, 272, 273, 274, 275, 276, 277, 278, 279, 280, 281, 282, and 283. In particular embodiments, a polynucleotide

encodes a chimeric antigen receptor comprising an amino acid sequence set forth in any one of SEQ ID NOs: 256, 258, 259, 262, 263, 266, 268, 270, and 272. In particular embodiments, a polynucleotide encodes a chimeric antigen receptor comprising an amino acid sequence set forth in any one of SEQ ID NOs: 259, 263, 266, and 270. In particular embodiments, a polynucleotide encodes a chimeric antigen receptor comprising an amino acid sequence set forth in any one of SEQ ID NO: 273 or 277. In particular embodiments, a polynucleotide encoding a chimeric antigen receptor comprises a polynucleotide sequence set forth in any one of SEQ ID NOs: 295, 296, 297, 298, 299, 300, 301, 302, 303, 304, 305, 306, 307, 308, 309, 310, 311, 312, 313, and 314.

Table 13 sets forth the SEQ ID NOs. and associated nucleic acid sequences encoding chimeric antigen receptor components and chimeric antigen receptors and the corresponding amino acid SEQ ID NO (AA SEQ ID NO.) encoded by the nucleic acid sequence.

TABLE 13

SEQ ID NO.	AA SEQ ID NO.	NUCLEIC ACID SEQUENCE
284	225	ACCACAACACCTGCTCCAGGCCCCCACCCGCTCCAACATAGCCAGCCAA CCATTGAGCCTCAGACCTGAAGCTTGACGGCCCGCAGCAGGAGGCGCGCTCCAT ACGCGAGGCTGGACTTCGCGTGTGAT
285	226	ATTGAAGTTATGTATCCTCCTTACCTAGACAATGAGAAGAGCAATGGAACC ATTATCCATGTGAAAGGAAACACCTTTGTCCAGTCCCTATTTCCTCGACCT TCTAAGCCC
286	228	GAGTCCAATATGGTCCCCCGTGCCACCATGCCCA
287	230	CTTGAGCCCCAATCTTGTGACAAAACCTCACACATGCCACCGTGCCCA
288	231	ATTTATATTTGGGCACCTTTGGCCGGAACATGTGGGGTGTGCTTCTCTCCCTT GTGATCACTCTGTATTGT
289	233	TTTTGGGTGCTGGTGGTGGTGGTGGAGTCTGGCTTGCTATAGCTTGCTAGTA ACATGGCCCTTTATTATTTTCTGGGTG
290	239	AAGCGCGGGAGAAAGAAGCTCCTGTACATCTTCAAGCAGCCTTTTATGCGACCT GTGCAACCACTCAGGAAGAAGATGGGTGTTTCATGCCGCTTCCCCGAGGAGGAA GAAGGAGGTGTGAACCTG
291	240	AGGAGTAAGAGGAGCAGGCTCCTGCACAGTGACTACATGAACATGACTCCCCGC CGCCCCGGGCCACCCGCAAGCATTACCAGCCCTATGCCCCACCACGCGACTTC GCAGCCTATCGCTCC
292	242	ACAAAAAGAAGTATTCATCCAGTGTGACGACCCTAACGGTGAATACATGTTT ATGAGAGCAGTGAACACAGCCAAAAATCTAGACTCACAGATGTGACCCTA

TABLE 13-continued

SEQ ID NO.	AA SEQ ID NO.	NUCLEIC ACID SEQUENCE
293	238	AGGGTGAAATTTCTAGAAGCGCCGATGCTCCCGCATATCAGCAGGGTCAGAAAT CAGCTCTCAATGAATTGAATCTCGGCAGGCGAGAAGAGTACGATGTTCTGGAC AAAAGACGGGCGAGGGATCCCGAGATGGGGGAAAGCCCCGGAGAAAAATCCT CAGGAGGGGTTGTACAATGAGCTGCAGAAGGACAAGATGGCTGAAGCCTATAGC GAGATCGGAATGAAAGGCGAAAGACGAGAGGCAAGGGCATGACGGTCTGTAC CAGGGTCTCTACAGCCACCAAGGACACTTATGATGCGTTGCATATGCAAGCC TTGCCACCCCGC
294	245	ATGGCTCTTCCCGTAACAGCCCTTTTGTGCCCCCTTGCACTCCTTCTGCATGCA GCACGACCG
295	256	GAGATCGTGTGACACAGTCTCCCGCCACACTGTCACTGTCTCCAGGCGAAAGA GCCACACTGAGCTGTAGAGCCAGCCAGAGCGTGTCTTACCTGGCCTGGTAT CAGCAGAAGCCTGGACAGGCTCCCGGCTGCTGATCTACGATGCCAGCAATAGA GCCACAGGCATCCCGCCAGATTTCTGGCAGCGGCTCTGGCACCGATTTCAACC CTGACCATAAGCAGCCTGGAACCTGAGGACTTCGCCGTGTACTACTGCCAGCAG AGAGTGGTGTACCCCATCACCTTTGGCGGAGGCAACCAAGGTGGAATCAAGGCG GGCGGAGGAAGCGGAGGCGGAGGATCTGGTGGTGGTGGATCTGGCGGAGGTGGC AGCCAGATCACACTGAAAGAGTCTGGCCCCACACTGGTCAAGCCCCACACAGACC CTGACACTGACCTGCACCTTCAGCGGCTTTAGCCTGAGCACATCTGGCGCTCGGC GTTGCTGGATTAGACAGCTCCTGGAAAGGCCCTGGAATGGCTGGCCCTGATC TACTGGAACGACGAGAAGAGATACAGCCCCAGCCTGAAGTCCCGGTGACCATC ACCAAGGACACCCAGCAAGAACCAGGTGGTGTGCTGACCATGACAAACATGGACCCC GTGGACACCGCGTGTATTATTGCGCCAGAGATGAGTACGGCGGCTTCGACATT TGGGGCCAGGGCACAAATGGTCACCGTGTCTAGTACCACAACACCTGTCTCAAGG CCCCCACACCCGCTCCAACTATAGCCAGCCAACCATTGAGCCTCAGACCTGAA GCTTGCAGGCCCCGAGCAGGAGGCGCCGTCCATACGCGAGGCCTGGACTTCGCG TGTGATATTTATATTTGGGCACCTTTGGCCGGAACATGTGGGGTGTGCTTCTC TCCCTTGTGATCACTCTGTATTGTAAAGCGCGGAGAAAGAGCTCCTGTACATC TTCAAGCAGCCTTTTATGCGACCTGTGCAAACCACTCAGGAAGAAGATGGGTGT TCATGCCGCTTCCCCGAGGAGGAAGAAGGAGGGTGTGAAGTGAAGGGTGAAATTT TCTAGAAGCGCCGATGCTCCCGCATATCAGCAGGGTCAGAATCAGCTCTACAAAT GAATTGAATCTCGGCAGGCGAGAAGAGTACGATGTTCTGGACAAAAGACGGGGC AGGGATCCCCAGATGGGGGAAAGCCCCGGAGAAAAATCCTCAGGAGGGGTTG TACAAATGAGCTGCAGAAGGACAAGATGGCTGAAGCCTATAGCAGATCGGAATG AAAGGCGAAAGACGAGAGGCAAGGGGCATGACGGTCTGTACAGGGTCTCTCT ACAGCCACCAAGGACACTTATGATGCGTTGCATATGCAAGCCTTGCCACCCCGC
296	258	GAGATCGTGTGACCCAGTCCCTTGCTACCTTGAGCCTGTCTCCAGGCGAGCGG GCCACACTGAGCTGTAGAGCTTCTCAGAGCGTGTCCAGCTACCTGGCCTGGTAT CAGCAGAAGCCTGGCCAGGCCCTAGACTGCTGATCTACGACGCCAGCAACCGG GCCACCGGCATCCCGCCAGATTCAGCGGATCTGGCAGCGGCACAGATTTTACC CTCACCATCAGCAGCCTGGAACCTGAGGACTTCGCCGTCTACTACTGCCAGCAA AGATTTCGACTACCCCATCACCTTCGCGCGCGAACAAGGTGGAATTAAGGGT GGTGGGGGCAGCGGTGGAGGTGGGAGCGGAGGCGGGGTAGCGGAGGCGGGGT AGCCAAATCACACTGAAAGAGAGCGGCCCTACACTCGTGAACCTACCCAGACC CTGACACTGACATGTACCTTCAGCGGCTTCTCCCTGAGCACCTCTGGCGCTCGGC GTTGGATGGATCAGACAGCCTCCAGGCAAGGCCCTGGAATGGCTGGCTCTGATC TATTGGAACGACGACAAGCGGTACAGCCCCAGCCTGAAGTCTAGACTGACCATC ACAAAGGACACCCAGCAAGAACCAGGTGGTGTGCTGACCATGACAAATATGGACCCC GTGGACACCGCGTGTACTACTGCGCCAGAGATGAGTACGGCGGATTGATATC TGGGGCCAGGGCACCATGGTGACCGTGTCCAGCACCACAACACCTGTCTCAAGG CCCCCACACCCGCTCCAACTATAGCCAGCCAACCATTGAGCCTCAGACCTGAA GCTTGCAGGCCCCGAGCAGGAGGCGCCGTCCATACGCGAGGCCTGGACTTCGCG TGTGATATTTATATTTGGGCACCTTTGGCCGGAACATGTGGGGTGTGCTTCTC TCCCTTGTGATCACTCTGTATTGTAAAGCGCGGAGAAAGAGCTCCTGTACATC TTCAAGCAGCCTTTTATGCGACCTGTGCAAACCACTCAGGAAGAAGATGGGTGT TCATGCCGCTTCCCCGAGGAGGAAGAAGGAGGGTGTGAAGTGAAGGGTGAAATTT TCTAGAAGCGCCGATGCTCCCGCATATCAGCAGGGTCAGAATCAGCTCTACAAAT GAATTGAATCTCGGCAGGCGAGAAGAGTACGATGTTCTGGACAAAAGACGGGGC AGGGATCCCCAGATGGGGGAAAGCCCCGGAGAAAAATCCTCAGGAGGGGTTG TACAAATGAGCTGCAGAAGGACAAGATGGCTGAAGCCTATAGCAGATCGGAATG AAAGGCGAAAGACGAGAGGCAAGGGGCATGACGGTCTGTACAGGGTCTCTCT ACAGCCACCAAGGACACTTATGATGCGTTGCATATGCAAGCCTTGCCACCCCGC
297	259	CAAGTGCAGCTCGTGGAAAGCGGCGGAGTGGTGCAGCCCCGGCCGAGCCTG AGACTGTCTTGCCTCGCTTCTGGATTACCTTCAGCAGCTACGGCATGCACTGG GTCAGACAGGCCCCCTGGCAAAGGCTGGAGTGGGTGGCCGTTATCAGCTACGAG GGCAGCAACAAGTATTACGCCGACAGCGTGAAGGGCCGCTTCAAACTCTCTAGA GATAATAGCAAGAACACCCCTGTACCTGCAGATGAACAGCCTGCGGGCCGAAGAT ACCGCCGTGTACTACTGTGCTAGAGAGCTGGGCGACGGCATGGACGTGTGGGA CAGGGCACCAACCGTGACCGTGTCTCTGGTGGTGGGGCAGCGGTGGAGGTGGG AGCGGAGGCGGGGGTAGCGGAGGCGGGGGTAGCGAGATCGTGTGACCCAGTCC CCTGCTACACTGAGCCTGTCTCCAGGCGAGCGGGCCACACTGAGCTGTAGAGCT TCTCAGAGCGTGTCCAGCTATCTGGCTGGTTCAGCAGAAACCTGGCCAGGCC

TABLE 13-continued

SEQ ID NO.	AA SEQ ID NO.	NUCLEIC ACID SEQUENCE
		CCTAGACTGCTGATCTACGACGCCAGCAACCGGGCCACCGGCATCCCCGCCAGA TTCAGCGGCTCTGGCAGCGGCACCGACTTCACCCCTCACCATCAGCAGCCTGGAA CCCGAGGATTTTGCCTCTACTACTGCCAGCAAAGAGTGGACCTGTGGACCTTC GGCGGAGGAACAAAGGTGGAATCAAGACCACAACACCTGCTCCAAGGCCCCC ACACCCGCTCCAACATATAGCCAGCCAACCATTTGAGCCTCAGACCTGAAGCTTGC AGGCCCGCAGCAGGAGGCGCCGTCCATACGCGAGGCTGGACTTCGCGTGTGAT ATTTATATTTTGGGCACCTTTGGCCGGAACATGTGGGGTGTGTCTCTCCCTT GTGATCACTCTGTATTGTAAGCGCGGGAGAAAGAGCTCCTGTACATCTTCAAG CAGCCTTTTATGCGACCTGTGCAAACTCAGGAAGAAGATGGGTGTTTCATGC CGCTTCCCCGAGGAGGAAGAAGGAGGTGTGAACCTGAGGGTGAAATTTCTAGA AGCGCCGATGCTCCCGCATATCAGCAGGGTGAAGCTCAGCTCTACAATGAATTG AATCTCGGCAGGCGAGAAGAGTACGATGTTCTGGACAAAAGACGGGCGAGGGAT CCCGAGATGGGGGAAAGCCCCGAGAAAAATCCTCAGGAGGGGTGTACAAT GAGCTGCAGAAGGACAAGATGGTGAAGCTTATAGCGAGATCGGAATGAAAGGC GAAAGACGAGAGGCAAGGGGCATGACGGTCTGTACAGGGTCTCTCTACAGCC ACCAAGGACACTTATGATGCGTTGCATATGCAAGCCTTGCCACCCCGC
298	262	GACATCCAGATGACCCAGAGCCCTTCGACCCATCCGCTTCGCTGGGTGACCGT GTGACCATCACCCTGTCGCGCTCGCAGAGCATCTCCTCCTGGCTCGCGTGGTAC CAACAGAAGCCTGGCAAGGCCCCAAGCTGCTGATTTACGACGCCAGTTCCTTG GAGCTCTGGCGTGGCATCCCGCTTCTCGGCAGCGGCAGCGGTACCGAGTTCACC CTGACGATCAGCTCCCTGCAGCCGGATGACTTTGCTACCTACTACTGTCAGCAG GTCTCCTCCCTCCCCCACCTTCGCTGGCGGTACCAAGGTGGAGATCAAGGGC GGGCGCGCTCTGTTGGCGGAGGTTCTGGCGGGGAGGTTCTGGGGGGGGAGGC TCCGAGGTGCAACTGGTAGAGAGCGCGGGGGACTGGTAACCCCGCGGCTCC CTGCGGCTGTATGCGCTGTAGCGGCTTACGTTTACGCGATTACTACATGAGT TGGATCCGCCAGGCCCCCGGGAAGGTTTGGAGTGGGTCTCGTATATCTCTTC AGCGGATCTACCATTTACTATGCGGACAGCGTGAAGGGGCGCTTCACCATATCT CGGGACAACGCCAAGAACTCCCTGTACCTGCAGATGAATTCCTGCGTGCCGAG GACAGGCGCGTGTATTACTGTGCCGCGACAGGGCAACTACGGCGTCGACGTG TGGGGCCAGGGTACAACCGTACCGTGTCCAGTACCACAACACCTGCTCCAAGG CCCCCACACCCGCTCCAACATATAGCCAGCCAACCATTTGAGCCTCAGACCTGAA GCTTGCAGGCCCGCAGCAGGAGGCGCCGTCCATACGCGAGGCTGGACTTCGCG TGTGATATTTATATTGGGCACCTTTGGCCGGAACATGTGGGGTGTGTCTCTC TCCCTTGTGATCACTCTGTATTGTAAGCGCGGGAGAAAGAGCTCCTGTACATC TTCAGCAGCCCTTTTATGCGACCTGTGCAAACTCAGGAAGAAGATGGGTGT TCATGCCGCTTCCCCGAGGAGGAAGAAGGAGGTGTGAACCTGAGGGTGAAATTT TCTAGAAGCGCCGATGCTCCCGCATATCAGCAGGGTGAAGCTCAGCTCTACAAT GAATTGAATCTCGGCAGGCGAGAAGAGTACGATGTTCTGGACAAAAGACGGGGC AGGGATCCCCGAGATGGGGGAAAGCCCCGAGAAAAATCCTCAGGAGGGGTG TACAATGAGCTGCAGAAGGACAAGATGGCTGAAGCCTTATAGCGAGATCGGAATG AAAGCGAAAGACGCGAGGGCAAGGGGCATGACGGTCTGTACAGGGTCTCTCT ACAGCCACCAAGGACACTTATGATGCGTTGCATATGCAAGCCTTGCCACCCCGC
299	263	CAAGTGCAGCTGGTCGAGAGCGGAGGAGCCTGGTTAAGCCCGCGGATCTCTC AGACTGAGCTGCGCCGCTAGCGGCTTTACATTACGCGACTACTACATGAGCTGG ATCCGGCAGGCCCCCTGGCAAGGGCCTGGAATGGGTGTCCTACATCAGCTCCTCC GGCAGCACATCTACTACGCCGACAGCGTGAAAGGCAGATTCACAATCTCTAGA GATAATGCGCAAGAACAGCCTGTACCTGCAGATGAACAGCCTGCGGGCCGAGGAC ACCGCCGTGTACTATTGTGTAGAGATCAGGGCAACTACGCGCTGGACGTGTGG GGCCAGGGCACACCGTGACCGTGTCTAGCGGTGGTGGGGCAGCGGTGGAGGT GGGAGCGGAGGCGGGGTAGCGGAGGCGGGGTAGCGATATCCAGATGACCCAG TCCCCATCTACACTGAGCGCCTCTGTGGGCGACCGGGTGACCATTACATGTAGA GCCAGCCAGAGCATCAGCAGCTGGCTGGCTTGGTATCAGCAGAAACCTGGCAG GCCCTTAAGCTGCTGATCTACGAGGCCAGCAGCCTGGAAGCGGCGTCCCCAGC AGATTACGCGGCGAGCGCTCTGGAAACAGAGTTACCCCTGACCATCTCCTCCCTG CAGCCTGACGACTTCGCCACCTACTACTGCCAGCAATCTGATAGCCACCCATC ACCTTTGGCGGAGGCACCAAGGTGGAATCAAGACCACAACCTGCTCCAAGG CCCCCACACCCGCTCCAACATATAGCCAGCCAACCATTTGAGCCTCAGACCTGAA GCTTGCAGGCCCGCAGCAGGAGGCGCCGTCCATACGCGAGGCTGGACTTCGCG TGTGATATTTATATTGGGCACCTTTGGCCGGAACATGTGGGGTGTGTCTCTC TCCCTTGTGATCACTCTGTATTGTAAGCGCGGGAGAAAGAGCTCCTGTACATC TTCAGCAGCCCTTTTATGCGACCTGTGCAAACTCAGGAAGAAGATGGGTGT TCATGCCGCTTCCCCGAGGAGGAAGAAGGAGGTGTGAACCTGAGGGTGAAATTT TCTAGAAGCGCCGATGCTCCCGCATATCAGCAGGGTGAAGCTCAGCTCTACAAT GAATTGAATCTCGGCAGGCGAGAAGAGTACGATGTTCTGGACAAAAGACGGGGC AGGGATCCCCGAGATGGGGGAAAGCCCCGAGAAAAATCCTCAGGAGGGGTG TACAATGAGCTGCAGAAGGACAAGATGGCTGAAGCCTTATAGCGAGATCGGAATG AAAGCGAAAGACGCGAGGGCAAGGGGCATGACGGTCTGTACAGGGTCTCTCT ACAGCCACCAAGGACACTTATGATGCGTTGCATATGCAAGCCTTGCCACCCCGC
300	266	GATATCCAGATGACCCAGTCCCCATCTACACTGAGCGCCTCTGTGGGCGACCGG GTGACAAATACCTGTAGAGCTAGCCAGAGCATCTCCTCCTGGCTGGCTGGTAC CAGCAAAAACCTGGCAAGGCCCTTAAGCTGCTGATCTACGAGGCCAGCAGCCTG GAAAGCGCGCTCCCTCTAGATTACGCGGACGCGGCTCTGGAACCGAGTTTACC CTGACAATCAGCAGCCTGCAGCCTGACGACTTCGCCACCTATTACTGCCAGCAG

TABLE 13-continued

SEQ ID NO.	AA SEQ ID NO.	NUCLEIC ACID SEQUENCE
		GCCAACAGCCACCCCATCACCTTTGGCGGAGGCACCAAGGTGGAATCAAGGGT GGTGGGGGCAGCGGTGGAGGTGGGAGCGGAGGCGGGGTAGCGGAGGCGGGGT AGCGAGGTGCAGCTGGTGGAAAGCGGCGGAGGACTCGTTAAGCCCGCGGCAGC CTGAGACTGAGCTGCGCCGTAGCGGATTTACCTTCAGCGACTACTACATGAGC TGGATCCGGCAGGCCCCCTGGCAAGGGCTTGGAAATGGGTGAGCTACATCAGCTCC TCTGGCTCTACAATCTACTACGCCGACAGCGTGAAAGGCAGATTACCATCTCT AGAGATAATGCCAAGAACAGCCTGTACCTGCAATGAACAGCCTGCGGGCCGAG GACACCGCCGTGTAATTTGTGCTAGAGATCAGGGCAACTACGGCGTGGACGTG TGGGGCCAGGGCACCACCGTGACAGTGTCTCCACCACAACACCTGCTCCAAGG CCCCCACACCGCTCCAACCTATAGCCAGCCAACCTTGAGCCTCAGACCTGAA GCTTGACAGGCCCCGACGAGGAGGCGCCGTCCATACGCGAGGCTGGACTTCGCG TGTGATATTTATATTTGGGCACCTTTGGCGGAACATGTGGGGTGTGCTTCTC TCCCTTGTGATCACTCTGTATTTGAAGCGCGGAGAAAGAGCTCCTGTACATC TTCAAGCAGCCTTTTATGCGACCTGTGCAAACTCAGGAAGAAGATGGGTGT TCTGGCTTCCCGAGGAGGAAGAAGGAGGGTGTGAAGTGAAGGGTGAATTT TCTAGAAGCGCCGATGCTCCCGCATATCAGCAGGGTGAATCAGCTCTACAAT GAATTGAATCTCGCAGGCGAGAAGAGTACGATGTTCTGGACAAAAGACGGGGC AGGGATCCCGAGATGGGGGAAAGCCCGGAGAAAAATCCTCAGGAGGGGTG TACATGAGCTGCAGAAGGACAAGATGGCTGAAGCCTATAGCGAGATCGGAATG AAAGGCGAAAGACGAGAGGCAAGGGGCATGACGGTCTGTACAGGGTCTCTCT ACAGCCACAAGGACACTTATGATGCGTTGCATATGCAAGCCTTGCCACCCCGC
301	268	GACATCCAGATGACCCAGAGCCCTAGCTCCCTGAGCGCCAGCGTGGGCGATAGA GTGACCATTACCTGTAGAGCCTCTCAGAGCATCTCCTCTACCTGAACCTGGTAT CAGCAGAAACCCGGCAAGGCCCTTAAGCTGCTGATCTACGCCGTAGCAGCCTG CAGTCTGGCGTCCCCAGCCGGTTCAGCGGCAGCGGATCTGGCACCGACTTCACC CTGACAACTCAGCAGCCTGCAACCTGAGGACTTTGCTACATACTACTCGCAGCAG GCCCAGACTCTCCAATCACCTTCGGCGGCGGAACAAGGTGGAATCAAGGGT GGTGGGGGCAGCGGTGGAGGTGGGAGCGGAGGCGGGGTAGCGGAGGCGGGGT AGCGAGGTGCAGCTGCTGGAAGCGGAGGCGGACTCGTTCAACCTGCGCGCAGC CTGAGACTGAGCTGCGCCGTCTTGGATTACCTTCAGCAACTACGCCATGAGC TGGGTGCGGCAGGCCCCCTGGCAAGGCCTGGAATGGGTCTCCGCCATCAGCGGC TCTGGCGCTCCACCTACTACGCCGACAGCGTGAAAGGCAGATTACCATCTCT AGAGATAATAGCAAGAACACCTGTACCTGCAGATGAACAGCCTGCGGGCCGAG GACACCGCCGTGTAATTTGTGCTAGACCCGGAGATGGCTACTACGAGGCGGTG TACTTCGACTACTGGGGCCAGGGCACACTGGTGACAGTGTCCAGCACCAACA CCTGCTCCAAGGCCCCCACACCCGCTCCAACCTATAGCCAGCCAACCTTGAGC CTCAGACCTGAAGCTTGCAGGCCCGCAGCAGGAGGCGCCGTCCATACGCGAGGC CTGGACTTCGCGTGTGATATTTATATTTGGGCACCTTTGGCCGAACATGTGGG GTGTTGCTTCTCTCCCTTGTGATCACTCTGATTGTGAAGCGCGGAGAAAGAAG CTCTGTACATCTTCAAGCAGCCTTTTATGCGACCTGTGCAAACTCAGGAA GAAGATGGGTGTTTATGCGCTTCCCGAGGAGGAAGAAGGAGGGTGTGAATG AGGGTGAATTTTCTAGAAGCGCGATGCTCCCGCATATCAGCAGGGTCAAGT CAGCTCTACAATGAATGAATCTCGGCAGGCGAGAAGTACGATGTTCTGGAC AAAAGACGGGCGAGGATCCCGAGATGGGGGAAAGCCCGGAGAAAAATCCT CAGGAGGGGTGTACAATGAGCTGCAGAAGGACAAGATGGCTGAAGCCTATAGC GAGATCGGAATGAAGGCGAAAGACGAGAGGCAAGGGGCATGACGGTCTGTAC CAGGGTCTCTACAGCCACCAAGGACACTTATGATGCGTTGCATATGCAAGCC TTGCCACCCCGC
302	270	GATATTCAGATGACCCAGAGCCCATCTAGCCTGAGCGCCAGCGTGGGCGATAGA GTGACCATTACCTGTAGGCTCTCAGGACATCGCTAATTACCTGAACCTGGTAT CAGCAGAAACCCGGCAAGGCCCTTAAGCTGCTGATCTACGACGCTCCAACCTG GAAACCGCGTGGCCAGCCGTTTCCAGCGCAGCGGATCTGGCACAGACTTCACC TTTACCATCAGCTCCCTCCAGCCTGAGGACATCGCCACATACTACTGCCAGCAA CACTTCAACCTGCTCTGACCTTCGGCGGCGGAACAAGGTGAGATCAAGGGT GGTGGGGGCAGCGGTGGAGGTGGGAGCGGAGGCGGGGTAGCGGAGGCGGGGT AGCCAAATCACCTGAAAGAGAGCGGACCTACACTGGTCAAGCCTACCCAGACA CTGACCCTCACATGTACATTAGCGGCTTTAGCCTGAGCACCTCCGCGTGGGA GTGGGCTGGATCAGACAGCCCCCGCAAGGCCCTGGAATGGCTGGCTCTGATC TATTGGAATGACGAGAAGCGGTACAGCCCTAGCCTGAAATCTAGACTGACAATC ACCAAGGACACAGCAAGAACAGGTGGTGTGACCATGACCAACATGGATCCT TGTGATACCGCGTGTACTACTGCGCCAGAGAAGGCTCTCAGCACTACAAGAGC TCCAAGTGGTTCGACCATGGGGCCAGGGCACCTGGTTACAGTGTCTAGCACC ACAACACCTGCTCCAAGGCCCCACACCCGCTCCAACCTATAGCCAGCCAACCA TTGAGCCTCAGACCTGAAGCTTGCAGGCCCGCAGCAGGAGGCGCCGTCCATACG CGAGGCTTGGACTTCGCGTGTGATATTTATATTTGGGCACCTTTGGCCGGAACA TGTGGGGTGTGCTTCTCTCCCTTGTGATCACTCTGTATTGTGAAGCGCGGAGA AAGAAGCTCTGTACATCTTCAAGCAGCCTTTTATGCGACCTGTGCAAACTCACT CAGGAAGAAGATGGGTGTTTATGCGCTTCCCGAGGAGGAAGAAGGAGGGTGT GAACTGAGGGTGAATTTTCTAGAAGCGCGATGCTCCCGCATATCAGCAGGGT CAGAACTAGCTCTACAATGAATTGAATCTCGGCAGGCGAGAAGAGTACGATGTT CTGGACAAAAGACGGGGCAGGGATCCCGAGATGGGGGAAAGCCCGGAGAAAA AATCTCAGGAGGGGTGTACAATGAGCTGCAGAAGGACAAGATGGCTGAAGCC TATAGCGAGATCGAATGAAGGCGAAAGACGAGAGGCAAGGGGCATGACGGT

TABLE 13-continued

SEQ ID NO.	AA SEQ ID NO.	NUCLEIC ACID SEQUENCE
		CTGTACCAGGGTCTCTCTACAGCCACCAAGGACACTTATGATGCGTTGCATATG CAAGCCTTGCCACCCCGC
303	272	GATATCGTGATGACCCAATCTCCACTGAGCCTGCCTGTGACACCTGGCGAGCCT GCTTCTATCAGCTGTAGAAGCAGCCAGTCCCTGCTGCACAGCAACGGCTACAAC TACCTGGACTGGTATCTGCAGAAACCCGGCCAGAGCCCCAGCTGCTGATCTAC CTCGGCTCTAATCGGGCCAGCGGAGTGCTGATAGATTACGCGGAAGCGGCTCC GGCACCAGCTTCACCCTGAAGATCAGCAGAGTGGAGCCGAGGACGTGGGCGTC TACTCTTCATGCAGGCCCTGGGCTGATTACATTTGGCGGCGGAACCAAGGTG GAAATCAAGGGTGGTGGGGGAGCGGTGGAGGTGGAGCGGAGGCGGGGTAGC GGAGGCGGGGTAGCGAAGTGCAGCTGGTTGAGAGCGGCGGCGGACTGGTGAAG CCCCGAGGCAGCCTCAGACTGAGCTGTGCTGCTTCTGGCTTACCTTCAGCTCT TATAGCATGAACCTGGGTGCGCAGGCCCTGGCAAGGGCTTGAATGGGTGAGC TCCATCAGCTCTTCTAGCAGCTACATCTACTACGCCGACAGCGTGAAGGGCAGA TTCACCATCAGCAGAGATAACGCCAAGAACAGCCTGTACCTGCAGATGAATAGC CTGCGGGCCGAGGACACCGCGGTGTACTACTGCGCCAGAGCCGCGACACCTAC AGCGCCGCCGATTACTACTACATGGACGTGTGGGGCAAAGGAACAACCGTGACA GTGTCTCTCCACCACAACACCTGCTCCAAGGCCCCACACCCGCTCCAACCTATA GCCAGCCAACCATTTAGCCTCAGACCTGAAGCTTGCAGGCCCGCAGCAGGAGGC GCCCTCCATACGCGAGGCCTGGACTTCGCGTGTGATATTTATATTTGGGCACCT TTGGCCGGAAACATGTGGGGTGTGCTTCTCTCCCTTGTGATCACTCTGTATTGT AAGCGCGGGAAGAAAGCTCCTGTACATCTTCAAGCAGCCTTTTATGCGACCT GTGCAAAACCACTCAGGAAGAAGATGGGTGTTTATGCCGCTTCCCGAGGAGGAA GAAGGAGGGTGTGAACGTAGGGTGAATTTTCTAGAAGCGCCGATGCTCCCGCA TATCAGCAGGGTCAGAATCAGCTCTACATGAATGAATTCGCGCAGGCGAGAA GAGTACGATGTTCTGGACAAAAGACGGGGCAGGGATCCCGAGATGGGGGAAAG CCCCGGAGAAAAATCCTCAGGAGGGGTGTACAAATGAGCTGCAGAAAGGACAA ATGGCTGAAGCCTATAGCAGATCGGAATGAAGGCGAAAGACGCAGAGGCAAG GGGCATGACGGTCTGTACCAGGGTCTCTCTACAGCCACCAGGACACTTATGAT GCGTTGCATATGCAAGCCTTGCCACCCCGC
304	273	GAAGTGCAACTGCTGGAAGCGGCGAGGCCTGGTCCAGCCCGGCGGCTCTCTG CGGCTCAGCTGCGCCGCTTCTGGATTTACCTTCGCGCAGCGAGGCTATGAGCTGG GTGCGGCAGGCCCTTGGAAGAGAGAGAGCTGGTGTCCGCCATCAGCGGCAGC GGCAGGTGACCTACTACGCCGACAGCGTGAAGGGCAGATTACCATCTCTAGA GATAATAGCAAGAACACCTGTACCTGCAGATGAACAGCCTGAGAGCCGAGGAC ACCGCGCTGTACTATTGTGAGAGCTGGTGAAGCCAAAGCGGCACTGGGGCCAG GGCACACAGGTTACAGTGTCCAGCACCAACACCTGCTCCAAGGCCCCCAACA CCCGCTCCAACCTATAGCCAGCCAACCATTTAGCCTCAGACCTGAAGCTTGCAGG CCCCGAGCAGGAGGCGCCGTCCATACGCGAGGCTTGGACTTCGCGTGTGATATT TATATTTGGGCACCTTTGGCCGGAACATGTGGGGTGTGCTTCTCTCCCTTGTG ATCATCTGTATTGTAAGCGCGGAGAAAGAGCTCCTGTACATCTTCAAGCAG CCTTTTATGCGACCTGTGCAAAACCACTCAGGAAGAAGATGGGTGTTTATGCCG TTCCCCGAGGAGGAAGAGGAGGGTGTGAACGTAGGGTGAATTTCTAGAACG GCCGATGCTCCCGCATATCAGCAGGGTCAGAATCAGCTCTACAATGAATTGAAT CTCGGCAGGCGAGAAGAGTACGATGTTCTGGACAAAAGACGGGGCAGGGATCCC GAGATGGGGGAAAGCCCCGAGAAAAATCCTCAGGAGGGTGTGACAATGAG CTGCAGAAAGGACAAGATGGCTGAAGCCTATAGCAGATCGGAATGAAGGCGAA AGACGCAGAGGCAAGGGCATGACGGTCTGTACCAGGGTCTCTCTACAGCCACC AAGGACACTTATGATGCGTTGCATATGCAAGCCTTGCCACCCCGC
305	274	GAAGTGCAACTGCTGGAATCTGGCGGAGGACTGGTGACGCCCGGCGGCAGCCTG CGGCTGAGCTGTGCTGCTTCTGGCTTTACCTTCGAGTCTGAGGCCATGAGCTGG TATAGACAGGCCCTTGCAAGGAAAGAGAGCTGGTACGCGTGATCACCAGCGAG GGCTCCACCTACTACGCCGACAGCGTGAAGGCAGATTACAATCAGCCGGGAC AATAGCAAGAACACCTGTACCTGCAGATGAACAGCCTGCGCGCCGAAGATACA GCCGTGTACTACTGCGCCACATCGAGTGGGAGACAAGACTCAACTGGGGCCAG GGCACCAGGTGACCGTGTCCAGCACCAACACCTGCTCCAAGGCCCCCAACA CCCGCTCCAACCTATAGCCAGCCAACCATTTAGCCTCAGACCTGAAGCTTGCAGG CCCCGAGCAGGAGGCGCCGTCCATACGCGAGGCTTGGACTTCGCGTGTGATATT TATATTTGGGCACCTTTGGCCGGAACATGTGGGGTGTGCTTCTCTCCCTTGTG ATCACTCTGTATTGTAAGCGCGGAGAAAGAGCTCCTGTACATCTTCAAGCAG CCTTTTATGCGACCTGTGCAAAACCACTCAGGAAGAAGATGGGTGTTTATGCCG TTCCCCGAGGAGGAAGAGGAGGGTGTGAACGTAGGGTGAATTTCTAGAACG GCCGATGCTCCCGCATATCAGCAGGGTCAGAATCAGCTCTACAATGAATTGAAT CTCGCAGGCGAGAAGAGTACGATGTTCTGGACAAAAGACGGGGCAGGGATCCC GAGATGGGGGAAAGCCCCGAGAAAAATCCTCAGGAGGGTGTGACAATGAG CTGCAGAAAGGACAAGATGGCTGAAGCCTATAGCAGATCGGAATGAAGGCGAA AGACGCAGAGGCAAGGGCATGACGGTCTGTACCAGGGTCTCTCTACAGCCACC AAGGACACTTATGATGCGTTGCATATGCAAGCCTTGCCACCCCGC
306	275	GAGGTGCAGCTGCTGGAAGCGGAGGGGGCTGGTCCAACCCGCGGGTCTCTT CGCTAAGCTGTGCGCTTCTGGCTTACCTTCGACGAGTACACCATGCACTGG TTCAGACAGGCCCCCGCAAGGAGCGGAGTTCGTGAGTGAATCAGCGGAGGC GGTACGAGACTTATTACGCGGACTCCGTGAAGGGCCGCTTACCATTAGCCGC GACAACTCCAAGAACACGCTGTACCTGCAGATGAATTCGTGCGCGCCGAAGT GAGTACGAGTGTGCTGCTGCTTCTGGCTTACCTTCGAGTCTGAGGCCATGAGCTGG TATAGACAGGCCCTTGCAAGGAAAGAGAGCTGGTACGCGTGATCACCAGCGAG GGCTCCACCTACTACGCCGACAGCGTGAAGGCAGATTACAATCAGCCGGGAC AATAGCAAGAACACCTGTACCTGCAGATGAACAGCCTGCGCGCCGAAGATACA GCCGTGTACTACTGCGCCACATCGAGTGGGAGACAAGACTCAACTGGGGCCAG GGCACCAGGTGACCGTGTCCAGCACCAACACCTGCTCCAAGGCCCCCAACA CCCGCTCCAACCTATAGCCAGCCAACCATTTAGCCTCAGACCTGAAGCTTGCAGG CCCCGAGCAGGAGGCGCCGTCCATACGCGAGGCTTGGACTTCGCGTGTGATATT TATATTTGGGCACCTTTGGCCGGAACATGTGGGGTGTGCTTCTCTCCCTTGTG ATCACTCTGTATTGTAAGCGCGGAGAAAGAGCTCCTGTACATCTTCAAGCAG CCTTTTATGCGACCTGTGCAAAACCACTCAGGAAGAAGATGGGTGTTTATGCCG TTCCCCGAGGAGGAAGAGGAGGGTGTGAACGTAGGGTGAATTTCTAGAACG GCCGATGCTCCCGCATATCAGCAGGGTCAGAATCAGCTCTACAATGAATTGAAT CTCGCAGGCGAGAAGAGTACGATGTTCTGGACAAAAGACGGGGCAGGGATCCC GAGATGGGGGAAAGCCCCGAGAAAAATCCTCAGGAGGGTGTGACAATGAG CTGCAGAAAGGACAAGATGGCTGAAGCCTATAGCAGATCGGAATGAAGGCGAA AGACGCAGAGGCAAGGGCATGACGGTCTGTACCAGGGTCTCTCTACAGCCACC AAGGACACTTATGATGCGTTGCATATGCAAGCCTTGCCACCCCGC

TABLE 13-continued

SEQ ID NO.	AA SEQ ID NO.	NUCLEIC ACID SEQUENCE
		ACGGCCGTGTACTACTGTGCCGCTGGTGGGGAGGAGGCTGGCGTGGGCTATTGG GGCCAGGGCACCAGGTACCGTGTCTCCACCACAACACCTGCTCCAAGGCC CCCACACCCGCTCCAACCTATAGCCAGCCAACCATAGAGCTCAGACCTGAAGCT TGCAGGCCCGCAGCAGGAGGCGCCGTCATACGCGAGGCTGGACTTCGCGTGT GATATTTATATTGGGCACCTTTGGCCGGAACATGTGGGGTGTGTGCTTCTCTCC CTTGTGATCACTCTGTATTGTAAGCGCGGGAGAAAGAGCTCCTGTACATCTTC AAGCAGCCTTTTATGCGACCTGTGCAACCACTCAGGAAGAAGATGGGTGTTC TGCCGCTTCCCCGAGGAGGAAGAAGGAGGGTGTGAAGTGGGGTGAAATTTCT AGAAGCGCCGATGCTCCCGCATATCAGCAGGGTCAGAATCAGCTCTACAATGAA TTGAATCTCGCAGGCGAGAAGAGTACGATGTTCTGGACAAAAGACGGGGCAGG GATCCCCGAGATGGGGGAAAGCCCGGAGAAAAATCCTCAGGAGGGGTGTAC AATGAGCTGCAGAAGGACAAGATGGCTGAAGCCTATAGCGAGATCGGAATGAAA GGCGAAAGACGAGAGGCAAGGGCATGACGGTCTGTACCAGGGTCTCTTACA GCCACCAAGGACACTTATGATGCGTTGCATATGCAAGCCTTGCCACCCCGC
307	276	GAGGTGCAGCTGCTGGAGAGCGGAGGCGGCCCTCGTGCAGCCAGGAGGTTCCCTA CGACTCTCCTGTGCCGCCAGCGGCTTACCTTCGAGGACTACGCCATGAGTTGG TTCGCGCAGGCCCGGGGAAGGAGCGCAGGGCGTGAGCGGATTTCTGGAAAG GGCGGCTCCACCTATTACGCGGACTCCGTGAAGGGTCGCTTACCATCTCTCGC GACAACCTCAAGAACAACGCTGTACCTGCAGATGAATAGCCTGCGCGCTGAGGAC ACTGCGGTGTACTACTGTCTGTCTTGGACGAGGAGGCGCGCGCAGAGGGCGGC TATTGGGGCCAGGGTACCCAGGTACCGTGTCTGCCACCAACACCTGTCTCCA AGGCCCCCACCACCGCTCCAACCTATAGCCAGCCAACCATTGAGCCTCAGACCT GAAGCTTCAGGCCCGCAGCAGGAGGCGCGTCCATACGCGAGGCTGGACTTC GCGTGTGATATTTATATTGGGCACCTTTGGCCGGAACATGTGGGGTGTGTCTT CTCTCCCTTGTGATCACTCTGTATTGTAAGCGCGGGAGAAAGAGCTCCTGTAC ATCTTCAAGCAGCCTTTTATGCGACCTGTGCAAAACCACTCAGGAAGAAGATGGG TGTTTCATGCCCTTCCCCGAGGAGGAAGAAGGAGGGTGTGAACCTGAGGGTGAAA TTTTCTAGAAGCGCCGATGCTCCCGCATATCAGCAGGGTCAGAATCAGCTCTAC AATGAATTGAATCTCGGCAGGCGAGAAGAGTACGATGTTCTGGACAAAAGACGG GGCAGGGATCCCGAGATGGGGGAAAGCCCGGAGAAAAATCCTCAGGAGGGG TTGTACAATGAGCTGCAGAAGGACAAGATGGCTGAAGCCTATAGCGAGATCGGA ATGAAAGGCGAAAGACGAGAGGCAAGGGCATGACGGTCTGTACCAGGGTCTC TCTACAGCCACCAAGGACACTTATGATGCGTTGCATATGCAAGCCTTGCCACCC CGC
308	277	GAGGTGCAACTGCTGGAAAGCGGCGGTGGACTGGTGCAGCCCGGCGGCAGCCTG AGACTGTCTTGTGCTGCTTCTGGATTTACATTCGACAGATACGCCATGAGCTGG TTCCGCCAGGCCCTTGCCAAAGAGCGGGAAGGCGTGTCCGCCATCTCCACAAGC GGAGATAGCACATACTATGCCGACAGCGTGAAGGGCAGATTACCATCAGCAGA GATAATAGCAAGAACCCTGTACCTGCAGATGAACAGCCTCCGGGCCGAGGAC ACCGCCGTCTACTACTGCGCGTGTCTGGACGAGGAAGCGCGCGCCGAGGGCGGC TACTGGGGCCAGGGCACCAGGTGACCGTGTCTAGCACCAACACCTGTCTCCA AGGCCCCCACCACCGCTCCAACCTATAGCCAGCCAACCATTGAGCCTCAGACCT GAAGCTTCAGGCCCGCAGCAGGAGGCGCGTCCATACGCGAGGCTGGACTTC GCGTGTGATATTTATATTGGGCACCTTTGGCCGGAACATGTGGGGTGTGTCTT CTCTCCCTTGTGATCACTCTGTATTGTAAGCGCGGGAGAAAGAGCTCCTGTAC ATCTTCAAGCAGCCTTTTATGCGACCTGTGCAAAACCACTCAGGAAGAAGATGGG TGTTTCATGCCCTTCCCCGAGGAGGAAGAAGGAGGGTGTGAACCTGAGGGTGAAA TTTTCTAGAAGCGCCGATGCTCCCGCATATCAGCAGGGTCAGAATCAGCTCTAC AATGAATTGAATCTCGGCAGGCGAGAAGAGTACGATGTTCTGGACAAAAGACGG GGCAGGGATCCCGAGATGGGGGAAAGCCCGGAGAAAAATCCTCAGGAGGGG TTGTACAATGAGCTGCAGAAGGACAAGATGGCTGAAGCCTATAGCGAGATCGGA ATGAAAGGCGAAAGACGAGAGGCAAGGGCATGACGGTCTGTACCAGGGTCTC TCTACAGCCACCAAGGACACTTATGATGCGTTGCATATGCAAGCCTTGCCACCC CGC
309	278	GAGGTGCAACTGCTGGAAAGCGGCGGAGGACTCGTCCAGCCCGGCGGCAGCCTG CGGCTGAGCTGTGCTGCTTCTGGATTTACCTTCGCCAGCGACGCCATGAGCTGG TATAGCAGGCCCTTGCCAAAGAGCGGGAACCTGGTGTCCGCCATCAGCGGCTCT GGCGGCTCCACCTACTACGCCGATAGCGTGAAGGGCAGATTACAATCTCTAGA GATAATAGCAAGAACCCTGTACCTGCAGATGAACAGCCTGAGAGCCGAGGAC ACCGCCGTGTACTACTGCGCCGCTACGACAGCGCGGAGGCTTACCTGGCCCTT GACTACTGGGGCAGGGCACACAGGTGACCGTGTCTAGCACCAACACCTGCT CCAAGGCCCCCCACACCGCTCCAACCTATAGCCAGCCAACCATTGAGCCTCAGA CCTGAAGCTTGCAAGGCCCGCAGCAGGAGGCGCGTCCATACGCGAGGCTGGAC TTCGCGTGTGATATTTATATTGGGCACCTTTGGCCGGAACATGTGGGGTGTG CTTCTCTCCCTTGTGATCACTCTGTATTGTAAGCGCGGGAGAAAGAGCTCCTG TACATCTTCAAGCAGCCTTTTATGCGACCTGTGCAAAACCACTCAGGAAGAAGAT GGGTGTTCATGCCCTTCCCCGAGGAGGAAGAAGGAGGGTGTGAACCTGAGGGTG AAATTTTCTAGAAGCGCCGATGCTCCCGCATATCAGCAGGGTCAGAATCAGCTC TACAATGAATTGAATCTCGGCAGGCGAGAAGAGTACGATGTTCTGGACAAAAGA CGGGGCGAGGGATCCCGAGATGGGGGAAAGCCCGGAGAAAAATCCTCAGGAG GGGTTGTACAATGAGCTGCAGAAGGACAAGATGGCTGAAGCCTATAGCGAGATC GGAATGAAAGGCGAAAGACGAGAGGCAAGGGCATGACGGTCTGTACCAGGGT

TABLE 13-continued

SEQ ID NO.	AA SEQ ID NO.	NUCLEIC ACID SEQUENCE
		CTCTCTACAGCCACCAAGGACACTTATGATGCGTTGCATATGCAAGCCTTGCCA CCCCGC
310	279	GAGGTGCAACTGCTGGAAGCGGCGGAGGACTCGTCCAGCCCCGGCGCAGCCTG AGGCTGAGCTGTGCTGCTTCTGGCTTTACCTTCGACTCCTACACAATGAGCTGG TATAGACAGGCCCCCTGGCAAGGAGCGGGAACGTGTTCGCCATCAGCGGCCAC GGCGACTCTACATACTACGCCGACAGCGTGAAAGGCAGATTACAATCTCTAGA GATAATAGCAAGAACACCCCTGTACCTGCAGATGAACAGCCTGCGGGCCGAGGAC ACCGCCGTGTACTACTGCACCAGAATCAGCATCACACCGAGTGGCTGGCCGGA GATTACTGGGCCAGGGCACCCAGGTGACAGTGTCCAGCACCAACACCTGCT CCAAGGCCCCCACCACCCGCTCCAACCTATAGCCAGCCAACATTGAGCCTCAGA CCTGAAGCTTGCAGGCCCGCAGCAGGAGGCGCCGTCCATACGCGAGGCCCTGGAC TTCGCGTGTGATATTATATTGGGCACCTTTGGCCGGAACATGTGGGGTGTG CTTCTCTCCCTTGTGATCACTCTGTATTGTAAGCGCGGGAGAAAGAAGCTCCTG TACATCTTCAAGCAGCCTTTTATGCGACCTGTGCAAACTCAGGAAGAAGAT GGGTGTTTATGCCGCTTCCCCGAGGAGGAAGAAGGAGGGTGTGAAGTGGGGTG AAATTTTCTAGAAGCGCCGATGCTCCCGCATATCAGCAGGGTCAGAATCAGCTC TACAAATGAATTGAATCTCGGCAGGCGAGAAGAGTACGATGTTCTGGACAAAAGA CGGGGCAGGGATCCCGAGATGGGGGGAAGCCCCGGAGAAAAATCCTCAGGAG GGGTGTGACAATGAGCTGCAGAAGGACAAGATGGCTGAAGCCTATAGCGAGATC GGAATGAAAGGCGAAAGACGCGAGAGGCAAGGGGCATGACGGTCTGTACCGGGT CTCTCTACAGCCACCAAGGACACTTATGATGCGTTGCATATGCAAGCCTTGCCA CCCCGC
311	280	GAGGTGCAGCTGCTGGAAGCGGAGGAGGCTGGTCCAACCTGGCGGCAGCCTG CGGCTGAGCTGCGCCGCTTCTGGCTTCACCTTCAGCAGCTACGCCATGAGCTGG TTCGGCAGGCCCCCTGGCAAGGAAGAGAGTTCGTGTCTTTTATCAGCGGATCT GGCGACTCCACTACTACGCTGATAGCGTGAAAGGCAGATTACCATCTCTAGA GATAATAGCAAGAACACCCCTGTACCTCCAGATGAACAGCCTGCGCGCCGAGGAC ACAGCCGTGTACTATTGTACCAGATGGCTTACGACTTCGAGGAACCAAGCGAG CCCCGCGTGTACTGGGGCCAGGGCACACAGGTGACAGTGTCTCCACCACAACA CCTGCTCCAAGGCCCCCACCACCCGCTCCAACCTATAGCCAGCCAACATTGAGC CTCAGACCTGAAGCTTGCAGGCCCGCAGCAGGAGGCGCCGTCCATACGCGAGGC CTGGACTTCGCGTGTGATATTATATTGGGCACCTTTGGCCGGAACATGTGGG GTGTTGCTTCTCTCCCTTGTGATCACTCTGTATTGTAAGCGCGGGAGAAAGAAG CTCTGTACATCTTCAAGCAGCCTTTTATGCGACCTGTGCAAACTCAGGAA GAAGATGGGTGTTTATGCCGCTTCCCCGAGGAGGAAGAAGGAGGGTGTGAAGT AGGGTGAAATTTTCTAGAAGCGCCGATGCTCCCGCATATCAGCAGGGTCAGAAT CAGCTCTACAATGAATTGAATCTCGGCAGGCGAGAAGAGTACGATGTTCTGGAC AAAAGACGGGGCAGGGATCCCGAGATGGGGGGAAGCCCCGGAGAAAAATCCT CAGGAGGGGTGTGACAATGAGCTGCAGAAGGACAAGATGGCTGAAGCCTATAGC GAGATCGGAATGAAAGGCGAAAGACGCGAGAGGCAAGGGGCATGACGGTCTGTAC CAGGGTCTCTCTACAGCCACCAAGGACACTTATGATGCGTTGCATATGCAAGCC TTGCCACCCCCGC
312	281	GAGGTGCAGCTGCTGGAAGCGGCGGAGGCTGGTGCAACCTGGCGGATCTCTC AGACTGAGCTGTGCTGCTTCTGGCTTCACATTACCGACTACGACATGAGCTGG TATAGACAGGCCCCCTGGAAGAGAGCGGGAACGTGTTCCTGTATCCACAGCGGC GGCTCCACTACTACGCCGATAGCGTGAAAGGCAGATTACCATCAGCAGAGAT AATAGCAAGAACACCCCTGTACCTGCAGATGAACAGCCTGCGGGCCGAGGACACC GCCGTGTACTACTGCGCCCCCGGCTACTACAGCGACCTGTCTTTGATTATTAC AACTTCGACTACTGGGGCCAGGGCACACAGGTGACAGTGTCCAGCACCAACA CCTGCTCCAAGGCCCCCACCACCCGCTCCAACCTATAGCCAGCCAACATTGAGC CTCAGACCTGAAGCTTGCAGGCCCGCAGCAGGAGGCGCCGTCCATACGCGAGGC CTGGACTTCGCGTGTGATATTATATTGGGCACCTTTGGCCGGAACATGTGGG GTGTTGCTTCTCTCCCTTGTGATCACTCTGTATTGTAAGCGCGGGAGAAAGAAG CTCCTGTACATCTTCAAGCAGCCTTTTATGCGACCTGTGCAAACTCAGGAA GAAGATGGGTGTTTATGCCGCTTCCCCGAGGAGGAAGAAGGAGGGTGTGAAGT AGGGTGAAATTTTCTAGAAGCGCCGATGCTCCCGCATATCAGCAGGGTCAGAAT CAGCTCTACAATGAATTGAATCTCGGCAGGCGAGAAGAGTACGATGTTCTGGAC AAAAGACGGGGCAGGGATCCCGAGATGGGGGGAAGCCCCGGAGAAAAATCCT CAGGAGGGGTGTGACAATGAGCTGCAGAAGGACAAGATGGCTGAAGCCTATAGC GAGATCGGAATGAAAGGCGAAAGACGCGAGAGGCAAGGGGCATGACGGTCTGTAC CAGGGTCTCTCTACAGCCACCAAGGACACTTATGATGCGTTGCATATGCAAGCC TTGCCACCCCCGC
313	282	GAGGTGCAGCTGCTGGAAGAGCGGTGGAGGGTGGTGCAAGCCCGGGGTAGCCTG CGTCTGTGTCGCGCCGCTTCCGGCTTCACGTTCTCTGATTACGCCATGCACTGG TTCGGCAGGCCCCCGTAAGGAGCGGTGCTGGTGTCTATTGACTCCGGC GGCTCCACTTACTACGCAGACAGTGTCAAGGGCCGTTTACCATCAGCCGCGAC AACAGCAAGAACACGCTGTACCTGCAGATGAATCCCTTCGAGCAGAGGACACC GCGGTGTACTACTGTAATGCGGGCTTCAAGGGCGATCACCCACCCCAAGGAT GCCCTTCGACATTGGGGCCAGGGCACCCAGGTACCCGTGTCTCCACCACAACA CCTGCTCCAAGGCCCCCACCACCCGCTCCAACCTATAGCCAGCCAACATTGAGC CTCAGACCTGAAGCTTGCAGGCCCGCAGCAGGAGGCGCCGTCCATACGCGAGGC CTGGACTTCGCGTGTGATATTATATTGGGCACCTTTGGCCGGAACATGTGGG GTGTTGCTTCTCTCCCTTGTGATCACTCTGTATTGTAAGCGCGGGAGAAAGAAG CTCCTGTACATCTTCAAGCAGCCTTTTATGCGACCTGTGCAAACTCAGGAA GAAGATGGGTGTTTATGCCGCTTCCCCGAGGAGGAAGAAGGAGGGTGTGAAGT AGGGTGAAATTTTCTAGAAGCGCCGATGCTCCCGCATATCAGCAGGGTCAGAAT CAGCTCTACAATGAATTGAATCTCGGCAGGCGAGAAGAGTACGATGTTCTGGAC AAAAGACGGGGCAGGGATCCCGAGATGGGGGGAAGCCCCGGAGAAAAATCCT CAGGAGGGGTGTGACAATGAGCTGCAGAAGGACAAGATGGCTGAAGCCTATAGC GAGATCGGAATGAAAGGCGAAAGACGCGAGAGGCAAGGGGCATGACGGTCTGTAC CAGGGTCTCTCTACAGCCACCAAGGACACTTATGATGCGTTGCATATGCAAGCC TTGCCACCCCCGC

TABLE 13-continued

SEQ ID NO.	AA SEQ ID NO.	NUCLEIC ACID SEQUENCE
		GTGTTGCTTCTCTCCCTTGTGATCACTCTGTATTGTAAGCGCGGAGAAAGAAG CTCCTGTACATCTTCAAGCAGCCTTTTATGCGACCTGTGCAAACCACTCAGGAA GAAGATGGGTGTTTCATGCCGCTTCCCGAGGAGGAAGAAGGAGGGTGTGAATG AGGGTGAAATTTCTAGAAGCGCCGATGCTCCCGCATATCAGCAGGGTCAGAAT CAGCTCTACAATGAATTGAATCTCGGCAGGCGAGAAGAGTACGATGTTCTGGAC AAAAGACGGGCGAGGATCCCGAGATGGGGGGAAGCCCGGAGAAAAATCCT CAGGAGGGGTGTACAATGAGCTGCAGAAGGACAAGATGGCTGAAGCCTATAGC GAGATCGGAATGAAAGGCGAAAGACGCGAGAGGCAAGGGGCATGACGGTCTGTAC CAGGGTCTCTACAGCCACCAAGGACACTTATGATGCGTTGCATATGCAAGCC TTGCCACCCCGC
314	283	GAGGTGCAACTGCTGGAATCCGGCGGAGGCCCTGGTGCAGCCCGGCGCAGCCTC AGACTGAGCTGTGCCGCTTCTGGCTTTACCTTCAGCAGCGAGGGCATGAGCTGG GTGCGGCGAGGCCCTGGCAAGGAAAGAGAGCTGGTCTCCGCCATCAGCGGATCT GGCGACCAACCTACTATGCCGATAGCGTGCAGGGAAGATTCAACAATCTCTAGA GATAATAGCAAGAACACCTGTACCTGCAGATGAACAGCCTGCGGGCCGAGGAC ACCGCCGTGTACTACTGCAACGCCCTGGAAGGCGGCCCTACAACAGCTATCCAG CCAGGAGGCCCTGACTACTGGGGCCAGGGCACCCAGGTGACCGTGTCCAGCACC ACAACACCTGTCTCAAGGCCCCACACCCGCTCCAATATAGCCAGCCAAACCA TTGAGCCTCAGACCTGAAGCTTGCAGGCCCGCAGCAGGAGCGCCGTCCTATACG CGAGGCTGGACTTCGCGTGTGATATTATATTTGGGCACCTTTGGCCGGAACA TGTGGGGTGTGCTTCTCTCCCTTGTGATCACTCTGTATTGTAAGCGCGGAGAA AAGAAGCTCCTGTACATCTTCAAGCAGCCTTTTATGCGACCTGTGCAAACCACT CAGGAAGAAGATGGGTGTTTCATGCCGCTTCCCGAGGAGGAAGAAGGAGGGTGT GAACCTGAGGGTGAATTTTCTAGAAGCGCCGATGCTCCCGCATATCAGCAGGGT CAGAATCAGCTCTACAATGAATTGAATCTCGGCAGGCGAGAAGAGTACGATGTT CTGGCAAAAAGACGGGGCAGGGATCCCGAGATGGGGGGAAGCCCGGAGAAAA AATCCTCAGGAGGGGTGTACAATGAGCTGCAGAAGGACAAGATGGCTGAAGCC TATAGCGAGATCGGAATGAAAGGCGAAGACGCGAGAGGCAAGGGGCATGACGGT CTGTACCAGGGTCTCTCTACAGCCACCAAGGACACTTATGATGCGTTGCATATG CAAGCCTTGCCACCCCGC

In particular embodiments, polynucleotides encoding a chimeric antigen receptor may be codon-optimized. As used herein, the term “codon-optimized” refers to substituting codons in a polynucleotide encoding a polypeptide in order to modulate polypeptide expression, stability and/or activity. Factors that influence codon optimization include, but are not limited to one or more of: (i) variation of codon biases between two or more organisms or genes or synthetically constructed bias tables, (ii) variation in the degree of codon bias within an organism, gene, or set of genes, (iii) systematic variation of codons including context, (iv) variation of codons according to their decoding tRNAs, (v) variation of codons according to GC %, either overall or in one position of the triplet, (vi) variation in degree of similarity to a reference sequence for example a naturally occurring sequence, (vii) variation in the codon frequency cutoff, (viii) structural properties of mRNAs transcribed from the DNA sequence, (ix) prior knowledge about the function of the DNA sequences upon which design of the codon substitution set is to be based, (x) systematic variation of codon sets for each amino acid, and/or (xi) isolated removal of spurious translation initiation sites.

A “nucleic acid cassette,” “expression cassette” or “nucleic acid expression cassette” refers to polynucleotide sequences sufficient to transcribe an RNA, which is ultimately translated to a polypeptide. In particular embodiments, a nucleic acid cassette comprises a polynucleotide-of-interest, a polynucleotide that encodes a polypeptide, e.g., a CAR. Nucleic acid expression cassettes contemplated in particular embodiments comprise one or more expression control sequences, e.g., a promoter, enhancer, poly(A) sequence, and one or more polynucleotide(s)-of-interest. In particular embodiments, a vector contemplated herein comprises one or more nucleic acid cassettes. In particular

embodiments, a nucleic acid cassette is oriented in a vector to enable transcription of a polynucleotide-of-interest.

In particular embodiments, a polynucleotide encoding a polypeptide may be combined with other polynucleotide sequences, such as expression control sequences, promoters and/or enhancers, untranslated regions (UTRs), polynucleotides encoding signal peptides, Kozak sequences, polyadenylation signals, restriction enzyme sites, multiple cloning sites, internal ribosomal entry sites (IRES), recombinase recognition sites, termination codons, transcriptional termination signals, and polynucleotides encoding self-cleaving polypeptides or epitope tags, as disclosed elsewhere herein or as known in the art.

Polynucleotides can be prepared, manipulated, expressed and/or delivered using any of a variety of well-established techniques known and available in the art. In order to express a desired polypeptide, a nucleotide sequence encoding the polypeptide, can be inserted into an appropriate vector, e.g., a lentiviral vector.

In particular embodiments, a vector comprises a polynucleotide comprising or encoding one or more exogenous, endogenous, or heterologous expression control sequences operably linked to a polynucleotide encoding one or more polynucleotides and/or polypeptides contemplated herein.

“Expression control sequences,” “control elements,” or “regulatory sequences” contemplated in particular embodiments include but not limited to promoters, enhancers, translation initiation signals (Shine Dalgarno sequence or Kozak sequence), introns, polyadenylation signals, 5' and 3' untranslated regions, all of which may interact with host cell proteins to carry out transcription and translation.

The term “promoter” as used herein refers to a recognition site of a polynucleotide (DNA or RNA) to which an RNA polymerase binds. An RNA polymerase initiates and tran-

scribes polynucleotides operably linked to the promoter. In particular embodiments, promoters operative in mammalian cells comprise an AT-rich region located approximately 25 to 30 bases upstream from the site where transcription is initiated and/or another sequence found 70 to 80 bases upstream from the start of transcription, a CNCAAT region where N may be any nucleotide. The term “enhancer” refers to a segment of DNA which contains sequences capable of providing enhanced transcription and in some instances can function independent of their orientation relative to another control sequence. An enhancer can function cooperatively or additively with promoters and/or other enhancer elements. The term “promoter/enhancer” refers to a segment of DNA which contains sequences capable of providing both promoter and enhancer functions.

The term “operably linked”, refers to a juxtaposition wherein the components described are in a relationship permitting them to function in their intended manner. In one embodiment, the term refers to a functional linkage between an expression control sequence (such as a promoter, and/or enhancer) and a second polynucleotide sequence encoding a polypeptide, wherein the expression control sequence directs transcription of the nucleic acid corresponding to the second sequence.

Illustrative ubiquitous expression control sequences suitable for use in particular embodiments include, but are not limited to, a 3-actin promoter, a cytomegalovirus (CMV) immediate early promoter, a simian virus 40 (SV40) (e.g., early or late) promoter, a Moloney murine leukemia virus (MoMLV) promoter, a Rous sarcoma virus (RSV) promoter, a herpes simplex virus (HSV) (thymidine kinase) promoter, an SV40/CD43 promoter, a spleen focus forming virus (SFFV) promoter, an elongation factor 1-alpha (EF1 α) short promoter (intronless), an EF1 α long promoter containing an intron, a Ubiquitin C (UBC) promoter, a phosphoglycerate kinase-1 (PGK) promoter, a cytomegalovirus enhancer/chicken β -actin (CAG) promoter, and a myeloproliferative sarcoma virus enhancer, negative control region deleted, dl587rev primer-binding site substituted (MND) U3 promoter (Haas et al., *Journal of Virology*, 2003; 77(17): 9439-9450).

Illustrative examples of ubiquitous expression control sequences suitable for use in particular embodiments contemplated herein include those comprising polynucleotide sequences set forth in Table 14.

TABLE 14

SEQ ID NO:	NUCLEIC ACID SEQUENCE
318	GCGTTACATAAAGTACGGTAAATGGCCCGCCTGGCTGACCGCCCAACGACCCCGCCCATTTGAC GTCAATAATGACGTATGTTCCCATAGTAACGCCAATAGGGAATTTCCATTGACGTCAATGGGTG GAGTATTTACGGTAACTGCCCACTTGGCAGTACATCAAGTGTATCATATGCCAAGTACGCCCC CTATTGACGTCAATGACGGTAAATGGCCCGCCTGGCATTATGCCAGTACATGACCTTATGGGA CTTTCCTACTTGGCAGTACATCTACGTATTAGTCATCGCTATTACCATGGTACAGGTGAGCCCC ACGTTCTGCTTCACTCTCCCCATCTCCCCCCCCCTCCCCACCCCAATTTTGTATTTATTTATTT TTTAATTATTTTGTGTCAGCGATGGGGGCGGGGGGGGGGGGGGGGGGGGGGGGGGGGGGGGGGG GCGGGGCGAGGGGCGGGGCGGGGCGAGGCGGAGAGGTGCGGCGGAGGCAATCGAGGCGGCGCG CTCCGAAAGTTTCTTTTATGGCGAGGCGGCGGCGGCGGCGGCGGCGGCGGCGGCGGCGGCGG GCGGGGCG
319	CGTGAGGCTCCGGTGCCCGTCAGTGGGCGAGCGCACATCGCCACAGTCCCCGAGAAGTTGGG GGGAGGGGTCCGCAATTGAACCGGTGCCTAGAGAAGGTGGCGCGGGGTAACTGGGAAAGTGAT GTCGTGTACTGGCTCCGCTTTTCCGAGGGTGGGGGAGAACCCTATATAAGTCAGTGTAGTCG CCGTGAACGTTCTTTTTCGCAACGGGTTTGGCCGAGAACACAGGTAAGTGGCGTGTGGTTTC CCGCGGGCCTGGCCTCTTTACGGGTTATGGCCCTTGCCTGCTTGAATTAATCTCCACCTGGCTG CAGTACGTGATTTGATCCCGAGCTTCGGGTTGGAAGTGGGTGGGAGAGTTCGAGGCGCTTGGC CTTAAGGAGCCCTTCGCCTCGTGCTTGAGTTGAGGCTTGGCCTGGGCGCTGGGGCGCGCGCGT GCGAATCTGGTGGCACCTTCGCGCCTGTCTCGCTGCTTTCGATAAGTCTCTAGCCATTAAAAA TTTTGATGACCTGCTGCGACGCTTTTTCGCAAGATAGTCTTGTAAATGCGGCGCAAGATC TGCACACTGGTATTTTCGTTTTCGGGCGCGGGCGGCGACGGGCGGCGGCGGCGGCGGCGGCGG ATGTTTCGGCGAGGCGGGGCTGCGAGCGCGGCGCACGAGAATCGGACGGGGGTAGTCTCAAGCT GGCCGGCCTGCTCTGGTGCCTGGCTCGCGCCGCGTGTATCGCCCCGCCCTGGGCGGCAAGGC TGGCCCGGTGCGCACAGTTGCGTGAGCGGAAAGATGGCCGCTTCCCGGCCCTGCTGCGAGGAG CTCAAAATGAGGACGCGCGCTCGGAGAGCGGGCGGGTGAATCACCACACAAGGAAAAAGG GCCTTTCCGTCCTCAGCCGTCGCTTCATGTGACTCCACGGAGTACCGGCGCGGTCACGACACC TCGATTAGTTCTCGAGCTTTTGGAGTACGTGCTCTTTAGGTTGGGGGGAGGGGTTTATGCGAT GGAGTTTCCCACTGAGTGGGTGGAGACTGAAGTTAGGCCAGCTTGGCACTTGATGTAATTC TCCTTGAATTTGCCCTTTTTCGTTTGGATTTGGTATCTGGTTCATTCTCAAGCCTCAGACAGTGGTT AAGTTTTCCTTCATTTCAGGTGTCGTGA
320	AATGAAAGACCCACCTGTAGGTTTGGCAAGCTAGGATCAAGGTTAGGAACAGAGAGACAGCAG AATATGGGCCAAACAGGATATCTGTGTAAGCAGTTCTGCCCCGGCTCAGGGCCAAGAACAGT TGAACAGCAGAAATATGGGCCAAACAGGATATCTGTGTAAGCAGTTCTGCCCCGGCTCAGGG CCAAGAACAGATGGTCCCGAGATGCGGTCCCGCCCTCAGCAGTTTCTAGAGAACCATCAGATGT TTCCAGGGTGCCCCAAGGACCTGAATGACCTGTGCCTTATTTGAACAAACCATCAGTTCCG TTCTCGCTTCTGTTCGCGCCTTCTGCTCCCGAGCTCAATAAAGAGGCCACAACCCCTCACT CGGC
321	GGGTAGGGGAGGCGCTTTTCCCAAGGAGTCTGGAGCATGCGCTTTAGCAGCCCCGCTGGGCAC TTGGCGCTACACAAGTGGCCTCTGGCCTCGCACACATTCACATCCACCGGTAGGCGCCAACCG GCTCCGTTCTTTGGTGGCCCCCTTCGCGCACCTTCTACTCTCCCTAGTACAGGAAGTTCCCCC CCGCCCCCGAGCTCGCTGCTGAGGACGTGACAAATGGAAGTAGCACGTCTCACTAGTCTCGT CGAGATGGACGACCGCTGAGCAATGGAAGCGGGTAGGCCTTTGGGGCAGCGGCCAATAGCAG CTTTGTCTCTTCGCTTTCTGGGCTCAGAGGCTGGGAGGGGTGGGTCCGGGGGCGGGCTCAGGG

SEQ
ID
NO: NUCLEIC ACID SEQUENCE

CGCGGGCTCAGGGGCGGGGCGGGCGCCGAAGGTCCTCCGGAGGCCCGGCATTCTGCACGCTTCA
AAAGCGCAGCTCTGCCGCGCTGTTTCTCTCTTCTCATCTCCGGGCCCTTTCG

322 TGAAGAGCCCCACCTGTAGGTTTGGCAAGATAGCTGCAGTAACGCCATTTTGAAGGCATGGAA
AAATACCAAACCAAGATAGAGAAGTTTCAGATCAAGGGCGGGTACATGAAAAAGCTAACGTTG
CGCCAAACAGGATATCTCGGTTGAGCAGTTTCGGCCCCGGCCCCGGGGCCAAAGAACAGATGGTCA
CGCGAGTTTCGGCCCGGGCCGAGGCCAAGAACAGATGGTCCCCAGATATGGCCCCAACCCCTCAG
CAGTTTCTTAAGCCCATCAGATGTTTCCAGGCTCCCCAAGGACCTGAAATGACCTTGCGCCT
TATTTGAATTAAACCAATCAGCCTGCTTCTCGCTTCTGTTTCGCGCGCTTCTGCTTCCGAGCTCT
ATAAAGAGCTCACAAACCCTCACTCGGCGCGCCAGTCTCTCGATTGACTGAGTCGCC

323 GGCTCTCCGCGCGGGTTTGGCGCTCTCCGCGGGGCGCCCCCTCTCACGGCGAGCGCTGCCAC
GTACAGCAAGGGCGCAGCGAGCTCTGATCTCTCCGCCGGAGCTCAGGACAGCGGCCCGCC
TGCTCATAAGACTCGGCCCTAGAACCCAGTATCAGCAGAAGGACATTTTAGGACGGGACTTGG
GTGACTCTAGGGCAGCTGGTTTTCTTCAGAGAGCGGAAACAGGCGAGGAAAAAGTAGTCCCTTCT
CGCGGATTCTCGGAGGGATCTCCGTGGGGCGGTGAACCGGATGATTATTAAGACAGCGCGCG
GGTGTGGCAGAGCTAGTTCGTCGCAGCGGGGATTGGGTTCGCGGTTCTTGTTTGTGGATCGCT
GTGATCGTCACTTGGTGAGTAGCGGGCTGCTGGGCTGGCCGGGGCTTTCGTGGCGCGCGGGCCG
CTCGGTGGGACGGAAGCGTGTGGAGAGCACCCGCAAGGCTGTAGTCTGGGTCGCGCAGCAAGGT
TGCCCTGAACCTGGGGGTTGGGGGAGCGCAGCAAAATGGCCGCTGTCCCGAGTCTTGAATGGA
AGACGCTTGTGAGGCGGGCTGTGAGGTCGTTGAAACAAGGTGGGGGGATGGTGGGCGGCAGAA
ACCAAGGCTTGTAGGCCCTTCGTAATGCGGGAAAGCTCTTATTCCGGTGTAGATGGGCTGGGGC
ACCATCTGGGGACCCTGACGTGAAGTTGTGTACTAGCTGGAGACCTCGGTTTGTCTGTGTTC
GGGGGCGGCAGTTATGCGGTGCCGTTGGGCAGTGACCCGTAACCTTTGGGAGCGCGCGCCCTC
GTCGTGTCTGTGAGCTCACCCGTTCTGTTGGCTTATAAGTCAAGGGTGGGGCCACCTGCCGCTAGG
GTGCGGTAGGCTTTTCTCCGTCGAGGACGAGGCTCGGGCCTAGGAGAGGCTCTGTAAT
CGACAGGCGCGGACCTCTGGTGAGGGGAGGATAAGTGAGCGCTACGCTCTTTGGTTCGGTTT
TAGTGACCTATCTTCTTAAGTAGCTGAAGCTCCGGTTTGAACATATGCGCTTCGGGGTGTGGCAG
TGTGTTTGTGAAGTTTGTAGCACTTTGAATGTAACTGTAATTTGGGTCAATATGTAATTTT
CAGTGTTAGACTAGTAAATTGTCCGCTAAATCTGGCCGTTTTTGGCTTTTTTGTAGAC

In particular embodiments, a vector comprises a promoter comprising a polynucleotide sequence set forth in any one of SEQ ID NOs: 318, 319, 320, 321, 322, and 323 operably linked to a polynucleotide sequence encoding a CAR set forth in any one of SEQ ID NOs: 255, 256, 257, 258, 259, 260, 261, 262, 263, 264, 265, 266, 267, 268, 269, 270, 271, 272, 273, 274, 275, 276, 277, 278, 279, 280, 281, 282, and 283. In particular embodiments, a vector comprises a promoter comprising a polynucleotide sequence set forth in any one of SEQ ID NOs: 318, 319, 320, 321, 322, and 323 operably linked to a polynucleotide sequence encoding a signal peptide set forth in SEQ ID NO: 245 and a polynucleotide encoding a CAR set forth in any one of SEQ ID NOs: 259, 263, 266, 270, 273, and 277. In particular embodiments, a vector comprises a promoter comprising a polynucleotide sequence set forth in any one of SEQ ID NOs: 318, 319, 320, 321, 322, and 323 operably linked to a polynucleotide comprising a polynucleotide sequence set forth in any one of SEQ ID NOs: 295, 296, 297, 298, 299, 300, 301, 302, 303, 304, 305, 306, 307, 308, 309, 310, 311, 312, 313, and 314. In particular embodiments, a vector comprises a promoter comprising a polynucleotide sequence set forth in any one of SEQ ID NOs: 318, 319, 320, 321, 322, and 323 operably linked to a polynucleotide sequence set forth in SEQ ID NO: 294 and a polynucleotide sequence set forth in any one of SEQ ID NOs: 295, 296, 297, 298, 299, 300, 301, 302, 303, 304, 305, 306, 307, 308, 309, 310, 311, 312, 313, and 314.

In particular embodiments, a vector comprises a promoter comprising a polynucleotide sequence set forth in SEQ ID NO: 319 operably linked to a polynucleotide sequence encoding a CAR set forth in any one of SEQ ID NOS: 259, 263, 266, 270, 273 or 277. In particular embodiments, a vector comprises a promoter comprising a polynucleotide sequence set forth in SEQ ID NO: 319 operably linked to a

polynucleotide comprising a polynucleotide sequence set forth in any one of SEQ ID NOs: 297, 299, 300, 302, 304, and 308. In particular embodiments, a vector comprises a promoter comprising a polynucleotide sequence set forth in SEQ ID NO: 319 operably linked to a polynucleotide sequence set forth in SEQ ID NO: 294 and a polynucleotide sequence set forth in any one of SEQ ID NOs: 297, 299, 300, 302, 304, and 308.

In particular embodiments, a vector comprises a promoter comprising a polynucleotide sequence set forth in SEQ ID NO: 320 operably linked to a polynucleotide sequence encoding a CAR set forth in any one of SEQ ID NOs: 259, 263, 266, 270, 273 or 277. In particular embodiments, a vector comprises a promoter comprising a polynucleotide sequence set forth in any one of SEQ ID NO: 320 operably linked to a polynucleotide comprising a polynucleotide sequence set forth in any one of SEQ ID NOs: 297, 299, 300, 302, 304, and 308. In particular embodiments, a vector comprises a promoter comprising a polynucleotide sequence set forth in SEQ ID NO: 320 operably linked to a polynucleotide sequence set forth in SEQ ID NO: 294 and a polynucleotide sequence set forth in any one of SEQ ID NOs: 297, 299, 300, 302, 304, and 308.

In particular embodiments, expression of polynucleotide sequences may be modulated by incorporating posttranscriptional regulatory elements into vectors. A variety of posttranscriptional regulatory elements may increase expression of a heterologous nucleic acid, e.g., woodchuck hepatitis virus posttranscriptional regulatory element (WPRE; Zufferey et al., 1999, *J. Virol.*, 73: 2886); the posttranscriptional regulatory element present in hepatitis B vims (HPRE) (Huang et al., *Mol. Cell. Biol.*, 5:3864); and the like (Liu et al., 1995, *Genes Dev.*, 9:1766).

Illustrative examples of posttranscriptional control sequences suitable for use in particular embodiments con-

templated herein include those comprising polynucleotide sequences set forth in Table 15.

encodes an MNDU3 promoter (e.g., SEQ ID NO: 320) operably linked to a polynucleotide sequence set forth in

TABLE 15

SEQ ID NO:	NUCLEIC ACID SEQUENCE
315	AATCAACCTCTGGATTACAAAATTTGTGAAAGATTGACTGGTATTCTTAACATATGTTGCTC CTTTTACGCTATGTGGATACGCTGCTTTAATGCCTTTGTATCATGCTATTGCTTCCCGTAT GGCTTTCATTTTCTCCTTGTATAAATCCTGGTTGCTGTCTCTTTATGAGGAGTTGTGG CCCGTTGTCAAGCAACGTGGCGTGGTGTGCACTGTGTTTGTGACGCAACCCCCACTGGTT GGGGCATTGCCACCACCTGTGAGTCCTTTCCGGGACTTTCGCTTTCCCCCTCCCTATTGC CACGGCGGAACATCATCGCGCTGCTTGCCTGCTGGACAGGGGCTCGGCTGTTGGGC ACTGACAATTCGTTGTTGTGCGGGAAGCTGACGTCCTTTCCATGGCTGCTCGCCTGTG TTGCCACCTGGATTCTGCGCGGACGTCCTTCTGCTACGTCCTTCGGCCCTCAATCCAGC GGACCTTCCTTCCGCGGCTGCTGCGGCTCTGCGGCTCTTCGCGTCTTCGCCTTCGC CCTCAGACGAGTCGATCTCCCTTTGGGCCGCTCCCGGCTG
316	AATCAACCTCTGGATTACAAAATTTGTGAAAGATTGACTGGTATTCTTAACATATGTTGCTC CTTTTACGCTATGTGGATACGCTGCTTTAATGCCTTTGTATCATGCTATTGCTTCCCGTAT GGCTTTCATTTTCTCCTTGTATAAATCCTGGTTGCTGTCTCTTTATGAGGAGTTGTGG CCCGTTGTCAAGCAACGTGGCGTGGTGTGCACTGTGTTTGTGACGCAACCCCCACTGGTT GGGGCATTGCCACCACCTGTGAGTCCTTTCCGGGACTTTCGCTTTCCCCCTCCCTATTGC CACGGCGGAACATCATCGCGCTGCTTGCCTGCTGGACAGGGGCTCGGCTGTTGGGC ACTGACAATTCGTTGTTGTGCGGGAAGTCTGCTGAGACTCGGGCTGCTCGCCTGTG TTGCCACCTGGATTCTGCGCGGACGTCCTTCTGCTACGTCCTTCGGCCCTCAATCCAGC GGACCTTCCTTCCGCGGCTGCTGCGGCTCTGCGGCTCTTCGCGTCTTCGCCTTCGC CCTCAGACGAGTCGATCTCCCTTTGGGCCGCTCCCGGCTG
317	AATCAACCTCTGGATTACAAAATTTGTGAAAGATTGACTGATATTCTTAACATATGTTGCTC CTTTTACGCTGTGTGGATATGCTGCTTTAATGCCTCTGTATCATGCTATTGCTTCCCGTAT GGCTTTCGTTTCTCCTTGTATAAATCCTGGTTGCTGTCTCTTTATGAGGAGTTGTGG CCCGTTGTCCGTCAACGTGGCGTGGTGTGCTCTGTGTTTGTGACGCAACCCCCACTGGCT GGGCATTGCCACCACCTGTCAACTCCTTTCTGGGACTTTCGCTTTCCCCCTCCGATCGC CACGGCAGAACTCATCGCGCTGCTTGCCTGCTGGACAGGGCTAGGTTGTGTTGGC ACTGATAATTCGTTGGTGTGTC

In particular embodiments, a vector comprises or encodes (in the case of an RNA vector, e.g., a retroviral vector) an MNDU3 promoter (e.g., SEQ ID NO: 320) operably linked to a polynucleotide encoding a chimeric antigen receptor comprising an amino acid set forth in any one of SEQ ID NOs: 259, 263, 266, 270, 273 or 277, and a polynucleotide comprising a posttranscriptional regulatory element set forth in any one of SEQ ID NOs: 315, 316, and 317. In particular embodiments, a vector comprises or encodes an MNDU3 promoter (e.g., SEQ ID NO: 320) operably linked to a polynucleotide sequence encoding a signal peptide set forth in SEQ ID NO: 245 and a polynucleotide encoding a chimeric antigen receptor comprising an amino acid set forth in any one of SEQ ID NOs: 259, 263, 266, 270, 273 or 277, and a polynucleotide comprising a posttranscriptional regulatory element set forth in any one of SEQ ID NOs: 315, 316, and 317. In particular embodiments, a vector comprises or encodes an MNDU3 promoter (e.g., SEQ ID NO: 320) operably linked to a polynucleotide sequence set forth in SEQ ID NO: 294, a polynucleotide encoding a chimeric antigen receptor comprising an amino acid set forth in any one of SEQ ID NOs: 259, 263, 266, 270, 273 or 277, and a polynucleotide comprising a posttranscriptional regulatory element set forth in any one of SEQ ID NOs: 315, 316, and 317.

In particular embodiments, a vector comprises or encodes an MNDU3 promoter (e.g., SEQ ID NO: 320) operably linked to a polynucleotide encoding a chimeric antigen receptor comprising a polynucleotide sequence set forth in any one of SEQ ID NOs: 297, 299, 300, 302, 304, and 308, and a polynucleotide comprising a posttranscriptional regulatory element set forth in any one of SEQ ID NOs: 315, 316, and 317. In particular embodiments, a vector comprises or

SEQ ID NO: 294, a polynucleotide sequence set forth in any one of SEQ ID NOs: 297, 299, 300, 302, 304, and 308, and a polynucleotide comprising a posttranscriptional regulatory element set forth in any one of SEQ ID NOs: 315, 316, and 317.

In particular embodiments, a vector comprises or encodes (in the case of an RNA vector, e.g., a retroviral vector) an EF1 α promoter (e.g., SEQ ID NO: 319) operably linked to a polynucleotide encoding a chimeric antigen receptor comprising an amino acid set forth in any one of SEQ ID NOs: 259, 263, 266, 270, 273 or 277, and a polynucleotide comprising a posttranscriptional regulatory element set forth in any one of SEQ ID NOs: 315, 316, and 317. In particular embodiments, a vector comprises or encodes an EF1 α promoter (e.g., SEQ ID NO: 319) operably linked to a polynucleotide sequence encoding a signal peptide set forth in SEQ ID NO: 245 and a polynucleotide encoding a chimeric antigen receptor comprising an amino acid set forth in any one of SEQ ID NOs: 259, 263, 266, 270, 273 or 277, and a polynucleotide comprising a posttranscriptional regulatory element set forth in any one of SEQ ID NOs: 315, 316, and 317. In particular embodiments, a vector comprises or encodes an EF1 α promoter (e.g., SEQ ID NO: 319) operably linked to a polynucleotide sequence set forth in SEQ ID NO: 294, a polynucleotide encoding a chimeric antigen receptor comprising an amino acid set forth in any one of SEQ ID NOs: 259, 263, 266, 270, 273 or 277, and a polynucleotide comprising a posttranscriptional regulatory element set forth in any one of SEQ ID NOs: 315, 316, and 317.

In particular embodiments, a vector comprises or encodes an EF1 α promoter (e.g., SEQ ID NO: 319) operably linked to a polynucleotide encoding a chimeric antigen receptor comprising a polynucleotide sequence set forth in any one of

SEQ ID NOs: 297, 299, 300, 302, 304, and 308, and a polynucleotide comprising a posttranscriptional regulatory element set forth in any one of SEQ ID NOs: 315, 316, and 317. In particular embodiments, a vector comprises or encodes an EF1 α promoter (e.g., SEQ ID NO: 319) operably linked to a polynucleotide sequence set forth in SEQ ID NO: 294, a polynucleotide sequence set forth in any one of SEQ ID NOs: 297, 299, 300, 302, 304, and 308, and a polynucleotide comprising a posttranscriptional regulatory element set forth in any one of SEQ ID NOs: 315, 316, and 317.

Efficient expression of polynucleotides can also be increased in some embodiments, by using sequences that increase translational efficiency, e.g., through an increase in mRNA ribosomal binding or an increase in mRNA stability. In certain embodiments, polynucleotides encoding a chimeric antigen receptor comprise a short recognition sequence, i.e., a Kozak sequence, that greatly facilitates the initial binding of mRNA to the small subunit of the ribosome and increases translation. The consensus Kozak sequence is (GCC)RCCATGG, where R is a purine (A or G) (Kozak, *Cell*. 44:283-92 (1986), and Kozak, *Nucleic Acids Res.* 15:8125-48 (1987)).

Elements directing the efficient termination and polyadenylation of heterologous nucleic acid transcripts may also increase heterologous gene expression. Transcription termination signals are generally found downstream of the polyadenylation signal. In particular embodiments, vectors comprise a polyadenylation sequence 3' to a sequence to be transcribed and/or expressed. The term "polyadenylation (or poly(A)) signal" refers to a DNA sequence which directs both the termination and polyadenylation of the nascent RNA transcript by RNA polymerase II. Polyadenylation signals can promote mRNA stability by addition of a poly(A) tail to the 3' end of the coding sequence and thus, contribute to increased translational efficiency. Cleavage and polyadenylation are directed by a poly(A) signal in the RNA. The core poly(A) signal for mammalian pre-mRNAs has two recognition elements flanking a cleavage-polyadenylation site. Typically, an almost invariant AAUAAA hexamer lies 20-50 nucleotides upstream of a more variable element rich in U or GU residues. Cleavage of the nascent transcript occurs between these two elements and is coupled to the addition of up to 250 adenosines to the 5' cleavage product. In particular embodiments, the core poly(A) signal is an ideal poly(A) signal (e.g., AATAAA, ATTAAA, AGTAAA). In particular embodiments, the poly(A) signal is an SV40 poly(A) signal, a bovine growth hormone poly(A) signal (BGHpA), a rabbit β -globin poly(A) signal (r β gpA), variants thereof, or another suitable heterologous or endogenous poly(A) signal known in the art. In particular embodiments, the poly(A) signal is synthetic.

In particular embodiments, a polynucleotide comprises or encodes a promoter operably a polynucleotide sequence encoding a chimeric antigen receptor comprising a signal peptide isolated from a polypeptide selected from the group consisting of CD8 α , murine IgGk, human IgGk, CD33, tPA, SEAP, hGM-CSF, gaussian luciferase, CSF2R, B2M, and CD80, wherein the signal peptide is subsequently cleaved from the translated chimeric antigen receptor. In particular embodiments, a polynucleotide comprises or encodes a promoter operably linked to a polynucleotide sequence encoding a chimeric antigen receptor comprising a signal peptide comprising an amino acid sequence set forth in any one of SEQ ID NOs: 245, 246, 247, 248, 249, 250, 251, 252, 253, and 254. An illustrative example of a polynucleotide encoding a signal peptide is set forth in SEQ ID NO: 294 (5'

ATGGCTCTTCCCGTAACAGCCCTTTGTTGCCCTTGCCTCTCTCTGCATGCA GCACGACCG 3').

H. Vectors

Recombinant lentiviral particles contemplated in particular embodiments comprise (i) a viral envelope comprising (a) a mutated vesiculovirus envelope glycoprotein, e.g., VSIV-G, COCV-G, that does not bind its cognate receptor, e.g., LDLR, and (b) a non-viral membrane-bound tropism polypeptide that redirects the particle to immune effector cells; and (ii) a lentiviral vector comprising a polynucleotide encoding a promoter operably linked to a polynucleotide encoding an anti-BCMA chimeric antigen receptor comprising an amino acid sequence set forth in any one of SEQ ID NOs: 255, 256, 257, 258, 259, 260, 261, 262, 263, 264, 265, 266, 267, 268, 269, 270, 271, 272, 273, 274, 275, 276, 277, 278, 279, 280, 281, 282, and 283.

In various embodiments, a lentiviral vector (lentivector) is engineered or derived from a lentivirus genome selected from the group consisting of HIV-1, HIV-2, VMV, CAEV, ELAV, FIV, BIV, and SIV. In particular embodiments, lentiviral vectors are derived from HIV viral genomes, preferably HIV-1 or HIV-2 viral genomes and more preferably, HIV-1 viral genomes (i.e., HIV-1 cis-acting sequence elements are preferred).

In various embodiments, a recombinant lentiviral particle comprises (i) a viral envelope comprising (a) a mutated vesiculovirus envelope glycoprotein, e.g., VSIV-G, COCV-G, that does not bind its cognate receptor, e.g., LDLR, and (b) a non-viral membrane-bound tropism polypeptide that redirects the particle to immune effector cells that express CD3; and (ii) two copies of a lentiviral vector-based RNA genome comprising a 5' long terminal repeat (LTR) comprising R and U5 regions; a Psi (Ψ) packaging signal; a cPPT/FLAP, an export element; a polynucleotide comprising or encoding a promoter operably linked to a polynucleotide encoding an anti-BCMA CAR; optionally a WPRE or HPRE; a 3' LTR comprising U3 and R regions; a polyadenylation signal and a poly(A) tail.

In various embodiments, a recombinant lentiviral particle comprises (i) a viral envelope comprising (a) a mutated vesiculovirus envelope glycoprotein, e.g., VSIV-G, COCV-G, that does not bind its cognate receptor, e.g., LDLR, and (b) a non-viral membrane-bound tropism polypeptide that redirects the particle to immune effector cells that express CD3; and (ii) two copies of a lentiviral vector-based RNA genome comprising a 5' long terminal repeat (LTR) comprising R and U5 regions; a Psi (Ψ) packaging signal; a cPPT/FLAP, an export element; a polynucleotide comprising or encoding a promoter operably linked to a polynucleotide encoding an anti-BCMA CAR comprising an amino acid sequence set forth in any one of SEQ ID NOs: 259, 263, 266, 270, 273, and 277; optionally a WPRE or HPRE; a 3' LTR comprising U3 and R regions; a polyadenylation signal and a poly(A) tail.

The term "long terminal repeat (LTR)," as used herein, refers to elements located at the ends of lentiviral polynucleotides which, in their natural sequence context, are direct repeats and contain U3, R and U5 regions. LTRs generally provide functions fundamental to the expression of lentiviral genes (e.g., promotion, initiation and polyadenylation of gene transcripts) and to lentiviral replication. The LTR contains numerous regulatory signals including transcriptional control elements, polyadenylation signals and sequences needed for replication and integration of the lentiviral genome. The lentiviral LTR is divided into three

regions called U3, R and U5. The U3 region contains the enhancer and promoter elements. The U5 region is the sequence between the primer binding site and the R region and contains the polyadenylation signal. The R (repeat) region is flanked by the U3 and U5 regions. A transfer plasmid, which is used to package a lentiviral vector genome comprises a 5' LTR comprising U3, R and/or U5 regions and a 3' LTR comprising U3, R and/or U5 regions. Adjacent to the 5' LTR are sequences necessary for reverse transcription of the genome (the tRNA primer binding site) and for efficient packaging of lentiviral RNA into particles (the Psi "Ψ" site). A lentiviral vector-based genome packaged in a particle comprises a 5' LTR comprising R and U5 regions and a 3' LTR comprising U3 and R regions. The lentiviral vector-based genome is reverse transcribed and integrated into the host cell genome as a provector. Through reverse transcription and second strand synthesis of the lentiviral vector genome, provectors comprise two copies of the 3' LTR, one copy that replaces the 5' LTR and the 3' LTR.

A "TAR" element as used herein, refers to the "trans-activation response" genetic element located in the R region of lentiviral vector LTRs. This element interacts with the lentiviral trans-activator (tat) genetic element to enhance lentiviral vector genome replication. In third generation lentiviral vectors, this element is not usually present because lentiviral vector transfer plasmids comprise a 5' LTR U3 region replaced by a heterologous promoter.

An "R region," as used herein, refers to the region within LTRs beginning at the start of the capping group (i.e., the start of transcription) and ending immediately prior to the start of the polyA signal. The R region is also defined as being flanked by the U3 and U5 regions. The R region plays a role during reverse transcription in permitting the transfer of nascent DNA from one end of the genome to the other.

As used herein, a "packaging signal" or "packaging sequence" refers to sequences located within the lentiviral genome which are required for insertion of the lentiviral RNA into the lentiviral capsid or particle, see e.g., Clever et al., 1995. *J. of Virology*, Vol. 69, No. 4; pp. 2101-2109. Several lentiviral vectors use the minimal packaging signal (also referred to as the psi [Ψ] or [Ψ+] sequence) needed for encapsidation of the lentiviral genome. Thus, as used herein, the terms "packaging sequence," "packaging signal," "psi" and the symbol "Ψ," are used in reference to the non-coding sequence required for encapsidation of lentiviral RNA strands during lentiviral particle formation.

A "FLAP element" or "cPPT/FLAP," as used herein refers to a nucleic acid whose sequence includes the central polypurine tract and central termination sequences (cPPT and CTS) of a lentivirus, e.g., HIV-1 or HIV-2. "FLAP element" and "cPPT/FLAP" may be used interchangeably to refer to the foregoing FLAP element. Suitable FLAP elements are described in U.S. Pat. No. 6,682,907 and in Zennou, et al., 2000, *Cell*, 101:173. During HIV-1 reverse transcription, central initiation of the plus-strand DNA at the central polypurine tract (cPPT) and central termination at the central termination sequence (CTS) lead to the formation of a three-stranded DNA structure: the HIV-1 central DNA flap. While not wishing to be bound by any particular theory, the DNA flap may act as a cis-active determinant of lentiviral vector nuclear import and/or may increase lentiviral titer.

As used herein, an "export element" refers to a cis-acting post-transcriptional regulatory element which regulates the transport of an RNA transcript from the nucleus to the cytoplasm of a cell. Examples of RNA export elements include, but are not limited to, the human immunodeficiency virus (HIV) rev response element (RRE) (see e.g., Cullen et

al., 1991. *J. Virol.* 65: 1053; and Cullen et al., 1991. *Cell* 58: 423), the woodchuck hepatitis virus posttranscriptional regulatory element (WPRE), and the hepatitis B virus post-transcriptional regulatory element (HPRE).

Lentiviral vectors may contain one or more safety enhancements to reduce the risk of replication, insertional mutagenesis, and off-target transduction and/or expression. In particular embodiments, a lentiviral vector comprises one or more of the following safety enhancements: one or more modifications of the 5' and 3' LTRs, cell or tissue specific expression control sequences, e.g., promoters, enhancers. A "modified LTR," as used herein, refers to one or more nucleotide additions, deletions or substitutions in the native HIV-1 5' LTR and/or 3' LTR. The skilled artisan would be able to determine whether an LTR is modified by comparison to a reference LTR.

"Self-inactivating" (SIN) vectors, as used herein, refer to replication-defective vectors, e.g., lentiviral vectors, in which the right (3') LTR enhancer-promoter region, known as the U3 region, has been modified (e.g., by deletion or substitution) to prevent viral transcription beyond the first round of lentiviral replication. Self-inactivation is achieved through a deletion in the U3 region of the 3' LTR of the lentiviral vector transfer plasmid that removes the LTR TATA box (e.g., deletions from -418 to -18), without significant reductions in titers.

An additional safety enhancement is provided by replacing the U3 region of the 5' LTR of the lentiviral vector transfer plasmid with a heterologous promoter to drive transcription of the lentiviral genome during production of recombinant lentiviral particles. Examples of heterologous promoters which can be used include, for example, viral simian virus 40 (SV40) (e.g., early or late), cytomegalovirus (CMV) (e.g., immediate early), Moloney murine leukemia virus (MoMLV), Rous sarcoma virus (RSV), and herpes simplex virus (HSV) (thymidine kinase) promoters.

In particular embodiments, a lentiviral vector is engineered to integrate into an immune effector cell genome.

In certain embodiments, a lentiviral vector is engineered to be integration defective, episomal, and not integrate in the cell genome. As used herein, the term "integration defective lentivirus" or "IDLV" refers to a lentivirus having an integrase that lacks the capacity to integrate the viral vector into the host cell genome. Integration-incompetent lentiviral vectors have been described in patent application WO 2006/010834, which is herein incorporated by reference in its entirety. Illustrative mutations in HIV-1 integrase suitable to reduce integrase activity include, but are not limited to: H12N, H12C, H16C, H16V, S81R, D41A, K42A, H51A, Q53C, D55V, D64E, D64V, E69A, K71A, E85A, E87A, D116N, D116I, D116A, N120G, N120I, N120E, E152G, E152A, K156E, K156A, E157A, K159E, K159A, K160A, R166A, D167A, E170A, H171A, K173A, K186Q, K186T, K188T, E198A, R199C, R199T, R199A, D202A, K211A, Q214L, Q216L, Q221L, W235F, W235E, K236S, K236A, K246A, G247W, D253A, R262A, R263A and K264H. In particular embodiments, an HIV-1 integration deficient integrase comprises a D64V, D161I, D116A, E152G, or E152A mutation; D64V, D116A, and E152G mutations; D64V, D116A, and E152A mutations; or a D64V mutation.

In particular embodiments, a recombinant lentiviral particle comprises (i) a viral envelope comprising (a) a mutated VSIV-G or COCV-G polypeptide comprising amino acid substitutions at positions 47 and 354 of the mature polypeptide and (b) a non-viral membrane-bound tropism polypeptide comprising an anti-CD3 scFv and a CD8α hinge and transmembrane domain; and (ii) two copies of an HIV-1

acid sequence set forth in any one of SEQ ID NOs: 324, 325, 326, 327, 328, 329, 330, and 331; and (ii) two copies of an HIV-1 lentiviral vector-based RNA genome comprising a 5' LTR comprising R and U5 regions; a Psi (Ψ) packaging signal; a cPPT/FLAP, an RRE export element; a promoter comprising or encoding a polynucleotide sequence set forth in SEQ ID NO: 320 operably linked to a polynucleotide encoding a signal peptide encoded by a polynucleotide sequence set forth in SEQ ID NO: 294 and an anti-BCMA CAR encoded by a polynucleotide sequence set forth in any one of SEQ ID NOs: 297, 299, 300, 302, 304, and 308, and a polynucleotide comprising a posttranscriptional regulatory element set forth in any one of SEQ ID NOs: 315, 316, and 317; a 3' LTR comprising U3 and R regions; a polyadenylation signal and a poly(A) tail.

In particular embodiments, a recombinant lentiviral particle comprises (i) a viral envelope comprising (a) a fusogen comprising an amino acid sequence set forth in any one of SEQ ID NOs: 336, 337, 338, and 339 and (b) a non-viral membrane-bound tropism polypeptide comprising an amino acid sequence set forth in any one of SEQ ID NOs: 324, 325, 326, 327, 328, 329, 330, and 331; and (ii) two copies of an HIV-1 lentiviral vector-based RNA genome comprising a 5' LTR comprising R and U5 regions; a Psi (Ψ) packaging signal; a cPPT/FLAP, an RRE export element; a promoter comprising or encoding a polynucleotide sequence set forth in SEQ ID NO: 319 operably linked to a polynucleotide encoding a signal peptide encoded by a polynucleotide sequence set forth in SEQ ID NO: 294 and an anti-BCMA CAR encoded by a polynucleotide sequence set forth in any one of SEQ ID NOs: 297, 299, 300, 302, 304, and 308, and a polynucleotide comprising a posttranscriptional regulatory element set forth in any one of SEQ ID NOs: 315, 316, and 317; a 3' LTR comprising U3 and R regions; a polyadenylation signal and a poly(A) tail.

I. Cells

Recombinant particles contemplated herein are engineered to bind and transduce an immune effector cell. In particular embodiments, a recombinant lentiviral particle engineered to bind and transduce and immune effector cell comprises a viral envelope comprising a mutated COCV-G polypeptide or a mutated VSIV-G polypeptide, wherein the mutated COCV-G or VSIV-G polypeptide comprises amino acid substitutions at positions 47 and 354; a non-viral membrane-bound tropism polypeptide comprising an anti-CD3 ϵ scFv and a human CD8 α hinge and transmembrane domain; and a recombinant lentiviral vector comprising a polynucleotide encoding a myeloproliferative sarcoma virus enhancer, negative control region deleted, dl587rev primer-binding site substituted (MND) U3 promoter or an EF1 α promoter operably linked to a polynucleotide encoding an anti-BCMA chimeric antigen receptor comprising an anti-BCMA scFv or anti-BCMA VHH, a CD8 α hinge and transmembrane domain, a CD137 costimulatory domain, a CD3 ζ primary signaling domain.

An "immune effector cell" is any cell of the immune system that has one or more effector functions (e.g., cytotoxic cell killing activity, secretion of cytokines, induction of ADCC and/or CDC). Illustrative types of immune effector cells contemplated in particular embodiments include, without limitation, T lymphocytes, dendritic cells (DC), Treg cells, natural killer (NK) cells, natural killer T (NKT) cells, and macrophages. The terms "T cell" or "T lymphocyte" are art-recognized and are intended, in particular embodiments, to include thymocytes, immature T lymphocytes, mature T lymphocytes, resting T lymphocytes, and/or activated T lymphocytes. Illustrative examples of T lymphocytes suit-

able for use in particular embodiments, include but not limited to cytotoxic T cells (CTLs; CD8 $^{+}$ T cells), TILs, helper T cells (HTLs; CD4 $^{+}$ T cells), CD4 $^{+}$ CD8 $^{+}$ T cells, CD4 $^{-}$ CD8 $^{-}$ T cells, or any other subset of T cells that has an effector function. In a particular embodiment, the cells comprise $\alpha\beta$ T cells. In a particular embodiment, the cells comprise $\gamma\delta$ T cells.

J. Compositions and Formulations

Compositions contemplated herein comprise a recombinant particle and/or an immune effector cell modified ex vivo formulated with a pharmaceutically acceptable or physiologically-acceptable carrier for administration to a cell, tissue, organ, or an animal, either alone, or in combination with one or more other modalities of therapy.

In particular embodiments, a composition comprises a recombinant lentiviral particle comprising a viral envelope comprising a mutated VSIV-G or COCV-G polypeptide comprising amino acid substitutions at positions 47 and 354 of the mature polypeptide and a non-viral membrane-bound tropism polypeptide comprising an anti-CD3 scFv and a CD8 α hinge and transmembrane domain; and one or more copies of a lentiviral vector comprising a polynucleotide comprising or encoding an MNDU3 or EF1 α promoter operably linked to a polynucleotide encoding an anti-BCMA CAR, and optionally a WPRE.

In particular embodiments, a composition comprises a recombinant lentiviral particle comprising a viral envelope comprising a mutated VSIV-G or COCV-G polypeptide comprising amino acid substitutions at positions 47 and 354 of the mature polypeptide and a non-viral membrane-bound tropism polypeptide comprising an anti-CD3 scFv and a CD8 α hinge and transmembrane domain; and one or more copies of a recombinant lentiviral vector comprising a polynucleotide comprising or encoding an MNDU3 or EF1 α promoter operably linked to a polynucleotide encoding a signal peptide, an anti-BCMA chimeric antigen receptor comprising an anti-BCMA scFv or anti-BCMA VHH, a CD8 α hinge and transmembrane domain, a CD137 costimulatory domain, a CD3 ζ primary signaling domain, and optionally a WPRE.

In particular embodiments, a composition comprises a recombinant lentiviral particle comprising a viral envelope comprising a fusogen comprising an amino acid sequence set forth in any one of SEQ ID NOs: 332, 333, 334, and 335 and a non-viral membrane-bound tropism polypeptide comprising an amino acid sequence set forth in any one of SEQ ID NOs: 324, 325, 326, 327, 328, 329, 330, and 331; and two copies of a lentiviral vector comprising a promoter comprising or encoding a polynucleotide sequence set forth in SEQ ID NO: 320 operably linked to a polynucleotide encoding a signal peptide comprising an amino acid sequence set forth in SEQ ID NO: 245 and an anti-BCMA CAR comprising an amino acid sequence set forth in any one of SEQ ID NOs: 259, 263, 266, 270, 273, and 277, and optionally, a polynucleotide comprising a posttranscriptional regulatory element set forth in any one of SEQ ID NOs: 315, 316, and 317.

In particular embodiments, a composition comprises a recombinant lentiviral particle comprising a viral envelope comprising a fusogen comprising an amino acid sequence set forth in any one of SEQ ID NOs: 332, 333, 334, and 335 and a non-viral membrane-bound tropism polypeptide comprising an amino acid sequence set forth in any one of SEQ ID NOs: 324, 325, 326, 327, 328, 329, 330, and 331; and two copies of a lentiviral vector comprising a promoter com-

prising or encoding a polynucleotide sequence set forth in SEQ ID NO: 320 operably linked to a polynucleotide encoding a signal peptide encoded by a polynucleotide sequence set forth in SEQ ID NO: 294 and an anti-BCMA CAR encoded by a polynucleotide sequence set forth in any one of SEQ ID NOs: 297, 299, 300, 302, 304, and 308, and optionally, a polynucleotide comprising a posttranscriptional regulatory element set forth in any one of SEQ ID NOs: 315, 316, and 317.

In particular embodiments, a composition comprises a recombinant lentiviral particle comprising a viral envelope comprising a fusogen comprising an amino acid sequence set forth in any one of SEQ ID NOs: 332, 333, 334, and 335 and a non-viral membrane-bound tropism polypeptide comprising an amino acid sequence set forth in any one of SEQ ID NOs: 324, 325, 326, 327, 328, 329, 330, and 331; and two copies of a lentiviral vector comprising a promoter comprising or encoding a polynucleotide sequence set forth in SEQ ID NO: 319 operably linked to a polynucleotide encoding a signal peptide comprising an amino acid sequence set forth in SEQ ID NO: 245 and an anti-BCMA CAR comprising an amino acid sequence set forth in any one of SEQ ID NOs: 259, 263, 266, 270, 273, and 277, and optionally, a polynucleotide comprising a posttranscriptional regulatory element set forth in any one of SEQ ID NOs: 315, 316, and 317.

In particular embodiments, a composition comprises a recombinant lentiviral particle comprising a viral envelope comprising a fusogen comprising an amino acid sequence set forth in any one of SEQ ID NOs: 332, 333, 334, and 335 and a non-viral membrane-bound tropism polypeptide comprising an amino acid sequence set forth in any one of SEQ ID NOs: 324, 325, 326, 327, 328, 329, 330, and 331; and two copies of a lentiviral vector comprising a promoter comprising or encoding a polynucleotide sequence set forth in SEQ ID NO: 319 operably linked to a polynucleotide encoding a signal peptide encoded by a polynucleotide sequence set forth in SEQ ID NO: 294 and an anti-BCMA CAR encoded by a polynucleotide sequence set forth in any one of SEQ ID NOs: 297, 299, 300, 302, 304, and 308, and optionally, a polynucleotide comprising a posttranscriptional regulatory element set forth in any one of SEQ ID NOs: 315, 316, and 317.

In particular embodiments, the composition is a pharmaceutical composition. A "pharmaceutical composition" refers to a composition formulated in a pharmaceutically-acceptable or physiologically-acceptable solution for administration to a cell or a subject, either alone, or in combination with one or more other modalities of therapy.

"Pharmaceutically acceptable" refers to molecular entities and compositions that do not produce excessive toxicity, irritation, allergic response, or other problems or complications, commensurate with a reasonable benefit/risk ratio when administered to a human.

In particular embodiments, a composition comprises a pharmaceutically acceptable carrier and a recombinant particle contemplated herein. The term "pharmaceutically acceptable carrier" refers to a diluent, adjuvant, excipient, vehicle and the like with which a recombinant particle, e.g., a recombinant retroviral or lentiviral particle, is physiologically compatible with administration to a human, including but not limited to pharmaceutically acceptable cell culture media, Dulbecco's phosphate buffered saline (PBS), Ringer's solution, 5% dextrose in water (D5W), and normal/physiologic saline (0.9% NaCl).

In particular embodiments, a composition comprises a pharmaceutically acceptable carrier and a recombinant len-

tiviral particle comprising a viral envelope comprising a mutated VSIV-G or COCV-G polypeptide comprising amino acid substitutions at positions 47 and 354 of the mature polypeptide and a non-viral membrane-bound tropism polypeptide comprising an anti-CD3 scFv and a CD8 α hinge and transmembrane domain; and one or more copies of a lentiviral vector comprising a polynucleotide comprising or encoding an MNDU3 or EF1 α promoter operably linked to a polynucleotide encoding an anti-BCMA CAR, and optionally a WPRE.

In particular embodiments, a composition comprises a pharmaceutically acceptable carrier and a recombinant lentiviral particle comprising a viral envelope comprising a mutated VSIV-G or COCV-G polypeptide comprising amino acid substitutions at positions 47 and 354 of the mature polypeptide and a non-viral membrane-bound tropism polypeptide comprising an anti-CD3 scFv and a CD8 α hinge and transmembrane domain; and one or more copies of a recombinant lentiviral vector comprising a polynucleotide comprising or encoding an MNDU3 or EF1 α promoter operably linked to a polynucleotide encoding a signal peptide, an anti-BCMA chimeric antigen receptor comprising an anti-BCMA scFv or anti-BCMA VHH, a CD8 α hinge and transmembrane domain, a CD137 costimulatory domain, a CD3 ζ primary signaling domain, and optionally a WPRE.

In particular embodiments, a composition comprises a pharmaceutically acceptable carrier and a recombinant lentiviral particle comprising a viral envelope comprising a fusogen comprising an amino acid sequence set forth in any one of SEQ ID NOs: 332, 333, 334, and 335 and a non-viral membrane-bound tropism polypeptide comprising an amino acid sequence set forth in any one of SEQ ID NOs: 324, 325, 326, 327, 328, 329, 330, and 331; and two copies of a lentiviral vector comprising a promoter comprising or encoding a polynucleotide sequence set forth in SEQ ID NO: 320 operably linked to a polynucleotide encoding a signal peptide comprising an amino acid sequence set forth in SEQ ID NO: 245 and an anti-BCMA CAR comprising an amino acid sequence set forth in any one of SEQ ID NOs: 259, 263, 266, 270, 273, and 277, and optionally, a polynucleotide comprising a posttranscriptional regulatory element set forth in any one of SEQ ID NOs: 315, 316, and 317.

In particular embodiments, a composition comprises a pharmaceutically acceptable carrier and a recombinant lentiviral particle comprising a viral envelope comprising a fusogen comprising an amino acid sequence set forth in any one of SEQ ID NOs: 332, 333, 334, and 335 and a non-viral membrane-bound tropism polypeptide comprising an amino acid sequence set forth in any one of SEQ ID NOs: 324, 325, 326, 327, 328, 329, 330, and 331; and two copies of a lentiviral vector comprising a promoter comprising or encoding a polynucleotide sequence set forth in SEQ ID NO: 320 operably linked to a polynucleotide encoding a signal peptide encoded by a polynucleotide sequence set forth in SEQ ID NO: 294 and an anti-BCMA CAR encoded by a polynucleotide sequence set forth in any one of SEQ ID NOs: 297, 299, 300, 302, 304, and 308, and optionally, a polynucleotide comprising a posttranscriptional regulatory element set forth in any one of SEQ ID NOs: 315, 316, and 317.

In particular embodiments, a composition comprises a pharmaceutically acceptable carrier and a recombinant lentiviral particle comprising a viral envelope comprising a fusogen comprising an amino acid sequence set forth in any one of SEQ ID NOs: 332, 333, 334, and 335 and a non-viral membrane-bound tropism polypeptide comprising an amino acid sequence set forth in any one of SEQ ID NOs: 324, 325,

326, 327, 328, 329, 330, and 331; and two copies of a lentiviral vector comprising a promoter comprising or encoding a polynucleotide sequence set forth in SEQ ID NO: 319 operably linked to a polynucleotide encoding a signal peptide comprising an amino acid sequence set forth in SEQ ID NO: 245 and an anti-BCMA CAR comprising an amino acid sequence set forth in any one of SEQ ID NOs: 259, 263, 266, 270, 273, and 277, and optionally, a polynucleotide comprising a posttranscriptional regulatory element set forth in any one of SEQ ID NOs: 315, 316, and 317.

In particular embodiments, a composition comprises a pharmaceutically acceptable carrier and a recombinant lentiviral particle comprising a viral envelope comprising a fusogen comprising an amino acid sequence set forth in any one of SEQ ID NOs: 332, 333, 334, and 335 and a non-viral membrane-bound tropism polypeptide comprising an amino acid sequence set forth in any one of SEQ ID NOs: 324, 325, 326, 327, 328, 329, 330, and 331; and two copies of a lentiviral vector comprising a promoter comprising or encoding a polynucleotide sequence set forth in SEQ ID NO: 319 operably linked to a polynucleotide encoding a signal peptide encoded by a polynucleotide sequence set forth in SEQ ID NO: 294 and an anti-BCMA CAR encoded by a polynucleotide sequence set forth in any one of SEQ ID NOs: 297, 299, 300, 302, 304, and 308, and optionally, a polynucleotide comprising a posttranscriptional regulatory element set forth in any one of SEQ ID NOs: 315, 316, and 317.

In particular embodiments, a composition comprises a recombinant particle and a pharmaceutically acceptable carrier suitable for enteral or parenteral, e.g., intravascular (intravenous or intraarterial), intraosseous, intraperitoneal, intraventricular, intracerebral, intracranial, intraspinal, intrathecal, intramuscular, and intramedullary, administration and formulation.

In particular embodiments, a composition is substantially free of *mycoplasma*, endotoxin, and microbial contamination. By “substantially free” with respect to endotoxin is meant that there is less endotoxin per dose of cells than is allowed by the FDA for a biologic, which is a total endotoxin of 5 EU/kg body weight per day, which for an average 70 kg person is 350 EU per total dose of cells. In particular embodiments, compositions contemplated herein contain about 0.5 EU/mL to about 5.0 EU/mL, or about 0.5 EU/mL, 1.0 EU/mL, 1.5 EU/mL, 2.0 EU/mL, 2.5 EU/mL, 3.0 EU/mL, 3.5 EU/mL, 4.0 EU/mL, 4.5 EU/mL, or 5.0 EU/mL.

In particular embodiments, compositions contemplated herein are used in the treatment of a cancer. In particular embodiments, a composition comprises a recombinant particle contemplated herein and one or more cytokines, growth factors, steroids, NSAIDs, DMARDs, anti-inflammatories, chemotherapeutics, radiotherapeutics, therapeutic antibodies, or other active and ancillary agents, either alone or in combination.

It would be understood by the skilled artisan that particular embodiments contemplated herein may comprise other formulations, such as those that are well known in the pharmaceutical art, and are described, for example, in *Remington: The Science and Practice of Pharmacy*, Volume I and Volume II. 23rd Edition. Edited by Adeboye Adejare. Academic Press, 2020, which is incorporated by reference herein, in its entirety.

K. Methods of Making

The manufacturing processes contemplated herein comprise an upstream process that produces a recombinant

lentiviral particle and a downstream process that purifies the recombinant lentiviral particle. Methods of manufacturing lentiviral particles are described in WO2023/003844, which is hereby incorporated by reference in its entirety. See, also Kutner et al., *BMC Biotechnol.* 2009; 9:10. doi: 10.1186/1472-6750-9-10 and Kutner et al. *Nat. Protoc.* 2009; 4(4): 495-505. doi: 10.1038/nprot.2009.22.

In particular embodiments, a method of manufacturing recombinant lentiviral particles comprises transfecting a host cell culture with packaging plasmids and a transfer plasmid, culturing transfected host cells to produce lentiviral particles; and collecting and processing the culture supernatant that contains the crude lentiviral particles to remove impurities and concentrate and formulate the particles for clinical use.

In particular embodiments, lentiviral particles are manufactured by transfecting host cells with a multi-plasmid system comprising (i) an envelope plasmid encoding a mutated VSIV-G or COCV-G polypeptide comprising amino acid substitutions at positions 47 and 354 of the mature polypeptide and a non-viral membrane-bound tropism polypeptide comprising an anti-CD3 scFv and a CD8 α hinge and transmembrane domain, (ii) a packaging plasmid encoding lentiviral gag-pol, (iii) a packaging plasmid encoding lentiviral rev, and (iv) a transfer plasmid comprising a polynucleotide encoding a lentiviral vector contemplated herein.

In particular embodiments, lentiviral particles are manufactured by transfecting host cells with a multi-plasmid system comprising (i) an envelope plasmid encoding a mutated VSIV-G or COCV-G polypeptide comprising amino acid substitutions at positions 47 and 354 of the mature polypeptide and a non-viral membrane-bound tropism polypeptide comprising an anti-CD3 scFv and a CD8 α hinge and transmembrane domain, (ii) a packaging plasmid encoding lentiviral gag-pol, (iii) a packaging plasmid encoding lentiviral rev, and (iv) a transfer plasmid comprising a polynucleotide encoding a lentiviral vector comprising or encoding an MNDU3 or EF1 α promoter operably linked to a polynucleotide encoding a signal peptide, an anti-BCMA chimeric antigen receptor comprising an anti-BCMA scFv or anti-BCMA VHH, a CD8 α hinge and transmembrane domain, a CD137 costimulatory domain, a CD3 ζ primary signaling domain, and optionally a WPRE.

In particular embodiments, lentiviral particles are manufactured by transfecting host cells with a multi-plasmid system comprising (i) an envelope plasmid encoding a mutated VSIV-G or COCV-G polypeptide comprising amino acid substitutions at positions 47 and 354 of the mature polypeptide and a non-viral membrane-bound tropism polypeptide comprising an anti-CD3 scFv and a CD8 α hinge and transmembrane domain, (ii) a packaging plasmid encoding lentiviral gag-pol, (iii) a packaging plasmid encoding lentiviral rev, and (iv) a transfer plasmid comprising a polynucleotide encoding a lentiviral vector comprising a promoter comprising or encoding a polynucleotide sequence set forth in SEQ ID NO: 320 operably linked to a polynucleotide encoding a signal peptide comprising an amino acid sequence set forth in SEQ ID NO: 245 and an anti-BCMA CAR comprising an amino acid sequence set forth in any one of SEQ ID NOs: 259, 263, 266, 270, 273, and 277, and optionally, a polynucleotide comprising a posttranscriptional regulatory element set forth in any one of SEQ ID NOs: 315, 316, and 317.

In particular embodiments, lentiviral particles are manufactured by transfecting host cells with a multi-plasmid system comprising (i) an envelope plasmid encoding a mutated VSIV-G or COCV-G polypeptide comprising amino

acid substitutions at positions 47 and 354 of the mature polypeptide and a non-viral membrane-bound tropism polypeptide comprising an anti-CD3 scFv and a CD8 α hinge and transmembrane domain, (ii) a packaging plasmid encoding lentiviral gag-pol, (iii) a packaging plasmid encoding lentiviral rev, and (iv) a transfer plasmid comprising a polynucleotide encoding a lentiviral vector comprising a promoter comprising or encoding a polynucleotide sequence set forth in SEQ ID NO: 320 operably linked to a polynucleotide encoding a signal peptide encoded by a polynucleotide sequence set forth in SEQ ID NO: 294 and an anti-BCMA CAR encoded by a polynucleotide sequence set forth in any one of SEQ ID NOs: 297, 299, 300, 302, 304, and 308, and optionally, a polynucleotide comprising a posttranscriptional regulatory element set forth in any one of SEQ ID NOs: 315, 316, and 317.

In particular embodiments, lentiviral particles are manufactured by transfecting host cells with a multi-plasmid system comprising (i) an envelope plasmid encoding a mutated VSIV-G or COCV-G polypeptide comprising amino acid substitutions at positions 47 and 354 of the mature polypeptide and a non-viral membrane-bound tropism polypeptide comprising an anti-CD3 scFv and a CD8 α hinge and transmembrane domain, (ii) a packaging plasmid encoding lentiviral gag-pol, (iii) a packaging plasmid encoding lentiviral rev, and (iv) a transfer plasmid comprising a polynucleotide encoding a lentiviral vector comprising a promoter comprising or encoding a polynucleotide sequence set forth in SEQ ID NO: 319 operably linked to a polynucleotide encoding a signal peptide comprising an amino acid sequence set forth in SEQ ID NO: 245 and an anti-BCMA CAR comprising an amino acid sequence set forth in any one of SEQ ID NOs: 259, 263, 266, 270, 273, and 277, and optionally, a polynucleotide comprising a posttranscriptional regulatory element set forth in any one of SEQ ID NOs: 315, 316, and 317.

In particular embodiments, lentiviral particles are manufactured by transfecting host cells with a multi-plasmid system comprising (i) an envelope plasmid encoding a mutated VSIV-G or COCV-G polypeptide comprising amino acid substitutions at positions 47 and 354 of the mature polypeptide and a non-viral membrane-bound tropism polypeptide comprising an anti-CD3 scFv and a CD8 α hinge and transmembrane domain, (ii) a packaging plasmid encoding lentiviral gag-pol, (iii) a packaging plasmid encoding lentiviral rev, and (iv) a transfer plasmid comprising a polynucleotide encoding a lentiviral vector comprising a promoter comprising or encoding a polynucleotide sequence set forth in SEQ ID NO: 319 operably linked to a polynucleotide encoding a signal peptide encoded by a polynucleotide sequence set forth in SEQ ID NO: 294 and an anti-BCMA CAR encoded by a polynucleotide sequence set forth in any one of SEQ ID NOs: 297, 299, 300, 302, 304, and 308, and optionally, a polynucleotide comprising a posttranscriptional regulatory element set forth in any one of SEQ ID NOs: 315, 316, and 317.

A “transfer plasmid” or “transfer vector” encodes a lentiviral genomic RNA modified to package a lentiviral vector that comprises a polynucleotide sequence delivered by recombinant lentiviral particle to a cell. In particular embodiments, a transfer plasmid comprises one or more polynucleotide sequences of interest flanked by LTR sequences, which facilitate packaging, reverse transcription and integration of the lentiviral vector and associated polynucleotide sequences into the host genome. Lentiviral vectors contemplated herein are replication incompetent, i.e., lack the genetic elements necessary for generation of infec-

tive particles in the host cell. For example, a lentiviral vector may comprise a deletion of the 3' LTR, rendering the virus “self-inactivating” (SIN).

Illustrative examples of host cells suitable for transfection with the aforementioned plasmid systems include but are not limited to HEK293 cells, HEK293S cells, HEK293T cells adapted for suspension culture (HEK293Ts), HEK293F cells, HEK293FT cells, HEK293FTM cells, and HEK293E cells.

Once host cells are transfected and produce lentiviral particles, the cell culture is subjected to a downstream process that yields particles sufficient for clinical use. In particular embodiments, a downstream process comprises treating the contents of the bioreactor with a DNA endonuclease, e.g., benzonase; harvesting and clarifying the suspension culture supernatant by filtration; capturing and concentrating the lentiviral particles in the resultant filtrate using affinity chromatography or cation exchange chromatography; filtering the eluate comprising the lentiviral particles; ultrafiltering and diafiltering the lentiviral particles using tangential flow filtration (TFF); and formulating the lentiviral particles in a physiologically acceptable diluent to produce a formulated bulk lentiviral particles. In one embodiment, the formulated bulk lentiviral particles are sterile filtered, filled, and frozen; and subsequently thawed, sterile filtered, subjected to a final fill finish, and frozen. In another embodiment, the bulk lentiviral particles are sterile filtered, subjected to a final fill finish, and frozen.

L. Methods of Use

Recombinant particles contemplated herein are engineered to modify an immune cell that expresses CD3 *in vivo* to express an anti-BCMA CAR, which redirects the CD3-expressing immune cell to a target cell expressing BCMA, thereby preventing, treating, or ameliorating at least one symptom associated with a disease, disorder, or condition associated therewith.

In particular embodiments, a method of preventing, treating, or ameliorating at least one symptom of a cancer comprises administering the subject an amount of recombinant particle contemplated herein. The term “amount” as used herein, refers to “an amount effective” or “an effective amount” of a recombinant particle contemplated herein comprising a payload contemplated herein, etc., to achieve a beneficial or desired prophylactic or therapeutic result, including clinical results. A “prophylactically effective amount” refers to an amount of recombinant particle contemplated herein comprising payload contemplated herein, effective to achieve the desired prophylactic result. A “therapeutically effective amount” refers to an amount of recombinant particle contemplated herein comprising a lentiviral vector encoding an anti-BCMA CAR, that is effective to “treat” a subject (e.g., a patient). When a therapeutic amount is indicated, the precise amount of the compositions to be administered can be determined by a physician with consideration of individual differences in age, weight, tumor size, extent of infection or metastasis, and condition of the patient (subject).

In particular embodiments, a recombinant lentiviral particle comprising a viral envelope comprising a mutated VSIV-G or COCV-G polypeptide comprising amino acid substitutions at positions 47 and 354 of the mature polypeptide and a non-viral membrane-bound tropism polypeptide comprising an anti-CD3 scFv and a CD8 α hinge and transmembrane domain; and one or more copies of a lentiviral vector comprising a polynucleotide comprising or encoding

an MNDU3 or EF1 α promoter operably linked to a polynucleotide encoding an anti-BCMA CAR, and optionally a WPRE, is administered to a subject to treat, prevent, or ameliorate at least one symptom of multiple myeloma in the subject.

In particular embodiments, a recombinant lentiviral particle comprising a viral envelope comprising a mutated VSIV-G or COCV-G polypeptide comprising amino acid substitutions at positions 47 and 354 of the mature polypeptide and a non-viral membrane-bound tropism polypeptide comprising an anti-CD3 scFv and a CD8 α hinge and transmembrane domain; and one or more copies of a recombinant lentiviral vector comprising a polynucleotide comprising or encoding an MNDU3 or EF1 α promoter operably linked to a polynucleotide encoding a signal peptide, an anti-BCMA chimeric antigen receptor comprising an anti-BCMA scFv or anti-BCMA VHH, a CD8 α hinge and transmembrane domain, a CD137 costimulatory domain, a CD3 ζ primary signaling domain, and optionally a WPRE, is administered to a subject to treat, prevent, or ameliorate at least one symptom of multiple myeloma in the subject.

In particular embodiments, a recombinant lentiviral particle comprising a viral envelope comprising a fusogen comprising an amino acid sequence set forth in any one of SEQ ID NOs: 332, 333, 334, and 335 and a non-viral membrane-bound tropism polypeptide comprising an amino acid sequence set forth in any one of SEQ ID NOs: 324, 325, 326, 327, 328, 329, 330, and 331; and two copies of a lentiviral vector comprising a promoter comprising or encoding a polynucleotide sequence set forth in SEQ ID NO: 320 operably linked to a polynucleotide encoding a signal peptide comprising an amino acid sequence set forth in SEQ ID NO: 245 and an anti-BCMA CAR comprising an amino acid sequence set forth in any one of SEQ ID NOs: 259, 263, 266, 270, 273, and 277, and optionally, a polynucleotide comprising a posttranscriptional regulatory element set forth in any one of SEQ ID NOs: 315, 316, and 317, is administered to a subject to treat, prevent, or ameliorate at least one symptom of multiple myeloma in the subject.

In particular embodiments, a recombinant lentiviral particle comprising a viral envelope comprising a fusogen comprising an amino acid sequence set forth in any one of SEQ ID NOs: 332, 333, 334, and 335 and a non-viral membrane-bound tropism polypeptide comprising an amino acid sequence set forth in any one of SEQ ID NOs: 324, 325, 326, 327, 328, 329, 330, and 331; and two copies of a lentiviral vector comprising a promoter comprising or encoding a polynucleotide sequence set forth in SEQ ID NO: 320 operably linked to a polynucleotide encoding a signal peptide encoded by a polynucleotide sequence set forth in SEQ ID NO: 294 and an anti-BCMA CAR encoded by a polynucleotide sequence set forth in any one of SEQ ID NOs: 297, 299, 300, 302, 304, and 308, and optionally, a polynucleotide comprising a posttranscriptional regulatory element set forth in any one of SEQ ID NOs: 315, 316, and 317, is administered to a subject to treat, prevent, or ameliorate at least one symptom of multiple myeloma in the subject.

In particular embodiments, a recombinant lentiviral particle comprising a viral envelope comprising a fusogen comprising an amino acid sequence set forth in any one of SEQ ID NOs: 332, 333, 334, and 335 and a non-viral membrane-bound tropism polypeptide comprising an amino acid sequence set forth in any one of SEQ ID NOs: 324, 325, 326, 327, 328, 329, 330, and 331; and two copies of a lentiviral vector comprising a promoter comprising or encoding a polynucleotide sequence set forth in SEQ ID

NO: 319 operably linked to a polynucleotide encoding a signal peptide comprising an amino acid sequence set forth in SEQ ID NO: 245 and an anti-BCMA CAR comprising an amino acid sequence set forth in any one of SEQ ID NOs: 259, 263, 266, 270, 273, and 277, and optionally, a polynucleotide comprising a posttranscriptional regulatory element set forth in any one of SEQ ID NOs: 315, 316, and 317, is administered to a subject to treat, prevent, or ameliorate at least one symptom of multiple myeloma in the subject.

In particular embodiments, a recombinant lentiviral particle comprising a viral envelope comprising a fusogen comprising an amino acid sequence set forth in any one of SEQ ID NOs: 332, 333, 334, and 335 and a non-viral membrane-bound tropism polypeptide comprising an amino acid sequence set forth in any one of SEQ ID NOs: 324, 325, 326, 327, 328, 329, 330, and 331; and two copies of a lentiviral vector comprising a promoter comprising or encoding a polynucleotide sequence set forth in SEQ ID NO: 319 operably linked to a polynucleotide encoding a signal peptide encoded by a polynucleotide sequence set forth in SEQ ID NO: 294 and an anti-BCMA CAR encoded by a polynucleotide sequence set forth in any one of SEQ ID NOs: 297, 299, 300, 302, 304, and 308, and optionally, a polynucleotide comprising a posttranscriptional regulatory element set forth in any one of SEQ ID NOs: 315, 316, and 317, is administered to a subject to treat, prevent, or ameliorate at least one symptom of multiple myeloma in the subject.

In particular embodiments, a recombinant lentiviral particle contemplated herein is administered to a subject to treat, prevent, or ameliorate at least one symptom of a multiple myeloma or precursor thereof selected from the group consisting of: monoclonal gammopathy of undetermined significance (MGUS), active multiple myeloma, smoldering multiple myeloma, light chain myeloma, non-secretory myeloma, IgD myeloma, IgE myeloma, osteosclerotic myeloma, solitary plasmacytoma of bone, and extramedullary plasmacytoma.

In particular embodiments, a recombinant lentiviral particle is administered to a subject in combination with one or more anti-cancer therapies including, but not limited to, an autologous stem cell transplant (ASCT), radiation, surgery, a chemotherapeutic agent, an immunomodulatory agent and a targeted cancer therapy.

In particular embodiments, the one or more anti-cancer therapies is selected from the group consisting of 6-mercaptopurine, abiraterone, alemtuzumab, all-trans retinoic acid, anastrozole, aprepitant, arsenic trioxide, atezolizumab, avelumab, azacytidine, bafetinib, bavituximab, bevacizumab, bivatuzumab, bleomycin, blinatumomab, bortezomib, bosutinib, cabazitaxel, capecitabine, carboplatin, carfilzomib, cetuximab, cisplatin, cladribine, conatumumab, corticosteroid, crizotinib, cyclophosphamide, cytarabine, dacetuzumab, dalotuzumab, daratumumab, dasatinib, daunorubicin, danusertib, decitabine, denosumab, dexamethasone, docetaxel, doxorubicin, duligotumab, durvalumab, elotuzumab, eribulin, erlotinib, etoposide, everolimus, exemestane, filgrastim, fludarabine, fluorouracil, fulvestrant, gemcitabine, gemtuzumab, hydroxyurea, ibritumomab, idarubicin, imatinib, imiquimod, indatuximab, inotuzumab, ipilimumab, ixabepilone, ixazomib, lapatinib, lenalidomide, letrozole, leuprolide, lorvotuzumab, lucatumumab, melphalan, methotrexate, milatuzumab, mitoxantrone, moxetumomab, nilotinib, nivolumab, ocaratuzumab, ofatumumab, oxaliplatin, paclitaxel, palonosetron, pembrolizumab, pemetrexed, pomalidomide, ponatinib, prednisone, radium-223, rituximab, saracatinib, siltuximab, sip-

uleucel-T, sorafenib, sunitinib, tamoxifen, temozolomide, temsirolimus, teprotumumab, thalidomide, tinorelbine, topotecan, tozasertib, trastuzumab, ublituximab, vincristine, and zoledronic acid.

M. Enumerated Embodiments

Embodiment 1: A recombinant lentiviral particle comprising:

- (a) a viral envelope comprising (i) a mutated cocal virus envelope glycoprotein (COCV-G) or a mutated vesicular stomatitis Indiana virus envelope glycoprotein (VSIV-G), wherein the mutated COCV-G or VSIV-G comprises amino acid substitutions at positions 47 and 354; and (ii) a non-viral membrane-bound tropism polypeptide comprising an anti-CD3 ϵ scFv and a human CD8 α hinge and transmembrane domain; and
- (b) a recombinant lentiviral vector comprising a polynucleotide encoding a myeloproliferative sarcoma virus enhancer, negative control region deleted, dl587rev primer-binding site substituted (MND) U3 promoter or an EF1 α promoter operably linked to a polynucleotide encoding an anti-BCMA chimeric antigen receptor comprising an anti-BCMA scFv or anti-BCMA VHH, a CD8 α hinge and transmembrane domain, a CD137 costimulatory domain, a CD3 ζ primary signaling domain.

Embodiment 2: The particle of embodiment 1, wherein the mutated COCV-G or the mutated VSIV-G comprises amino acid substitutions selected from the group consisting of: K47A and R354A; K47A and R354Q; K47Q and R354A; and K47Q and R354Q.

Embodiment 3: The particle of embodiment 1 or embodiment 2, wherein the mutated COCV-G or the mutated VSIV-G comprises the amino acid substitutions K47A and R354A.

Embodiment 4: The particle of embodiment 1 or embodiment 2, wherein the mutated COCV-G or the mutated VSIV-G comprises the amino acid substitutions K47A and R354Q.

Embodiment 5: The particle of embodiment 1 or embodiment 2, wherein the mutated COCV-G or the mutated VSIV-G comprises the amino acid substitutions K47Q and R354A.

Embodiment 6: The particle of embodiment 1 or embodiment 2, wherein the mutated COCV-G or the mutated VSIV-G comprises the amino acid substitutions K47Q and R354Q.

Embodiment 7: The particle of any one of embodiments 1 to 6, wherein the mutated COCV-G or the mutated VSIV-G comprises the amino acid sequence set forth in any one of SEQ ID NOs: 332, 333, 334, 335, 336, 337, 338, and 339.

Embodiment 8: The particle of any one of embodiments 1 to 7, wherein the mutated COCV-G or the mutated VSIV-G comprises the amino acid sequence set forth in any one of SEQ ID NOs: 332, 333, 334, and 335.

Embodiment 9: The particle of any one of embodiments 1 to 7, wherein the mutated COCV-G or the mutated VSIV-G comprises the amino acid sequence set forth in any one of SEQ ID NOs: 336, 337, 338, and 339.

Embodiment 10: The particle of any one of embodiments 1 to 9, wherein the anti-CD3 ϵ scFv is isolated from an antibody selected from the group consisting of: OKT3, UCHT1, YTH12.5, TR66, and variants thereof.

Embodiment 11: The particle of any one of embodiments 1 to 10, wherein the anti-CD3 ϵ scFv is isolated from OKT3.

Embodiment 12: The particle of any one of embodiments 1 to 10, wherein the anti-CD3 ϵ scFv is isolated from UCHT1.

Embodiment 13: The particle of any one of embodiments 1 to 10, wherein the anti-CD3 ϵ scFv is isolated from YTH12.5.

Embodiment 14: The particle of any one of embodiments 1 to 10, wherein the anti-CD3 ϵ scFv is isolated from TR66.

Embodiment 15: The particle of any one of embodiments 1 to 10, wherein the anti-CD3 ϵ scFv comprises an amino acid sequence set forth in any one of SEQ ID NOs: 153, 154, 163, 164, 173, 174, 183, 184, 193, 194, 203, 204, 213, 214, 223, and 224.

Embodiment 16: The particle of any one of embodiments 1 to 10, wherein the non-viral membrane-bound tropism polypeptide comprises an amino acid sequence set forth in any one of SEQ ID NOs: 324, 325, 326, 327, 328, 329, 330, and 331.

Embodiment 17: The particle of any one of embodiments 1 to 16, wherein the MND U3 promoter comprises the polynucleotide sequence set forth in SEQ ID NO: 320.

Embodiment 18: The particle of any one of embodiments 1 to 16, wherein the EF1 α promoter comprises the polynucleotide sequence set forth in SEQ ID NO: 319.

Embodiment 19: The particle of any one of embodiments 1 to 18, wherein the anti-BCMA CAR comprises an anti-BCMA scFv comprising an amino acid sequence selected from the group consisting of: 19, 20, 29, 30, 39, 40, 49, 50, 59, 60, 69, 70, 79, 80, 89, 90, 99, and 100.

Embodiment 20: The particle of any one of embodiments 1 to 19, wherein the anti-BCMA CAR comprises an anti-BCMA scFv comprising an amino acid sequence selected from the group consisting of: 20, 30, 39, 50, 59, 70, 80, 90, and 100.

Embodiment 21: The particle of any one of embodiments 1 to 20, wherein the anti-BCMA CAR comprises an anti-BCMA scFv comprising an amino acid sequence selected from the group consisting of: 39, 59, 70, and 90.

Embodiment 22: The particle of any one of embodiments 1 to 18, wherein the anti-BCMA CAR comprises an anti-BCMA VHH comprising an amino acid sequence selected from the group consisting of: 101, 105, 109, 113, 117, 121, 125, 129, 133, 137, and 141.

Embodiment 23: The particle of any one of embodiments 1 to 18, wherein the anti-BCMA CAR comprises an anti-BCMA VHH comprising an amino acid sequence selected from the group consisting of: 101 and 117.

Embodiment 24: The particle of any one of embodiments 1 to 23, wherein the anti-BCMA CAR comprises the amino acid sequence set forth in any one of SEQ ID NOs: 259, 263, 266, 270, 273 and 277.

Embodiment 25: The particle of any one of embodiments 1 to 24, wherein the polynucleotide encoding the anti-BCMA CAR comprises the polynucleotide sequence set forth in any one of SEQ ID NOs: 297, 299, 300, 302, 304, and 308.

Embodiment 26: The particle of any one of embodiments 1 to 25, wherein the polynucleotide encoding the anti-BCMA CAR further comprises a polynucleotide sequence encoding a signal peptide.

Embodiment 27: The particle of any one of embodiments 1 to 25, wherein the polynucleotide encoding the anti-BCMA CAR further comprises a polynucleotide sequence encoding a signal peptide isolated from a polypeptide selected from the group consisting of: CD8 α , mIgGk, hlgGk, CD33, tPA, SEAP, hGM-CSF, CSF2R, and B2M.

101

Embodiment 28: The particle of any one of embodiments 1 to 27, wherein the polynucleotide encoding the anti-BCMA CAR further comprises a polynucleotide sequence encoding a signal peptide comprising an amino acid sequence set forth in any one of SEQ ID NOs: 245, 246, 247, 248, 249, 250, 251, 252, 253, and 254.

Embodiment 29: The particle of any one of embodiments 21 to 23, wherein the polynucleotide encoding the signal peptide comprises the polynucleotide sequence set forth in SEQ ID NO: 294.

Embodiment 30: The particle of any one of embodiments 1 to 24, wherein the lentiviral vector further comprises a WPRE operably linked to the 3' end of the polynucleotide encoding the anti-BCMA CAR.

Embodiment 31: The particle of any one of embodiments 1 to 25, wherein the lentiviral vector further comprises a WPRE that comprises, consists essentially of, or consists of a polynucleotide sequence set forth in any one of SEQ ID NOs: 315, 316, and 317.

Embodiment 32: A recombinant lentiviral particle comprising:

(a) a viral envelope comprising (i) a mutated viral envelope glycoprotein comprising an amino acid sequence set forth in any one of SEQ ID NOs: 332, 333, 334, and 335 and (ii) a non-viral membrane-bound tropism polypeptide comprising an amino acid sequence set forth in any one of SEQ ID NOs: 324, 325, 326, 327, 328, 329, 330, and 331; and

(b) a recombinant lentiviral vector comprising a 5' long terminal repeat (LTR) comprising R and U5 regions; a Psi (Ψ) packaging signal, a cPPT/FLAP, a rev response element (RRE); a polynucleotide encoding an MND promoter or an EF1 α promoter operably linked to a polynucleotide encoding an anti-BCMA chimeric antigen receptor (CAR) comprising an amino acid sequence set forth in any one of SEQ ID NOs: 259, 263, 266, 270, 273, and 277 or an amino acid sequence at least 95%, at least 96%, at least 97%, at least 98%, at least 99% identical thereto; optionally a WPRE; a 3' LTR comprising U3 and R regions; a polyadenylation signal and a poly(A) tail.

Embodiment 33: The particle of embodiment 32, wherein the mutated viral envelope glycoprotein comprises an amino acid sequence set forth in SEQ ID NO: 332.

Embodiment 34: The particle of embodiment 32, wherein the mutated viral envelope glycoprotein comprises an amino acid sequence set forth in SEQ ID NO: 333.

Embodiment 35: The particle of embodiment 32, wherein the mutated viral envelope glycoprotein comprises an amino acid sequence set forth in SEQ ID NO: 334.

Embodiment 36: The particle of embodiment 32, wherein the mutated viral envelope glycoprotein comprises an amino acid sequence set forth in SEQ ID NO: 335.

Embodiment 37: A recombinant lentiviral particle comprising:

(a) a viral envelope comprising (i) a mutated viral envelope glycoprotein comprising an amino acid sequence set forth in any one of SEQ ID NOs: 336, 337, 338, and 339 and (ii) a non-viral membrane-bound tropism polypeptide comprising an amino acid sequence set forth in any one of SEQ ID NOs: 324, 325, 326, 327, 328, 329, 330, and 331; and

(b) a recombinant lentiviral vector comprising a 5' long terminal repeat (LTR) comprising R and U5 regions; a Psi (Ψ) packaging signal, a cPPT/FLAP, a rev response element (RRE); a polynucleotide encoding an MND promoter or an EF1 α promoter operably linked to a

102

polynucleotide encoding an anti-BCMA chimeric antigen receptor (CAR) comprising an amino acid sequence set forth in any one of SEQ ID NOs: 259, 263, 266, 270, 273, and 277 or an amino acid sequence at least 95%, at least 96%, at least 97%, at least 98%, at least 99% identical thereto; optionally a WPRE; a 3' LTR comprising U3 and R regions; a polyadenylation signal and a poly(A) tail.

Embodiment 38: The particle of embodiment 37, wherein the mutated viral envelope glycoprotein comprises an amino acid sequence set forth in SEQ ID NO: 336.

Embodiment 39: The particle of embodiment 37, wherein the mutated viral envelope glycoprotein comprises an amino acid sequence set forth in SEQ ID NO: 337.

Embodiment 40: The particle of embodiment 37, wherein the mutated viral envelope glycoprotein comprises an amino acid sequence set forth in SEQ ID NO: 338.

Embodiment 41: The particle of embodiment 37, wherein the mutated viral envelope glycoprotein comprises an amino acid sequence set forth in SEQ ID NO: 339.

Embodiment 42: The particle of any one of embodiments 32 to 41, wherein the non-viral membrane-bound tropism polypeptide comprises an amino acid sequence set forth in SEQ ID NO: 324.

Embodiment 43: The particle of any one of embodiments 32 to 41, wherein the non-viral membrane-bound tropism polypeptide comprises an amino acid sequence set forth in SEQ ID NO: 325.

Embodiment 44: The particle of any one of embodiments 32 to 41, wherein the non-viral membrane-bound tropism polypeptide comprises an amino acid sequence set forth in SEQ ID NO: 326.

Embodiment 45: The particle of any one of embodiments 32 to 41, wherein the non-viral membrane-bound tropism polypeptide comprises an amino acid sequence set forth in SEQ ID NO: 327.

Embodiment 46: The particle of any one of embodiments 32 to 41, wherein the non-viral membrane-bound tropism polypeptide comprises an amino acid sequence set forth in SEQ ID NO: 328.

Embodiment 47: The particle of any one of embodiments 32 to 41, wherein the non-viral membrane-bound tropism polypeptide comprises an amino acid sequence set forth in SEQ ID NO: 329.

Embodiment 48: The particle of any one of embodiments 32 to 41, wherein the non-viral membrane-bound tropism polypeptide comprises an amino acid sequence set forth in SEQ ID NO: 330.

Embodiment 49: The particle of any one of embodiments 32 to 41, wherein the non-viral membrane-bound tropism polypeptide comprises an amino acid sequence set forth in SEQ ID NO: 331.

Embodiment 50: The particle of any one of embodiments 32 to 49, wherein the recombinant lentiviral vector is derived from HIV-1 or HIV-2.

Embodiment 51: The particle of any one of embodiments 32 to 50, wherein the MND U3 promoter comprises the polynucleotide sequence set forth in SEQ ID NO: 320.

Embodiment 52: The particle of any one of embodiments 32 to 50, wherein the EF1 α promoter comprises the polynucleotide sequence set forth in SEQ ID NO: 319.

Embodiment 53: The particle of any one of embodiments 32 to 52, wherein the polynucleotide encoding the anti-BCMA CAR comprises the polynucleotide sequence set forth in any one of SEQ ID NOs: 297, 299, 300, 302, 304, and 308.

Embodiment 54: The particle of any one of embodiments 32 to 53, wherein the polynucleotide encoding the anti-BCMA CAR further comprises a polynucleotide sequence encoding a signal peptide.

Embodiment 55: The particle of any one of embodiments 32 to 54, wherein the polynucleotide encoding the anti-BCMA CAR further comprises a polynucleotide sequence encoding a signal peptide isolated from a polypeptide selected from the group consisting of: CD8 α , mIgGk, hIgGk, CD33, tPA, SEAP, hGM-CSF, CSF2R, and B2M.

Embodiment 56: The particle of any one of embodiments 32 to 55, wherein the polynucleotide encoding the anti-BCMA CAR further comprises a polynucleotide sequence encoding a signal peptide comprising an amino acid sequence set forth in any one of SEQ ID NOs: 245, 246, 247, 248, 249, 250, 251, 252, 253, and 254.

Embodiment 57: The particle of any one of embodiments 32 to 56, wherein the lentiviral vector further comprises a WPRE operably linked to the 3' end of the polynucleotide encoding the anti-BCMA CAR.

Embodiment 58: The particle of any one of embodiments 32 to 56, wherein the lentiviral vector further comprises a WPRE operably linked to the 3' end of the polynucleotide encoding the anti-BCMA CAR, wherein the WPRE comprises, consists essentially of, or consists of the polynucleotide sequence set forth in any one of SEQ ID NOs: 315, 316, and 317.

Embodiment 59: A recombinant lentiviral particle comprising:

- (a) a viral envelope comprising (i) a mutated viral envelope glycoprotein comprising an amino acid sequence set forth in SEQ ID NO: 335 and (ii) a non-viral membrane-bound tropism polypeptide comprising an amino acid sequence set forth in SEQ ID NO: 324; and
- (b) a recombinant lentiviral vector comprising a 5' long terminal repeat (LTR) comprising R and U5 regions; a Psi (Ψ) packaging signal, a cPPT/FLAP, a rev response element (RRE); a polynucleotide encoding an EF1 α promoter operably linked to a polynucleotide encoding an anti-BCMA chimeric antigen receptor (CAR) comprising an amino acid sequence set forth in SEQ ID NO: 266 or an amino acid sequence at least 95%, at least 96%, at least 97%, at least 98%, at least 99% identical thereto; optionally a WPRE; a 3' LTR comprising U3 and R regions; a polyadenylation signal and a poly(A) tail.

Embodiment 60: A cell transduced with the particle of any one of embodiments 1 to 59.

Embodiment 61: The cell of embodiment 60, wherein the cell is an immune effector cell.

Embodiment 62: The cell of embodiment 60 or embodiment 61, wherein the cell is a T cell or a natural killer T (NKT) cell.

Embodiment 63: A composition comprising the particle of any one of embodiments 1 to 58 or the cell of any one of embodiments 60 to 62.

Embodiment 64: A pharmaceutical composition comprising a pharmaceutically acceptable carrier and the particle of any one of embodiments 1 to 59, the cell of any one of embodiments 60 to 62, or the composition of embodiment 63.

Embodiment 65: A method of treating, preventing, or ameliorating at least one symptom of a disease, disorder or condition associated therewith in a subject, comprising administering to the subject an effective amount of the particle of any one of embodiments 1 to 59, the cell of any

one of embodiments 60 to 62, the composition of embodiment 63, or the pharmaceutical composition of embodiment 64.

Embodiment 66: The method of embodiment 65, wherein the disease, disorder, or condition is a cancer.

Embodiment 67: The method of embodiment 65 or embodiment 66, wherein the cancer is a multiple myeloma (MM).

Embodiment 68: The method of embodiment 66 or embodiment 67, wherein the cancer is MM selected from the group consisting of: active multiple myeloma, smoldering multiple myeloma, light chain myeloma, non-secretory myeloma, IgD myeloma, IgE myeloma, osteosclerotic myeloma, solitary plasmacytoma of bone, and extramedullary plasmacytoma.

Embodiment 69: The method of any one of embodiments 66 to 68, wherein the cancer is relapsed and/or refractory.

Embodiment 70: A method of treating a subject that has, or has been diagnosed with, a multiple myeloma, comprising administering the subject an effective amount of the particle of any one of embodiments 1 to 59, the cell of any one of embodiments 60 to 62, the composition of embodiment 63, or the pharmaceutical composition of embodiment 64.

Embodiment 71: The method of embodiment 70, wherein the administration is parenteral administration.

Embodiment 72: The methods of embodiment 70 or embodiment 71, wherein the administration is intravenous.

Embodiment 73: A method of transducing an immune effector cell in vivo, comprising administering to a subject a pharmaceutical composition comprising a pharmaceutically acceptable carrier and an effective amount of the particle of any one of embodiments 1 to 59, the cell of any one of embodiments 60 to 62, the composition of embodiment 63, or the pharmaceutical composition of embodiment 64.

Embodiment 74: A method of making a recombinant lentivirus comprising (a) transfecting a host cell with four polynucleotides: a first polynucleotide that encodes lentiviral gag-pol, a second polynucleotide that encodes lentiviral rev, a third polynucleotide that encodes the mutated viral envelope glycoprotein set forth in any one of embodiments 1 to 59 and the non-viral membrane-bound tropism polypeptide set forth in any one of embodiments 1 to 59, and a fourth polynucleotide that is a transfer plasmid encoding the recombinant lentiviral vector of any one of embodiments 1 to 59; and b) culturing the transduced cell for about 1 to 3 days to produce the recombinant lentivirus.

Embodiment 75: A kit comprising the particle of any one of embodiments 1 to 59, a pharmaceutically acceptable carrier, and instructions for use.

All publications, patent applications, and issued patents cited in this specification are herein incorporated by reference as if each individual publication, patent application, or issued patent were specifically and individually indicated to be incorporated by reference.

Although the foregoing embodiments have been described in some detail by way of illustration and example for purposes of clarity of understanding, it will be readily apparent to one of ordinary skill in the art in light of the teachings contemplated herein that certain changes and modifications may be made thereto without departing from the spirit or scope of the appended claims. The following examples are provided by way of illustration only and not by way of limitation. Those of skill in the art will readily

recognize a variety of noncritical parameters that could be changed or modified to yield essentially similar results.

N. EXAMPLES

Example 1

Recombinant Lentivirus Delivers Functional Anti-BCMA CARs to T Cells

Recombinant T cell specific lentiviral particles with a viral envelope expressing a mutated viral envelope glycoprotein(fusogen) and a non-viral membrane bound tropism molecule and harboring a lentiviral vector encoding an anti-BCMA CAR were generated. FIG. 1.

HEK293T cells were transfected with plasmids encoding a non-viral membrane bound tropism molecule comprising an anti-CD3 scFv fused to a CD8 α hinge and transmembrane domain (e.g., SEQ ID NO: 324); a mutant VSIV-G fusogen (e.g., SEQ ID NOs: 335); lentiviral GAG/POL; lentiviral REV; and a transfer plasmid encoding a lentiviral vector comprising an MNDU3 promoter operably linked to a polynucleotide encoding a CD8 α signal peptide and an anti-BCMA CAR and a WPRE element operably linked to the 3' end of the polynucleotide encoding the anti-BCMA CAR.

Table 16 lists the recombinant lentivirus (LV) reference number and the corresponding SEQ ID NOs of the amino acid sequences of the anti-BCMA CARs and their CAR architectures.

TABLE 16

LV Ref.	SEQ ID NO.	anti-BCMA binding domain	Hinge	TM	Costim	Primary
LV 1	256	scFv	CD8 α	CD8 α	CD137	CD3 ζ
LV 2	258	scFv	CD8 α	CD8 α	CD137	CD3 ζ
LV 3	259	scFv	CD8 α	CD8 α	CD137	CD3 ζ
LV 4	262	scFv	CD8 α	CD8 α	CD137	CD3 ζ
LV 5	263	scFv	CD8 α	CD8 α	CD137	CD3 ζ
LV 6	266	scFv	CD8 α	CD8 α	CD137	CD3 ζ
LV 7	268	scFv	CD8 α	CD8 α	CD137	CD3 ζ
LV 8	270	scFv	CD8 α	CD8 α	CD137	CD3 ζ
LV 9	273	VHH	CD8 α	CD8 α	CD137	CD3 ζ
LV 10	274	VHH	CD8 α	CD8 α	CD137	CD3 ζ
LV 11	275	VHH	CD8 α	CD8 α	CD137	CD3 ζ
LV 12	276	VHH	CD8 α	CD8 α	CD137	CD3 ζ
LV 13	277	VHH	CD8 α	CD8 α	CD137	CD3 ζ
LV 14	278	VHH	CD8 α	CD8 α	CD137	CD3 ζ
LV 15	279	VHH	CD8 α	CD8 α	CD137	CD3 ζ
LV 16	281	VHH	CD8 α	CD8 α	CD137	CD3 ζ
LV 17	282	VHH	CD8 α	CD8 α	CD137	CD3 ζ
LV 18	283	VHH	CD8 α	CD8 α	CD137	CD3 ζ
LV 19	NA	scFv	CD8 α	CD8 α	CD137	CD3 ζ

Jurkat Functional Titer

1 $\times 10^5$ Jurkat cells were plated in each well of a 96-well plate. Cells were transduced with recombinant lentiviruses LV 1 to LV 18 that harbor novel anti-BCMA CARs and LV 19, which harbors a control anti-BCMA CAR obtained from the literature. Seven days post-transduction, Jurkat cells were harvested and stained with a recombinant, phycoerythrin (PE) labeled, BCMA extracellular domain-FC fusion protein (BCMA-PE) and analyzed by flow cytometry. Functional titer, expressed as the number of transducing units (TU) per mL, was determined by measuring the number of transduced Jurkat cells. FIG. 2A.

Anti-BCMA CAR Expression

5 $\times 10^5$ human PBMCs were plated in each well of a 24-well plate. Cells were transduced with recombinant len-

tiviruses LV 1 to LV 19 at a MOI 2 based on the Jurkat functional titer, or a 0.5 mL volumetric transduction if MOI 2 was not achievable. Seven days post-transduction, PBMCs were harvested and stained with BCMA-PE and analyzed by flow cytometry to assess the percentage of anti-BCMA CAR expressing cells. FIG. 2B.

Anti-BCMA CAR Activity

5 $\times 10^4$ PBMCs transduced with recombinant lentiviruses LV 1 to LV 19 were co-cultured with 5 $\times 10^4$ high BCMA-expressing tumor cells (RPMI-8226) or 5 $\times 10^4$ low BCMA-expressing tumor cells (Daudi) for 24 hours. Anti-BCMA CAR activity was assessed by harvesting co-culture supernatants and measuring IFN γ levels using a Meso Scale Discovery (MSD®) assay. The percentage of anti-BCMA CAR positive cells was plotted against IFN γ levels produced in co-culture. FIG. 2D.

Summary

These data indicate that the recombinant T cell specific lentiviral particles harboring anti-BCMA CARs (LV 1 to LV 18) are able to transduce CD3 expressing cells, that anti-BCMA CARs are expressed on PBMCs transduced with LV 1 to LV 18 and that the transduced PBMCs express anti-BCMA CARs that recognize high or low BCMA-expressing cells and produce IFN γ in response to binding antigen.

Example 2

Lentiviral Vector Architecture and Anti-BCMA CAR Expression and Function

Recombinant T cell specific lentiviral particles with a viral envelope expressing a mutated viral envelope glycoprotein(fusogen) and a non-viral membrane bound tropism molecule and harboring a lentiviral vector encoding various promoters, anti-BCMA CARs, and either no posttranscriptional response element (PRE) or a wild-type WPRE, or a mutated WPRE.

HEK293T cells were transfected with plasmids encoding a non-viral membrane bound tropism molecule comprising an anti-CD3 scFv fused to a CD8 α hinge and transmembrane domain (e.g., SEQ ID NO: 324); a mutant VSIV-G fusogen (e.g., SEQ ID NOs: 335); lentiviral GAG/POL; lentiviral REV; and a transfer plasmid encoding a lentiviral vector comprising either an MNDU3 promoter (SEQ ID NO: 319), an SFFV promoter (SEQ ID NO: 322), or an EF1 α promoter (SEQ ID NO: 320) operably linked to a polynucleotide encoding a CD8 α signal peptide and an anti-BCMA CAR and either no posttranscriptional response element or a wild-type WPRE (SEQ ID NO: 315) or a mutated WPRE (SEQ ID NO: 316) operably linked to the 3' end of the polynucleotide encoding the anti-BCMA CAR.

Table 17 lists the recombinant lentivirus reference number and the corresponding SEQ ID NOs of the amino acid sequences of the anti-BCMA CARs and the different lentiviral vector architectures.

TABLE 17

Ref.	SEQ ID NO.	Promoter	WPRE
LV 3.1	259	MNDU3	wild-type
LV 3.2	259	MNDU3	no PRE
LV 3.3	259	MNDU3	mutant WPRE
LV 3.4	259	SFFV	wild-type
LV 3.5	259	SFFV	no PRE
LV 3.6	259	SFFV	mutant WPRE
LV 3.7	259	EF1 α	wild-type
LV 3.8	259	EF1 α	no PRE

TABLE 17-continued

Ref.	SEQ ID NO.	Promoter	WPRES
LV 3.9	259	EF1 α	mutant WPRES
LV 5.1	263	MNDU3	wild-type
LV 5.2	263	MNDU3	no PRE
LV 5.3	263	MNDU3	mutant WPRES
LV 5.4	263	SFFV	wild-type
LV 5.5	263	SFFV	no PRE
LV 5.6	263	SFFV	mutant WPRES
LV 5.7	263	EF1 α	wild-type
LV 5.8	263	EF1 α	no PRE
LV 5.9	263	EF1 α	mutant WPRES
LV 6.1	266	MNDU3	wild-type
LV 6.2	266	MNDU3	no PRE
LV 6.3	266	MNDU3	mutant WPRES
LV 6.4	266	SFFV	wild-type
LV 6.5	266	SFFV	no PRE
LV 6.6	266	SFFV	mutant WPRES
LV 6.7	266	EF1 α	wild-type
LV 6.8	266	EF1 α	no PRE
LV 6.9	266	EF1 α	mutant WPRES
LV 8.1	270	MNDU3	wild-type
LV 8.2	270	MNDU3	no PRE
LV 8.3	270	MNDU3	mutant WPRES
LV 8.4	270	SFFV	wild-type
LV 8.5	270	SFFV	no PRE
LV 8.6	270	SFFV	mutant WPRES
LV 8.7	270	EF1 α	wild-type
LV 8.8	270	EF1 α	no PRE
LV 8.9	270	EF1 α	mutant WPRES
LV 9.1	273	MNDU3	wild-type
LV 9.2	273	MNDU3	no PRE
LV 9.3	273	MNDU3	mutant WPRES
LV 9.4	273	SFFV	wild-type
LV 9.5	273	SFFV	no PRE
LV 9.6	273	SFFV	mutant WPRES
LV 9.7	273	EF1 α	wild-type
LV 9.8	273	EF1 α	no PRE
LV 9.9	273	EF1 α	mutant WPRES
LV 13.1	277	MNDU3	wild-type
LV 13.2	277	MNDU3	no PRE
LV 13.3	277	MNDU3	mutant WPRES
LV 13.4	277	SFFV	wild-type
LV 13.5	277	SFFV	no PRE
LV 13.6	277	SFFV	mutant WPRES
LV 13.7	277	EF1 α	wild-type
LV 13.8	277	EF1 α	no PRE
LV 13.9	277	EF1 α	mutant WPRES
LV19		MNDU3	no PRE

Infectious Titer

1 $\times 10^5$ Jurkat cells were plated in each well of a 96-well plate and transduced with the recombinant lentiviruses listed in Table 17 including LV 19, which harbors a lentiviral vector encoding a control anti-BCMA CAR obtained from the literature. Three days post-transduction, the cells were passaged. Seven days post-transduction the cells were harvested. Genomic DNA was isolated and purified from the harvested cells and used in a quantitative PCR (qPCR) assay to determine vector copy number (VCN) and subsequently, IU/mL. FIG. 3A.

All lentiviral vector architectures examined produced infectious titers and were subsequently used to transduce PBMCs.

VCN and Anti-BCMA CAR Expression

5 $\times 10^5$ human PBMCs were plated in each well of a 24-well plate and transduced with volume matched recombinant lentiviruses listed in Table 17.

Four days post-transduction, PBMCs were passaged to a 24-well GREX plate. Seven days post-transduction, PBMCs were harvested, one aliquot of cells was stained with BCMA-PE and analyzed by flow cytometry to assess the percentage of anti-BCMA CAR expressing cells and another

aliquot was used to isolate and purify genomic DNA for a quantitative PCR (qPCR) assay to determine vector copy number (VCN). FIG. 3B.

These data show that different lentiviral vector architectures tested in combination with different anti-BCMA CARs result in a spectrum of transduction and anti-BCMA CAR expression.

Anti-BCMA CAR Activity

5 $\times 10^5$ human PBMCs were plated in each well of a 24-well plate and transduced with recombinant lentiviruses listed in Table 17 that have the following lentiviral vector architectures: MNDU3 promoter and wild-type WPRES, MNDU3 promoter and a mutated WPRES, SFFV promoter and a mutated WPRES, and EF1 α promoter and no WPRES. PBMCs were transduced at an MOI of 1 (based on IU/mL determined in Jurkat cells), except for LV 3.6, LV 3.8, LV 9.8, and LV 13.8, in which volume matched lentivirus was used. Four days post-transduction, PBMCs were passaged to a 24-well GREX plate. Seven days post-transduction, PBMCs were harvested, one aliquot of cells was stained with BCMA-PE and analyzed by flow cytometry to assess the number of anti-BCMA CAR expressing cells and another aliquot was used in co-culture assays to assess anti-BCMA CAR function.

5 $\times 10^4$ transduced PBMCs were co-cultured with 5 $\times 10^4$ RPMI-8226 cells for 24 hours. Anti-BCMA CAR activity was assessed by harvesting PBMC/RPMI-8226 cell co-culture supernatants and measuring IFN γ and IL-2 levels using an MSD assay. IFN γ and IL-2 levels produced in co-culture were plotted against the percentage of anti-BCMA CAR positive cells. FIGS. 3C and 3D.

Antigen independent anti-BCMA CAR activity was assessed by culturing 5 $\times 10^4$ transduced PBMCs in the absence of target cells for 24 hours. After 24 hours, the supernatants were harvested and IFN γ levels measured using an MSD assay. IFN γ levels were plotted against lentiviral architectures used to express the anti-BCMA CARs. FIG. 3E.

These data indicate that combinations of different lentiviral architectures and anti-BCMA CARs can be selected to modulate anti-BCMA CAR expression and activity. Further, the data show that PBMCs expressing the anti-BCMA CARs set forth in SEQ ID NOS: 259, 263, 266, 270, 273, and 277 show comparable or increased cell expansion and comparable or increased activity compared to the control anti-BCMA CAR and that only three combinations showed high levels of antigen independent (tonic) signaling.

Off-Target Transduction

Off-target transduction of multiple myeloma cells was evaluated in two BCMA-expressing multiple myeloma cell lines, RPMI-8226 cells and KMS-11 cells. 1 $\times 10^5$ RPMI-8226 or 1 $\times 10^5$ KMS-11 cells were plated in each well of a 96-well plate and treated at an MOI of 1 with recombinant lentiviruses listed in Table 17 that have the following lentiviral vector architectures, MNDU3 promoter and wild-type WPRES, MNDU3 promoter and a mutated WPRES, SFFV promoter and a mutated WPRES, and EF1 α promoter and no WPRES; LV 19; and with LV 20. LV 20 is a recombinant lentiviral particle comprising a viral envelope that expresses a non-viral membrane bound tropism molecule comprising an anti-CD3 scFv fused to a CD8 α hinge and transmembrane domain (e.g., SEQ ID NO: 324); a mutant VSIV-G fusogen (e.g., SEQ ID NOS: 335); and a lentiviral vector comprising an MNDU3 promoter (SEQ ID NO: 319), operably linked to a polynucleotide encoding a

CD8 α signal peptide and GFP and a wild-type WPRE (SEQ ID NO: 315) operably linked to the 3' end of the polynucleotide encoding GFP.

Three days post-treatment, the cells were passaged. Seven days post-treatment, the cells were harvested and genomic DNA was isolated and purified for a qPCR assay to determine vector integration using VCN. VCN values for anti-BCMA CARs were normalized to VCN for LV 20, which expresses GFP rather than an anti-BCMA CAR.

The data show that differences in off-target multiple myeloma transduction were largely driven by the particular anti-BCMA CAR being expressed, rather than any particular lentiviral vector architecture. Several architectures used to express the anti-BCMA CARs in LV 3, LV 5, LV 6, LV 8, and LV 9 showed low levels of off-target transduction that were comparable to or less than LV 19, which expresses a control anti-BCMA CAR. In contrast, LV 13 exhibited the highest rates of off-target transduction compared to other LVs. FIG. 3F.

Example 3

In Vivo Administered Lentivirus Demonstrates Anti-Tumor Efficacy in a Multiple Myeloma Mouse Model

The anti-tumor efficacy of in vivo administered recombinant lentiviral particles comprising an envelope that expresses an anti-CD3-based tropism molecule and a mutant VSIV-G fusogen and a lentiviral vector encoding an anti-BCMA CAR was investigated in multiple myeloma mouse models.

Recombinant lentivirus for in vivo administration was produced by transient transfection of HEK293T cells with plasmids encoding an anti-CD3-based tropism molecule (SEQ ID NO: 324); a mutant VSIV-G fusogen (e.g., SEQ ID NO: 335); lentiviral GAG/POL; lentiviral REV; and a transfer plasmid encoding a lentiviral vector comprising: (i) an MNDU3 promoter (SEQ ID NO: 319) operably linked to a polynucleotide encoding a CD8 α signal peptide and an anti-BCMA CAR, and a wild-type WPRE (SEQ ID NO: 315) operably linked to the 3' end of the polynucleotide encoding the anti-BCMA CAR; (ii) an MNDU3 promoter (SEQ ID NO: 319) operably linked to a polynucleotide encoding a CD8 α signal peptide and an anti-BCMA CAR, and a mutated WPRE (SEQ ID NO: 316) operably linked to the 3' end of the polynucleotide encoding the anti-BCMA CAR; (iii) an SFFV promoter (SEQ ID NO: 322) operably linked to a polynucleotide encoding a CD8 α signal peptide and an anti-BCMA CAR, and a mutated WPRE (SEQ ID NO: 316) operably linked to the 3' end of the polynucleotide encoding the anti-BCMA CAR; or (iv) an EF1 α promoter (SEQ ID NO: 320) operably linked to a polynucleotide encoding a CD8 α signal peptide and an anti-BCMA CAR without a PRE.

The recombinant lentivirus reference number, the SEQ ID NO of the anti-BCMA CAR polypeptide and the corresponding lentiviral architectures shown in Table 18 were used in this Example.

TABLE 18

Ref.	SEQ ID NO.	Promoter	WPRE
LV 3.1	259	MNDU3	wild-type
LV 3.3	259	MNDU3	mutant WPRE
LV 3.6	259	SFFV	mutant WPRE

TABLE 18-continued

Ref.	SEQ ID NO.	Promoter	WPRE
5	LV 3.8	259	EF1 α
	LV 5.1	263	MNDU3
	LV 5.3	263	MNDU3
	LV 5.6	263	SFFV
	LV 5.8	263	EF1 α
10	LV 6.1	266	MNDU3
	LV 6.3	266	MNDU3
	LV 6.6	266	SFFV
	LV 6.8	266	EF1 α
	LV 8.1	270	MNDU3
	LV 8.3	270	MNDU3
	LV 8.6	270	SFFV
	LV 8.8	270	EF1 α
15	LV 9.1	273	MNDU3
	LV 9.3	273	MNDU3
	LV 9.6	273	SFFV
	LV 9.8	273	EF1 α
	LV 13.1	277	MNDU3
	LV 13.3	277	MNDU3
20	LV 13.6	277	SFFV
	LV 13.8	277	EF1 α

Ex vivo anti-BCMA CAR T cells were also prepared. Briefly, HEK293T cells were transiently transfected with plasmids encoding a wild-type VSIV-G fusogen; lentiviral GAG/POL; lentiviral REV; and a transfer plasmid encoding a lentiviral vector comprising an MNDU3 promoter operable linked to a CD8 α signal peptide and a control anti-BCMA CAR obtained from the literature, and a wild-type WPRE (SEQ ID NO: 315) operably linked to the 3' end of the polynucleotide encoding the anti-BCMA CAR. PBMCs were then transduced with the recombinant lentivirus and cultured for 7 days to generate anti-BCMA CAR T cells. First Daudi Model Study

NSG mice were intravenously injected with 2×10^6 Daudi cells labeled with firefly luciferase. After four days, four out of five groups of mice were intravenously administered 1×10^6 human PBMCs. The next day mice that received the 1×10^6 human PBMCs were administered vehicle control (DMEM); or 2.2×10^8 IU of LV 3.1, LV 6.1, LV 8.1, or LV 13.1. Mice that were not administered PBMCs were administered 5×10^6 ex vivo anti-BCMA CAR T cells. All groups of mice then received three doses of 2×10^5 IU recombinant human IL-2 at 6, 24, and 48 hours post LV administration. Tumor volume was measured by using a bioluminescence imaging system.

Tumor size increased in mice treated with vehicle. Mice treated with ex vivo anti-BCMA CAR T cells and in vivo with LV anti-BCMA CAR experienced tumor regression. FIG. 4A.

Second Daudi Model Study

NSG mice were intravenously injected with 2×10^6 Daudi cells labeled with firefly luciferase. After four days, eight out of nine groups of mice were intravenously administered 1×10^6 human PBMCs. The next day mice that received the 1×10^6 human PBMCs were administered vehicle control (DMEM); 1.25×10^8 IU of LV 3.1, LV 6.1, LV 6.3, LV 8.1, LV 9.3, LV 9.6, or LV 13.8; or 5.6×10^7 IU of LV 6.8. Mice that were not administered PBMCs were administered 5×10^6 ex vivo anti-BCMA CAR T cells. All groups of mice then received three doses of 2×10^5 IU recombinant human IL-2 at 6, 24, and 48 hours post LV administration. Tumor volume was measured by using a bioluminescence imaging system.

Tumor size increased in mice treated with vehicle. Mice treated with ex vivo anti-BCMA CAR T cells and in vivo with some LV anti-BCMA CARs experienced mild control

of tumor growth, whereas LV 6.8 and LV 13.8 experienced durable tumor regression. FIG. 4B.

Third Daudi Model Study

NSG mice were intravenously injected with 2×10^6 Daudi cells labeled with firefly luciferase. After four days, eight out of nine groups of mice were intravenously administered 1×10^6 human PBMCs. The next day mice that received the 1×10^6 human PBMCs were administered vehicle control (DMEM); 1.25×10^8 IU of LV 3.3, LV 3.6, LV 8.3, LV 8.6, LV 8.8, LV 13.3, or LV 13.6; or 5.6×10^7 IU of LV 6.8. Mice that were not administered PBMCs were administered 5×10^6 ex vivo anti-BCMA CAR T cells. All groups of mice then received three doses of 2×10^5 IU recombinant human IL-2 at 6, 24, and 48 hours post LV administration. Tumor volume was measured by using a bioluminescence imaging system.

Tumor size increased in mice treated with vehicle. Mice treated with ex vivo anti-BCMA CAR T cells and in vivo with some LV anti-BCMA CARs experienced mild control of tumor growth, whereas LV 6.8 and LV 8.8 experienced durable tumor regression. FIG. 4C.

First RPMI Model Study

NOD scid gamma (NSG) mice were subcutaneously injected with 1×10^6 RPMI-8226 cells (a BCMA positive tumor cell line). Tumors were allowed to grow to a size of about 110 mm^3 to 140 mm^3 (about two and a half weeks).

Five out of six groups of mice were then intravenously administered 1×10^6 human PBMCs. The next day, mice that received the 1×10^6 human PBMCs were administered vehicle control (DMEM); 5.0×10^7 IU of LV 6.3, LV 6.8, LV 8.3, or LV 8.8. The sixth group of mice was administered 2×10^6 unmodified ex vivo anti-BCMA CAR T cells. All groups of mice then received three doses of 2×10^5 IU recombinant human IL-2 at 6, 24, and 48 hours post LV administration. Tumor volume was measured externally using calipers and mice were euthanized at pre-determined humane endpoints based on tumor size and body condition.

Tumor size increased in mice treated with vehicle control. Mice treated with LV 6.3 experienced moderate tumor regression, whereas mice treated with ex vivo anti-BCMA CAR T cells or in vivo with LV 6.8, LV 8.3, or LV 8.8 experienced complete and durable tumor regression. FIG. 4D.

Mice that were not administered PBMCs were administered 5×10^6 ex vivo anti-BCMA CAR T cells. All groups of mice then received three doses of 2×10^5 IU recombinant human IL-2 at 6, 24, and 48 hours post LV administration.

Second RPMI Model Study

NOD scid gamma (NSG) mice were subcutaneously injected with 1×10^6 RPMI-8226 cells (a BCMA positive tumor cell line). Tumors were allowed to grow to a size of about 110 mm^3 to 140 mm^3 (about two and a half weeks).

Four out of five groups of mice were then intravenously administered 1×10^6 human PBMCs. The next day, mice that received the 1×10^6 human PBMCs were administered vehicle control (DMEM); 1.25×10^7 IU of LV 6.8, 5.0×10^7 IU of LV 6.8, or 1.25×10^8 IU of LV 6.8. The fifth group of mice was administered 2×10^6 unmodified ex vivo anti-BCMA CAR T cells. All groups of mice then received three doses of 2×10^5 IU recombinant human IL-2 at 6, 24, and 48 hours post LV administration. Tumor volume was measured externally using calipers and mice were euthanized at pre-determined humane endpoints based on tumor size and body condition.

Tumor size increased in mice treated with vehicle control. Mice treated with all three doses of LV 6.8 experienced dose-dependent but complete and durable tumor regression.

Mice treated with ex vivo anti-BCMA CAR T cells also experienced complete and durable tumor regression. FIG. 4E.

Third RPMI Model Study

NOD scid gamma (NSG) mice were subcutaneously injected with 1×10^6 RPMI-8226 cells (a BCMA positive tumor cell line). Tumors were allowed to grow to a size of about 110 mm^3 to 140 mm^3 (about two and a half weeks).

Three out of four groups of mice were then intravenously administered 1×10^6 human PBMCs. The next day, mice that received the 1×10^6 human PBMCs were administered vehicle control (DMEM); 5.0×10^7 IU of LV 6.3 or 1.25×10^8 IU of LV 6.3. The fourth group of mice was administered 2×10^6 unmodified ex vivo anti-BCMA CAR T cells. All groups of mice then received three doses of 2×10^5 IU recombinant human IL-2 at 6, 24, and 48 hours post LV administration. Tumor volume was measured externally using calipers and mice were euthanized at pre-determined humane endpoints based on tumor size and body condition.

Tumor size increased in mice treated with vehicle control. Mice treated with both doses of LV 6.3 experienced dose-dependent tumor regression. Mice treated with ex vivo anti-BCMA CAR T cells experienced complete and durable tumor regression. FIG. 4F.

Fourth Daudi Model Study

NSG mice were intravenously injected with 2×10^6 Daudi cells labeled with firefly luciferase. After four days, four out of five groups of mice were intravenously administered 1×10^6 human PBMCs. The next day mice that received the 1×10^6 human PBMCs were administered vehicle control (DMEM); 1.25×10^8 IU of LV 6.1 or LV 6.3; or 5.6×10^7 IU of LV 6.8. Mice that were not administered PBMCs were administered 5×10^6 ex vivo anti-BCMA CAR T cells. All groups of mice then received three doses of 2×10^5 IU recombinant human IL-2 at 6, 24, and 48 hours post LV administration. Tumor volume was measured by using a bioluminescence imaging system.

Tumor size increased in mice treated with vehicle. Mice treated with ex vivo anti-BCMA CAR T cells and in vivo with LV 6.1 and LV 6.3 experienced mild control of tumor growth, whereas LV 6.8 experienced complete and durable tumor regression. FIG. 4G.

Example 4

Comparative Anti-Tumor Efficacy in a Multiple Myeloma Mouse Model in Both In Vivo and Ex Vivo Formats

The anti-tumor efficacy of recombinant lentiviral particles comprising an envelope that expresses an anti-CD3-based tropism molecule and a mutant VSIV-G fusogen and a lentiviral vector encoding various anti-BCMA CARs was investigated in multiple myeloma mouse models. The recombinant lentiviruses were formulated as in vivo administered lentiviral particles and were also used to manufacture ex vivo anti-BCMA CAR T cells.

Recombinant lentivirus was produced by transient transfection of HEK293T cells with plasmids encoding a non-viral membrane bound tropism molecule comprising an anti-CD3 scFv fused to a CD8 α hinge and transmembrane domain; a mutant VSIV-G fusogen comprising K47Q and R354A amino acid substitutions; lentiviral GAG/POL; lentiviral REV; and a transfer plasmid encoding a lentiviral vector encoding an anti-BCMA CAR set forth in SEQ ID NO: 266, SEQ ID NO: 340, or SEQ ID NO: 341 or a GFP control.

TABLE 19

SEQ ID NO:	NUCLEIC ACID SEQUENCE
340	DIVLTQSPPSLAMSGLGKRATISCRASESVTILGSHLIHWYQQKPGQPPTLLIQLASNVQTG VPARFSGSGSRDFTLTIDPVEEDDVAVYYCLQSRITPRTFGGGTKLEIKGSTSGSGKPGS GEGSTKGQIQLVQSGPELKKPGETVKISCKASGYTFTDYSINWVKRAPGKGLKWMGWINTE TREPAYAYDERGRFAFSLTASTAYLQINNLYEDTATYFPCALDYSYAMDYWGQTSVTV SSAAATTTTPAPRPPTPAPTIASQPLSLRPEACRPAAGGAVHTRGLDFACDIYIWAFLAGTC GVLLLSLVI TLYCKRGRKKLLYIFKQPFMRPVQTTQEEDGCS CRFPEEEEGGCEL RVKFSR SADAPAYQQGQNLVYELNLRREEYDVLDRGRDPEMGGKPRRKNPQEGLYNELQKDKM AEAYSEIGMKGERRRGKGHDGLYQGLSTATKDTYDALHMQALPPR
341	QVKLEESGGGLVQAGRSRLRLSCAASEHTFSSHVMGWFRQAPGKERESVAVIGWRDISTSYA DSVKGRFTISRDNAKKTLYLQMNLSLKPEDTAVYYCAARRIDAADFDSWGQGTQVTVSSGGG GSEVQLVESGGGLVQAGGSLRLSCAASGRTFTMGWFRQAPGKEREFVAAISLSPTLAYYAE SVKGRFTISRDNAKNTVVLQMNLSLKPEDTALYYCAADRKSVMISIRPDYWGQGTQVTVSSST TTTPAPRPPTPAPTIASQPLSLRPEACRPAAGGAVHTRGLDFACDIYIWAFLAGTCGVLLL SLVITLYCKRGRKKLLYIFKQPFMRPVQTTQEEDGCS CRFPEEEEGGCEL RVKFSRSADAP AYQQGQNLVYELNLRREEYDVLDRGRDPEMGGKPRRKNPQEGLYNELQKDKMAEAYS EIGMKGERRRGKGHDGLYQGLSTATKDTYDALHMQALPPR

The recombinant lentivirus reference number, the SEQ ID NO of the anti-BCMA CAR polypeptide and the corresponding lentiviral architectures shown in Table 12 were used in this Example.

TABLE 20

Ref.	SEQ ID NO.	Promoter	WPRE
LV 6.8	266	EF1 α	none
LV A	340	MNDU3	WT WPRE
LV B	341	EF1 α	none
LV19	GFP	MNDU3	none

Ex vivo anti-BCMA CAR T cells were also prepared by transducing PBMCs with the recombinant lentivirus and culturing the transduced cell for 7 days to generate anti-BCMA CAR T cells.

In Vivo Daudi Model Study

NSG mice were intravenously injected with 2×10^6 Daudi cells labeled with firefly luciferase. After four days, four out of five groups of mice were intravenously administered 1×10^6 human PBMCs. The next day, mice that did not receive PBMCs were administered vehicle control (DMEM) and mice that received the PBMCs were administered 5.0×10^7 IU of LV 6.8, LV A, LV B, or LV 19 (GFP control). Tumor volume was measured by using a bioluminescence imaging system.

Tumor size increased in mice treated with vehicle, mice treated with the GFP control, and mice treated with a lentivirus expressing an anti-BCMA CAR comprising the binding domain used in idecabtagene vicleucel. Mice treated with a lentivirus expressing an anti-BCMA CAR comprising

the binding domains like those used in ciltacabtagene autoleucel experienced suppression of tumor growth. Only mice treated with an anti-BCMA CAR comprising SEQ ID NO: 266 experienced tumor regression. FIG. 5A.

Ex Vivo Daudi Model Study

NSG mice were intravenously injected with 2×10^6 Daudi cells labeled with firefly luciferase. After five days, three out of five groups of mice were intravenously administered 2×10^6 human anti-BCMA CAR T cells. Mice that did not receive anti-BCMA CAR T cells were administered vehicle control (DMEM) or 2×10^6 untransduced control human T cells (UTD) and mice that received the anti-BCMA CAR T cells were administered 2×10^6 anti-BCMA CAR T cells expressing the CAR encoded by SEQ ID NO: 266, SEQ ID NO: 340 or SEQ ID NO: 341. Tumor volume was measured by using a bioluminescence imaging system.

Tumor size increased in mice treated with vehicle and with untransduced control T cells. Mice treated with CAR T cells expressing an anti-BCMA CAR comprising the binding domain used in idecabtagene vicleucel showed a transient decrease in tumor burden whereas mice treated with CAR T cells expressing an anti-BCMA CAR comprising SEQ ID NO: 266 or an anti-BCMA CAR comprising the binding domains like those used in ciltacabtagene autoleucel experienced comparable and complete tumor regression. FIG. 5B.

In general, in the following claims, the terms used should not be construed to limit the claims to the specific embodiments disclosed in the specification and the claims, but should be construed to include all possible embodiments along with the full scope of equivalents to which such claims are entitled. Accordingly, the claims are not limited by the disclosure.

SEQUENCE LISTING

Sequence total quantity: 374
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 mol_type = protein
 organism = synthetic construct
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 ASKWVTTCD F RYGGPKYI TH SIRSFTPSVE QCKESIEQTK QGTWLNPGFP PQSCGYATVT 120
 DAEAVIVQVT PHHVLVDEYT GEWVDSQFIN GKCSNYICPT VHNSTTWHSD YKVKGLCDSN 180

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LISMDITFFS EDGELSSSLGK EGTGFRSNYP AYETGGKACK MQYCKHWGVR LPSGVWFEMA 240
DKDLFAAARF PECPEGSSIS APSQTSVDVS LIQDVERILD YSLCQETWSK IRAGLPISPV 300
DLSYLAPKNP GTGPAFTIIN GTLKYFETRY IRVDIAAPIL SRMVGMIISGT TTERELWDDW 360
APYEDVEIGP NGVLRITSSGY KFPLYMIGHG MLDSDLHLSS KAQVFEHPHI QDAASQLPDD 420
ESLFFGDTGL SKNPIELVEG WFSSWKSSIA SFFFIIGLII GLFLVLRVGI HLCIKLKHTK 480
KRQIYTDIEM NRLGK 495

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SEQ ID NO: 2      moltype = AA length = 495
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DAEAVIVQVT PHHVLVDEYT GEWVDSQFIN GKCSNYICPT VHNSTTWHSY YKVKGLCDSN 180
LISMDITFFS EDGELSSSLGK EGTGFRSNYP AYETGGKACK MQYCKHWGVR LPSGVWFEMA 240
DKDLFAAARF PECPEGSSIS APSQTSVDVS LIQDVERILD YSLCQETWSK IRAGLPISPV 300
DLSYLAPKNP GTGPAFTIIN GTLKYFETRY IRVDIAAPIL SRMVGMIISGT TTERELWDDW 360
APYEDVEIGP NGVLRITSSGY KFPLYMIGHG MLDSDLHLSS KAQVFEHPHI QDAASQLPDD 420
ESLFFGDTGL SKNPIELVEG WFSSWKSSIA SFFFIIGLII GLFLVLRVGI HLCIKLKHTK 480
KRQIYTDIEM NRLGK 495

```

```

SEQ ID NO: 3      moltype = AA length = 495
FEATURE          Location/Qualifiers
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                 mol_type = protein
                 organism = synthetic construct

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SEQUENCE: 3
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DAEAVIVQVT PHHVLVDEYT GEWVDSQFIN GKCSNYICPT VHNSTTWHSY YKVKGLCDSN 180
LISMDITFFS EDGELSSSLGK EGTGFRSNYP AYETGGKACK MQYCKHWGVR LPSGVWFEMA 240
DKDLFAAARF PECPEGSSIS APSQTSVDVS LIQDVERILD YSLCQETWSK IRAGLPISPV 300
DLSYLAPKNP GTGPAFTIIN GTLKYFETRY IRVDIAAPIL SRMVGMIISGT TTERELWDDW 360
APYEDVEIGP NGVLRITSSGY KFPLYMIGHG MLDSDLHLSS KAQVFEHPHI QDAASQLPDD 420
ESLFFGDTGL SKNPIELVEG WFSSWKSSIA SFFFIIGLII GLFLVLRVGI HLCIKLKHTK 480
KRQIYTDIEM NRLGK 495

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SEQ ID NO: 4      moltype = AA length = 495
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                 mol_type = protein
                 organism = synthetic construct

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SEQUENCE: 4
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ASKWVTTCDP RWYGPKYITQ SIRSFTPSVE QCKESIEQTK QGTWLNPGFP PQSCGYATVT 120
DAEAVIVQVT PHHVLVDEYT GEWVDSQFIN GKCSNYICPT VHNSTTWHSY YKVKGLCDSN 180
LISMDITFFS EDGELSSSLGK EGTGFRSNYP AYETGGKACK MQYCKHWGVR LPSGVWFEMA 240
DKDLFAAARF PECPEGSSIS APSQTSVDVS LIQDVERILD YSLCQETWSK IRAGLPISPV 300
DLSYLAPKNP GTGPAFTIIN GTLKYFETRY IRVDIAAPIL SRMVGMIISGT TTERELWDDW 360
APYEDVEIGP NGVLRITSSGY KFPLYMIGHG MLDSDLHLSS KAQVFEHPHI QDAASQLPDD 420
ESLFFGDTGL SKNPIELVEG WFSSWKSSIA SFFFIIGLII GLFLVLRVGI HLCIKLKHTK 480
KRQIYTDIEM NRLGK 495

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SEQ ID NO: 5      moltype = AA length = 495
FEATURE          Location/Qualifiers
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                 mol_type = protein
                 organism = synthetic construct

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SEQUENCE: 5
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ASKWVTTCDP RWYGPKYITQ SIRSFTPSVE QCKESIEQTK QGTWLNPGFP PQSCGYATVT 120
DAEAVIVQVT PHHVLVDEYT GEWVDSQFIN GKCSNDICLT VHNSTTWHSY YKVKGLCDSN 180
LISMDITFFS EDGELSSSLGK AGTGFRSNYP AYETGDKACK MQYCKHWGVR LPSGVWFEMA 240
DKDLFAAARF PECPEGSSIS APSQTSVDVS LIQDVERILD YSLCQETWSK IRAGLPISPV 300
DLSYLAPKNP GTGPAFTIIN GTLKYFETRY IRVDIAAPIL SRMVGMIISGT NTERELWEDW 360
APYEDVEIGP NGVLRITSSGY KFPLYMIGHG MLDSDLHLSS KVQVFEHPHI QDAASQLPDD 420
ETLFFGDTGL SKNPIELVEG WFGWKSSIA SFFFIIGLII GLFLVLRVGI YLCIKLKHTK 480
KRKIYADIEM NRLGK 495

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SEQ ID NO: 6      moltype = AA length = 495
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                 mol_type = protein
                 organism = synthetic construct

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SEQUENCE: 6
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ASKWVTTCDP RWYGPKYITQ SIRSFTPSVE QCKESIEQTK QGTWLNPGFP PQSCGYATVT 120

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-continued

DAAEAVIVQVT	PHHVLVDEYT	GEWVDSQFIN	GKCSDDICPT	VHNSTTWHSD	YKVKGLCDSN	180
LISMDITFFS	EDGELSSLGK	EGTGFRSNYF	AYETGDKACK	MQYCKHWGVR	LPSGVWFEMA	240
DNLFPAARF	PECPEGSSIS	APSQTSVDVS	LIQDVERILD	YSLCQETWSK	IRAGLPISPV	300
DLSYLAPKNP	GTGPAFTIIN	GTLKYFETRY	IRVDIAAPIL	SRMVGMISGT	TTERELWEDW	360
APYEDVEIGP	NGVLRSSGY	KFPLYMIGHG	MLDSDLHLSS	KAQVFEHPHI	PDATSQLPDD	420
ETLFFGDTGL	SKDPIELVEG	WFSGWKSSIA	SFFFIIGLII	GLFFVLIRIGV	YLCIKLKHTN	480
KRQIYTDIEM	NRLGK					495

SEQ ID NO: 7 moltype = AA length = 495
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 mol_type = protein
 organism = synthetic construct

SEQUENCE: 7

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ASKWVTTCDF	RWYGPKYITH	SIRSFTPSVE	QCRSIEQTK	QGTWLNPGFP	PQSCGYATVT	120
DAAEAVIVQVT	PHHVLVDEYT	GEWVDSQFIN	GKCSNDICPT	VHNSTTWHSD	YKVKGLCDSN	180
LISMDITFFS	EDGELSSLGK	EGTGFRSNHF	AYETGDKACK	MQYCKHWGVR	LPSGVWFEMA	240
DQDLFAARF	PECPEGSSIS	APSQTSVDVS	LIQDVERILD	YSLCQETWSK	IGAGLPISPV	300
DLSYLAPKNP	GTGPAFTIIN	GTLKYFETRY	IRVDIAAPIL	SRMVGMISGT	TTERELWDDW	360
APYEDVEIGP	NGVLRSSGY	KFPLYMIGHG	MLDSDLHLSS	KAQVFEHPHI	QDAASQLPDD	420
ETLFFGDTGL	SKNPIELVEG	WFSGWKSSIA	SFFFIIGLII	GLFLVLRVGI	YLCIKLKHTK	480
KRQIYTDIEM	NRLGK					495

SEQ ID NO: 8 moltype = AA length = 495
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 mol_type = protein
 organism = synthetic construct

SEQUENCE: 8

KFTIVFPNHQ	KGNWKNVPSN	YHYCPSSSDL	NWHNDLIGTG	LQVKMPKSHK	AIQADGWMCH	60
ASKWVTTCDF	RWYGPKYITH	SIRSFTPSVE	QCKESIEQTK	QGTWLNPGFP	PQSCGYATVT	120
DAAEAVIVQVT	PHHVLVDEYT	GEWVDSQFIN	GKCSNDICPT	VHNSTTWHSD	YKVKGLCDSN	180
LISTDITFFS	EDRELSSLGK	EGTGFRSNYF	AYETGDKACK	MQYCKHWGVR	LPSGVWFEMA	240
DKDLFAARF	PECPEGSSIS	APSQTSVDVS	LIQDVERILD	YSLCQETWSK	IRAGLPISPV	300
DLSYLAPKNP	GTGPAFTIIN	GTLKYFETRY	IRVDIAAPIL	SRMVGMISGT	TTERELWDDW	360
APYEDVEIGP	NGVLRSSGY	KFPLYMIGHG	MLDSGLHLSS	KAQVFEHPHI	QDAASQLPDD	420
EILFFGDTGL	SKNPIDFVEG	WFSSWKSSIA	SFFFIIGLII	GLFLVLRVGI	YLYIKLKHTK	480
KRQIYTDIEM	NRLGR					495

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SEQUENCE: 9

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DSVAVVQAT	PHHVLVDEYT	GEWIDSQFPN	GKCETEECT	VHNSTVWYSD	YKVTGLCDAT	180
LVDTEITFFS	EDGKESIGK	PNTGYRSNYF	AYEKGDVKCK	MNYCKHAGVR	LPSGVWFEFV	240
DQDVYAAAKL	PECVPGATIS	APTQTSVDVS	LILDVERILD	YSLCQETWSK	IRSKQPVSPV	300
DLSYLAPKNP	GTGPAFTIIN	GTLKYFETRY	IRIDIDNPPI	SKMVGKISGS	QTERELWTEW	360
FPYEGVEIGP	NGILKTPTGY	KFPLFMIGHG	MLDSDLHKTS	QAEVFEHPHL	AEAPKQLPEE	420
ETLFFGDTGI	SKNPVELIEG	WFSSWKSTVV	TFFFAIGVFI	LLYVVARIVI	AVRYRYQGSN	480
NKRIYNDIEM	SRFRK					495

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 organism = Homo sapiens

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GLSLIISLAV	FVLMFLRKI	NSEPLKDEFK	NTGSGLLGMA	NIDLEKSRTG	DEIILPRGLE	120
YTVEECTCED	CIKSKPKVDS	DHCFPLPAME	EGATILVTTK	TNDYCKSLPA	ALSATEIEKS	180
ISAR						184

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 organism = synthetic construct

SEQUENCE: 11

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YSPSLKSRLT	ITKDTSKNQV	VLTMTNMDPV	DTAVVYCARD	EYGGFDIWGQ	GTMVTVSS	118

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YSPSLKSRLT ITKDTSKNQV VLTMTNMDPV DTAVYYCARD EYGGFDIWGQ GTMVTVSSGG		120
GGSGGGSGG GSGGGGSEI VLTQSPATLS LSPGERATLS CRASQSVSSY LAWYQQKPGQ		180
APRLLIYDAS NRATGIPARF SGSGSGTDFT LTISSLEPED FAVYVCQQRV VYPITFGGT		240
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GSGGGGSQIT LKESGPTLVK PTQTLTLTCT FSGFSLSTSG VGVGWIRQPP GKALEWLALI		180
YWNDEKRYSP SLKSRLTITK DTSKNQVVL TMTNMDPVDTA VYCARDEYG GFDIWQGTM		240
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SEQ ID NO: 22	moltype = AA length = 7	
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FEATURE	Location/Qualifiers	
source	1..16	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 23		
LIYWNDKRY SPSLKS		16
SEQ ID NO: 24	moltype = AA length = 8	
FEATURE	Location/Qualifiers	
source	1..8	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 24		
DEYGGFDI		8
SEQ ID NO: 25	moltype = AA length = 107	
FEATURE	Location/Qualifiers	
source	1..107	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 25		
EIVLTQSPAT LSLSPGERAT LSCRASQSVS SYLAWYQQKP GQAPRLLIYD ASNRATGIPA		60
RFGSGSGGTD FTLTISSLEP EDFAVYCCQ RFDYPITFGG GTKVEIK		107
SEQ ID NO: 26	moltype = AA length = 11	
FEATURE	Location/Qualifiers	
source	1..11	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 26		
RASQSVSSYL A		11
SEQ ID NO: 27	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
source	1..7	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 27		
DASNRAT		7
SEQ ID NO: 28	moltype = AA length = 9	
FEATURE	Location/Qualifiers	
source	1..9	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 28		
QQRFDYPIT		9
SEQ ID NO: 29	moltype = AA length = 245	
FEATURE	Location/Qualifiers	
source	1..245	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 29		
QITLKESGPT LVKPTQTLTL TCTFSGFSLT TSGVGVGWIR QPPGKALEWL ALIYWNDDKR		60
YSPSLKSRLT ITKDTSKNQV VLTMTNMDPV DTAIVYCARD EYGGFDIWGQ GTMVTVSSGG		120
GGSGGGGSGG GGSGGGGSEI VLTQSPATLS LSPGERATLS CRASQSVSSY LAWYQQKPGQ		180
APRLLIYDAS NRATGIPARF SGSGSGTDFT LTISSLEPED FAVYCCQRF DYPITFGGGT		240
KVEIK		245
SEQ ID NO: 30	moltype = AA length = 245	
FEATURE	Location/Qualifiers	

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source                1..245
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 30
EIVLTQSPAT LSLSPGERAT LSCRASQSVS SYLAWYQQKP GQAPRLLIYD ASNRATGIPA 60
RFSGSGSGTD FTLTISSLEP EDFAVYVCQQ RFDYPITFGG GTKVEIKGGG GSGGGGSGGG 120
GSGGGGSGIT LKESGPTLVK PTQTLTLTCT FSGFSLSTSG VGVGWIRQPP GKALEWLALI 180
YWNDDKRYSP SLKSRLTITK DTSKNQVVLV MTNMDPVDTA VYYCARDEYG GFDIWGQGTM 240
VTVSS 245

SEQ ID NO: 31          moltype = AA length = 117
FEATURE               Location/Qualifiers
source                1..117
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 31
QVQLVESGGG VVQPGRLRL SCAASGFTFS SYGMHWVRQA PGKGLEWVAV ISYEGSNKYY 60
ADSVKGRFTI SRDNSKNTLY LQMNSLRAED TAVYYCAREL GDGMDVWQGQ TTVTVSS 117

SEQ ID NO: 32          moltype = AA length = 5
FEATURE               Location/Qualifiers
source                1..5
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 32
SYGMH 5

SEQ ID NO: 33          moltype = AA length = 17
FEATURE               Location/Qualifiers
source                1..17
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 33
VISYEGSNKY YADSVKG 17

SEQ ID NO: 34          moltype = AA length = 8
FEATURE               Location/Qualifiers
source                1..8
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 34
ELGDGMDV 8

SEQ ID NO: 35          moltype = AA length = 106
FEATURE               Location/Qualifiers
source                1..106
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 35
EIVLTQSPAT LSLSPGERAT LSCRASQSVS SYLAWFQQKP GQAPRLLIYD ASNRATGIPA 60
RFSGSGSGTD FTLTISSLEP EDFAVYVCQQ RVDLWTFGGG TKVEIK 106

SEQ ID NO: 36          moltype = AA length = 11
FEATURE               Location/Qualifiers
source                1..11
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 36
RASQSVSSYL A 11

SEQ ID NO: 37          moltype = AA length = 7
FEATURE               Location/Qualifiers
source                1..7
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 37
DASNRAT 7

SEQ ID NO: 38          moltype = AA length = 8
FEATURE               Location/Qualifiers
source                1..8
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 38
QQRVDLWT 8

SEQ ID NO: 39          moltype = AA length = 243
FEATURE               Location/Qualifiers
source                1..243

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mol_type = protein
organism = synthetic construct

SEQUENCE: 39
QVQLVESGGG VVQPGKSLRL SCAASGFTFS SYGMHWVRQA PGKGLEWVAV ISYEGSNKYY 60
ADSVKGRFTI SRDNSKNTLY LQMNSLRAED TAVYYCAREL GDGMDVWGQG TTVTVSSGGG 120
SGGGGSGGGG GSGGGGSEIV LTQSPATLSL SPGERATLSC RASQSVSSYL AWFQKPGQA 180
PRLLIYDASN RATGIPARFS GSGSGTDFTL TISSLEPEDF AVYYCQQRVD LWFQGGGTKV 240
EIK 243

SEQ ID NO: 40      moltype = AA length = 243
FEATURE           Location/Qualifiers
source            1..243
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 40
EIVLTQSPAT LSLSPGERAT LSCRASQSVS SYLAWFQQKP GQAPRLLIYD ASNRATGIPA 60
RFGSGSGGTD FTLTISSLEP EDFAVYYCQQ RVDLWTFGGG TKVEIKGGGG SGGGSGGGG 120
SGGGGSQVQL VESGGGVVQP GRSLRLSCAA SGFTFSSYGM HWVRQAPKGK LEWVAVISYE 180
GSNKYYADSV KGRFTISRDN SKNTLYLQMN SLRAEDTAVY YCARELGDGM DVWGQGTITV 240
VSS 243

SEQ ID NO: 41      moltype = AA length = 118
FEATURE           Location/Qualifiers
source            1..118
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 41
EVQLVESGGG LVKPGGSLRL SCAASGFTFS DYYMSWIRQA PGKGLEWVSY ISSSGSTIYY 60
ADSVKGRFTI SRDNAKNSLY LQMNSLRAED TAVYYCARDQ GNYGVDVWGQ GTTITVSS 118

SEQ ID NO: 42      moltype = AA length = 5
FEATURE           Location/Qualifiers
source            1..5
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 42
DYYMS 5

SEQ ID NO: 43      moltype = AA length = 17
FEATURE           Location/Qualifiers
source            1..17
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 43
YISSSGSTIY YADSVKG 17

SEQ ID NO: 44      moltype = AA length = 9
FEATURE           Location/Qualifiers
source            1..9
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 44
DQGNYGVDV 9

SEQ ID NO: 45      moltype = AA length = 107
FEATURE           Location/Qualifiers
source            1..107
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 45
DIQMTQSPST LSASVGRVT ITCRASQSIG SWLAWYQQKPK GKAPKLLIYD ASSLESGVPS 60
RFGSGSGGTE FTLTISSLPQ DDFATYYCQQ VSSLPPTFGG GTKVEIK 107

SEQ ID NO: 46      moltype = AA length = 11
FEATURE           Location/Qualifiers
source            1..11
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 46
RASQSISWLA A 11

SEQ ID NO: 47      moltype = AA length = 7
FEATURE           Location/Qualifiers
source            1..7
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 47
DASSLES 7

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SEQ ID NO: 48      moltype = AA length = 9
FEATURE           Location/Qualifiers
source            1..9
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 48
QQVSSLPPT                                     9

SEQ ID NO: 49      moltype = AA length = 245
FEATURE           Location/Qualifiers
source            1..245
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 49
EVQLVESGGG LVKPGGSLRL SCAASGFTFS DYYMSWIRQA PGKGLEWVSY ISSSGSTIYY 60
ADSVKGRFTI SRDIAKNSLY LQMNSLRAED TAVYYCARDQ GNYGVDVWGQ GTTIVTVSSG 120
GGSGGGGSGG GSGGGGSDI QMTQSPSTLS ASVGRVTIT CRASQSISSW LAWYQQKPGK 180
APKLLIYDAS SLESGVPSRF SGSGSGTEFT LTISLQPD FATYQCQVS SLPPTFGGT 240
KVEIK                                           245

SEQ ID NO: 50      moltype = AA length = 245
FEATURE           Location/Qualifiers
source            1..245
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 50
DIQMTQSPST LSASVGRVT ITCRASQSI SWLAWYQQK GKAPKLLIYD ASSLESGVPS 60
RFSGSGSGTE FTLTISSLQ DDFATYQCQ VSSLPPTFGG GTKVEIKGGG GSGGGGSGGG 120
GSGGGGSEVQ LVESGGGLVK PGGSLRLSCA ASGFTFSDYY MSWIRQAPGK GLEWVSYISS 180
SGSTIYADS VKGRFTISR NAKNSLYLQM NSLRAEDTAV YYCARDQGN YGVDVWGQGT 240
VTVSS                                           245

SEQ ID NO: 51      moltype = AA length = 118
FEATURE           Location/Qualifiers
source            1..118
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 51
QVQLVESGGG LVKPGGSLRL SCAASGFTFS DYYMSWIRQA PGKGLEWVSY ISSSGSTIYY 60
ADSVKGRFTI SRDIAKNSLY LQMNSLRAED TAVYYCARDQ GNYGVDVWGQ GTTIVTVSS 118

SEQ ID NO: 52      moltype = AA length = 5
FEATURE           Location/Qualifiers
source            1..5
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 52
DYYMS                                           5

SEQ ID NO: 53      moltype = AA length = 17
FEATURE           Location/Qualifiers
source            1..17
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 53
YISSSGSTIY YADSVKG                               17

SEQ ID NO: 54      moltype = AA length = 9
FEATURE           Location/Qualifiers
source            1..9
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 54
DQGNYGVDV                                           9

SEQ ID NO: 55      moltype = AA length = 107
FEATURE           Location/Qualifiers
source            1..107
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 55
DIQMTQSPST LSASVGRVT ITCRASQSI SWLAWYQQK GKAPKLLIYD ASSLESGVPS 60
RFSGSGSGTE FTLTISSLQ DDFATYQCQ SDSPITFGG GTKVEIK 107

SEQ ID NO: 56      moltype = AA length = 11
FEATURE           Location/Qualifiers
source            1..11
                  mol_type = protein
                  organism = synthetic construct

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SEQUENCE: 56
RASQSISSWL A 11

SEQ ID NO: 57 moltype = AA length = 7
FEATURE Location/Qualifiers
source 1..7
mol_type = protein
organism = synthetic construct

SEQUENCE: 57
EASSLES 7

SEQ ID NO: 58 moltype = AA length = 9
FEATURE Location/Qualifiers
source 1..9
mol_type = protein
organism = synthetic construct

SEQUENCE: 58
QQSDSHPIT 9

SEQ ID NO: 59 moltype = AA length = 245
FEATURE Location/Qualifiers
source 1..245
mol_type = protein
organism = synthetic construct

SEQUENCE: 59
QVQLVESGGG LVKPGGSLRL SCAASGFTFS DYYMSWIRQA PGKGLEWVSY ISSSGSTIYY 60
ADSVKGRFTI SRDIAKNSLY LQMNSLRAED TAVYYCARDQ GNYGVDVWGQ GTTIVTVSSGG 120
GSGGGGSGGG GSGGGGSDI QMTQSPSTLS ASVGDRTIT CRASQSISSW LAWYQQKPGK 180
APKLLIYEAS SLESGVPSRF SGSGSGTEFT LTISSLQPDG FATYYCQQSD SHPITFGGGT 240
KVEIK 245

SEQ ID NO: 60 moltype = AA length = 245
FEATURE Location/Qualifiers
source 1..245
mol_type = protein
organism = synthetic construct

SEQUENCE: 60
DIQMTQSPST LSASVGDRVT ITCRASQSI SWLAWYQQKPGKAPKLLIYE ASSLESGVPS 60
RFSGSGSGTE FTLTISSLQP DDFATYYCQQ SDSPITFGG GTKVEIKGGG GSGGGGSGGG 120
GSGGGGSGVQ LVESGGGLVK PGGSLRLSQA ASGFTFSDYY MSWIRQAPGK GLEWVSYISS 180
SGSTIYYADS VKGRFTISR DAKNSLYLQM NSLRAEDTAV YYCARDQGNV GVDVWGQGT 240
TVSS 245

SEQ ID NO: 61 moltype = AA length = 118
FEATURE Location/Qualifiers
source 1..118
mol_type = protein
organism = synthetic construct

SEQUENCE: 61
EVQLVESGGG LVKPGGSLRL SCAASGFTFS DYYMSWIRQA PGKGLEWVSY ISSSGSTIYY 60
ADSVKGRFTI SRDIAKNSLY LQMNSLRAED TAVYYCARDQ GNYGVDVWGQ GTTIVTVSS 118

SEQ ID NO: 62 moltype = AA length = 5
FEATURE Location/Qualifiers
source 1..5
mol_type = protein
organism = synthetic construct

SEQUENCE: 62
DYYMS 5

SEQ ID NO: 63 moltype = AA length = 17
FEATURE Location/Qualifiers
source 1..17
mol_type = protein
organism = synthetic construct

SEQUENCE: 63
YISSSGSTIY YADSVKG 17

SEQ ID NO: 64 moltype = AA length = 9
FEATURE Location/Qualifiers
source 1..9
mol_type = protein
organism = synthetic construct

SEQUENCE: 64
DQGNVGVVDV 9

SEQ ID NO: 65 moltype = AA length = 107
FEATURE Location/Qualifiers
source 1..107

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mol_type = protein
organism = synthetic construct

SEQUENCE: 65
DIQMTQSPST LSASVGDVRT ITCRASQSIG SWLAWYQQKP GKAPKLLIYE ASSLESGVPS 60
RFGSGSGSTE FTLTISSLQP DDFATYYCQQ ANSHPITFGG GTKVEIK 107

SEQ ID NO: 66      moltype = AA length = 11
FEATURE           Location/Qualifiers
source            1..11
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 66
RASQSISSWL A 11

SEQ ID NO: 67      moltype = AA length = 7
FEATURE           Location/Qualifiers
source            1..7
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 67
EASSLES 7

SEQ ID NO: 68      moltype = AA length = 9
FEATURE           Location/Qualifiers
source            1..9
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 68
QQANSHPIT 9

SEQ ID NO: 69      moltype = AA length = 245
FEATURE           Location/Qualifiers
source            1..245
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 69
EVQLVESGGG LVKPGGSLRL SCAASGFTFS DYYMSWIRQA PGKGLEWVSY ISSSGSTIYY 60
ADSVKGRFTI SRDPAKNSLY LQMNSLRAED TAVYYCARDQ GNYGVDVWGQ GTTIVTVSSGG 120
GGSGGGGSGG GSGGGGGSDI QMTQSPSTLS ASVGDVRTIT CRASQSISW LAWYQQKPGK 180
APKLLIYEAS SLESGVPSRF SSGSGGTEFT LTISSLQPDG FATYYCQQAN SHPITFGGGT 240
KVEIK 245

SEQ ID NO: 70      moltype = AA length = 245
FEATURE           Location/Qualifiers
source            1..245
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 70
DIQMTQSPST LSASVGDVRT ITCRASQSIG SWLAWYQQKP GKAPKLLIYE ASSLESGVPS 60
RFGSGSGSTE FTLTISSLQP DDFATYYCQQ ANSHPITFGG GTKVEIKGGG GSGGGGSGGG 120
GSGGGGSEVQ LVESGGGLVK PGGSLRLSCA ASGFTFSDYY MSWIRQAPGK GLEWVSYISS 180
SGSTIYYADS VKGRFTISR DPAKNSLYLQM NSLRAEDTAV YYCARDQGNV GVDVWGQGT 240
TVTSS 245

SEQ ID NO: 71      moltype = AA length = 122
FEATURE           Location/Qualifiers
source            1..122
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 71
EVQLLESVGGG LVQPGGSLRL SCAASGFTFS NYAMSWVRQA PGKGLEWVSA ISGSGGSTYY 60
ADSVKGRFTI SRDPAKNTLY LQMNSLRAED TAVYYCARPG DGYVEGVYFD YWGQGLVTV 120
SS 122

SEQ ID NO: 72      moltype = AA length = 5
FEATURE           Location/Qualifiers
source            1..5
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 72
NYAMS 5

SEQ ID NO: 73      moltype = AA length = 17
FEATURE           Location/Qualifiers
source            1..17
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 73
AISGSGGSTY YADSVKG 17

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SEQ ID NO: 74	moltype = AA length = 13		
FEATURE	Location/Qualifiers		
source	1..13		
	mol_type = protein		
	organism = synthetic construct		
SEQUENCE: 74			
PGDGYEYGVY FDY			13
SEQ ID NO: 75	moltype = AA length = 107		
FEATURE	Location/Qualifiers		
source	1..107		
	mol_type = protein		
	organism = synthetic construct		
SEQUENCE: 75			
DIQMTQSPSS LSASVGDRVT	ITCRASQSSIS	SYLNWYQQKP	GKAPKLLIYA
ASSLQSGVPS	60		
RFSGSGSGTD FTLTISSLQP	EDFATYYCQQ	AHSSPITFGG	GTKVEIK
			107
SEQ ID NO: 76	moltype = AA length = 11		
FEATURE	Location/Qualifiers		
source	1..11		
	mol_type = protein		
	organism = synthetic construct		
SEQUENCE: 76			
RASQSISSYL N			11
SEQ ID NO: 77	moltype = AA length = 7		
FEATURE	Location/Qualifiers		
source	1..7		
	mol_type = protein		
	organism = synthetic construct		
SEQUENCE: 77			
AASSLQS			7
SEQ ID NO: 78	moltype = AA length = 9		
FEATURE	Location/Qualifiers		
source	1..9		
	mol_type = protein		
	organism = synthetic construct		
SEQUENCE: 78			
QQAHSSPIT			9
SEQ ID NO: 79	moltype = AA length = 249		
FEATURE	Location/Qualifiers		
source	1..249		
	mol_type = protein		
	organism = synthetic construct		
SEQUENCE: 79			
EVQLLESQGGG LVQPQGSRLR	SCAASGFTFS	NYAMSWVRQA	PGKGLEWVSA
ISGSGGSTYY	60		
ADSVKGRFTI SRDNSKNTLY	LQMNSLRAED	TAVYYCARPG	DGYEYGVYFD
YWGQGTLLTV	120		
SSGGGGSGGG GSGGGSGGG	GSDIQMTQSP	SSLSASVGDR	VTITCRASQS
ISSYLNWYQQ	180		
KPGKAPKLLI YAASSLQSGV	PSRFSGSGSG	TDFTLTISSL	QPEDFATYYC
QQAHSSPITF	240		
GGGTKVEIK	249		
SEQ ID NO: 80	moltype = AA length = 249		
FEATURE	Location/Qualifiers		
source	1..249		
	mol_type = protein		
	organism = synthetic construct		
SEQUENCE: 80			
DIQMTQSPSS LSASVGDRVT	ITCRASQSSIS	SYLNWYQQKP	GKAPKLLIYA
ASSLQSGVPS	60		
RFSGSGSGTD FTLTISSLQP	EDFATYYCQQ	AHSSPITFGG	GTKVEIKGGG
GSGGGSGGG	120		
SGGSTYYADS VKGRFTISR	NSKNTLYLQM	NSLRAEDTAV	YYCARPGDGY
YEGVYFDYWG	240		
QGTLLTVSS	249		
SEQ ID NO: 81	moltype = AA length = 124		
FEATURE	Location/Qualifiers		
source	1..124		
	mol_type = protein		
	organism = synthetic construct		
SEQUENCE: 81			
QITLKESGPT LVKPTQTLTL	TCTFSGFSLS	TSGVGVGWIR	QPPGKALEWL
ALIWYNDEKR	60		
YSPSLKSRLT ITKDTSKNQV	VLTMNTNMDPV	DTAVYYCARE	GSHDYKSSNW
FDPWQGGTLV	120		
TVSS	124		
SEQ ID NO: 82	moltype = AA length = 7		
FEATURE	Location/Qualifiers		
source	1..7		

-continued

SEQUENCE: 82	mol_type = protein	
TSGVGVG	organism = synthetic construct	7
SEQ ID NO: 83	moltype = AA length = 16	
FEATURE	Location/Qualifiers	
source	1..16	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 83		
LIYWNDEKRY SPSLKS		16
SEQ ID NO: 84	moltype = AA length = 14	
FEATURE	Location/Qualifiers	
source	1..14	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 84		
EGSHDYKSSN WFDP		14
SEQ ID NO: 85	moltype = AA length = 107	
FEATURE	Location/Qualifiers	
source	1..107	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 85		
DIQMTQSPSS LSASVGDVRT ITCQASQDIA NYLNWYQQKP GKAPKLLIYD ASNLETGVPS		60
RFSGSGSGTD FTFTISLQP EDIATYYCQQ HFNPLPLTFGG GTKVEIK		107
SEQ ID NO: 86	moltype = AA length = 11	
FEATURE	Location/Qualifiers	
source	1..11	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 86		
QASQDIANYL N		11
SEQ ID NO: 87	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
source	1..7	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 87		
DASNLET		7
SEQ ID NO: 88	moltype = AA length = 10	
FEATURE	Location/Qualifiers	
source	1..10	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 88		
QQHFNPLPLTF		10
SEQ ID NO: 89	moltype = AA length = 251	
FEATURE	Location/Qualifiers	
source	1..251	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 89		
QITLKESGPT LVKPTQTLTL TCTFSGFSL S TSGVGVGWIR QPPGKALEWL ALIYWNDEKR		60
YSPSLKSRLT ITKDTSKNQV VLTMTNMDPV DTAVYYCARE GSHDYKSSNW FDPWGQGLTV		120
TVSSGGGGSG GGGSGGGSG GGGSDIQMTQ SPSSLSASVG DRVITTCQAS QDIANYLNWY		180
QQKPGKAPKL LIYDASNLET GVPSRFSGSG SGTDFFTIS SLQPEDIATY YCQQHFNPLPL		240
TFGGGTKVEI K		251
SEQ ID NO: 90	moltype = AA length = 251	
FEATURE	Location/Qualifiers	
source	1..251	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 90		
DIQMTQSPSS LSASVGDVRT ITCQASQDIA NYLNWYQQKP GKAPKLLIYD ASNLETGVPS		60
RFSGSGSGTD FTFTISLQP EDIATYYCQQ HFNPLPLTFGG GTKVEIKGGG GSGGGSGGG		120
GSGGGGSQIT LKESGPTLVK PTQTLTLTCT FSGFSLSTSG VGVGWIRQPP GKALEWLALI		180
YWNDEKRYSP SLKSRLTITK DTSKNQVVL TMTNMDPVDTA VYYCAREGSH DYKSSNWFDP		240
WGQGLVTVS S		251
SEQ ID NO: 91	moltype = AA length = 124	

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FEATURE	Location/Qualifiers	
source	1..124	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 91		
EVQLVESGGG LVKPGGSLRL SCAASGFTFS SYSMNWVRQA PGKGLEWVSS ISSSSSYIYY		60
ADSVKGRFTI SRDIAKNSLY LQMNSLRAED TAVYYCARAG DTYSADYY MDVWGKGT		120
TVSS		124
SEQ ID NO: 92	moltype = AA length = 5	
FEATURE	Location/Qualifiers	
source	1..5	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 92		
SYSMN		5
SEQ ID NO: 93	moltype = AA length = 17	
FEATURE	Location/Qualifiers	
source	1..17	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 93		
SISSSSSYIY YADSVKG		17
SEQ ID NO: 94	moltype = AA length = 15	
FEATURE	Location/Qualifiers	
source	1..15	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 94		
AGDTYSADY YYMDV		15
SEQ ID NO: 95	moltype = AA length = 111	
FEATURE	Location/Qualifiers	
source	1..111	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 95		
DIVMTQSPIS LPVTPGEPAS ISCRSSQSLH HSNQYNYLDW YLQKPGQSPQ LLIYLGSRNA		60
SGVPDRFSGS GSGTDFTLKI SRVEAEDVGV YYCMQALGLI TFGGGTKVEI K		111
SEQ ID NO: 96	moltype = AA length = 16	
FEATURE	Location/Qualifiers	
source	1..16	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 96		
RSSQSLHNS GYNYLD		16
SEQ ID NO: 97	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
source	1..7	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 97		
LGSNRAS		7
SEQ ID NO: 98	moltype = AA length = 8	
FEATURE	Location/Qualifiers	
source	1..8	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 98		
MQALGLIT		8
SEQ ID NO: 99	moltype = AA length = 255	
FEATURE	Location/Qualifiers	
source	1..255	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 99		
EVQLVESGGG LVKPGGSLRL SCAASGFTFS SYSMNWVRQA PGKGLEWVSS ISSSSSYIYY		60
ADSVKGRFTI SRDIAKNSLY LQMNSLRAED TAVYYCARAG DTYSADYY MDVWGKGT		120
TVSSGGGGSG GGGGGGGSG GGGSDIVMTQ SPLSLPVTGP EPASISCRSS QSLHNSNGYN		180
YLDWYLQKPG QSPQLLIYLG SNRASGVPDR FSGSGSGTDF TLKISRVEAE DVGVIYCMQA		240
LGLITFGGGT KVEIK		255
SEQ ID NO: 100	moltype = AA length = 255	

-continued

FEATURE	Location/Qualifiers	
source	1..255	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 100		
DIVMTQSPLS LPVTPGEPAS	ISCRSSQSLL HSENGYNYLDW YLQKPGQSPQ LLIYLGSNRA	60
SGVPDRFSGS GSGTDFTLKI	SRVEAEDVG VYCMQALGLI TFGGGTKVEI KGGGSGGGG	120
SGGGSGGGG SEVQLVESGG	GLVKPGGSLR LSCAASGFTF SSYSMNWVRQ APGKGLEWVS	180
SISSSSYIY YADSVKGRFT	ISRDNAKNSL YLQMNSLRAE DTAVYVCARA GDTYSAADYY	240
YMDVWGKGT VTVSS		255
SEQ ID NO: 101	moltype = AA length = 116	
FEATURE	Location/Qualifiers	
source	1..116	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 101		
EVQLLESGGG LVQPGGSLRL	SCAASGFTFG SEAMSWVRQA PGKERELVSA ISGSGEVTTY	60
ADSVKGRFTI SRDNSKNTLY	LQMNSLRAED TAVYVCQLV EAKRHWGQGT QVTVSS	116
SEQ ID NO: 102	moltype = AA length = 5	
FEATURE	Location/Qualifiers	
source	1..5	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 102		
SEAMS		5
SEQ ID NO: 103	moltype = AA length = 17	
FEATURE	Location/Qualifiers	
source	1..17	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 103		
AISGSGEVTY YADSVKG		17
SEQ ID NO: 104	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
source	1..7	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 104		
LVEAKRH		7
SEQ ID NO: 105	moltype = AA length = 116	
FEATURE	Location/Qualifiers	
source	1..116	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 105		
EVQLLESGGG LVQPGGSLRL	SCAASGFTFE SEAMSWYRQA PGKERELVSV ITSEGSTYYA	60
DSVKGRFTIS RDNSKNTLYL	QMNSLRAEDT AVYYCAHIEW ETRLNWGQGT QVTVSS	116
SEQ ID NO: 106	moltype = AA length = 5	
FEATURE	Location/Qualifiers	
source	1..5	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 106		
SEAMS		5
SEQ ID NO: 107	moltype = AA length = 16	
FEATURE	Location/Qualifiers	
source	1..16	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 107		
VITSEGSTYY ADSVKG		16
SEQ ID NO: 108	moltype = AA length = 8	
FEATURE	Location/Qualifiers	
source	1..8	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 108		
IEWETRLN		8
SEQ ID NO: 109	moltype = AA length = 118	
FEATURE	Location/Qualifiers	

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source                1..118
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 109
EVQLLES GGG LVQPGGSLRL SCAASGFTFD EYTMHWFRQA PGKEREFVSA ISGGGSETYY 60
ADSVKGRFTI SRDNSKNTLY LQMNSLRAED TAVYYCAAGG EEAGVGWYWGQ GTQVTVSS 118

SEQ ID NO: 110        moltype = AA length = 5
FEATURE              Location/Qualifiers
source               1..5
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 110
EYTMH                                                         5

SEQ ID NO: 111        moltype = AA length = 17
FEATURE              Location/Qualifiers
source               1..17
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 111
AISGGGSETY YADSVKG                                         17

SEQ ID NO: 112        moltype = AA length = 9
FEATURE              Location/Qualifiers
source               1..9
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 112
GGEEAGVGY                                                  9

SEQ ID NO: 113        moltype = AA length = 120
FEATURE              Location/Qualifiers
source               1..120
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 113
EVQLLES GGG LVQPGGSLRL SCAASGFTFE DYAMSWFRQA PGKEREGVSA ISGKGGSTYY 60
ADSVKGRFTI SRDNSKNTLY LQMNSLRAED TAVYYCAVLD EEAGAEGGYW GQGTQVTVSS 120

SEQ ID NO: 114        moltype = AA length = 5
FEATURE              Location/Qualifiers
source               1..5
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 114
DYAMS                                                         5

SEQ ID NO: 115        moltype = AA length = 17
FEATURE              Location/Qualifiers
source               1..17
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 115
AISGKGGSTY YADSVKG                                         17

SEQ ID NO: 116        moltype = AA length = 11
FEATURE              Location/Qualifiers
source               1..11
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 116
LDDEAGAEGG Y                                               11

SEQ ID NO: 117        moltype = AA length = 120
FEATURE              Location/Qualifiers
source               1..120
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 117
EVQLLES GGG LVQPGGSLRL SCAASGFTFD RYAMSWFRQA PGKEREGVSA ISTSGDSTYY 60
ADSVKGRFTI SRDNSKNTLY LQMNSLRAED TAVYYCAVLD EEAGAEGGYW GQGTQVTVSS 120

SEQ ID NO: 118        moltype = AA length = 5
FEATURE              Location/Qualifiers
source               1..5
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 118

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RYAMS	5
SEQ ID NO: 119	moltype = AA length = 17
FEATURE	Location/Qualifiers
source	1..17
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 119	
AISTSGDSTY YADSVKG	17
SEQ ID NO: 120	moltype = AA length = 11
FEATURE	Location/Qualifiers
source	1..11
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 120	
LDEEAGAEGG Y	11
SEQ ID NO: 121	moltype = AA length = 121
FEATURE	Location/Qualifiers
source	1..121
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 121	
EVQLLESGGG LVQPGGSLRL SCAASGFTFA SDAMSWYRQA PGKERELVSA ISGSGGSTYY	60
ADSVKGRFTI SRDNSKNTLY LQMNSLRAED TAVYYCAHD SGEAYLAPDY WGQGTQVTVS	120
S	121
SEQ ID NO: 122	moltype = AA length = 5
FEATURE	Location/Qualifiers
source	1..5
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 122	
SDAMS	5
SEQ ID NO: 123	moltype = AA length = 17
FEATURE	Location/Qualifiers
source	1..17
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 123	
AISGSGGSTY YADSVKG	17
SEQ ID NO: 124	moltype = AA length = 12
FEATURE	Location/Qualifiers
source	1..12
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 124	
HDSGEAYLAF DY	12
SEQ ID NO: 125	moltype = AA length = 121
FEATURE	Location/Qualifiers
source	1..121
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 125	
EVQLLESGGG LVQPGGSLRL SCAASGFTFD SYTMSWYRQA PGKERELVSA ISGHGDSTYY	60
ADSVKGRFTI SRDNSKNTLY LQMNSLRAED TAVYYCTRIS ITTEWLAGDY WGQGTQVTVS	120
S	121
SEQ ID NO: 126	moltype = AA length = 5
FEATURE	Location/Qualifiers
source	1..5
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 126	
SYTMS	5
SEQ ID NO: 127	moltype = AA length = 17
FEATURE	Location/Qualifiers
source	1..17
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 127	
AISGHGDSTY YADSVKG	17
SEQ ID NO: 128	moltype = AA length = 12

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FEATURE	Location/Qualifiers	
source	1..12	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 128		
ISITTEWLAG DY		12
SEQ ID NO: 129	moltype = AA length = 123	
FEATURE	Location/Qualifiers	
source	1..123	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 129		
EVQLLESGGG LVQPGGSLRL SCAASGFTFS SYAMSWFRQA PGKEREFVSF ISGSGDSTYY		60
ADSVKGRFTI SRDNSKNTLY LQMNSLRAED TAVYYCTRWP YDFEEPSEPG VYWGQGTQVT		120
VSS		123
SEQ ID NO: 130	moltype = AA length = 5	
FEATURE	Location/Qualifiers	
source	1..5	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 130		
SYAMS		5
SEQ ID NO: 131	moltype = AA length = 17	
FEATURE	Location/Qualifiers	
source	1..17	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 131		
FISGSGDSTY YADSVKG		17
SEQ ID NO: 132	moltype = AA length = 14	
FEATURE	Location/Qualifiers	
source	1..14	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 132		
WPYDFEEPSE PGVY		14
SEQ ID NO: 133	moltype = AA length = 123	
FEATURE	Location/Qualifiers	
source	1..123	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 133		
EVQLLESGGG LVQPGGSLRL SCAASGFTFT DYDMSWYRQA PGKERELVSV IHSGGSTYYA		60
DSVKGRFTIS RDNSKNTLYL QMNSLRAEDT AVYYCAPGY SDLSFDYYNF DYWGQGTQVT		120
VSS		123
SEQ ID NO: 134	moltype = AA length = 5	
FEATURE	Location/Qualifiers	
source	1..5	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 134		
DYDMS		5
SEQ ID NO: 135	moltype = AA length = 16	
FEATURE	Location/Qualifiers	
source	1..16	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 135		
VIHSGGSTYY ADSVKG		16
SEQ ID NO: 136	moltype = AA length = 15	
FEATURE	Location/Qualifiers	
source	1..15	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 136		
GYSDLSFDY YNFDY		15
SEQ ID NO: 137	moltype = AA length = 123	
FEATURE	Location/Qualifiers	
source	1..123	
	mol_type = protein	

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                                organism = synthetic construct
SEQUENCE: 137
EVQLLESQGG LVQPGGSLRL SCAASGFTFS DYAMHWFRQA PGKERVLVSS IDSGGSTYYA 60
DSVKGGRFTIS RDNSKNTLYL QMNSLRAEDT AVYYCNAGFK GDHPHPKDAF DIWGQGTQVT 120
VSS 123

SEQ ID NO: 138      moltype = AA length = 5
FEATURE            Location/Qualifiers
source             1..5
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 138
DYAMH 5

SEQ ID NO: 139      moltype = AA length = 16
FEATURE            Location/Qualifiers
source             1..16
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 139
SIDSGGSTYY ADSVKG 16

SEQ ID NO: 140      moltype = AA length = 15
FEATURE            Location/Qualifiers
source             1..15
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 140
GFKGDHPHPK DAFDI 15

SEQ ID NO: 141      moltype = AA length = 125
FEATURE            Location/Qualifiers
source             1..125
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 141
EVQLLESQGG LVQPGGSLRL SCAASGFTFS SEGMSWVRQA PGKERELVSA ISGSGDHTYY 60
ADSVRGRFTI SRDNSKNTLY LQMNSLRAED TAVYYCNALE GGPTTAIQPG GPDYWGQGTQ 120
VTVSS 125

SEQ ID NO: 142      moltype = AA length = 5
FEATURE            Location/Qualifiers
source             1..5
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 142
SEGMS 5

SEQ ID NO: 143      moltype = AA length = 17
FEATURE            Location/Qualifiers
source             1..17
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 143
AISGSGDHTY YADSVRG 17

SEQ ID NO: 144      moltype = AA length = 16
FEATURE            Location/Qualifiers
source             1..16
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 144
LEGGPTTAIQ PGGPDY 16

SEQ ID NO: 145      moltype = AA length = 119
FEATURE            Location/Qualifiers
source             1..119
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 145
DVQLVQSGAE VKKPGASVKV SCKASGYTFT RYTMHWVRQA PGQGLEWIGY INPSRGYTN 60
ADSVKGRFTI TTDKSTSTAY MELSSLRSED TATYYCARYY DDHYCLDYWG QGTTVTVSS 119

SEQ ID NO: 146      moltype = AA length = 5
FEATURE            Location/Qualifiers
source             1..5
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 146

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-continued

RYTMH	5
SEQ ID NO: 147	moltype = AA length = 17
FEATURE	Location/Qualifiers
source	1..17
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 147	
YINPSRGYTN YADSVKG	17
SEQ ID NO: 148	moltype = AA length = 10
FEATURE	Location/Qualifiers
source	1..10
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 148	
YYDDHYCLDY	10
SEQ ID NO: 149	moltype = AA length = 106
FEATURE	Location/Qualifiers
source	1..106
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 149	
DIVLTQSPAT LSLSPGERAT LSCRASQSVS YMNWYQQKPG KAPKRWIYDT SKVASGVPAR	60
FSGSGSGTDY SLTINSLEAE DAATYYCQQW SSNPLTFGGG TKVEIK	106
SEQ ID NO: 150	moltype = AA length = 10
FEATURE	Location/Qualifiers
source	1..10
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 150	
RASQSVSYMN	10
SEQ ID NO: 151	moltype = AA length = 7
FEATURE	Location/Qualifiers
source	1..7
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 151	
DTSKVAS	7
SEQ ID NO: 152	moltype = AA length = 9
FEATURE	Location/Qualifiers
source	1..9
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 152	
QQWSSNPLT	9
SEQ ID NO: 153	moltype = AA length = 241
FEATURE	Location/Qualifiers
source	1..241
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 153	
DVQLVQSGAE VKKPGASVKV SCKASGYTFT RYTMHWVRQA PGQGLEWIGY INPSRGYTN	60
ADSVKGRFTI TTDKSTSTAY MELSSLRSED TATYYCARYY DDHYCLDYWG QGTTTVTVSS	120
GGGSGGGGGS GGGGSDIVLT QSPATLSLSP GERATLSCRA SQSVSYMNWY QQKPGKAPKR	180
WIYDTSKVAS GVPARFSGSG SGTDYSLTIN SLEAEDAATY YCQQWSSNPL TFGGGTKVEI	240
K	241
SEQ ID NO: 154	moltype = AA length = 241
FEATURE	Location/Qualifiers
source	1..241
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 154	
DIVLTQSPAT LSLSPGERAT LSCRASQSVS YMNWYQQKPG KAPKRWIYDT SKVASGVPAR	60
FSGSGSGTDY SLTINSLEAE DAATYYCQQW SSNPLTFGGG TKVEIKSGGG GSGGGGSGGG	120
GSDVQLVQSG AEVKKPGASV KVCKASGYT FTRYTMHWVR QAPGQGLEWI GYINPSRGYT	180
NYADSVKGRF TITTDKSTST AYMESSSLRS EDTATYYCAR YYDDHYCLDY WGQGTTVTVS	240
S	241
SEQ ID NO: 155	moltype = AA length = 119
FEATURE	Location/Qualifiers
source	1..119
	mol_type = protein

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                                organism = synthetic construct
SEQUENCE: 155
DVQLVQSGAE VKKPGASVKV SCKASGYTFT RYTMHWVRQA PGQGLEWIGY INPSRGYTN 60
ADSVKGRFTI TTDKSTSTAY MELSSLRSED TATYYCARYY DDPYCLDYWG QGTTVTVSS 119

SEQ ID NO: 156      moltype = AA length = 5
FEATURE            Location/Qualifiers
source              1..5
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 156
RYTMH 5

SEQ ID NO: 157      moltype = AA length = 17
FEATURE            Location/Qualifiers
source              1..17
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 157
YINPSRGYTN YADSVKG 17

SEQ ID NO: 158      moltype = AA length = 10
FEATURE            Location/Qualifiers
source              1..10
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 158
YYDDPYCLDY 10

SEQ ID NO: 159      moltype = AA length = 106
FEATURE            Location/Qualifiers
source              1..106
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 159
DIVLTQSPAT LSLSPGERAT LSCRASQSVS YMNWYQQKPG KAPKRWIYDT SKVASGVPAR 60
FSGSGSGTDY SLTINSLEAE DAATYYCQQW SSNPLTFGGG TKVEIK 106

SEQ ID NO: 160      moltype = AA length = 10
FEATURE            Location/Qualifiers
source              1..10
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 160
RASQSVSYMN 10

SEQ ID NO: 161      moltype = AA length = 7
FEATURE            Location/Qualifiers
source              1..7
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 161
DTSKVAS 7

SEQ ID NO: 162      moltype = AA length = 9
FEATURE            Location/Qualifiers
source              1..9
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 162
QQWSSNPLT 9

SEQ ID NO: 163      moltype = AA length = 241
FEATURE            Location/Qualifiers
source              1..241
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 163
DVQLVQSGAE VKKPGASVKV SCKASGYTFT RYTMHWVRQA PGQGLEWIGY INPSRGYTN 60
ADSVKGRFTI TTDKSTSTAY MELSSLRSED TATYYCARYY DDPYCLDYWG QGTTVTVSS 120
GGGSGGGGS GGGGSDIVLT QSPATLSLSP GERATLSLSCRA SQSVSYMNWY QQKPGKAPKR 180
WIYDTSKVAS GVPARFSGSG SGTDSLTLIN SLEAEDAATY YCQQWSSNPL TFGGGTKVEI 240
K 241

SEQ ID NO: 164      moltype = AA length = 241
FEATURE            Location/Qualifiers
source              1..241
                    mol_type = protein
                    organism = synthetic construct

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SEQUENCE: 164
 DIVLTQSPAT LSLSPGERAT LSCRASQSVS YMNWYQQKPG KAPKRWIYDT SKVASGVPAR 60
 FSGSGSGTDY SLTINSLEAE DAATYYCQQW SSNPLTFGGG TKVEIKSGGG GSGGGGSGGG 120
 GSDVQLVQSG AEVKKPGASV KVSCKASGYT FTRYTMHWVR QAPQGLEWI GYINPSRGYT 180
 NYADSVKGRF TITDKSTST AYMESSLRS EDTATYYCAR YYDDPYCLDY WQGTTVTVS 240
 S 241

SEQ ID NO: 165 moltype = AA length = 120
 FEATURE Location/Qualifiers
 source 1..120
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 165
 DVQLVQSGAE VKKPGASVKV SCKASGYTFT RYTMHWVRQA PGQGLEWIGY INPSRGYTNY 60
 ADSVKGRFTI TTDKSTSTAY MELSSLRSED TATYYCARYY NDQYCLDYWG QGTTVTVSSS 120

SEQ ID NO: 166 moltype = AA length = 5
 FEATURE Location/Qualifiers
 source 1..5
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 166
 RYTMH 5

SEQ ID NO: 167 moltype = AA length = 17
 FEATURE Location/Qualifiers
 source 1..17
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 167
 YINPSRGYTN YADSVKG 17

SEQ ID NO: 168 moltype = AA length = 10
 FEATURE Location/Qualifiers
 source 1..10
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 168
 YYNDQYCLDY 10

SEQ ID NO: 169 moltype = AA length = 106
 FEATURE Location/Qualifiers
 source 1..106
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 169
 DIVLTQSPAT LSLSPGERAT LSCRASQSVS YMNWYQQKPG KAPKRWIYDT SKVASGVPAR 60
 FSGSGSGTDY SLTINSLEAE DAATYYCQQW SSNPLTFGGG TKVEIK 106

SEQ ID NO: 170 moltype = AA length = 10
 FEATURE Location/Qualifiers
 source 1..10
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 170
 RASQSVSYMN 10

SEQ ID NO: 171 moltype = AA length = 7
 FEATURE Location/Qualifiers
 source 1..7
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 171
 DTSKVAS 7

SEQ ID NO: 172 moltype = AA length = 9
 FEATURE Location/Qualifiers
 source 1..9
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 172
 QQWSSNPLT 9

SEQ ID NO: 173 moltype = AA length = 242
 FEATURE Location/Qualifiers
 source 1..242
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 173

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DVQLVQSGAE	VKKPGASVKV	SCKASGYTFT	RYTMHWVRQA	PGQGLEWIGY	INPSRGYTN	60
ADSVKGRFTI	TTDKSTSTAY	MELSSLRSED	TATYYCARYY	NDQYCLDYWG	QGTITVTVSS	120
SGGGSGGGG	SGGGGSDIVL	TQSPATLSLS	PGERATLSCR	ASQSVSYMNW	YQKPGKAPK	180
RWIYDTSKVA	SGVPARFSGS	SGGTDYSLTI	NSLEAEDAAT	YYCQQWSSNP	LTFGGGTKVE	240
IK						242
SEQ ID NO: 174	moltype = AA length = 242					
FEATURE	Location/Qualifiers					
source	1..242					
	mol_type = protein					
	organism = synthetic construct					
SEQUENCE: 174						
DIVLTQSPAT	LSLSPGERAT	LSCRASQSVS	YMNWYQQKPG	KAPKRWIYDT	SKVASGVPAR	60
FSGSGSGTDY	SLTINSLEAE	DAATYYCQQW	SSNPLTFGGG	TKVEIKSGGG	GSGGGGSGGG	120
GSDVQLVQSG	AEVKKPGASV	KVSCKASGYT	FTRYTMHWVR	QAPGQGLEWI	GYINPSRGYT	180
NYADSVKGRF	TITTDKSTST	AYMELSSLR	EDTATYYCAR	YYNDQYCLDY	WGQGTITVTVS	240
SS						242
SEQ ID NO: 175	moltype = AA length = 119					
FEATURE	Location/Qualifiers					
source	1..119					
	mol_type = protein					
	organism = synthetic construct					
SEQUENCE: 175						
DVQLVQSGAE	VKKPGASVKV	SCKASGYTFT	RYTMHWVRQA	PGQGLEWIGY	INPSRGYTN	60
ADSVKGRFTI	TTDKSTSTAY	MELSSLRSED	TATYYCARYY	DAHNCLDYWG	QGTITVTVSS	119
SEQ ID NO: 176	moltype = AA length = 5					
FEATURE	Location/Qualifiers					
source	1..5					
	mol_type = protein					
	organism = synthetic construct					
SEQUENCE: 176						
RYTMH						5
SEQ ID NO: 177	moltype = AA length = 17					
FEATURE	Location/Qualifiers					
source	1..17					
	mol_type = protein					
	organism = synthetic construct					
SEQUENCE: 177						
YINPSRGYTN	YADSVKG					17
SEQ ID NO: 178	moltype = AA length = 10					
FEATURE	Location/Qualifiers					
source	1..10					
	mol_type = protein					
	organism = synthetic construct					
SEQUENCE: 178						
YYDAHNCLDY						10
SEQ ID NO: 179	moltype = AA length = 106					
FEATURE	Location/Qualifiers					
source	1..106					
	mol_type = protein					
	organism = synthetic construct					
SEQUENCE: 179						
DIVLTQSPAT	LSLSPGERAT	LSCRASQSVS	YMNWYQQKPG	KAPKRWIYDT	SKVASGVPAR	60
FSGSGSGTDY	SLTINSLEAE	DAATYYCQQW	SSNPLTFGGG	TKVEIK		106
SEQ ID NO: 180	moltype = AA length = 10					
FEATURE	Location/Qualifiers					
source	1..10					
	mol_type = protein					
	organism = synthetic construct					
SEQUENCE: 180						
RASQSVSYMN						10
SEQ ID NO: 181	moltype = AA length = 7					
FEATURE	Location/Qualifiers					
source	1..7					
	mol_type = protein					
	organism = synthetic construct					
SEQUENCE: 181						
DTSKVAS						7
SEQ ID NO: 182	moltype = AA length = 9					
FEATURE	Location/Qualifiers					
source	1..9					

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mol_type = protein
organism = synthetic construct

SEQUENCE: 182
QQWSSNPLT 9

SEQ ID NO: 183
FEATURE
source
moltype = AA length = 241
Location/Qualifiers
1..241
mol_type = protein
organism = synthetic construct

SEQUENCE: 183
DVQLVQSGAE VKKPGASVKV SCKASGYTFT RYTMHWVRQA PGQGLEWIGY INPSRGYTNY 60
ADSVKGRFTI TTDKSTSTAY MELSSLRSED TATYYCARYY DAHNCLDYWG QGTTVTVSSS 120
GGGSGGGGGS GGGGSDIVLT QSPATLSLSP GERATLSGRA SQSVSYMNWY QQKPGKAPKR 180
WYDTSKVAS GVPARFSGSG SGTDSLTLIN SLEAEDAATY YCQQWSSNPL TFGGGTKVEI 240
K 241

SEQ ID NO: 184
FEATURE
source
moltype = AA length = 241
Location/Qualifiers
1..241
mol_type = protein
organism = synthetic construct

SEQUENCE: 184
DIVLTQSPAT LSLSPGERAT LSCRASQSVS YMNWYQQKPG KAPKRWIYDT SKVASGVPAR 60
FSGSGSGTDY SLTINSLEAE DAATYYCQQW SSNPLTFGGG TKVEIKSGGG GSGGGGSGGG 120
GSDVQLVQSG AEVKKGASV KVSCKASGYT FTRYTMHWVR QAPGQGLEWI GYINPSRGYT 180
NYADSVKGRF TITDKSTST AYMESSLRS EDTATYYCAR YDAHNCLDY WGQGTITVTS 240
S 241

SEQ ID NO: 185
FEATURE
source
moltype = AA length = 119
Location/Qualifiers
1..119
mol_type = protein
organism = synthetic construct

SEQUENCE: 185
DVQLVQSGAE VKKPGASVKV SCKASGYTFT RYTMHWVRQA PGQGLEWIGY INPSRGYTNY 60
ADSVKGRFTI TTDKSTSTAY MELSSLRSED TATYYCARYS DDHYCLDYWG QGTTVTVSS 119

SEQ ID NO: 186
FEATURE
source
moltype = AA length = 5
Location/Qualifiers
1..5
mol_type = protein
organism = synthetic construct

SEQUENCE: 186
RYTMH 5

SEQ ID NO: 187
FEATURE
source
moltype = AA length = 17
Location/Qualifiers
1..17
mol_type = protein
organism = synthetic construct

SEQUENCE: 187
YINPSRGYTIN YADSVKG 17

SEQ ID NO: 188
FEATURE
source
moltype = AA length = 10
Location/Qualifiers
1..10
mol_type = protein
organism = synthetic construct

SEQUENCE: 188
YSDDHYCLDY 10

SEQ ID NO: 189
FEATURE
source
moltype = AA length = 106
Location/Qualifiers
1..106
mol_type = protein
organism = synthetic construct

SEQUENCE: 189
DIVLTQSPAT LSLSPGERAT LSCRASQSVS YMNWYQQKPG KAPKRWIYDT SKVASGVPAR 60
FSGSGSGTDY SLTINSLEAE DAATYYCQQW SSNPLTFGGG TKVEIK 106

SEQ ID NO: 190
FEATURE
source
moltype = AA length = 10
Location/Qualifiers
1..10
mol_type = protein
organism = synthetic construct

SEQUENCE: 190
RASQSVSYMN 10

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-continued

SEQ ID NO: 191	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
source	1..7	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 191		
DTSKVAS		7
SEQ ID NO: 192	moltype = AA length = 9	
FEATURE	Location/Qualifiers	
source	1..9	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 192		
QQWSSNPLT		9
SEQ ID NO: 193	moltype = AA length = 241	
FEATURE	Location/Qualifiers	
source	1..241	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 193		
DVQLVQSGAE VKKPGASVKV SCKASGYTFT RYTMHWVRQA PGQGLEWIGY INPSRGYTNV		60
ADSVKGRFTI TTDKSTSTAY MELSSLRSED TATYYCARYS DDHYCLDYWG QGTTTVTVSSS		120
GGGGSGGGGS GGGGSDIVLT QSPATLSLSP GERATLSLSCRA SQSVSYMNWY QKPGKAPKR		180
WIYDTSKVAS GVPARFSGSG SGTDSLTLIN SLEAEDAATY YCQQWSSNPL TFGGGTKVEI		240
K		241
SEQ ID NO: 194	moltype = AA length = 241	
FEATURE	Location/Qualifiers	
source	1..241	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 194		
DIVLTQSPAT LSLSPGERAT LSCRASQSVS YMNWYQQKPG KAPKRWIYDT SKVASGVPAR		60
FGSGSGTDY SLTINSLEAE DAATYYCQQW SSNPLTFGGG TKVEIKSGGG GSGGGGSGGG		120
GSDVQLVQSG AEVKKPGASV KVSCKASGYT FTRYTMHWVR QAPGQGLEWI GYINPSRGYT		180
NYADSVKGRF TITTDKSTST AYMESSLRS EDTATYYCAR YSDDHYCLDY WQGGTTVTVS		240
S		241
SEQ ID NO: 195	moltype = AA length = 119	
FEATURE	Location/Qualifiers	
source	1..119	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 195		
DVQLVQSGAE VKKPGASVKV SCKASGYTFT RYTMHWVRQA PGQGLEWIGY INPSRGYTNV		60
ADSVKGRFTI TTDKSTSTAY MELSSLRSED TATYYCARYS DDRYCLDYWG QGTTTVTVSS		119
SEQ ID NO: 196	moltype = AA length = 5	
FEATURE	Location/Qualifiers	
source	1..5	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 196		
RYTMH		5
SEQ ID NO: 197	moltype = AA length = 17	
FEATURE	Location/Qualifiers	
source	1..17	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 197		
YINPSRGYTN YADSVKG		17
SEQ ID NO: 198	moltype = AA length = 10	
FEATURE	Location/Qualifiers	
source	1..10	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 198		
YSDDRYCLDY		10
SEQ ID NO: 199	moltype = AA length = 106	
FEATURE	Location/Qualifiers	
source	1..106	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 199		

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DIVLTQSPAT LSLSPGERAT LSCRASQSVS YMNWYQQKPG KAPKRWIYDT SKVASGVPAR	60
FSGSGSGTDY SLTINSLEAE DAATYYCQQW SSNPLTFGGG TKVEIK	106
SEQ ID NO: 200 moltype = AA length = 10	
FEATURE Location/Qualifiers	
source 1..10	
mol_type = protein	
organism = synthetic construct	
SEQUENCE: 200	
RASQSVSYMN	10
SEQ ID NO: 201 moltype = AA length = 7	
FEATURE Location/Qualifiers	
source 1..7	
mol_type = protein	
organism = synthetic construct	
SEQUENCE: 201	
DTSKVAS	7
SEQ ID NO: 202 moltype = AA length = 9	
FEATURE Location/Qualifiers	
source 1..9	
mol_type = protein	
organism = synthetic construct	
SEQUENCE: 202	
QQWSSNPLT	9
SEQ ID NO: 203 moltype = AA length = 241	
FEATURE Location/Qualifiers	
source 1..241	
mol_type = protein	
organism = synthetic construct	
SEQUENCE: 203	
DVQLVQSGAE VKKPGASVKV SCKASGYTFT RYTMHWVRQA PGQGLEWIGY INPSRGYTNV	60
ADSVKGRFTI TTDKSTSTAY MELSSLRSED TATYYCARYS DRYCLDYWG QGTTTVTVSS	120
GGGSGGGGS GGGGSDIVLT QSPATLSLSP GERATLSLSCRA SQSVSYMNWY QQKPGKAPKR	180
WIYDTSKVAS GVPARFSGSG SGTDYSLTIN SLEAEDAATY YCQQWSSNPL TFGGGTKVEI	240
K	241
SEQ ID NO: 204 moltype = AA length = 241	
FEATURE Location/Qualifiers	
source 1..241	
mol_type = protein	
organism = synthetic construct	
SEQUENCE: 204	
DIVLTQSPAT LSLSPGERAT LSCRASQSVS YMNWYQQKPG KAPKRWIYDT SKVASGVPAR	60
FSGSGSGTDY SLTINSLEAE DAATYYCQQW SSNPLTFGGG TKVEIKSGGG GSGGGGSGGG	120
GSDVQLVQSG AEVKKPGASV KVSCKASGYT FTRYTMHWVR QAPGQGLEWI GYINPSRGYT	180
NYADSVKGRF TITTDKSTST AYMESSLRS EDTATYYCAR YSDDRYCLDY WQGTTTVTVS	240
S	241
SEQ ID NO: 205 moltype = AA length = 120	
FEATURE Location/Qualifiers	
source 1..120	
mol_type = protein	
organism = synthetic construct	
SEQUENCE: 205	
DVQLVQSGAE VKKPGASVKV SCKASGYTFT RYTMHWVRQA PGQGLEWIGY INPSRGYTNV	60
ADSVKGRFTI TTDKSTSTAY MELSSLRSED TATYYCARYY DDNYCLDYWG QGTTTVTVSS	120
SEQ ID NO: 206 moltype = AA length = 5	
FEATURE Location/Qualifiers	
source 1..5	
mol_type = protein	
organism = synthetic construct	
SEQUENCE: 206	
RYTMH	5
SEQ ID NO: 207 moltype = AA length = 17	
FEATURE Location/Qualifiers	
source 1..17	
mol_type = protein	
organism = synthetic construct	
SEQUENCE: 207	
YINPSRGYTN YADSVKG	17
SEQ ID NO: 208 moltype = AA length = 10	
FEATURE Location/Qualifiers	
source 1..10	

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mol_type = protein
organism = synthetic construct

SEQUENCE: 208
YYDDNYCLDY 10

SEQ ID NO: 209      moltype = AA length = 106
FEATURE            Location/Qualifiers
source             1..106
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 209
DIVLTQSPAT LSLSPGERAT LSCRASQSVS YMNWYQQKPG KAPKRWIYDT SKVASGVPAR 60
FSGSGSGTDY SLTINSLEAE DAATYYCQQW SSNPLTFGGG TKVEIK 106

SEQ ID NO: 210      moltype = AA length = 10
FEATURE            Location/Qualifiers
source             1..10
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 210
RASQSVSYMN 10

SEQ ID NO: 211      moltype = AA length = 7
FEATURE            Location/Qualifiers
source             1..7
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 211
DTSKVAS 7

SEQ ID NO: 212      moltype = AA length = 9
FEATURE            Location/Qualifiers
source             1..9
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 212
QQWSSNPLT 9

SEQ ID NO: 213      moltype = AA length = 242
FEATURE            Location/Qualifiers
source             1..242
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 213
DVQLVQSGAE VKKPGASVKV SCKASGYTFT RYTMHWVRQA PGQGLEWIGY INPSRGYTNY 60
ADSVKGRFTI TTDKSTSTAY MELSSLRSED TATYYCARYY DDNYCLDYWG QGTTTVTVSS 120
SGGGGSGGGG SGGGGSDIVL TQSPATLSLS PGERATLSLR ASQSVSYMNW YQQKPGKAPK 180
RWIYDTSKVA SGVPARFSGS GSGTDYSLTI NSLEAEDAAT YYCQQWSSNP LTFGGGTKVE 240
IK 242

SEQ ID NO: 214      moltype = AA length = 242
FEATURE            Location/Qualifiers
source             1..242
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 214
DIVLTQSPAT LSLSPGERAT LSCRASQSVS YMNWYQQKPG KAPKRWIYDT SKVASGVPAR 60
FSGSGSGTDY SLTINSLEAE DAATYYCQQW SSNPLTFGGG TKVEIKSGGG GSGGGGSGGG 120
GSDVQLVQSG AEVKKPGASV KVSCKASGYT FTRYTMHWVR QAPGQGLEWI GYINPSRGYT 180
NYADSVKGRF TITTDKSTST AYMESSLRS EDTATYYCAR YYDDNYCLDY WQGTTTVTVS 240
SS 242

SEQ ID NO: 215      moltype = AA length = 119
FEATURE            Location/Qualifiers
source             1..119
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 215
DVQLVQSGAE VKKPGASVKV SCKASGYTFT RYTMHWVRQA PGQGLEWIGY INPSRGYTNY 60
ADSVKGRFTI TTDKSTSTAY MELSSLRSED TATYYCARYY DDQYCLDYWG QGTTTVTVSS 119

SEQ ID NO: 216      moltype = AA length = 5
FEATURE            Location/Qualifiers
source             1..5
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 216
RYTMH 5

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SEQ ID NO: 217	moltype = AA length = 17	
FEATURE	Location/Qualifiers	
source	1..17	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 217		
YINPSRGYTN YADSVKG		17
SEQ ID NO: 218	moltype = AA length = 10	
FEATURE	Location/Qualifiers	
source	1..10	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 218		
YYDDQYCLDY		10
SEQ ID NO: 219	moltype = AA length = 106	
FEATURE	Location/Qualifiers	
source	1..106	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 219		
DIVLTQSPAT LSLSPGERAT LSCRASQSVS YMNWYQQKPG KAPKRWIYDT SKVASGVPAR		60
FSGSGSGTDY SLTINSLEAE DAATYYCQQW SSNPLTFGGG TKVEIK		106
SEQ ID NO: 220	moltype = AA length = 10	
FEATURE	Location/Qualifiers	
source	1..10	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 220		
RASQSVSYMN		10
SEQ ID NO: 221	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
source	1..7	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 221		
DTSKVAS		7
SEQ ID NO: 222	moltype = AA length = 9	
FEATURE	Location/Qualifiers	
source	1..9	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 222		
QQWSSNPLT		9
SEQ ID NO: 223	moltype = AA length = 241	
FEATURE	Location/Qualifiers	
source	1..241	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 223		
DVQLVQSGAE VKKPGASVKV SCKASGYTFT RYTMHWVRQA PGQGLEWIGY INPSRGYTN		60
ADSVKGRFTI TTDKSTSTAY MELSSLRSED TATYYCARY DDQYCLDYWG QGTTVTVSS		120
GGGSGGGGGS GGGGSDIVLT QSPATLSLSP GERATLSLSCRA SQSVSYMNWY QQKPGKAPKR		180
WIYDTSKVAS GVPARFSGSG SGTDSLTLIN SLEAEDAATY YCQQWSSNPL TFGGGTKVEI		240
K		241
SEQ ID NO: 224	moltype = AA length = 241	
FEATURE	Location/Qualifiers	
source	1..241	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 224		
DIVLTQSPAT LSLSPGERAT LSCRASQSVS YMNWYQQKPG KAPKRWIYDT SKVASGVPAR		60
FSGSGSGTDY SLTINSLEAE DAATYYCQQW SSNPLTFGGG TKVEIKSGGG GSGGGGSGGG		120
GSDVQLVQSG AEVKKPGASV KVSCASGYT FTRYTMHWVR QAPGQGLEWI GYINPSRGYT		180
NYADSVKGRF TITTDKSTST AYMESSLR S EDTATYYCAR YYDDQYCLDY WQGTTVTVS		240
S		241
SEQ ID NO: 225	moltype = AA length = 45	
FEATURE	Location/Qualifiers	
source	1..45	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 225		

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TTTPAPRPPT PAPTIASQPL SLRPEACRPA AGGAVHTRGL DFACD	45
SEQ ID NO: 226	moltype = AA length = 39
FEATURE	Location/Qualifiers
source	1..39
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 226	
IEVMYPPPYL DNEKSNGTII HVKGKHLCPs PLFPGPSKP	39
SEQ ID NO: 227	moltype = AA length = 22
FEATURE	Location/Qualifiers
source	1..22
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 227	
SGQVLLESNI KVLPTWSTPV QP	22
SEQ ID NO: 228	moltype = AA length = 12
FEATURE	Location/Qualifiers
source	1..12
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 228	
ESKYGPPCPP CP	12
SEQ ID NO: 229	moltype = AA length = 228
FEATURE	Location/Qualifiers
source	1..228
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 229	
ESKYGPPCPP CPAPPVAGPS VFLEPPKPKD TLMISRTPEV TCVVVDVSQE DPEVQFNWYV	60
DGVEVHNAKT KPREEQFQST YRVSVLTVL HQDWLNGKEY KCKVSNKGLP SSIEKTISKA	120
KGQPREPQVY TLPPSQEEMT KNQVSLTCLV KGFYPSDIAV EWESNGQPEN NYKTTTPVLD	180
SDGSFPLYSR LTVDKSRWQE GNVFSCSVMH EALHNHYTQK SLSLSLKG	228
SEQ ID NO: 230	moltype = AA length = 16
FEATURE	Location/Qualifiers
source	1..16
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 230	
LEPKSCDKTH TCPPCP	16
SEQ ID NO: 231	moltype = AA length = 24
FEATURE	Location/Qualifiers
source	1..24
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 231	
IYIWAPLAGT CGVLLLSLVI TLYC	24
SEQ ID NO: 232	moltype = AA length = 27
FEATURE	Location/Qualifiers
source	1..27
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 232	
IISFPLALTS TALLFLLFFL TLRFSVV	27
SEQ ID NO: 233	moltype = AA length = 27
FEATURE	Location/Qualifiers
source	1..27
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 233	
FWLVVVGGV LACYSLLVTV AFIFWV	27
SEQ ID NO: 234	moltype = AA length = 21
FEATURE	Location/Qualifiers
source	1..21
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 234	
VAAILGLGLV LGLLGPLAIL L	21
SEQ ID NO: 235	moltype = AA length = 23
FEATURE	Location/Qualifiers

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source	1..23	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 235		
WLPIGCAAFV VVCILGCILI CWL		23
SEQ ID NO: 236	moltype = AA length = 26	
FEATURE	Location/Qualifiers	
source	1..26	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 236		
VMSVATIVIV DICITGGLLL LVYYWS		26
SEQ ID NO: 237	moltype = AA length = 22	
FEATURE	Location/Qualifiers	
source	1..22	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 237		
MALIVLGGVA GLLLFILGLI FF		22
SEQ ID NO: 238	moltype = AA length = 112	
FEATURE	Location/Qualifiers	
source	1..112	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 238		
RVKFSRSADA PAYQQGQNQL YNELNLGRRE EYDVLDKRRG RDPENGGKPR RKNPQEGLYN		60
ELQKDKMAEA YSEIGMKGER RRGKGHDGLY QGLSTATKDT YDALHMQALP PR		112
SEQ ID NO: 239	moltype = AA length = 42	
FEATURE	Location/Qualifiers	
source	1..42	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 239		
KRGRKKLLYI FKQPFMRPVQ TTQEEDGCSC RPEEEEGGC EL		42
SEQ ID NO: 240	moltype = AA length = 41	
FEATURE	Location/Qualifiers	
source	1..41	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 240		
RSKRSRLLS DYMNMTPRRP GPTRKHYQPY APPRDFAAAYR S		41
SEQ ID NO: 241	moltype = AA length = 42	
FEATURE	Location/Qualifiers	
source	1..42	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 241		
ALYLLRRDQR LPPDAHKKPPG GGSFRTPIQE EQADAHSTLA KI		42
SEQ ID NO: 242	moltype = AA length = 35	
FEATURE	Location/Qualifiers	
source	1..35	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 242		
TKKKYSSSVH DPNGEYMFMR AVNTAKKSRL TDVTL		35
SEQ ID NO: 243	moltype = AA length = 48	
FEATURE	Location/Qualifiers	
source	1..48	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 243		
QRRKYRSNKG ESPVEPAEPC HYSCPREEEG STIPIQEDYR KPEPACSP		48
SEQ ID NO: 244	moltype = AA length = 174	
FEATURE	Location/Qualifiers	
source	1..174	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 244		
KKKPLCLQRE AKVPHLPADK ARGTOGPEQQ HLLITAPSSS SSSLESSASA LDRRAPTRNQ		60
PQAPGVEASG AGEARASTGS SDSSPGGHGT QVNVTCIVNV CSSSDHSSQC SSQASSTMGD		120

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TDSSPSESPK DEQVFFSKEE CAPRSQLETP ETLGSTEELK PLPLGVDPAG MKPS	174
SEQ ID NO: 245	moltype = AA length = 21
FEATURE	Location/Qualifiers
source	1..21
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 245	
MALPVTALLL PLALLHAAR P	21
SEQ ID NO: 246	moltype = AA length = 20
FEATURE	Location/Qualifiers
source	1..20
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 246	
METDTLLLVV LLLWVPGSTG	20
SEQ ID NO: 247	moltype = AA length = 22
FEATURE	Location/Qualifiers
source	1..22
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 247	
MDMRVPAQLL GLLLLWLRGA RC	22
SEQ ID NO: 248	moltype = AA length = 16
FEATURE	Location/Qualifiers
source	1..16
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 248	
MPLLLLLPLL WAGALA	16
SEQ ID NO: 249	moltype = AA length = 23
FEATURE	Location/Qualifiers
source	1..23
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 249	
MDAMKRGSLCC VLLLCGAVFV SPS	23
SEQ ID NO: 250	moltype = AA length = 17
FEATURE	Location/Qualifiers
source	1..17
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 250	
MLLLLLLLGL RLQLSLG	17
SEQ ID NO: 251	moltype = AA length = 17
FEATURE	Location/Qualifiers
source	1..17
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 251	
MWLQSLLLLG TVACIS	17
SEQ ID NO: 252	moltype = AA length = 22
FEATURE	Location/Qualifiers
source	1..22
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 252	
MLLLVTSLLL CELPHPAFL IP	22
SEQ ID NO: 253	moltype = AA length = 20
FEATURE	Location/Qualifiers
source	1..20
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 253	
MSRSVALAVL ALLSLSGLEA	20
SEQ ID NO: 254	moltype = AA length = 16
FEATURE	Location/Qualifiers
source	1..16
	mol_type = protein
	organism = synthetic construct

-continued

SEQUENCE: 254
 MLLLLLLLLL LALALA 16

SEQ ID NO: 255 moltype = AA length = 468
 FEATURE Location/Qualifiers
 source 1..468
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 255
 QITLKESGPT LVKPTQTTLT TCTFSGFSL S TSGVGVGWIR QPPGKALEWL ALIYWNDK 60
 YSPSLKSRIT ITKDTSKNQV VLTMTNMDPV DTAVYYCARD EYGGFDIQQ GTMTVSSGG 120
 GSGGGGSGG GSGGGGSEI VLTQSPATLS LSPGERATLS CRASQSVSSY LAWYQKPGQ 180
 APRLLIYDAS NRATGIPARF SSGSGTDFT LTISSEPED FAVYCCQQRV VYPITFGGT 240
 KVEIKTTTPA PRPPTPAPTI ASQPLSLRPE ACRPAAGGAV HTRGLDFACD IYIWAPLAGT 300
 CGVLLSLVI TLYCKRGRKK LLYIFKQPFM RPVQTTQED GCSCRFPEEE EGGCELRVKF 360
 SRSADAPAYQ QGQNQLYNEL NLGRREEYDV LDKRRGRDPE MGGKPRRKNP QEGLYNELQK 420
 DKMAEAYSEI GMKGERRRGK GHDGLYQGLS TATKDTYDAL HMQALPPR 468

SEQ ID NO: 256 moltype = AA length = 468
 FEATURE Location/Qualifiers
 source 1..468
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 256
 EIVLTQSPAT LSLSPGERAT LSCRASQSVS SYLAWYQKQP QAPRLLIYD ASNRATGIPA 60
 RFSGSGSGTD FTLTISSELEP EDFAVYYCQQ RVVYPTITFGG GTKVEIKGGG GSGGGGSGGG 120
 GSGGGGSGIT LKESGPTLVK PTQTTLTCT FSGFSLSTSG VGVGWIRQPP GKALEWLALI 180
 YWNDKRYSP SLKSRITITK DTSKNQVLT MTNMDPVDTA VYCARDEYG GFDIWQGT 240
 VTSSTTTTPA PRPPTPAPTI ASQPLSLRPE ACRPAAGGAV HTRGLDFACD IYIWAPLAGT 300
 CGVLLSLVI TLYCKRGRKK LLYIFKQPFM RPVQTTQED GCSCRFPEEE EGGCELRVKF 360
 SRSADAPAYQ QGQNQLYNEL NLGRREEYDV LDKRRGRDPE MGGKPRRKNP QEGLYNELQK 420
 DKMAEAYSEI GMKGERRRGK GHDGLYQGLS TATKDTYDAL HMQALPPR 468

SEQ ID NO: 257 moltype = AA length = 468
 FEATURE Location/Qualifiers
 source 1..468
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 257
 QITLKESGPT LVKPTQTTLT TCTFSGFSL S TSGVGVGWIR QPPGKALEWL ALIYWNDDK 60
 YSPSLKSRIT ITKDTSKNQV VLTMTNMDPV DTAVYYCARD EYGGFDIQQ GTMTVSSGG 120
 GSGGGGSGG GSGGGGSEI VLTQSPATLS LSPGERATLS CRASQSVSSY LAWYQKPGQ 180
 APRLLIYDAS NRATGIPARF SSGSGTDFT LTISSEPED FAVYCCQQRV DYPITFGGT 240
 KVEIKTTTPA PRPPTPAPTI ASQPLSLRPE ACRPAAGGAV HTRGLDFACD IYIWAPLAGT 300
 CGVLLSLVI TLYCKRGRKK LLYIFKQPFM RPVQTTQED GCSCRFPEEE EGGCELRVKF 360
 SRSADAPAYQ QGQNQLYNEL NLGRREEYDV LDKRRGRDPE MGGKPRRKNP QEGLYNELQK 420
 DKMAEAYSEI GMKGERRRGK GHDGLYQGLS TATKDTYDAL HMQALPPR 468

SEQ ID NO: 258 moltype = AA length = 468
 FEATURE Location/Qualifiers
 source 1..468
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 258
 EIVLTQSPAT LSLSPGERAT LSCRASQSVS SYLAWYQKQP QAPRLLIYD ASNRATGIPA 60
 RFSGSGSGTD FTLTISSELEP EDFAVYYCQQ RFDYPITFGG GTKVEIKGGG GSGGGGSGGG 120
 GSGGGGSGIT LKESGPTLVK PTQTTLTCT FSGFSLSTSG VGVGWIRQPP GKALEWLALI 180
 YWNDKRYSP SLKSRITITK DTSKNQVLT MTNMDPVDTA VYCARDEYG GFDIWQGT 240
 VTSSTTTTPA PRPPTPAPTI ASQPLSLRPE ACRPAAGGAV HTRGLDFACD IYIWAPLAGT 300
 CGVLLSLVI TLYCKRGRKK LLYIFKQPFM RPVQTTQED GCSCRFPEEE EGGCELRVKF 360
 SRSADAPAYQ QGQNQLYNEL NLGRREEYDV LDKRRGRDPE MGGKPRRKNP QEGLYNELQK 420
 DKMAEAYSEI GMKGERRRGK GHDGLYQGLS TATKDTYDAL HMQALPPR 468

SEQ ID NO: 259 moltype = AA length = 466
 FEATURE Location/Qualifiers
 source 1..466
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 259
 QVQLVESGGG VVQPGSLRL SCAASGFTFS SYGMHWVRQA PGKGLEWVAV ISYEGSNKYY 60
 ADSVKGRFTI SRDNSKNTLY LQMSLRAED TAVYYCAREL GDGMDVWQGG TTVTVSSGGG 120
 GSGGGGSGG GSGGGGSEIV LTQSPATLSL SPGERATLSC RASQSVSSYL AWFQKPGQA 180
 PRLLIYDASN RATGIPARFS GSGSGDFTL TISSLEPEDF AVYCCQQRVD LWTFGGKT 240
 EIKTTTPAPR PPTPAPTIAS QPLSLRPEAC RPAAGGAVHT RGLDFACDIY IWAPLAGTCG 300
 VLLSLVITL YCKRGRKKLL YIFKQPFMRP VQTTQEDGDC SCRFPEEEEG GCELRVKFSR 360
 SADAPAYQGG QNQLYNELNL GRREEYDVL DKKRRDPEMG GKPRRKNPQE GLYNELQKDK 420
 MAEAYSEIGM KGERRRGKGH DGLYQGLSTA TKDTYDALHM QALPPR 466

SEQ ID NO: 260 moltype = AA length = 466

FEATURE	Location/Qualifiers				
source	1..466				
	mol_type = protein				
	organism = synthetic construct				
SEQUENCE: 260					
EIVLTQSPAT LSLSPGERAT	LSCRASQSVS	SYLAWFQQKP	GQAPRLLIYD	ASN RATGIPA	60
RFGSGSGGTD FTLTISLEP	EDFAVYVCQQ	RVDLWTFGGG	TKVEIKGGGG	SGGGSGGGG	120
SGGGGSGVQL VESGGGVQVP	GRSLRLSCAA	SGFTPSSYGM	HWVRQAPGKG	LEWVAVISYE	180
GSNKYYADVSL KGRFTISRDN	SKNTLYLQMN	SLRAEDTAVY	YCARLEGDGM	DVWGQGTTVT	240
VSTTTTPAPR PTPAPTIAS	QPLSLRPEAC	PRAAGGAVHT	RGLDFACDIY	IWAPLAGTCG	300
VLLLSLVITL YCKRGRKKLL	YIFQPFMPMR	VQTQTQEDQG	SCRFPNEEBG	GCELRVKFSR	360
SADAPAYQQG QNQLYNELNL	GRGEEYDVLD	KRRGRDPEMG	GKPRRKNPQG	GLYNELQKDK	420
MAEAYSIQGM KGERRRGKGH	DGLYQGLSTA	TKDITYDALHM	QALPPR		466

SEQ ID NO: 261	moltype = AA	length = 468	
FEATURE	Location/Qualifiers		
source	1..468		
	mol_type = protein		
	organism = synthetic construct		
SEQUENCE: 261			
EVQLVESGGG	LVPKGGSLRL	SCAASGFTFS	DYYMSWIRQA PGKGLEWVS
ADSVKGRFTI	SRDNAKNSLY	LQMNSLRAED	TAVVYCARDQ GNYGVDVWGQ
GGSGGGGSGG	GGSGGGGSDI	QMTQSPSTLS	ASVGDVRVIT CRASQSISW
APKLLIYDAS	SLESGVPSRF	SGSGSGTEFTL	LTISSLQPDV FATYYCQQVS
KVEIKTTTPA	PRPPTPAPTI	ASQPLSLRPE	ACRPAAGAVG HTRGLDFACD
CGVLLLSLVI	TLYCKRGRKK	LLYIFKQPFM	RPVQTTQEDE GCSCRPFEE
SRSDAPAYQ	QKQQLLYNEL	NLGRREYVD	LDKRRGRDPE MGKPKRPNP
DKMAEAYSEI	GMKGERRRGK	GHGDLVGLGS	TATKDTYDAL HMAALPPR

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SEQ ID NO: 262          moltype = AA   length = 468
FEATURE                 Location/Qualifiers
source                  1..468
                        mol_type = protein
                        organism = synthetic construct

SEQUENCE: 262
DIQM TQSPST  LSA SVGDRV T  ITCRASQ SIS  SWLAWYQ QKP  GKAPKLLIYD ASSLES GVP S  60
RFSGSGSGSTE  FTLTISSLQP DDFATYYCQ Q  VSSLPPTFGG GTKVEIKGGG GSGGGGSGGG  120
SGSGGGGSEVQ  LVESGGLGVK  PGGSRLRLSCA ASGFTFS DIY  MSWIRQAPGK GLEWVS YISS  180
GSGSTIYYADS  VKGRFTISR D  NAKNSLYLQM  NSLRADETA V  YYCARDQGN Y  GVDVWGQGT T  240
VTVSSSTTPPA PRPPTPAPT ASQPLSLRPE ACRAAAGAV  HTRGLDFAC D  IYIWAPLAG T  300
CGVLLLSLVI  TLYCKRGRK K  LLYIFKQPFM  RPVQT TQEE D  GCSCRPFEE E  EGGCEL RVKF  360
SRSDADAPAYQ QQQNGLYNEL NLGRREYDV D  LDKRRGRDE P  MGKGRPRKN Y  QEGLYNELQ K  420
DKMAEAYSEI  GMKGERRRKG GHGDIYGLS T  ATKDTYDAL  HMOALPPR  468

```

SEQ ID NO: 263	moltype = AA	length = 468	
FEATURE	Location/Qualifiers		
source	1..468		
	mol_type = protein		
	organism = synthetic construct		
SEQUENCE: 263			
QVQLVESGGG	LVPKPGSRL	SCAASGFTFS	DYMSWIRQA PGKGLEWSY ISSSGSIYY 60
ADSVKGRFTI	SRDNAKNSLY	LQMNSLRAED	TAVVYCARDQ GNYGVDVWGQ GTTVTVSSGG 120
GGSGGGGSGG	GGSGGGGSDI	QMTQSPSTLS	ASVGRDVTIT CRASQSISSW LAWYQQKPGK 180
APKLLIYEAS	SLESGVPSRF	SGSGSGTEFT	LTISSLQPDV FATYYCQQSD SHPITFGGGT 240
KVEIKTTTPA	PRPPTPAPTI	ASQPLSLRPE	ACRPAAGAVG HTRGELDFACD IYIWAPLAGT 300
CGVLLLSLVI	TLYCKRGRKK	LLYIFKQPFM	RPVQTTQEED GCSCRPFEEE EGGCELRVKF 360
SRADAPAYQ	QQQNGLYNEL	NLGRREEYDV	LDKRRGRDPE MGKGPRRKNQ QEGLYNELQK 420
DKMARAYSET	GMKGERRRRK	GHGDIYQGLS	TATKTDYDAI HMOALPPR 468

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SEQ ID NO: 264           moltype = AA   length = 468
FEATURE                  Location/Qualifiers
source                   1..468
                           mol_type = protein
                           organism = synthetic construct

SEQUENCE: 264
DIQMTQSPST LSASVGDRVT ITCRASQSIG SWLAWYQQKP GKAPKLLIYE ASSLESQVPS 60
RFGSGSGSGTE FTLTISSLQP DDFATYYCQQ SDShpITFGG GTKVEIKGGG GSGGGGSGGG 120
GSGGGGSGVQK LVESGGGLVK PGGSLRLSCA ASGFTFSDDY MSWIRQAPGK GLEWVSYISS 180
SGSTIYYADS VKGRFTISR D NAKNLSYLQM NSLRADTAV YICARDQNGY GVDVWGQGT 240
VTVSSTTTPTA PRPPTPAPTI ASQPLSLRPE ACRAAGGAV HTGRGLDFACD IYIWAPLAGT 300
CGVLLLSLVI  TLYCKGRKKK LLYIFKQPFM RVQTTQBED GCSCRPEEEB EGGCELRVKF 360
SRSDAPAPYQ QGQNQLYNEL NLGRREEDYD LDKRRGRDPE MGKQPRRKNP QEGYLINELQK 420
DKMAEAYSEI GMKGERRRKG GHGDLGYGLS TATKDTYDAL HMOALPPR 468

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SEQ ID NO: 265      moltype = AA  length = 468
FEATURE             Location/Qualifiers
source              1..468
                   mol_type = protein
                   organism = synthetic construct
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SEQUENCE: 265
 EVQLVESGGG LVKPGGSLRL SCAASGFTFS DYYMSWIRQA PGKGLEWVSY ISSSGSTIYY 60
 ADSVKGRFTI SRDNAKNSLY LQMNSLRAED TAVYYCARDQ GNYGVDVWGQ GTTVTVSSGG 120
 GSGGGGSGSG GSGGGGSDI QMTQSPSTLS ASVGDRTIT CRASQSISSW LAWYQQKPGK 180
 APKLLIYEAS SLESGVPSRF SSGSGTEFT LTISLQPD FATYYCQAN SHPITFGGT 240
 KVEIKTTTPA PRPPTPPTI ASQPLSLRPE ACRPAAGGAV HTRGLDFACD IYIWAPLAGT 300
 CGVLLLSLVI TLYCKRGRKK LLYIFKQPFM RPVQTTQEED GCSCRFPEEE EGGCELRVKF 360
 SRSADAPAYQ QGQNQLYNEL NLGRREEYDV LDKRRGRDPE MGGKPRRKNP QEGLYNELQK 420
 DKMAEAYSEI GMKGERRRGK GHDGLYQGLS TATKDTYDAL HMQALPPR 468

SEQ ID NO: 266 moltype = AA length = 468
 FEATURE Location/Qualifiers
 source 1..468
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 266
 DIQMTQSPST LSASVGDRTV ITCRASQSI SWLAWYQQKP GKAPKLLIYE ASSLESQVPS 60
 RFSGSGSGTE FTLTISLQPD DDFATYYCQQ ANSHPITFGG GTKVEIKGGG GSGGGGSGGG 120
 GSGGGGSEVQ LVESGGGLVK PGGSLRLSCA ASGFTFSDYY MSWIRQAPGK GLEWVSYISS 180
 SGSTIYYADS VKGRFTISR NAKNSLYLQM NSLRAEDTAV YYCARDQGNV GVDVWGQGT 240
 VTVSSTTTTPA PRPPTPPTI ASQPLSLRPE ACRPAAGGAV HTRGLDFACD IYIWAPLAGT 300
 CGVLLLSLVI TLYCKRGRKK LLYIFKQPFM RPVQTTQEED GCSCRFPEEE EGGCELRVKF 360
 SRSADAPAYQ QGQNQLYNEL NLGRREEYDV LDKRRGRDPE MGGKPRRKNP QEGLYNELQK 420
 DKMAEAYSEI GMKGERRRGK GHDGLYQGLS TATKDTYDAL HMQALPPR 468

SEQ ID NO: 267 moltype = AA length = 472
 FEATURE Location/Qualifiers
 source 1..472
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 267
 EVQLLESGGG LVQPGGSLRL SCAASGFTFS NYAMSWVRQA PGKGLEWVSA ISGSGGSTYY 60
 ADSVKGRFTI SRDNSKNTLY LQMNSLRAED TAVYYCARPG DGYVEGVYFD YWQGTLVTV 120
 SSGGGGSGGG GSGGGGSGGG GSDIQMTQSP SLSASVGDRT VTITCRASQS ISSYLNWYQQ 180
 KPGKAPKLLI YAASSLQSGV PSRFSGSGSG TDFTLTISL QPEDFATYYC QQAHSPTITF 240
 GGGTKVEIKT TTPAPRPPTP APTIASQPLS LRPEACRPAA GGAVHTRGLD FACDIYIWAP 300
 LAGTCGVLLL SLVITLYCKR GRKKLLYIFK QPFMRPVQTT QEEDGCSCRF PEEEEGGCEL 360
 RVKFSRSADA PAYQQGQNL YNELNLGRRE EYDVLDKRRG RDPMEGKPR RKNPQEGLYN 420
 ELQKDKMAEA YSEIGMKGER RRGKGHDGLY QGLSTATKDT YDALHMQALP PR 472

SEQ ID NO: 268 moltype = AA length = 472
 FEATURE Location/Qualifiers
 source 1..472
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 268
 DIQMTQSPSS LSASVGDRTV ITCRASQSI SYLNWYQQKP GKAPKLLIYA ASSLQSGVPS 60
 RFSGSGSGTD FTLTISLQPD EDFATYYCQQ AHSSPITFGG GTKVEIKGGG GSGGGGSGGG 120
 GSGGGGSEVQ LLESGGGLVQ PGGSLRLSCA ASGFTFSNYA MSWVRQAPGK GLEWVSAISG 180
 SGGSTYYADS VKGRFTISR NSKNTLYLQM NSLRAEDTAV YYCARPGDGY YEGVYFDYWG 240
 QGTLVTVSST TTPAPRPPTP APTIASQPLS LRPEACRPAA GGAVHTRGLD FACDIYIWAP 300
 LAGTCGVLLL SLVITLYCKR GRKKLLYIFK QPFMRPVQTT QEEDGCSCRF PEEEEGGCEL 360
 RVKFSRSADA PAYQQGQNL YNELNLGRRE EYDVLDKRRG RDPMEGKPR RKNPQEGLYN 420
 ELQKDKMAEA YSEIGMKGER RRGKGHDGLY QGLSTATKDT YDALHMQALP PR 472

SEQ ID NO: 269 moltype = AA length = 474
 FEATURE Location/Qualifiers
 source 1..474
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 269
 QITLKESGPT LVKPTQTLTL TCTFSGFSLT TSGVGVGWIR QPPGKALEWL ALIYWNDEKR 60
 YSPSLKSRLT ITKDTSKNQV VLTMTNMDPV DTAVYYCARE GSHDYKSSNW FDPWQGQTLV 120
 TVSSGGGSGG GGGSGGGSGG GGGSDIQMTQ SPSSLSASVG DRVTITCQAS QDIANYLNWY 180
 QQKPGKAPKL LIYDASNLET GVPSRFSGSG SGTDFTFITIS SLQPEDIATY YCQQHFNLPL 240
 TFGGGTKVEI KTTTPAPRPP TPAPTIASQP LSLRPEACRP AAGGAVHTRG LDFACDIYIW 300
 APLAGTCGVLL LSLVITLYC KRGKKLLYI FKQPFMRPVQ TTQEEDGCSC RFEEEEGGGC 360
 ELRVKFSRSA DAPAYQQQN QLYNELNLGR REEYDVLDKR RGRDPEMGK PRRKNPQEG 420
 YNELQKDKMA EAYSEIGMKG ERRRGKGHDG LYQGLSTATK DTYDALHMQA LPPR 474

SEQ ID NO: 270 moltype = AA length = 474
 FEATURE Location/Qualifiers
 source 1..474
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 270
 DIQMTQSPSS LSASVGDRTV ITCQASQDIA NYLNWYQQKP GKAPKLLIYD ASNLETGVPS 60
 RFSGSGSGTD FTFITISLQPD EDIATYYCQQ HFNPLPTFGG GTKVEIKGGG GSGGGGSGGG 120
 GSGGGGSGQT LKESGPTLVK PTQTLTLTCT FSGFSLSTSG VGVGWIRQPP GKALEWLALI 180

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SEQ ID NO: 276      moltype = AA  length = 343
FEATURE             Location/Qualifiers
source              1..343
                   mol type = protein
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                                organism = synthetic construct
SEQUENCE: 276
EVQLLESQGGG LVQPGGSLRL SCAASGFTFE DYAMSWFRQA PGKEREGVSA ISGKGGSTYY 60
ADSVKGRFTI SRDNSKNTLY LQMNSLRAED TAVYYCAVL D EAGAEGGYW GQGTQVTVSS 120
TTTPAPRPPT PAPTIASQPL SLRPEACRPA AGGAVHTRGL DFACDIYIWA PLAGTCGVLL 180
LSLVITLYCK RGRKKLLYIF KQPFMRPVQT TQEDGDCSCR FPEEEEGGCE LRVKFSRSAD 240
APAYQQGQNG LYNELNLGRR EEYDVLDRR GRDPEMGKPK RRKNPQEGLY NELQKDKMAE 300
AYSEIGMKGE RRRGKGHDGL YQGLSTATKD TYDALHMQAL PPR 343

SEQ ID NO: 277      moltype = AA length = 343
FEATURE            Location/Qualifiers
source              1..343
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 277
EVQLLESQGGG LVQPGGSLRL SCAASGFTFD RYAMSWFRQA PGKEREGVSA ISTSGDSTYY 60
ADSVKGRFTI SRDNSKNTLY LQMNSLRAED TAVYYCAVL D EAGAEGGYW GQGTQVTVSS 120
TTTPAPRPPT PAPTIASQPL SLRPEACRPA AGGAVHTRGL DFACDIYIWA PLAGTCGVLL 180
LSLVITLYCK RGRKKLLYIF KQPFMRPVQT TQEDGDCSCR FPEEEEGGCE LRVKFSRSAD 240
APAYQQGQNG LYNELNLGRR EEYDVLDRR GRDPEMGKPK RRKNPQEGLY NELQKDKMAE 300
AYSEIGMKGE RRRGKGHDGL YQGLSTATKD TYDALHMQAL PPR 343

SEQ ID NO: 278      moltype = AA length = 344
FEATURE            Location/Qualifiers
source              1..344
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 278
EVQLLESQGGG LVQPGGSLRL SCAASGFTFA SDAMSWYRQA PGKERELVSA ISGSGGSTYY 60
ADSVKGRFTI SRDNSKNTLY LQMNSLRAED TAVYYCAHD SGEAYLAFDY WQGTQVTVS 120
STTTAPRPPT TPAPTIASQP LSLRPEACRP AAGGAVHTRG LDFACDIYIW APLAGTCGVL 180
LLSLVITLYC KRGRKKLLYI FKQPFMRPVQ TTQEDGDCSC RFPEEEEGGC ELRVKFSRSA 240
DAPAYQQGQN QLYNELNLGR REEYDVLDR RGRDPEMGK PRRKNPQEGL YNELQKDKMA 300
EAYSEIGMKG ERRRGKGHDG LYQGLSTATK DTYDALHMQA LPPR 344

SEQ ID NO: 279      moltype = AA length = 344
FEATURE            Location/Qualifiers
source              1..344
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 279
EVQLLESQGGG LVQPGGSLRL SCAASGFTFD SYTMSWYRQA PGKERELVSA ISGHGDSTYY 60
ADSVKGRFTI SRDNSKNTLY LQMNSLRAED TAVYYCTRIS ITTEWLAGDY WQGTQVTVS 120
STTTAPRPPT TPAPTIASQP LSLRPEACRP AAGGAVHTRG LDFACDIYIW APLAGTCGVL 180
LLSLVITLYC KRGRKKLLYI FKQPFMRPVQ TTQEDGDCSC RFPEEEEGGC ELRVKFSRSA 240
DAPAYQQGQN QLYNELNLGR REEYDVLDR RGRDPEMGK PRRKNPQEGL YNELQKDKMA 300
EAYSEIGMKG ERRRGKGHDG LYQGLSTATK DTYDALHMQA LPPR 344

SEQ ID NO: 280      moltype = AA length = 346
FEATURE            Location/Qualifiers
source              1..346
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 280
EVQLLESQGGG LVQPGGSLRL SCAASGFTFS SYAMSWFRQA PGKEREFVSF ISGSGDSTYY 60
ADSVKGRFTI SRDNSKNTLY LQMNSLRAED TAVYYCTRWP YDFEEPSEPG VYWGQGTQVT 120
VSSTTTAPRP PPTPAPTIAS QPLSLRPEAC RPAAGGAVHT RGLDFACDIY IWAPLAGTCG 180
VLLLSLVITL YCKRGRKKLL YIFKQPFMRP VQTTQEDGCG SCRFPEEEEG GCELRVKFSR 240
SADAPAYQQG QNQLYNELNL GRREEYDVL D KRRGRDPEMG GKPRRKNPQE GLYNELQKDK 300
MAEAYSEIGM KGERRRGKGH DGLYQGLSTA TKDTYDALHM QALPPR 346

SEQ ID NO: 281      moltype = AA length = 346
FEATURE            Location/Qualifiers
source              1..346
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 281
EVQLLESQGGG LVQPGGSLRL SCAASGFTFT DYDMSWYRQA PGKERELVSV IHSGGSTYYA 60
DSVKGRFTIS RDNSKNTLYL QMNSLRAEDT AVYYCAPGYI SDLSFDYYNF DYWGQGTQVT 120
VSSTTTAPRP PPTPAPTIAS QPLSLRPEAC RPAAGGAVHT RGLDFACDIY IWAPLAGTCG 180
VLLLSLVITL YCKRGRKKLL YIFKQPFMRP VQTTQEDGCG SCRFPEEEEG GCELRVKFSR 240
SADAPAYQQG QNQLYNELNL GRREEYDVL D KRRGRDPEMG GKPRRKNPQE GLYNELQKDK 300
MAEAYSEIGM KGERRRGKGH DGLYQGLSTA TKDTYDALHM QALPPR 346

SEQ ID NO: 282      moltype = AA length = 346
FEATURE            Location/Qualifiers
source              1..346
                    mol_type = protein
                    organism = synthetic construct

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SEQ ID NO: 290      moltype = DNA  length = 126
FEATURE            Location/Qualifiers
source             1..126
                  mol_type = other DNA
                  organism = synthetic construct
```


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SEQUENCE: 290
aagcgcgga gaaagaagct cctgtacatc ttcaagcagc cttttatgcg acctgtgcaa 60
accactcagg aagaagatgg gtgttcatgc cgcttccccg aggaggaaga aggagggtgt 120
gaactg 126

SEQ ID NO: 291 moltype = DNA length = 123
FEATURE Location/Qualifiers
source 1..123
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 291
aggagtaaga ggagcaggct cctgcacagt gactacatga acatgactcc ccgccgcccc 60
gggccccccc gcaagcatta ccagccctat gcccaccac gcgacttcgc agcctatcgc 120
tcc 123

SEQ ID NO: 292 moltype = DNA length = 105
FEATURE Location/Qualifiers
source 1..105
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 292
acaaaaaaga agtattcatc cagtgtgcac gaccctaacg gtgaatacat gttcatgaga 60
gcagtgaaca cagccaaaaa atctagactc acagatgtga cccta 105

SEQ ID NO: 293 moltype = DNA length = 336
FEATURE Location/Qualifiers
source 1..336
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 293
agggtgaaat tttctagaag cgccgatgct cccgcatac agcagggtca gaatcagctc 60
tacaatgaat tgaatctcgg caggcgagaa gactacgat ttctggacaa aagacggggc 120
agggatcccc agatgggggg aaagccccc agaaaaaac ctcaggagggt gttgtacaat 180
gagctgcaga aggacaagat ggctgaagcc tatagcgaga tcggaatgaa aggcgaaaga 240
cgagaggcca aggggcatga cggtctgtac cagggtctct ctacagccac caaggacact 300
tatgatgcgt tgcataatgca agccttgcca cccgcg 336

SEQ ID NO: 294 moltype = DNA length = 63
FEATURE Location/Qualifiers
source 1..63
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 294
atggctcttc ccgtaacagc ccttttgttg ccccttgcac tcctttgcga tgcagcacga 60
ccg 63

SEQ ID NO: 295 moltype = DNA length = 1404
FEATURE Location/Qualifiers
source 1..1404
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 295
gagatcgtgc tgacacagtc tccccccaca ctgtcactgt ctccaggcga aagagccaca 60
ctgagctgta gagccagcca gagcgtgtcc tcttacctgg cctgggtatca gcagaagcct 120
ggacaggctc cccggctgct gatctacgat gccagcaata gagccacagg catccccgcc 180
agattttctg gcagcggctc tggcaccgat ttacccttga ccataagcag cctggaacct 240
gaggacttcg ccgtgtacta ctgccagcag agagtgggtg accccatcac ctttggcgga 300
ggcaccacag tggaatcaaa agcggcgcca ggaagcggag gcggaggatc tgggtggtggt 360
ggatctggcg gaggtggcag ccagatcaca ctgaaagagt ctggcccccac actgggtcaag 420
cccacacaga ccttgacact gacctgcacc ttcagcggct ttagcctgag cacatctggc 480
gtcggcggtg gctggattag acagcctcct ggaaaggccc tggaaagggt ggccctgatc 540
tactggaacg acgagaagag atacagcccc agcctgaagt cccggctgac catcaccaag 600
gacaccagca agaaccaggt ggtgctgacc atgacaaaaca tggaccccggt ggacaccgcc 660
gtgtattatt gcgccaagaga tgagtagcgc ggcttcgaca tttggggcca gggcacaatg 720
gtcaccgtgt ctagtaccac aacacctgct ccaaggcccc ccacaccgcg tccaactata 780
gccagccaac cattgagcct cagacctgaa gcttgaggcg ccgcagcagg aggcgcgctc 840
catacgcgag gcttggaact cgcggtgtgat atttatatatt gggcaccttt ggccggaaca 900
tgtgggggtg tcttctcttc ccttggtgat actctgtatt gtaagcgcgg gagaaagaag 960
ctcctgtaca tctcaagca gccttttatg cgacctgtgc aaaccactca ggaagaagat 1020
gggtgttcat gccgcttccc cgaggaggaa gaaggagggt gtgaactgag ggtgaaattt 1080
tctagaagcg ccgatgctcc cgcatatcag cagggtcaga atcagctcta caatgaattg 1140
aatctcggca ggcgagaaga gtacgatgtt ctggacaaaa gacggggcag ggatccccgag 1200
atggggggaa agccccggag aaaaaatcct caggagggggt tgtacaatga gctgcagaag 1260
gacaagatgg ctgaagccta tagcgagatc ggaatgaaag gcgaaagacg cagaggcaag 1320
gggcatgacg gtctgtacca gggctctctc acagccacca aggacactta tgatgcgttg 1380
catatgcaag ccttgccacc ccgc 1404

SEQ ID NO: 296 moltype = DNA length = 1404
FEATURE Location/Qualifiers

-continued

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source          1..1404
                mol_type = other DNA
                organism = synthetic construct

SEQUENCE: 296
gagatcgtgc  tgaccacagtc  ccttgctacc  ctgagcctgt  ctccaggcga  gcgggccaca  60
ctgagctgta  gagcttctca  gagcgtgtcc  agctacctgg  cctgggtatca  gcagaaacct  120
ggccaggccc  ctagactgct  gatctacgac  gccagcaacc  gggccaccgg  catccccgcc  180
agattcagcg  gatctggcag  cggcacagat  tttaacctca  ccatacgcag  cctggaaacct  240
gaggacttcg  ccgtctacta  ctgccagcaa  agattcgact  accccatcac  cttcggcggc  300
ggaacaaagg  tggaaattaa  ggggtgtggg  ggcagcgggt  gaggtgggag  cggaggcggg  360
ggtagcggag  gcgggggtag  ccaaatcaca  ctgaaagaga  gcggccctac  actcgtgaaa  420
cctaccacga  ccctgacact  gacatgtacc  ttcagcggct  tctccctgag  cactcctggc  480
gtcggcgctg  gatggatcag  acagcctcca  ggcaaggccc  tggaaatggc  ggctctgac  540
tattggaaag  acgacaagcg  gtacagcccc  agcctgaagt  ctgactgac  catcacaaag  600
gacaccagca  agaaccagggt  ggtgctgacc  atgacaaata  tggaccccg  ggacaccgcc  660
gtgtactact  gcgccagaga  tgagtacggc  ggatttgata  tctggggcca  gggcaccatg  720
gtgaccgtgt  ccagcaccac  aacacctgct  ccaaggcccc  ccacaccgc  tccaactata  780
gccagccaac  cattgagcct  cagacctgaa  gcttgaggc  ccgcagcagg  aggcgcgcgt  840
catacgcgag  gcctggactt  cgcgtgtgat  atttatatt  gggcaccttt  ggccggaaca  900
tgtgggggtg  tgcttctctc  ccttgctgac  actctgtatt  gtaagcgcg  gagaaagaag  960
ctcctgtaca  tottcaagca  gccctttatg  cgacctgtgc  aaacctatca  ggaagaagat  1020
gggtgttcat  ccgcttccc  cgaggaggaa  gaaggagggt  gtgaactgag  ggtgaaattt  1080
tctagaagcg  ccatgctctc  cgcatatcag  cagggctcag  atcagctcta  caatgaattg  1140
aatctcggca  ggcgagaaag  gtacgatgtt  ctggacaaaa  gacggggcag  ggatcccgag  1200
atggggggaa  agccccggag  aaaaaatcct  caggaggggg  tgtacaatga  gctgcagaag  1260
gacaagatgg  ctgaagccta  tagcgagatc  ggaatgaaag  gcgaaagacg  cagaggcaag  1320
gggcatgacg  gtctgtacca  ggtgtctctc  acagccacca  aggacactta  tgatgcgttg  1380
catatgcaag  ccttgccacc  ccgc  1404

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SEQ ID NO: 297      moltype = DNA  length = 1398
FEATURE
source              Location/Qualifiers
                    1..1398
                    mol_type = other DNA
                    organism = synthetic construct

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SEQUENCE: 297
caagtgcagc  tcgtggaaag  cggcgccgga  gtgggtgcagc  ccggccggag  cctgagactg  60
tcctgcgcgc  cttctggatt  taccttcagc  agctacggca  tgcactgggt  cagacaggcc  120
cctggcaaa  gcctggagtg  ggtggccgtt  atcagctacg  agggcagcaa  caagtattac  180
gccgacagcg  tgaagggccg  cttcacaaatc  tctagagata  atagcaagaa  caccctgtac  240
ctgcagatga  acagcctgcg  ggcgaagat  accgcctgt  actactgtgc  tagagagctg  300
ggcgacggca  tggacgtgtg  gggacagggc  acaaccgtga  ccgtgtcttc  tgggtgggg  360
ggcagcgggt  gaggtggggg  cggaggcggg  ggtagcggag  gcgggggtag  cgagatcgtg  420
ctgacccagt  ccctgcctac  actgagcctg  tctccaggcg  agcgggccac  actgagctgt  480
agagcttctc  agagcgtgtc  cagctatctg  gcctgggttc  agcagaaacc  tggccaggcc  540
cctagactgc  tgatctacga  cgcacgaacc  cgggccaccg  gcctcccgcc  cagatcaccg  600
ggctctggca  gcggcaccga  cttcaccctc  accatcagca  gcctggaacc  cgaggatttt  660
gccgtctact  actgccagca  aagagtggac  ctgtggacct  tcggcggagg  aacaaagggt  720
gaaatcaaga  ccacaacacc  tgcctcaagg  cccccacac  ccgtcccaac  tatagccagc  780
caaccattga  gccctcagac  tgaagcttgc  agggccgcag  caggaggcgc  cgtccatacg  840
cgaggcctgg  acctgcgctg  tgatatttat  atttgggcac  ctttggccgg  aacatgtggg  900
gtgtgtcttc  tctccctgtg  gatcactctg  tattgtaagc  gcgggagaaa  gaagctcctg  960
tacatcttca  agcagccttt  tatgcagact  gtgcaaaacca  ctcaggaaaga  agatgggtgt  1020
tcatgccgct  tccccgagga  ggaagaagga  ggggtgtgaa  tgagggtgaa  attttctaga  1080
agcgcgatg  cctccgcata  tcagcagggt  cagaatcagc  tctacaatga  attgaatctc  1140
ggcaggcgag  aagagtacga  tgttctggac  aaaagacggg  gcagggatcc  cgatggggg  1200
ggaaagcccc  ggagaaaaaa  tcctcaggag  ggggtgtaca  atgagctgca  gaaggacaag  1260
atggctgaag  cctatagcga  gatcggaatg  aaaggcgaaa  gacgcagagg  caaggggcat  1320
gacggctgtg  accagggtct  ctctacagcc  accaaggaca  cttatgatgc  gttgcatatg  1380
caagccttgc  caccctgc  1398

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```

SEQ ID NO: 298      moltype = DNA  length = 1404
FEATURE
source              Location/Qualifiers
                    1..1404
                    mol_type = other DNA
                    organism = synthetic construct

```

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SEQUENCE: 298
gacatccaga  tgaccacagag  cccttcgacc  ctatccgctt  ccgtgggtga  ccgtgtgacc  60
atcacctgtc  gcgcgtcgca  gagcatctcc  tcttggctcg  cgtggtagca  acagaagcct  120
ggcaaggccc  ccaagctgct  gatttacgac  gccagttccc  tggagtctgg  cgtgccatcc  180
cgctctccg  gcagcggcag  cggtagcgag  ttcacacctga  cgatcagctc  cctgcagccg  240
gatgactttg  ctacctacta  ctgtcagcag  gtctcctccc  tccccccac  cttcgggtgg  300
ggtaccaagg  tggagatcaa  gggcgccggc  ggctctggtg  cgggaggttc  tggcggggga  360
ggttcggggg  ggggaggctc  cgagggtcaa  ctggtagaga  gcggcggggg  actggtaaaa  420
cccgcgctc  ccctgcggct  gtcagcgct  gctagcgct  tcacgttcag  cgattactac  480
atgagttgga  tccgccagcc  ccccggaag  ggttggagt  gggctctcgt  tatctcttcc  540
agcggatcta  ccatttacta  tcggacagc  gtgaaggggc  gcttcacat  atctcgggac  600
aacgccaaga  actccctgta  cctgcagatg  aattccctgc  gtgccgagga  cagggccgtg  660
tattactgtg  ccgcgcacca  gggcaactac  ggcgtcgacg  tgtggggcca  gggtagaacc  720
gtcaccgtgt  ccagtaccac  aacacctgct  ccaaggcccc  ccacaccgc  tccaactata  780

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gccagccaac cattgagcct cagacctgaa gcttgcaggc ccgcagcagg aggcgcgcgtc 840
catacgcgag gcctggactt cgcgtgtgat atttatattt gggcaccttt ggccggaaca 900
tgtgggggtgt tgcttctctc ccttgtgatc actctgtatt gtaagcgcgg gagaaagaag 960
ctcctgtaca tcttcaagca gccttttatg cgacctgtgc aaacctactca ggaagaagat 1020
gggtgttcatt gccgcttccc cgaggaggaa gaaggagggt gtgaactgag ggtgaaattt 1080
tctagaagcg ccgatgtccc cgcataatcag cagggtcaga atcagctcta caatgaattg 1140
aatctcggca ggcgagaaga gtacgatgtt ctggacaaaa gacggggcag ggatcccag 1200
atgggggggaa agcccccggag aaaaaatcct caggaggggt tgtacaatga gctgcagaag 1260
gacaagatgg ctgaagccta tagcgagatc ggaatgaaag gcgaaagacg cagaggcaag 1320
gggcatgacg gtctgtacca ggggtctctc acagccacca aggacactta tgatgcgttg 1380
catatgcaag ccttgccacc ccgc 1404

```

```

SEQ ID NO: 299      moltype = DNA length = 1404
FEATURE            Location/Qualifiers
source              1..1404
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 299
caagtgcagc tggctcgagag cggaggaggc ctggttaagc ccggcggatc tctcagactg 60
agctgcgcgc ctacgcgctt tacattcagc gactactaca tgagctggat ccggcaggcc 120
cctggcaagg gcctggaatg ggtgtcctac atcagctcct ccggcagcac catctactac 180
gccgacagcg tgaaggcgag attcacaatc tctagagata atgccaagaa cagcctgtac 240
ctgcagatga acagcctcgc ggcggaggac accgcctgt actatttgtc tagagatcag 300
ggcaactacg gcgtggagct gtggggccag ggcaccaccg tgaccgtgtc tagcggtggt 360
gggggcagcg gtggaggtgg gagcggaggc gggggtagcg gaggcggggg tagcgatatt 420
cagatgaccc agtcccctac tacactgagc gcctctgtgg gcgaccgggt gaccattaca 480
ttagagacca gccagagcat cagcagctgg ctggcttggt atcagcagaa acctggcaag 540
gcccctaagc tgctgtacta cagggccagc agcctggaaa gcggcgtccc cagcagattc 600
agcggcagcg gctctggaac agagttcacc ctgaccatct cctccctgca gctgacgac 660
ttcgccacct actactgcca gcaatctgat agccacccca tcacctttgg cggaggcacc 720
aaggtgaaaa tcaagaccac aacacctgct ccaaggcccc ccacaccgc tccaactata 780
gccagccaac cattgagcct cagacctgaa gcttgcaggc ccgcagcagg aggcgcgcgtc 840
catacgcgag gcctggactt cgcgtgtgat atttatattt gggcaccttt ggccggaaca 900
tgtgggggtgt tgcttctctc ccttgtgatc actctgtatt gtaagcgcgg gagaaagaag 960
ctcctgtaca tcttcaagca gccttttatg cgacctgtgc aaacctactca ggaagaagat 1020
gggtgttcatt gccgcttccc cgaggaggaa gaaggagggt gtgaactgag ggtgaaattt 1080
tctagaagcg ccgatgtccc cgcataatcag cagggtcaga atcagctcta caatgaattg 1140
aatctcggca ggcgagaaga gtacgatgtt ctggacaaaa gacggggcag ggatcccag 1200
atgggggggaa agcccccggag aaaaaatcct caggaggggt tgtacaatga gctgcagaag 1260
gacaagatgg ctgaagccta tagcgagatc ggaatgaaag gcgaaagacg cagaggcaag 1320
gggcatgacg gtctgtacca ggggtctctc acagccacca aggacactta tgatgcgttg 1380
catatgcaag ccttgccacc ccgc 1404

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SEQ ID NO: 300      moltype = DNA length = 1404
FEATURE            Location/Qualifiers
source              1..1404
                    mol_type = other DNA
                    organism = synthetic construct

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SEQUENCE: 300
gatatccaga tgaccagtc cccatctaca ctgagcgctt ctgtgggcga ccgggtgaca 60
attacctgta gagctagcca gagcatctcc tctgggctgg cttggtacca gcaaaaacct 120
ggcaaggccc taaagctgct gatctacgag gccagcagcc tggaaagcgg cgtcccctct 180
agattcagcg gcagcggctc tggaaaccgag ttcaacctga caatcagcag cctgcagcct 240
gacgacttcg ccacctatta ctgccagcag gccaacagcc accccatcac ctttggcgga 300
ggcaccgaag tggaaatcaa ggggtggtgg ggacgcggtg gaggtgggag cggagcgggg 360
ggtagcggag gcgggggtag cgaggtgcaa ctgggtgaaa gcggcggagg actcgtaaag 420
cccggcgcca gccgtgagact gagctgcgcc gctagcggat ttaccttcag cgactactac 480
atgagctgga tccggcaggc ccttgccaag ggctcggaat gggtcagcta catcagctcc 540
tctggctcta caatctacta cgccgacagc gtgaaaggca gattcaccat ctctagagat 600
aatgccaaga acagcctgta cctgcaaatg aacagcctgc gggccgagga caccgcctg 660
tactattgtg ctagagatca gggcaactac ggcgtggacg tgtggggcca gggcaccacc 720
gtgacagtgt cctccaccac aacacctgct ccaaggcccc ccacaccgc tccaactata 780
gccagccaac cattgagcct cagacctgaa gcttgcaggc ccgcagcagg aggcgcgcgtc 840
catacgcgag gcctggactt cgcgtgtgat atttatattt gggcaccttt ggccggaaca 900
tgtgggggtgt tgcttctctc ccttgtgatc actctgtatt gtaagcgcgg gagaaagaag 960
ctcctgtaca tcttcaagca gccttttatg cgacctgtgc aaacctactca ggaagaagat 1020
gggtgttcatt gccgcttccc cgaggaggaa gaaggagggt gtgaactgag ggtgaaattt 1080
tctagaagcg ccgatgtccc cgcataatcag cagggtcaga atcagctcta caatgaattg 1140
aatctcggca ggcgagaaga gtacgatgtt ctggacaaaa gacggggcag ggatcccag 1200
atgggggggaa agcccccggag aaaaaatcct caggaggggt tgtacaatga gctgcagaag 1260
gacaagatgg ctgaagccta tagcgagatc ggaatgaaag gcgaaagacg cagaggcaag 1320
gggcatgacg gtctgtacca ggggtctctc acagccacca aggacactta tgatgcgttg 1380
catatgcaag ccttgccacc ccgc 1404

```

```

SEQ ID NO: 301      moltype = DNA length = 1416
FEATURE            Location/Qualifiers
source              1..1416
                    mol_type = other DNA
                    organism = synthetic construct

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SEQUENCE: 301

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gacatccaga tgaccocagag cccatagctcc ctgagcgcca gcgtgggcca tagagtgacc 60
attacctgta gagcctctca gagcatctcc tccatcctga actgggtatca gcagaaaccc 120
ggcaaggccc ctaagctgct gatctacgcc gctagcagcc tgcagtctgg cgtccccagc 180
cgggttcagcg gcagcggatc tggcacgcac ttcacctga caatcagcag cctgcaccc 240
gaggactttg ctacatacta ctgccagcag gccacagct ctccaatcac cttcggcgcc 300
ggaacaaaagg tggaaatcaa ggggtggtggg ggcagcgggtg gagtgaggag cggaggcggg 360
ggtagcggag gcgggggtag caggtgagc ctgctggaaa gcggaggcgg actcgttcaa 420
cctggcgcca gccctgagact gagctgcgcc gcttctggat ttaccttcag caactacgcc 480
atgagctggg tgcggcaggg cccctggcaaa ggcttggat gggtctccgc catcagcgcc 540
tctggcggtt ccactacta cgcgcagcag gtgaaggcca gattcaccat cctagagat 600
aatagcaaga acacctgta cctgcagatg aacagcctcg gggccgagga caccgcctg 660
tactattgtg ctgacccgg agatggctac tacgaggcgg tgtacttcga ctactggggc 720
cagggcacac tggtagcag gtccagcacc acaacacctg ctccaaggcc cccacaccc 780
gtccaacta tagccagcca accattgagc ctacagacct aagcttgag gcccgagca 840
ggaggcgccg tccatagcg aggcctggac ttccgctgtg atatttatat ttgggcacct 900
ttggcggaa catgtgggt gttgcttct tccctgtga tcaactgta ttgtaagcg 960
gggagaaaga agctcctgta catctcaag cagccttcta tgcgacctgt gaaacccact 1020
caggaagaag atgggtgttc atgccccttc cccgaggagg aagaaggagg gtgtgaact 1080
agggtgaaat ttcttagaag cgcgatgct cccgcatac agcagggtca gaatcagct 1140
tacaatgaat tgaatctcgg caggcgagaa gactacgat ttctggacaa aagacgggg 1200
agggatcccg agatgggggg aaagcccccg agaaaaatc ctgaggagg gttgtacaat 1260
gagctgcaga aggacaagat ggctgaagcc tatagcgaga tcggaatgaa aggcgaaga 1320
cgagaggcca aggggcatga cggctgtgac cagggtctct ctacagccac caaggacact 1380
tatgatgcgt tgcatacgca agccttgcca ccccg 1416

```

SEQ ID NO: 302

moltype = DNA length = 1422

FEATURE

Location/Qualifiers

source

1..1422

mol_type = other DNA

organism = synthetic construct

SEQUENCE: 302

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gatattcaga tgaccocagag cccatctagc ctgagcgcca gcgtgggcca tagagtgacc 60
atcacctgtc aggcctctca ggacatcgct aattacctga actgggtatca gcagaaaccc 120
ggcaaggccc ctaagctgct gatctacgac gccctcaacc tggaaacggc cgtgcccagc 180
cgggttcagcg gcagcggatc tggcacagac ttcacctta ccacagctc cctccagcct 240
gaggacatcg ccacatacta ctgccagcaa cacttcaacc tgcctctgac cttcggcgcc 300
ggaacaaaagg tgcagatcaa ggggtggtggg ggcagcgggtg gagtgaggag cggaggcggg 360
ggtagcggag gcgggggtag ccaaatcacc ctgaaagaga gcggacctac actggtcaag 420
cctacccaga cactgacct ccatgtaca ttacgcggct ttagcctgag cactccggc 480
gtgggagtgg cctggatcag acagccccc ggcaaggccc tggaaatggc ggctctgac 540
tattggaatg acgagaagcg gtacagccct agcctgaaat ctgactgac aatcaccaag 600
gacaccagca agaaccaggt ggtgctgacc atgaccaaca tggatcctgt ggataccgcc 660
gtgtactact gcgccagaga aggcctctac gactacaaga gctccaactg gttcgaccca 720
tggggccagg gcacccctgt tacagtgtct agcaccacaa cacctgctcc aaggccccc 780
acaccgctc caactatagc cagccaacca ttgagcctca gacctgaagc ttgcaggccc 840
gcagcaggag gcgcgctcca tacgcgaggg ctggacttcg cgtgtgatat ttatatattg 900
gcacctttgg ccggaacatg tgggggtgtg cttctctccc ttgtgatcac tctgtattgt 960
aagcgcggga gaaagaagct cctgtacatc ttcaagcagc cttttatgcg acctgtgcaa 1020
accactcagg aagaagatgg gtgttcatgc cgttccccg agggaggaaga agggagggtg 1080
gaactgaggg tgaattttc tagaagcgcc gatgtctccg catatagca gggtcagaat 1140
cagctctaca atgaattgaa tctcggcagg cgagaagagt acgatgttct ggacaaaaga 1200
cggggcaggg atccccgat ggggggaaag ccccgagaaa aaaatcctca ggagggttg 1260
tacaatgagc tgcagaagga caagatggct gaagcctata gcgagatcg aatgaaagg 1320
gaaagacgca gaggcaagg gcatgacggt ctgtaccagg gtctctctac agccaccaag 1380
gacacttatg atgcgttgca tatgcaagcc ttgccacccc gc 1422

```

SEQ ID NO: 303

moltype = DNA length = 1434

FEATURE

Location/Qualifiers

source

1..1434

mol_type = other DNA

organism = synthetic construct

SEQUENCE: 303

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gatatcgtga tgacccaatc tccactgagc ctgcctgtga cacctggcca gctgcttct 60
atcagctgta gaagcagcca gtccctgtgt cacagcaacg gctacaacta cctggactgg 120
tatctgcaga aaccggcca gagccccag ctgctgatct acctcggtc taatcgggcc 180
agcggagtgc ctgatagatt cagcggagc ggctccggca ccgacttcac cctgaagatc 240
agcagagtgg aagcggagga cgtggcgctc tactactgca tgcaggccct gggcctgatt 300
acatttgccg gcggaaccaa ggtggaatc aagggtggtg ggggcagcgg ttgaggtggg 360
agcggaggcg ggggtagcgg aggcgggggt agcgaagtgc agctggttga gagcggcgcc 420
ggactggtga agcccgagg cagcctcaga ctgagctgtg ctgcttctgg cttaccttc 480
agctcttata gcatgaactg ggtgcggcag gccctggcca agggcctgga atgggtcagc 540
tccatcagct cttctagcag ctacatctac tacgcgcaga gcgtgaaggc cagattcacc 600
atcagcagag ataagcccaa gaacagcctg tacctgcaga tgaatagcct gcgggcggag 660
gacaccgccc tgtactactg gccagagacc ggcgacacct acagcgccgc cgattactac 720
tacatggacg tgtggggcaa aggaacaacc gtgacagtgt cctccaccac aacacctgct 780
ccaaggcccc ccacaccgc tccaactata gccagccaac cattgagcct cagacctgaa 840
gcttgacggc ccgagcagg aggcgcgctc catacgcgag gcctggactt cgcgtgtgat 900
atztatattt gggcaccttt ggcgggaaca tgtgggggtg tgcctctctc cctgtgtgat 960

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-continued

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actctgtatt gtaagcgcg gagaagaag ctectgtaca tcttcaagca gccttttatg 1020
cgacctgtgc aaaccactca ggaagaagat ggggtgttcat gccgcttccc cgaggaggaa 1080
gaaggagggt gtgaactgag ggtgaaatth tctagaagcg ccgatgctcc cgcataatcag 1140
caggggtcaga atcagctcta caatgaattg aatctcggca ggcgagaaga gtacgatgtt 1200
ctggacaaaa gacggggcgag ggatcccag atggggggaa agcccggag aaaaaatcct 1260
caggagggggt tgtacaatga gctgcagaag gacaagatgg ctgaagccta tagcgagatc 1320
ggaatgaaa gcgaaagacg cagaggcaag gggcatgacg gtctgtacca gggctctctc 1380
acagccacca aggacactta tgatgcgttg catatgcaag ccttgccacc ccgc 1434

```

```

SEQ ID NO: 304      moltype = DNA length = 1017
FEATURE            Location/Qualifiers
source             1..1017
                  mol_type = other DNA
                  organism = synthetic construct

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SEQUENCE: 304
gaagtgaac tgctggaaa gggcgaggc ctggtccagc ccggcggtc tctgcggctc 60
agctgcgcg cttctggatt taccttcggc agcagggcta tgagctgggt gcggcaggcc 120
cctggaaaag agagagagct ggtgtccgcc atcagcgcca ggcgagaggt gacctactac 180
gccgacagcg tgaaggcgag attcaccatc tctagagata atagcaagaa caccctgtac 240
ctgcagatga acagcctgag agccgaggac accgcctgt actattgtca gagactgggt 300
gaagccaagc ggcactgggg ccagggcaca caggttacag tgtccagcac cacaacacct 360
gctccaagcg ccccccacac cgctccaact atagccagcc aaccattgag cctcagacct 420
gaagcttgca ggcccgagc aggagggcgc gtccatacgc gaggcctgga cttcgcgtgt 480
gatatttata tttgggcacc ttggccggga acatgtgggg tgttgcctct ctccttctgt 540
atcactctgt attgtaagcg cgggagaaa aagctcctgt acatcttcaa gcagcctttt 600
atgcgacctg tgcaaacacc tcaggaagaa gatgggtgtt catgccgctt ccccgaggag 660
gaagaaggag ggtgtgaact gaggggtgaaa ttttctagaa gcgcccagtc tcccgcata 720
cagcagggtc agaatacagc ctacaatgaa ttgaatctcg gcaggcgaga agagtacgat 780
gttctggaca aaagacgggg cagggatccc gagatggggg gaaagccccc gagaaaaaat 840
cctcaggagg ggtgtgacaa tgagctgcag aaggacaaga tggctgaagc ctatagcgag 900
atcggaatga aaggcgaaag acgcagagcg aaggggcatg acggtctgta ccagggtctc 960
tctacagcca aaggagacac ttatgatgcg ttgcatatgc aagccttgcc accccgc 1017

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```

SEQ ID NO: 305      moltype = DNA length = 1017
FEATURE            Location/Qualifiers
source             1..1017
                  mol_type = other DNA
                  organism = synthetic construct

```

```

SEQUENCE: 305
gaagtgaac tgctggaaac tggcgaggga ctggtccagc ccggcggtc cctgcggctg 60
agctgtgctg cttctggctt taccttcgag tctgaggcca tgagctggta tagacaggcc 120
cctggcaagg aaagagagct ggtcagcgct atcaccagcg agggctccac ctactacgcc 180
gacagcgtga aaggcgatt cacaatcagc cgggacaata gcaagaacac cctgtacctg 240
cagatgaaca gccctgcgcg cgaagataca gccgtgtact actgcgccc ccatcgagtgg 300
gagacaagac tcaactgggg ccagggcacc caggtgaccg tgtccagcac cacaacacct 360
gctccaagcg ccccccacac cgctccaact atagccagcc aaccattgag cctcagacct 420
gaagcttgca ggcccgagc aggagggcgc gtccatacgc gaggcctgga cttcgcgtgt 480
gatatttata tttgggcacc ttggccggga acatgtgggg tgttgcctct ctccttctgt 540
atcactctgt attgtaagcg cgggagaaa aagctcctgt acatcttcaa gcagcctttt 600
atgcgacctg tgcaaacacc tcaggaagaa gatgggtgtt catgccgctt ccccgaggag 660
gaagaaggag ggtgtgaact gaggggtgaaa ttttctagaa gcgcccagtc tcccgcata 720
cagcagggtc agaatacagc ctacaatgaa ttgaatctcg gcaggcgaga agagtacgat 780
gttctggaca aaagacgggg cagggatccc gagatggggg gaaagccccc gagaaaaaat 840
cctcaggagg ggtgtgacaa tgagctgcag aaggacaaga tggctgaagc ctatagcgag 900
atcggaatga aaggcgaaag acgcagagcg aaggggcatg acggtctgta ccagggtctc 960
tctacagcca ccaaggacac ttatgatgcg ttgcatatgc aagccttgcc accccgc 1017

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SEQ ID NO: 306      moltype = DNA length = 1023
FEATURE            Location/Qualifiers
source             1..1023
                  mol_type = other DNA
                  organism = synthetic construct

```

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SEQUENCE: 306
gaggtgcagc tgctggaaa gggagggggc ctggtccaac ccggcggtc tcttcgcta 60
agctgtgccc cttctggctt caccttcgac gaggtaacca tgactgggt cagacaggcc 120
cccgcaagg agcgagagtt cgtcagtgca atcagcgga ggcgtagcga gacttattac 180
gcggactccg tgaaggcccg cttcaccatt agcccgaca actccaagaa cagctgtac 240
ctgcagatga attcgtgccc cgccgaagat acggcctgt actactgtgc cgtggtgtg 300
gaggaggctg gcgtgggcta ttggggccag ggcacccagc tcacgtgtc gtccaccaca 360
acactgtgc caagggcccc cacaccgct ccaactatag ccagccaacc attgagcttc 420
agactgaag cttgcaggcc cgagcagga ggcccgctcc atacgcgagg cctggacttc 480
gcgtgtgata tttatatttg ggcaccttg gccggaacat gtgggtgtt gcttctctcc 540
cttgtgatca ctctgtattg taagcgcggg agaaaagaag tcctgtacat cttaagcag 600
ccttttatgc gacctgtgca aaccactcag gaagaagatg ggtgttcatg ccgcttcccc 660
gaggaggaag aaggagggtg tgaactgagg gtgaaatth ctagaagcgc cgtgctccc 720
gcatatcagc aggtcagaa tcagctctac aatgaattga atctcgagc gcgagaagag 780
tacgatgttc tggacaaaag acggggcagg gatcccaga tggggggaaa gccccggaga 840
aaaaatcctc aggaggggtt gtacaatgag ctgcagaagg acaagatggc tgaagcctat 900
agcgagatcg gaatgaaagg cgaagacgc agaggcaagg ggcagacgg tctgtaccag 960

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gggtctctcta cagccaccaa ggacacttat gatgcgttgc atatgcaagc cttgccaccc 1020
cgc                                                    1023

```

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SEQ ID NO: 307      moltype = DNA length = 1029
FEATURE            Location/Qualifiers
source              1..1029
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 307
gaggtgcagc tgctggagag cggaggcggc ctctgtcagc caggaggttc cctacgactc 60
tcctgtgccc ccagcgggett caccttcgag gactacgcca tgagttgggt ccgccaggcc 120
ccggggaagg agcgcgaggg cgtgagcgcg atttctggaa agggcggctc cacctattac 180
gcggactccg tgaagggtcg ctttaccatc tctcgcgaca actccaagaa cacgctgtac 240
ctgcagatga atagcctgcg cgtgaggac actgcgctgt actactgtgc tgtcttggac 300
gaggaggccg gcgcagaggg cggctattgg ggccagggtta cccaggtcac cgtgtcgtcc 360
accacaacac ctgctccaag gccccccaca cccgtccaa ctatagccag ccaaccattg 420
agcctcagac ctgaagcttg caggcccgca gcaggaggcg ccgtccatac gcgaggcctg 480
gacttcgcgt gtgatattta tatttgggca cctttggccg gaacatgttg ggtgttgctt 540
ctctcccttg tgatcactct gtattgtaag cgcgggagaa agaagctcct gtacatcttc 600
aagcagcctt ttatgcgacc ttgtgcaaac actcagggaag aagatgggtg ttcattgccg 660
ttccccgagg aggaagaagg aggggtgtgaa ctgagggtga aattttctag aagcgccgat 720
gctccccgat atcagcaggg tcagaatcag ctctacaatg aattgaatct cggcaggcga 780
gaagagtacg atgttctgga caaaagacgg ggcagggtac ccgagatggg gggaaaagccc 840
cggagaaaaa atcctcagga ggggttgtag aatgagctgc agaaggacaa gatggctgaa 900
gcctatagcg agatcggaat gaaaggcgaa agacgcagag gcaaggggca tgacggctctg 960
taccagggtc tctctacagc caccaaggac acttatgatg cgttgcatat gcaagccttg 1020
ccaccccg 1029

```

```

SEQ ID NO: 308      moltype = DNA length = 1029
FEATURE            Location/Qualifiers
source              1..1029
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 308
gaggtgcaac tgctggaaag cggcgggtga ctggtgcagc ccggcggcag cctgagactg 60
tcttgtgctg cttctggatt tacattcgac agatacgcca tgagctgggt ccgccaggcc 120
cctggcaaaag agcgggaagg cgtgtccgcc atctccacaa gcggagatag cacatactat 180
gccgacagcg tgaaggcgag attcaccatc agcagagata atagcaagaa caccctgtac 240
ctgcagatga acagcctccg ggcgcaggac accgcgctct actactgcgc cgtgtcggac 300
gaggaaagcc gcgcgcaggg cggctactgg ggccagggca cccagggtgac cgtgtctagc 360
accacaacac ctgctccaag gccccccaca cccgtccaa ctatagccag ccaaccattg 420
agcctcagac ctgaagcttg caggcccgca gcaggaggcg ccgtccatac gcgaggcctg 480
gacttcgcgt gtgatattta tatttgggca cctttggccg gaacatgttg ggtgttgctt 540
ctctcccttg tgatcactct gtattgtaag cgcgggagaa agaagctcct gtacatcttc 600
aagcagcctt ttatgcgacc ttgtgcaaac actcagggaag aagatgggtg ttcattgccg 660
ttccccgagg aggaagaagg aggggtgtgaa ctgagggtga aattttctag aagcgccgat 720
gctccccgat atcagcaggg tcagaatcag ctctacaatg aattgaatct cggcaggcga 780
gaagagtacg atgttctgga caaaagacgg ggcagggtac ccgagatggg gggaaaagccc 840
cggagaaaaa atcctcagga ggggttgtag aatgagctgc agaaggacaa gatggctgaa 900
gcctatagcg agatcggaat gaaaggcgaa agacgcagag gcaaggggca tgacggctctg 960
taccagggtc tctctacagc caccaaggac acttatgatg cgttgcatat gcaagccttg 1020
ccaccccg 1029

```

```

SEQ ID NO: 309      moltype = DNA length = 1032
FEATURE            Location/Qualifiers
source              1..1032
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 309
gaggtgcaac tgctggaaag cggcgggagga ctctccagc ccggcggcag cctgcggctg 60
agctgtgctg cttctggatt taccttcgcc agcgcgcca tgagctggta tagacaggcc 120
cctggcaaaag agcgggaact ggtgtccgcc atcagcggtc ctggcggctc cacctactac 180
gccgatagcg tgaaggcgag attcacaatc tctagagata atagcaagaa caccctgtac 240
ctgcagatga acagcctgag agccgaggac accgcgctgt actactgcgc cgctcacgac 300
agcggcgagg cctacctggc cttcgactac tggggccagg gcacacaggt gaccgtgtct 360
agcaccacaa cacctgtccc aaggcccccc acaccgcctc caactatagc cagccaacca 420
ttgagcctca gacctgaagc ttgcaggccc gcagcaggag gcgcgcgtcca tacgcaggcc 480
ctggacttcg cgtgtgatat ttatatattg gcacctttgg ccggaacatg tggggtgttg 540
cttctctccc ttgtgatcac tctgtattgt aagcgcggga gaaagaagct cctgtacatc 600
ttcaagcagc cttttatgag acctgtgcaa accactcagg aagaagatgg gtgttcatgc 660
cgcttccccg aggaggaaga aggagggtgt gaactgaggg tgaatttttc tagaagcgcc 720
gatgctcccg catatcagca gggtcagaat cagctctaca atgaattgaa tctcggcagg 780
cgagaagagt acgatgttct ggacaaaaga cggggcaggg atcccagat ggggggaaaag 840
ccccggagaa aaaatcctca ggagggggtg tacaatgagc tgcagaagga caagatggct 900
gaagcctata gcgagatcgg aatgaaaagg gaaagacgca gaggcaaggg gcatgacggt 960
ctgtaccagg gtctctctac agccaccaag gacacttatg atgcgttgca tatgcaagcc 1020
ttgccacccc gc 1032

```

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SEQ ID NO: 310      moltype = DNA length = 1032

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-continued

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FEATURE          Location/Qualifiers
source           1..1032
                mol_type = other DNA
                organism = synthetic construct

SEQUENCE: 310
gaggtgcaac tgctggaaag cggcggagga ctgctccagc cggcggcag cctgaggctg 60
agctgtgctg cttctggctt taccttcgac tcctacacaa tgagctggta tagacaggcc 120
cctggcaagg agcgggaact ggtgtccgcc atcagcggcc acggcgactc tacatactac 180
gccgacagcg tgaaggcgag attcacaatc tctagagata atagcaagaa caccctgtac 240
ctgcagatga acagcctgcg ggcggaggac accgccgtgt actactgcac cagaatcagc 300
atcaccacag agtggctggc cggagattac tggggccagg gcaccagggt gacagtgtcc 360
agcaccacaa cacctgtccc aaggccccc acaccgcctc caactatagc cagccaacca 420
ttgagcctca cagctgaagc ttgcaggccc gcagcaggag gcgcccgtcc tacgcgaggc 480
ctggacttcg cgtgtgatat ttatatattg gcacctttgg ccggaacatg tgggggtgtg 540
ctctctctcc ttgtgatcac tctgtattgt aagcgcggga gaaagaagct cctgtacatc 600
ttcaagcagc cttttatgcg acctgtgcaa accactcagg aagaagatgg gtgttcatgc 660
cgcttccccg agggaggaag agggaggtgt gaactgaggg tgaaattttc tagaagcgcc 720
gatgtccccg catatcagca ggttcagaat cagctctaca atgaattgaa tctcggcagg 780
cgagaagagt accgatgttc ggacaaaaa cggggcaggg atcccagat ggggggaaag 840
ccccggagaa aaaatcctca ggaggggttg tacaatgagc tgacagaagg caagatggct 900
gaagcctata gcgagatcgg aatgaaaggg gaaagacgca gaggcaaggg gcatgacggt 960
ctgtaccagg gtctctctac agccaccaag gacacttatg atgcgttgca tatgcaagcc 1020
ttgccacccc gc                                     1032

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SEQ ID NO: 311      moltype = DNA length = 1038
FEATURE            Location/Qualifiers
source             1..1038
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 311
gaggtgcagc tgctggaaag cggaggaggc ctgggtccaac ctggcggcag cctgcggctg 60
agctgcgcgc cttctggctt caccttcagc agctacgcca tgagctgggt ccggcaggcc 120
cctggcaagg aaagagagtt cgtgtctttt atcagcggat ctggcgactc caccctactac 180
gctgatagcg tgaaggcgag atttaccatc tctagagata atagcaagaa caccctgtac 240
ctccagatga acagcctgcg cgccgaggac acagccgtgt actattgtac cagatggcct 300
tacgacttcg aggaaccaag cgagcccgcc gtgtactggg gccagggcac acaggtgaca 360
gtgtctctca ccacaaacac tgctccaagg cccccacac ccgctccaac tatagccagc 420
caaccattga gcctcagacc tgaagcttgc aggcccgagc caggaggcgc cgtccatacg 480
cgaggccttg acttcgcgtg tgatatattt atttgggcac ctttggccgg aacatgtggg 540
gtgttgcttc tctcccttgt gatcactctg tattgtaagc gcgggagaaa gaagctcctg 600
tacatcttca agcagccttt tatgcgacct gtgcaaaaca ctcaggaaga agatgggtgt 660
tcatgccgct tccccgagga ggaagaagga ggggtgtgaa tgaggggtgaa attttctaga 720
agcgcgatg ctcccgcata tcagcagggt cagaatcagc tctacaatga attgaatctc 780
ggcaggcgag aagagtacga tgttctggac aaaagacggg gcagggatcc cgagatgggg 840
ggaaagcccc ggagaaaaaa tcctcaggag ggggtgtgaca atgagctgca gaaggacaag 900
atggctgaag cctatagcga gatcggaatg aaaggcgaaa gacgcagagg caaggggcat 960
gacggctctg accagggtct ctctacagcc accaaggaca cttatgatgc gttgcatatg 1020
caagccttgc caccgccg                                     1038

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```

SEQ ID NO: 312      moltype = DNA length = 1038
FEATURE            Location/Qualifiers
source             1..1038
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 312
gaggtgcagc tgctggaaag cggcggaggg ctgggtgcaac ctggcggatc tctcagactg 60
agctgtgctg cttctggctt cacattcacc gactacgaca tgagctggta tagacaggcc 120
cctggaaaaa agcgggaact ggtctccgtg atccacagcg gcggtccac ctactacgcc 180
gatagcgtga agggcagatt caccatcagc agagataata gcaagaacac cctgtacctg 240
cagatgaaca gcctgcgggc cgaggacacc gccgtgtact actgcgccc cggctactac 300
agcgacctgt cttttgatta ttacaacttc gactactggg gccagggcac acaggtgaca 360
gtgtccagca ccacaacacc tgetccaagg cccccacac ccgctccaac tatagccagc 420
caaccattga gcctcagacc tgaagcttgc aggcccgagc caggaggcgc cgtccatacg 480
cgaggccttg acttcgcgtg tgatatattt atttgggcac ctttggccgg aacatgtggg 540
gtgttgcttc tctcccttgt gatcactctg tattgtaagc gcgggagaaa gaagctcctg 600
tacatcttca agcagccttt tatgcgacct gtgcaaaaca ctcaggaaga agatgggtgt 660
tcatgccgct tccccgagga ggaagaagga ggggtgtgaa tgaggggtgaa attttctaga 720
agcgcgatg ctcccgcata tcagcagggt cagaatcagc tctacaatga attgaatctc 780
ggcaggcgag aagagtacga tgttctggac aaaagacggg gcagggatcc cgagatgggg 840
ggaaagcccc ggagaaaaaa tcctcaggag ggggtgtgaca atgagctgca gaaggacaag 900
atggctgaag cctatagcga gatcggaatg aaaggcgaaa gacgcagagg caaggggcat 960
gacgggtctg accagggtct ctctacagcc accaaggaca cttatgatgc gttgcatatg 1020
caagccttgc caccgccg                                     1038

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```

SEQ ID NO: 313      moltype = DNA length = 1038
FEATURE            Location/Qualifiers
source             1..1038
                  mol_type = other DNA
                  organism = synthetic construct

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-continued

SEQUENCE: 313

gaggtgcagc	tgctggagag	cgggtggagg	ttggtgcagc	cgggggtag	cctgcgtctg	60
tcgtgcgcgc	cttcggcgtt	cacgttctct	gattacgcca	tgactcgtt	ccggcaggcc	120
cccgtaagg	agcgcgtgct	ggtgtcgtct	attgactccg	gcggtccac	ttactacgca	180
gacagtgtca	agggccgttt	caccatcagc	cgcgacaaca	gcaagaacac	gctgtacctg	240
cagatgaact	cccttcgagc	agaggacacc	gcggtgtact	actgtaatgc	gggcttcaag	300
ggcgatcacc	cccaccccaa	ggatgccttc	gacatttggg	gccagggcac	ccaggtcacc	360
gtgtcgtcca	ccacaacacc	tgctccaagg	ccccccacac	ccgctccaac	tatagccagc	420
caaccattga	gcctcagacc	tgaagcttgc	aggcccgag	caggaggcgc	cgtccatcac	480
cgaggcctgg	acttcgcgtg	tgatatttat	atttgggcac	ctttggccgg	aacatgtggg	540
gtgttgcttc	tctcccttgc	gatcactctg	tattgtaagc	gcgggagaaa	gaagctcctg	600
tacatcttca	agcagccttt	tatgcgacct	gtgcaaacca	ctcaggaaga	agatgggtgt	660
tcatgccgtc	tcccagagga	ggaagaagga	gggtgtgaac	tgagggtgaa	attttctaga	720
agcccgatg	ctccgcata	tcagcagggt	cagaatcagc	tctacaatga	attgaatctc	780
ggcaggcgag	aagagtacga	tgttctggac	aaaagacggg	gcagggtacc	cgagatgggg	840
ggaaagcccc	ggagaaaaaa	tcttcaggag	gggtgtgaca	atgagctgca	gaaggacaag	900
atggctgaag	cctatagcga	gatcggaatg	aaaggcgaaa	gacgcagagg	caaggggcat	960
gacggtctgt	accagggtct	ctctacagcc	accaaggaca	cttatgatgc	gttgcatatg	1020
caagccttgc	caccccg					1038

SEQ ID NO: 314 moltype = DNA length = 1044
 FEATURE Location/Qualifiers
 source 1..1044
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 314

gaggtgcaac	tgctggaatc	cggcgaggag	ctggtgcagc	cggcggcag	cctcagactg	60
agctgtgcgc	actctggctt	taccttcagc	agcaggggca	tgactcgtt	gcggcaggcc	120
cctggcaagg	aaagagagct	ggtctccgcc	atcagcggat	ctggcgacca	cacctactat	180
gccgatagcg	tgccgcgaag	attcaccaatc	tctagagata	atagcaagaa	cacctgttac	240
ctgcagatga	acagcctgcg	ggccgaggac	accgcccgtg	actactgcaa	cgccctggaa	300
ggcgccctca	caacagctat	ccagccagga	ggccctgact	actggggcca	gggcacccag	360
gtgaccgtgt	ccagcaccac	aacacctgct	ccaaggcccc	ccacaccgcg	tccaactata	420
gccagccaac	cattgagcct	cagacctgaa	gcttgcaggc	ccgcagcagg	aggcgccgtc	480
catacgccag	gcctggactt	cgcgtgtgat	atttatattt	gggcaccttt	ggccggaaca	540
tgtgggggtg	tgcttctctc	ccttgtgatc	actctgtatt	gtaagcgcgg	gagaaagaag	600
ctcctgtaca	tcttcaagca	gccttttatg	cgactgtgac	aaaccaactc	ggaagaagat	660
gggtgttcat	gccgcttccc	cgaggaggaa	gaaggagggt	gtgaactgag	ggtgaaattt	720
tctagaagcg	ccgatgtccc	cgcataatcag	cagggtcaga	atcagctcta	caatgaattg	780
aatctcggca	ggcgagaaga	gtacgatgtt	ctggacaaaa	gacggggcag	ggaatcccgag	840
atggggggaa	agcccccggg	aaaaaatcct	caggaggggg	tgtacaatga	gctgcagaag	900
gacaagatgg	ctgaagccta	tagcgagatc	ggaatgaaag	gcgaagacgc	cagaggcaag	960
gggcatgacg	gctctgtacc	gtgtctctct	acagccacca	aggacactta	tgatgcgttg	1020
catatgcaag	ccttgccacc	cgcg				1044

SEQ ID NO: 315 moltype = DNA length = 592
 FEATURE Location/Qualifiers
 source 1..592
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 315

aatcaacctc	tggattacaa	aatttgtgaa	agattgactg	gtattcttaa	ctatgttgct	60
ccttttacgc	tatgtggata	cgtctgttta	atgcctttgt	atcatgctat	tgttcccgt	120
atggctttca	ttttctcttc	cttgataaaa	tctcgttgcc	tgtctcttta	tgaggagtgt	180
tgcccccgtg	tcaggcaaac	tgccgtgggt	tgactgtgtg	ttgctgacgc	aacccccact	240
ggttggggca	ttgccaccac	ctgtcagctc	ctttccggga	ctttcgtttt	ccccctccct	300
attgccacgg	cggaaactat	cgcgcctgct	cttgcccgtc	gctggacagg	ggctcgggtg	360
ttgggcactg	acaattccgt	ggtgttgtcg	gggaagctga	cgtcctttcc	atggctgctc	420
gcctgtgttg	ccacctggat	tctgcgcggg	acgtccttct	gctacgtccc	ttcgcccttc	480
aatccagcgg	accttctctc	cgcgcggcct	ctgcgggctc	tgccgctctc	tccgctctct	540
cgccttcgcc	ctcagacgag	tcggatctcc	ctttgggccc	cctccccgcc	tg	592

SEQ ID NO: 316 moltype = DNA length = 592
 FEATURE Location/Qualifiers
 source 1..592
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 316

aatcaacctc	tggattacaa	aatttgtgaa	agattgactg	gtattcttaa	ctatgttgct	60
ccttttacgc	tatgtggata	cgtctgttta	atgcctttgt	atcatgctat	tgttcccgt	120
atggctttca	ttttctcttc	cttgataaaa	tctcgttgcc	tgtctcttta	tgaggagtgt	180
tgcccccgtg	tcaggcaaac	tgccgtgggt	tgactgtgtg	ttgctgacgc	aacccccact	240
ggttggggca	ttgccaccac	ctgtcagctc	ctttccggga	ctttcgtttt	ccccctccct	300
attgccacgg	cggaaactat	cgcgcctgct	cttgcccgtc	gctggacagg	ggctcgggtg	360
ttgggcactg	acaattccgt	ggtgttgtcg	gggaaggtct	gctgagactc	ggggctgctc	420
gcctgtgttg	ccacctggat	tctgcgcggg	acgtccttct	gctacgtccc	ttcgcccttc	480
aatccagcgg	accttctctc	cgcgcggcct	ctgcgggctc	tgccgctctc	tccgctctct	540
cgccttcgcc	ctcagacgag	tcggatctcc	ctttgggccc	cctccccgcc	tg	592

-continued

SEQ ID NO: 317 moltype = DNA length = 389
 FEATURE Location/Qualifiers
 source 1..389
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 317

aatcaacctc	tggattacaa	aatttgtgaa	agattgactg	atattcttaa	ctatgttgct	60
ccttttacgc	tgtgtggata	tgttgcttta	atgcctctgt	atcatgctat	tgettcccg	120
acggctttcg	ttttctctc	cttgataaaa	tcttggtg	tgtctcttta	tgaggagt	180
tggcccggtg	tccgtcaacg	tggcgtggtg	tgtctgtgtg	ttgtgacgc	aacccccact	240
ggctggggca	ttgccaccac	ctgtcaactc	ctttctggga	ctttcgcttt	ccccctccg	300
atcgccacgg	cagaactcat	cgcgcctgc	cttgcccgct	gctggacagg	ggctaggtg	360
ctgggcactg	ataattccgt	ggtgtgtgc				389

SEQ ID NO: 318 moltype = DNA length = 584
 FEATURE Location/Qualifiers
 source 1..584
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 318

gcgttacata	acttacggta	aatggcccg	ctgggtgacc	gccaacgac	ccccgccat	60
tgacgtcaat	aatgacgtat	gttcccatag	taacgccaat	agggactttc	cattgacgtc	120
aatgggtgga	gtatttacgg	taaactgccc	acttggcagt	acatcaagt	tatcatatgc	180
caagtacgcc	ccctattgac	gtcaatgacg	gtaaatggcc	cgctggcat	tatgccag	240
acatgacctt	atgggacttt	cctacttggc	agtaacatcta	cgtattagtc	atcgctatta	300
ccatgggtcga	ggtgagcccc	acgttctgct	tcactctccc	catctcccc	ccctccccc	360
ccccaatttt	gtatttattt	attttttaat	tattttgtgc	agcgatggg	gcgggggggg	420
ggggggggcg	cgcgccaggc	ggggcggggc	ggggcgaggg	gcgggggcg	gcgagggcga	480
gagggtcgcc	ggcagccaat	cagagcggcg	cgtccgaaa	gttcccttt	atggcgaggc	540
ggcgcgcgcg	gcggccctat	aaaaagcgaa	gcgcgcggcg	ggcg		584

SEQ ID NO: 319 moltype = DNA length = 1184
 FEATURE Location/Qualifiers
 source 1..1184
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 319

cgtgaggctc	cggtgcccg	cagtgggcag	agcgcacatc	gcccacagtc	cccgagaagt	60
tggggggagg	ggtcggaat	tgaaccggtg	cctagagaag	gtggcgcggg	gtaaaactgg	120
aaagtgatgt	cgtgtactgg	ctccgccttt	ttcccgaggg	tgggggagaa	ccgtatataa	180
gtgcagtagt	cgcgctgaac	gttctttttc	gcaacgggtt	tgccgcagaa	acacaggtaa	240
gtgccgtgtg	tggttcccg	gggcctggcc	tctttacggg	ttatggccct	tgctgtcctt	300
gaattacttc	cacttgctg	cagtacgtga	ttcttgatcc	cgagcttcgg	gttggaaagt	360
ggtgggagag	ttcgaggcct	tgcgcttaag	gagcccttc	gcctcgtgct	tgagttgagg	420
cctggcctgg	gcgctggggc	cgcgcgctgc	gaatctggtg	gcaccttcgc	gcctgtctcg	480
ctgccttcga	taagtctcta	gccatttaaa	atttttgatg	acctgctgcg	acgctttttt	540
tctggcaaga	tagtcttgta	aatgcggggc	aagatctgca	cactgggtatt	tcggtttttg	600
gggcccggcg	cggcgacggg	gcccgctgct	cccagcgcac	atgttcggcg	aggcgggggc	660
tgcgagcgcg	gccaccgaga	atcggacggg	ggtagttctca	agctggcccg	cctgctctgg	720
tgcctggcct	cgcgcgcgcg	tgtatcgccc	cgcctggggc	ggcaaggctg	gcccgggtcg	780
caccagttgc	gtgagcggaa	agatggccgc	ttcccggccc	tgtgacagg	agctcaaaat	840
ggaggacgcg	gcgctcgga	gagcgggcgg	gtgagtcacc	cacacaaaag	aaaagggcct	900
ttccgtcttc	agcgtcgct	tcattgtgact	ccacggagta	ccgggcgcgc	ttcaggcacc	960
tcgattagtt	ctcgagcttt	tggagtacgt	cgtcttttag	ttggggggag	gggtttttat	1020
cgatggagtt	ttcccacact	gagtggggtg	agactgaagt	taggccagct	tggaacttga	1080
tgtaatcttc	cttggaattt	gcctcttttg	agtttggatc	ttggttcatt	ctcaagcctc	1140
agacagtggt	tcaaagtttt	ttcttcccat	ttcagggtgc	gtga		1184

SEQ ID NO: 320 moltype = DNA length = 388
 FEATURE Location/Qualifiers
 source 1..388
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 320

aatgaaagac	cccacctgta	ggtttggcaa	gctaggatca	aggttaggaa	cagagagaca	60
gcagaatatg	ggccaaacag	gatattctgt	gtaagcagtt	cctgccccgg	ctcaggggcca	120
agaacagttg	gaacagcaga	atatggggca	aacaggatat	ctgtggtta	cagttcctgc	180
ccggctcag	ggccaagaac	agatgggtcc	cagatgcggg	cccgcctca	gcagtttcta	240
gagaaccatc	agatgtttcc	agggcgcccc	aaggacctga	aatgacctgc	tgctttat	300
gaactaacca	atcagttcgc	ttctcgcttc	tgttcgcgcg	cttctgctcc	ccgagctcaa	360
taaaagagcc	cacaaccctc	cactcggc				388

SEQ ID NO: 321 moltype = DNA length = 500
 FEATURE Location/Qualifiers
 source 1..500
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 321

gggtagggga	ggcgcttttc	ccaaggcagt	ctggagcatg	cgctttagca	gccccgctgg	60
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-continued

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gcacttggcg ctacacaagt ggcctctggc ctcgcacaca ttccacatcc accggtagge 120
gcaaacccgc tccgttcttt ggtggccctc tcgcgccacc ttctactcct cccctagtca 180
ggaagtcccc cccgcgcccg cagctcgcgt cgtgcaggac gtgacaaatg gaagtagcac 240
gtctcactag tctcgtgcag atggacagca ccgctgagca atggaagcgg gtaggccttt 300
ggggcagcgg ccaatagcag ctttgctcct tcgctttctg ggtcagagg ctgggaagg 360
gtgggtcccg gggcgggctc agggcggggc tcaggggcgg ggcggggccc cgaaggtcct 420
ccggaggccc ggcattctgc acgcttcaaa agcgcacgtc tgccgcgctg ttctctctt 480
ctcatctcc gggcctttcg                                     500

```

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SEQ ID NO: 322      moltype = DNA length = 443
FEATURE            Location/Qualifiers
source             1..443
                  mol_type = other DNA
                  organism = synthetic construct

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SEQUENCE: 322
tgaaagacc caccgttagg tttggcaaga tagctgcagt aacgccattt tgcaaggcat 60
ggaaaaaac caaaccaaga atagagaagt tcagatcaag ggcgggtaca tgaaaaatagc 120
taacgttggg ccaaacagga tatctcgggt gaggcagttc gggcccgccc cggggccaag 180
aacagatggt caccgcagtt tcggcccgcg cccgaggcca agaacagatg gtccccagat 240
atggcccaac cctcagcagt ttcttaagac ccatcagatg ttccaggctt ccccaagga 300
cctgaaatga cctcgcgctt tatttgaatt aaccaatcag cctgctcttc gcttctgttc 360
gcgcgcttct gcttcccag cctataaaa gagctcacaa cccctcactc ggcgcgccag 420
tcctccgatt gactgagtcg ccc                                     443

```

```

SEQ ID NO: 323      moltype = DNA length = 1212
FEATURE            Location/Qualifiers
source             1..1212
                  mol_type = other DNA
                  organism = synthetic construct

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SEQUENCE: 323
ggcctccgcg ccgggttttt ggcctcccg cggggccccc cctcctcacg gcgagcgctg 60
ccacgtcaga cgaaggggcg agcgagcgtc ctgactcttc cgcgccgacg ctcaggacag 120
cggcccgctg ctcataagac tcggccttag aaccccagta tcagcagaag gacattttag 180
gacgggactt ggggtactct agggcactgg ttttctttcc agagagcgga acaggcgagg 240
aaaagtatgc cctctcggc gattctgcgg agggatctcc gtggggcggt gaacgccgat 300
gattatataa ggacgcgcgg ggtgtggcac agctagtctc gtgcagcccg ggatttgggt 360
cgcggttctt gtttgtggat cgctgtgatc gtcacttggt gactagcggg ctgctgggct 420
ggccgggggt ttctgtggcg ccgggcccgt cgggtgggac gaagcgtgtg gagagaccgc 480
caagggtctg agtctgggtc cgcgagcaag gttgccctga actgggggtt ggggggagcg 540
cagcaaaatg ccggtctgtt ccgagtcctg aatggaagac gcttgtgagg cgggctgtga 600
ggtcgttgaa acaagggtgg gggcatgggt ggcggcaaga acccaaggtc ttgaggcctt 660
cgctaatacg ggaaagctct tatcgggtg agatgggctg gggcaccatc tggggaccct 720
gacgtgaagt ttgtcactga cttgagaact cggtttgtcg tctgttgagg gggcggcagt 780
tatggcgggt ccgttgggca gtgcacccgt acccttgagg cgcgcgcgcc tcgtcgtgtc 840
gtgacgtcac ccgttctgtt ggcttataat gcagggtggg gccacctgcc ggtaggtgtg 900
cggtaggctt ttctccgctc caggacgcag ggttcgggcc tagggtaggc tctcctgaat 960
cgacaggcgc cggacctctg gtgaggggag ggataagtga ggcgtcagtt tctttgggtc 1020
gttttatgta cctatctctt taagtagctg aaagtcgggt tttgaactat gcgctcgggg 1080
ttggcgagtg tgttttgtga agttttttag gcaccttttg aaatgtaate atttgggtca 1140
atatgtaatt ttccagtgtta gactagtaaa ttgtccgcta aattctgggc gtttttggtt 1200
tttttgtagt ac                                     1212

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SEQ ID NO: 324      moltype = AA length = 310
FEATURE            Location/Qualifiers
source             1..310
                  mol_type = protein
                  organism = synthetic construct

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```

SEQUENCE: 324
DVQLVQSGAE VKKPGASVKV SCKASGYTFT RYTMHWVRQA PGQGLEWIGY INPSRGYTNV 60
ADSVKGRFTI TTDKSTSTAY MELSSLRSED TATYYCARY DDHYCLDYWG QGTTVTVSSS 120
GGGSGGGGGS GGGGSDIVLT QSPATLSLSP GERATLSLSCRA SQSVSYMNWY QKPKGKAPKR 180
WIYDTSKVAS GVPARFSGSG SGTDSLTLIN SLEAEDAATY YCQQWSSNPL TFGGGTKVEI 240
KTTTPAPRPP TPAPTIASQP LSLRPEACRP AAGGAVHTRG LDFACDIYIW APLAGTCGVL 300
LLSLVITLYC                                     310

```

```

SEQ ID NO: 325      moltype = AA length = 310
FEATURE            Location/Qualifiers
source             1..310
                  mol_type = protein
                  organism = synthetic construct

```

```

SEQUENCE: 325
DVQLVQSGAE VKKPGASVKV SCKASGYTFT RYTMHWVRQA PGQGLEWIGY INPSRGYTNV 60
ADSVKGRFTI TTDKSTSTAY MELSSLRSED TATYYCARY DDPYCLDYWG QGTTVTVSSS 120
GGGSGGGGGS GGGGSDIVLT QSPATLSLSP GERATLSLSCRA SQSVSYMNWY QKPKGKAPKR 180
WIYDTSKVAS GVPARFSGSG SGTDSLTLIN SLEAEDAATY YCQQWSSNPL TFGGGTKVEI 240
KTTTPAPRPP TPAPTIASQP LSLRPEACRP AAGGAVHTRG LDFACDIYIW APLAGTCGVL 300
LLSLVITLYC                                     310

```

```

SEQ ID NO: 326      moltype = AA length = 311

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FEATURE
source Location/Qualifiers
1..311
mol_type = protein
organism = synthetic construct

SEQUENCE: 326
DVQLVQSGAE VKKPGASVKV SCKASGYTFT RYTMHWVRQA PGQGLEWIGY INPSRGYTNV 60
ADSVKGRFTI TTDKSTSTAY MELSSLRSED TATYYCARYY NDQYCLDYWG QGTTTVTVSSS 120
SGGGSGGGG SGGGSDIVL TQSPATLSLS PGERATLSCR ASQSVSYMNW YQKPGKAPK 180
RWIYDTSKVA SGVPARFSGS GSGTDYSLTI NSLEAEDAAT YCQWSSNP LTFGGGTKVE 240
IKTTTPAPRP PTPAPTASQ PLSLRPEACR PAAGGAVHTR GLDFACDIYI WAPLAGTCGV 300
LLSLVITLY C 311

SEQ ID NO: 327 moltype = AA length = 310
FEATURE Location/Qualifiers
source 1..310
mol_type = protein
organism = synthetic construct

SEQUENCE: 327
DVQLVQSGAE VKKPGASVKV SCKASGYTFT RYTMHWVRQA PGQGLEWIGY INPSRGYTNV 60
ADSVKGRFTI TTDKSTSTAY MELSSLRSED TATYYCARYY DAHNCCLDYWG QGTTTVTVSSS 120
GGGGSGGGG SGGGSDIVLT QSPATLSLSP GERATLSLSCRA SQSVSYMNWY QQKPGKAPKR 180
WIYDTSKVAS GVPARFSGS SGTDYSLTIN SLEAEDAATY YCQWSSNPL TFGGGTKVEI 240
KTTTPAPRPP TPAPTASQP LSLRPEACRP AAGGAVHTRG LDFACDIYIW APLAGTCGVL 300
LLSLVITLYC 310

SEQ ID NO: 328 moltype = AA length = 310
FEATURE Location/Qualifiers
source 1..310
mol_type = protein
organism = synthetic construct

SEQUENCE: 328
DVQLVQSGAE VKKPGASVKV SCKASGYTFT RYTMHWVRQA PGQGLEWIGY INPSRGYTNV 60
ADSVKGRFTI TTDKSTSTAY MELSSLRSED TATYYCARYS DDHYCLDYWG QGTTTVTVSSS 120
GGGGSGGGG SGGGSDIVLT QSPATLSLSP GERATLSLSCRA SQSVSYMNWY QQKPGKAPKR 180
WIYDTSKVAS GVPARFSGS SGTDYSLTIN SLEAEDAATY YCQWSSNPL TFGGGTKVEI 240
KTTTPAPRPP TPAPTASQP LSLRPEACRP AAGGAVHTRG LDFACDIYIW APLAGTCGVL 300
LLSLVITLYC 310

SEQ ID NO: 329 moltype = AA length = 310
FEATURE Location/Qualifiers
source 1..310
mol_type = protein
organism = synthetic construct

SEQUENCE: 329
DVQLVQSGAE VKKPGASVKV SCKASGYTFT RYTMHWVRQA PGQGLEWIGY INPSRGYTNV 60
ADSVKGRFTI TTDKSTSTAY MELSSLRSED TATYYCARYS DDRYCLDYWG QGTTTVTVSSS 120
GGGGSGGGG SGGGSDIVLT QSPATLSLSP GERATLSLSCRA SQSVSYMNWY QQKPGKAPKR 180
WIYDTSKVAS GVPARFSGS SGTDYSLTIN SLEAEDAATY YCQWSSNPL TFGGGTKVEI 240
KTTTPAPRPP TPAPTASQP LSLRPEACRP AAGGAVHTRG LDFACDIYIW APLAGTCGVL 300
LLSLVITLYC 310

SEQ ID NO: 330 moltype = AA length = 311
FEATURE Location/Qualifiers
source 1..311
mol_type = protein
organism = synthetic construct

SEQUENCE: 330
DVQLVQSGAE VKKPGASVKV SCKASGYTFT RYTMHWVRQA PGQGLEWIGY INPSRGYTNV 60
ADSVKGRFTI TTDKSTSTAY MELSSLRSED TATYYCARYY DDNYCLDYWG QGTTTVTVSSS 120
SGGGSGGGG SGGGSDIVL TQSPATLSLS PGERATLSCR ASQSVSYMNW YQKPGKAPK 180
RWIYDTSKVA SGVPARFSGS GSGTDYSLTI NSLEAEDAAT YCQWSSNP LTFGGGTKVE 240
IKTTTPAPRP PTPAPTASQ PLSLRPEACR PAAGGAVHTR GLDFACDIYI WAPLAGTCGV 300
LLSLVITLY C 311

SEQ ID NO: 331 moltype = AA length = 310
FEATURE Location/Qualifiers
source 1..310
mol_type = protein
organism = synthetic construct

SEQUENCE: 331
DVQLVQSGAE VKKPGASVKV SCKASGYTFT RYTMHWVRQA PGQGLEWIGY INPSRGYTNV 60
ADSVKGRFTI TTDKSTSTAY MELSSLRSED TATYYCARYY DDQYCLDYWG QGTTTVTVSSS 120
GGGGSGGGG SGGGSDIVLT QSPATLSLSP GERATLSLSCRA SQSVSYMNWY QQKPGKAPKR 180
WIYDTSKVAS GVPARFSGS SGTDYSLTIN SLEAEDAATY YCQWSSNPL TFGGGTKVEI 240
KTTTPAPRPP TPAPTASQP LSLRPEACRP AAGGAVHTRG LDFACDIYIW APLAGTCGVL 300
LLSLVITLYC 310

SEQ ID NO: 332 moltype = AA length = 495
FEATURE Location/Qualifiers

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source                1..495
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 332
KFTIVFPFHQ KGNWKNVPSN YHYCPSSSDL NWHNDLIGTA IQVKMPQSHK AIQADGWMCH 60
ASKWVTTCDF RWYGPKYITQ SIRSFTPSVE QCKESIEQTK QGTWLNPGFP PQSCGYATVT 120
DAEAVIVQVT PHHVLVDEYT GEWVDSQFIN GKCSNYICPT VHNSTTWHS YKVKGLCDSN 180
LISMDITFFS EDGELSSLGK EGTGFRSNYF AYETGGKACK MQYCKHWGVR LPSGVWFEMA 240
DKDLFAAARF PECPEGSSIS APSQTSVDVS LIQDVERILD YSLCQETWSK IRAGLPISPV 300
DLSYLAPKNP GTGPAFTIIN GTLKYFETRY IRVDIAAPIL SRMVGMISGT TTEAELWDDW 360
APYEDVEIGP NGVLR TSSGY KFPFLYMIGHG MLDSDLHLSS KAQVFEHPHI QDAASQLPDD 420
ESLFFGDTGL SKNPIELVEG WFSSWKSSIA SFFFIIGLII GLFLVLRVGI HLCIKLKHTK 480
KRQIYTDIEM NRLGK 495

SEQ ID NO: 333      moltype = AA length = 495
FEATURE            Location/Qualifiers
source              1..495
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 333
KFTIVFPFHQ KGNWKNVPSN YHYCPSSSDL NWHNDLIGTA LQVKMPQSHK AIQADGWMCH 60
ASKWVTTCDF RWYGPKYITQ SIRSFTPSVE QCKESIEQTK QGTWLNPGFP PQSCGYATVT 120
DAEAVIVQVT PHHVLVDEYT GEWVDSQFIN GKCSNYICPT VHNSTTWHS YKVKGLCDSN 180
LISMDITFFS EDGELSSLGK EGTGFRSNYF AYETGGKACK MQYCKHWGVR LPSGVWFEMA 240
DKDLFAAARF PECPEGSSIS APSQTSVDVS LIQDVERILD YSLCQETWSK IRAGLPISPV 300
DLSYLAPKNP GTGPAFTIIN GTLKYFETRY IRVDIAAPIL SRMVGMISGT TTEAELWDDW 360
APYEDVEIGP NGVLR TSSGY KFPFLYMIGHG MLDSDLHLSS KAQVFEHPHI QDAASQLPDD 420
ESLFFGDTGL SKNPIELVEG WFSSWKSSIA SFFFIIGLII GLFLVLRVGI HLCIKLKHTK 480
KRQIYTDIEM NRLGK 495

SEQ ID NO: 334      moltype = AA length = 495
FEATURE            Location/Qualifiers
source              1..495
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 334
KFTIVFPFHQ KGNWKNVPSN YHYCPSSSDL NWHNDLIGTA IQVKMPQSHK AIQADGWMCH 60
ASKWVTTCDF RWYGPKYITH SIRSFTPSVE QCKESIEQTK QGTWLNPGFP PQSCGYATVT 120
DAEAVIVQVT PHHVLVDEYT GEWVDSQFIN GKCSNYICPT VHNSTTWHS YKVKGLCDSN 180
LISMDITFFS EDGELSSLGK EGTGFRSNYF AYETGGKACK MQYCKHWGVR LPSGVWFEMA 240
DKDLFAAARF PECPEGSSIS APSQTSVDVS LIQDVERILD YSLCQETWSK IRAGLPISPV 300
DLSYLAPKNP GTGPAFTIIN GTLKYFETRY IRVDIAAPIL SRMVGMISGT TTEAELWDDW 360
APYEDVEIGP NGVLR TSSGY KFPFLYMIGHG MLDSDLHLSS KAQVFEHPHI QDAASQLPDD 420
ESLFFGDTGL SKNPIELVEG WFSSWKSSIA SFFFIIGLII GLFLVLRVGI HLCIKLKHTK 480
KRQIYTDIEM NRLGK 495

SEQ ID NO: 335      moltype = AA length = 495
FEATURE            Location/Qualifiers
source              1..495
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 335
KFTIVFPFHQ KGNWKNVPSN YHYCPSSSDL NWHNDLIGTA LQVKMPQSHK AIQADGWMCH 60
ASKWVTTCDF RWYGPKYITH SIRSFTPSVE QCKESIEQTK QGTWLNPGFP PQSCGYATVT 120
DAEAVIVQVT PHHVLVDEYT GEWVDSQFIN GKCSNYICPT VHNSTTWHS YKVKGLCDSN 180
LISMDITFFS EDGELSSLGK EGTGFRSNYF AYETGGKACK MQYCKHWGVR LPSGVWFEMA 240
DKDLFAAARF PECPEGSSIS APSQTSVDVS LIQDVERILD YSLCQETWSK IRAGLPISPV 300
DLSYLAPKNP GTGPAFTIIN GTLKYFETRY IRVDIAAPIL SRMVGMISGT TTEAELWDDW 360
APYEDVEIGP NGVLR TSSGY KFPFLYMIGHG MLDSDLHLSS KAQVFEHPHI QDAASQLPDD 420
ESLFFGDTGL SKNPIELVEG WFSSWKSSIA SFFFIIGLII GLFLVLRVGI HLCIKLKHTK 480
KRQIYTDIEM NRLGK 495

SEQ ID NO: 336      moltype = AA length = 495
FEATURE            Location/Qualifiers
source              1..495
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 336
KFSIVFPQSQ KGNWKNVPSS YHYCPSSSDQ NWHNDLLGIT MKVKMPATHK AIQADGWMCH 60
AAKWITTCDP RWYGPKYITH SIHSIQPTSE QCKESIKQTK QGTWMSPGFP PQNCGYATVT 120
DSVAVVQAT PHHVLVDEYT GEWIDSQFPN GKCEEEECET VHNSTTVWYS YKVTGLCDAT 180
LVDTEITFFS EDGKKESIGK PNTGYRSNYF AYEKGDKVCK MNYCKHAGVR LPSGVWFEPV 240
DQDVYAAAKL PECPVGATIS APTQTSVDVS LILDVERILD YSLCQETWSK IRSKQPVSPV 300
DLSYLAPKNP GTGPAFTIIN GTLKYFETRY IRIDIDNP II SKMVGKISGS QTEAELWTEW 360
FPYEGVEIGP NGILKTPTGY KFPFLFMIGHG MLDSDLHKTS QAEVFEHPHL AEAPKQLPEE 420
ETLFFGDTGI SKNPVELIEG WFSSWKSTVV TFFFAIGVFI LLYVVARIVI AVRYRYQGSN 480
NKRIYNDIEM SRFRK 495

SEQ ID NO: 337      moltype = AA length = 495

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FEATURE
source Location/Qualifiers
1..495
mol_type = protein
organism = synthetic construct

SEQUENCE: 337

KFSIVFPQSQ	KGNWKNVPSS	YHYCPSSSDQ	NWHNDLLGIT	MKVKMPATHK	AIQADGWMCH	60
AAKWITTCDF	RWYGPKYITH	SIHSIQPTSE	QCKESIKQTK	QGTWMSPGFP	PQNCGYATVT	120
DSVAVVVQAT	PHHVLVDEYT	GEWIDSQFPN	GKCETEECET	VHNSTVWYSD	YKVTGLCDAT	180
LVDTEITFFS	EDGKKESIGK	PNTGYRSNYF	AYEKGDKVCK	MNYCKHAGVR	LPSGVWFEFV	240
DQDVYAAAKL	PECPVGATIS	APTQTSVDVS	LILDVERILD	YSLCQETWSK	IRSKQPVSPV	300
DLSYLAPKNP	GTGPAFTIIN	GTLKYFETRY	IRIDIDNPPI	SKMVGKISGS	QTEQELWTEW	360
FPYEGVEIGP	NGILKTPTGY	KFPLFMIGHG	MLDSLHKT	QAEVFEHPL	AEAPKQLPEE	420
ETLFFPGDTGI	SKNPVELIEG	WFSSWKSTVV	TFFFAIGVFI	LLYVVARIVI	AVRIRYQGSN	480
NKRIYNDIEM	SRFRK					495

SEQ ID NO: 338 moltype = AA length = 495

FEATURE
source Location/Qualifiers
1..495
mol_type = protein
organism = synthetic construct

SEQUENCE: 338

KFSIVFPQSQ	KGNWKNVPSS	YHYCPSSSDQ	NWHNDLLGIT	MKVKMPQTHK	AIQADGWMCH	60
AAKWITTCDF	RWYGPKYITH	SIHSIQPTSE	QCKESIKQTK	QGTWMSPGFP	PQNCGYATVT	120
DSVAVVVQAT	PHHVLVDEYT	GEWIDSQFPN	GKCETEECET	VHNSTVWYSD	YKVTGLCDAT	180
LVDTEITFFS	EDGKKESIGK	PNTGYRSNYF	AYEKGDKVCK	MNYCKHAGVR	LPSGVWFEFV	240
DQDVYAAAKL	PECPVGATIS	APTQTSVDVS	LILDVERILD	YSLCQETWSK	IRSKQPVSPV	300
DLSYLAPKNP	GTGPAFTIIN	GTLKYFETRY	IRIDIDNPPI	SKMVGKISGS	QTEAELWTEW	360
FPYEGVEIGP	NGILKTPTGY	KFPLFMIGHG	MLDSLHKT	QAEVFEHPL	AEAPKQLPEE	420
ETLFFPGDTGI	SKNPVELIEG	WFSSWKSTVV	TFFFAIGVFI	LLYVVARIVI	AVRIRYQGSN	480
NKRIYNDIEM	SRFRK					495

SEQ ID NO: 339 moltype = AA length = 495

FEATURE
source Location/Qualifiers
1..495
mol_type = protein
organism = synthetic construct

SEQUENCE: 339

KFSIVFPQSQ	KGNWKNVPSS	YHYCPSSSDQ	NWHNDLLGIT	MKVKMPQTHK	AIQADGWMCH	60
AAKWITTCDF	RWYGPKYITH	SIHSIQPTSE	QCKESIKQTK	QGTWMSPGFP	PQNCGYATVT	120
DSVAVVVQAT	PHHVLVDEYT	GEWIDSQFPN	GKCETEECET	VHNSTVWYSD	YKVTGLCDAT	180
LVDTEITFFS	EDGKKESIGK	PNTGYRSNYF	AYEKGDKVCK	MNYCKHAGVR	LPSGVWFEFV	240
DQDVYAAAKL	PECPVGATIS	APTQTSVDVS	LILDVERILD	YSLCQETWSK	IRSKQPVSPV	300
DLSYLAPKNP	GTGPAFTIIN	GTLKYFETRY	IRIDIDNPPI	SKMVGKISGS	QTEQELWTEW	360
FPYEGVEIGP	NGILKTPTGY	KFPLFMIGHG	MLDSLHKT	QAEVFEHPL	AEAPKQLPEE	420
ETLFFPGDTGI	SKNPVELIEG	WFSSWKSTVV	TFFFAIGVFI	LLYVVARIVI	AVRIRYQGSN	480
NKRIYNDIEM	SRFRK					495

SEQ ID NO: 340 moltype = AA length = 472

FEATURE
source Location/Qualifiers
1..472
mol_type = protein
organism = synthetic construct

SEQUENCE: 340

DIVLTQSPPS	LAMSLGKRAT	ISCRASESVT	ILGSHLIHWY	QQKQPQPPTL	LIQLASNVQT	60
GVPARFSGSG	SRTDFTLTID	PVEEDDVAVY	YCLQSRITPR	TFGGKTLEI	KGSTSGSGKP	120
GSSEGSTKGQ	IQLVQSGPEL	KKPGETVKIS	CKASGYTFTD	YSINWVKRAP	GKGLKMWGI	180
NTETREPAYA	YDFRGRFAPS	LETSASTAYL	QINNLYEDT	ATYFCALDYS	YAMDYWGQGT	240
SVTVSSAAT	TPAPRPPTP	APTIASQPLS	LRPEACRPAA	GGAVHTRGLD	FACDIYIAP	300
LAGTCGVLLL	SLVITLYCKR	GRKKLLYIFK	QPFMRPVQTT	QEEDGCSCRF	PEEEEGGCEL	360
RVKFSRSADA	PAYQQGQNL	YNELNLGRRE	EYDVLDKRRG	RDPEMGKPR	RKNPQEGLYN	420
ELQDKMAEA	YSEIGMKGER	RRGKGHDGLY	QGLSTATKDT	YDALHMQALP	PR	472

SEQ ID NO: 341 moltype = AA length = 467

FEATURE
source Location/Qualifiers
1..467
mol_type = protein
organism = synthetic construct

SEQUENCE: 341

QVKLEESGGG	LVQAGSLRL	SCAASEHTFS	SHVMGWFRQA	PGKERESVAV	IGWRDISTSY	60
ADSVKGRFTI	SRDNAKTILY	LQMNSLKPED	TAVYYCAARR	IDAADFDSWG	QGTQVTVSSG	120
GGSEVQLVLE	SGGGLVQAGG	SLRLSCAASG	RTFTMGWFRQ	APGKEREFVA	AISLSPTLAY	180
YAESVKGRFT	ISRDNAKNTV	VLMNSLKPE	DTALYYCAAD	RKSVMSIRPD	YWGGTQVTV	240
SSTSTTPAP	RPPTPAPTIA	SQPLSLRPEA	CRPAAGGAVH	TRGLDFACDI	YIWAPLAGTC	300
GVLVLLSLVIT	LYCKRGRKKL	LYIFKQPFMR	PVQTTQEEDG	CSCRFPEEEE	GGCELRVKFS	360
RSADAPAYQQ	GQNQLYNELN	LGRREEYDVL	DKRRGRDPEM	GGKPRRKNPQ	EGLYNELQKD	420
KMAEAYSEIG	MKGERRRGKG	HDGLYQGLST	ATKDTYDALH	MQALPPR		467

SEQ ID NO: 342 moltype = AA length = 5

FEATURE
source Location/Qualifiers

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source	1..5 mol_type = protein organism = synthetic construct	
SEQUENCE: 342 TGEKP		5
SEQ ID NO: 343 FEATURE source	moltype = AA length = 5 Location/Qualifiers 1..5 mol_type = protein organism = synthetic construct	
SEQUENCE: 343 GGGGS		5
SEQ ID NO: 344 FEATURE source	moltype = AA length = 10 Location/Qualifiers 1..10 mol_type = protein organism = synthetic construct	
SEQUENCE: 344 GGGSGGGGS		10
SEQ ID NO: 345 FEATURE source	moltype = AA length = 15 Location/Qualifiers 1..15 mol_type = protein organism = synthetic construct	
SEQUENCE: 345 GGGSGGGGS GGGGS		15
SEQ ID NO: 346 FEATURE source	moltype = AA length = 20 Location/Qualifiers 1..20 mol_type = protein organism = synthetic construct	
SEQUENCE: 346 GGGSGGGGS GGGSGGGGS		20
SEQ ID NO: 347 FEATURE source	moltype = AA length = 25 Location/Qualifiers 1..25 mol_type = protein organism = synthetic construct	
SEQUENCE: 347 GGGSGGGGS GGGSGGGGS GGGGS		25
SEQ ID NO: 348 FEATURE source	moltype = AA length = 14 Location/Qualifiers 1..14 mol_type = protein organism = synthetic construct	
SEQUENCE: 348 EGKSSGSGSE SKVD		14
SEQ ID NO: 349 FEATURE source	moltype = AA length = 18 Location/Qualifiers 1..18 mol_type = protein organism = synthetic construct	
SEQUENCE: 349 KESGSVSSEQ LAQFRSLD		18
SEQ ID NO: 350 FEATURE source	moltype = AA length = 9 Location/Qualifiers 1..9 mol_type = protein organism = synthetic construct	
SEQUENCE: 350 LRQRDGERP		9
SEQ ID NO: 351 FEATURE source	moltype = AA length = 12 Location/Qualifiers 1..12 mol_type = protein organism = synthetic construct	
SEQUENCE: 351 LRQKGGGSE RP		12
SEQ ID NO: 352	moltype = AA length = 16	

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FEATURE	Location/Qualifiers	
source	1..16	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 352		
LRQKDGGS		16
SEQ ID NO: 353	moltype = AA length = 18	
FEATURE	Location/Qualifiers	
source	1..18	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 353		
GEGTSTGSGG		18
SEQ ID NO: 354	moltype = AA length = 18	
FEATURE	Location/Qualifiers	
source	1..18	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 354		
GSTSGSGKPG		18
SEQ ID NO: 355	moltype = AA length = 22	
FEATURE	Location/Qualifiers	
source	1..22	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 355		
GSGATNFSLL		22
SEQ ID NO: 356	moltype = AA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 356		
ATNFSLLKQA		19
SEQ ID NO: 357	moltype = AA length = 14	
FEATURE	Location/Qualifiers	
source	1..14	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 357		
LLKQAGDVEE		14
SEQ ID NO: 358	moltype = AA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 358		
GSGEGRGSLL		21
SEQ ID NO: 359	moltype = AA length = 18	
FEATURE	Location/Qualifiers	
source	1..18	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 359		
EGRGSLLTCG		18
SEQ ID NO: 360	moltype = AA length = 13	
FEATURE	Location/Qualifiers	
source	1..13	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 360		
LLTCGDVEEN		13
SEQ ID NO: 361	moltype = AA length = 23	
FEATURE	Location/Qualifiers	
source	1..23	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 361		
GSGQCTNYAL		23

-continued

SEQ ID NO: 362	moltype = AA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 362		
QCTNYALLKL AGDVESNPGP		20
SEQ ID NO: 363	moltype = AA length = 14	
FEATURE	Location/Qualifiers	
source	1..14	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 363		
LLKLAGDVES NPGP		14
SEQ ID NO: 364	moltype = AA length = 25	
FEATURE	Location/Qualifiers	
source	1..25	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 364		
GSGVKQTLNF DLLKLAGDVE SNPGP		25
SEQ ID NO: 365	moltype = AA length = 22	
FEATURE	Location/Qualifiers	
source	1..22	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 365		
VKQTLNFDLL KLAGDVESNP GP		22
SEQ ID NO: 366	moltype = AA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 366		
LLNFDLLKLA GDVESNPGP		19
SEQ ID NO: 367	moltype = AA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 367		
TLNFDLLKLA GDVESNPGP		19
SEQ ID NO: 368	moltype = AA length = 17	
FEATURE	Location/Qualifiers	
source	1..17	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 368		
NFDLLKLAGD VESNPGP		17
SEQ ID NO: 369	moltype = AA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 369		
QLLNFDLLKL AGDVESNPGP		20
SEQ ID NO: 370	moltype = AA length = 24	
FEATURE	Location/Qualifiers	
source	1..24	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 370		
APVKQTLNFD LLKLAGDVES NPGP		24
SEQ ID NO: 371	moltype = AA length = 40	
FEATURE	Location/Qualifiers	
source	1..40	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 371		
VTELLYRMKR AETYCPRPLL AIHPTEARHK QKIVAPVKQT		40

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SEQ ID NO: 372      moltype = AA  length = 18
FEATURE            Location/Qualifiers
source              1..18
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 372
LNFDLLKLAG DVESNPGP                                18

SEQ ID NO: 373      moltype = AA  length = 40
FEATURE            Location/Qualifiers
source              1..40
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 373
LLAIHPTEAR HKQKIVAPVK QTLNFDLLKL AGDVESNPGP        40

SEQ ID NO: 374      moltype = AA  length = 33
FEATURE            Location/Qualifiers
source              1..33
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 374
EARHKQKIVA PVKQTLNFDL LKLAGDVESN PGP                33

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What is claimed is:

1. A recombinant lentiviral particle comprising:
 - (a) a viral envelope comprising (i) a vesicular stomatitis Indiana virus envelope glycoprotein (VSIV-G) comprising amino acid substitutions selected from the group consisting of: K47A and R354A; K47A and R354Q; K47Q and R354A; and K47Q and R354Q; and (ii) a non-viral membrane-bound tropism polypeptide comprising an anti-CD3 ϵ scFv and a human CD8 α hinge and transmembrane domain; and
 - (b) a recombinant lentiviral vector comprising a polynucleotide encoding a myeloproliferative sarcoma virus enhancer, negative control region deleted, d1587rev primer-binding site substituted (MND) U3 promoter or an EF1 α promoter operably linked to a polynucleotide encoding a signal peptide and an anti-BCMA chimeric antigen receptor comprising an anti-BCMA scFv comprising a heavy chain variable region (VH) comprising a CDRH1, a CDRH2, and a CDRH3 comprising the amino acid sequences of SEQ ID NOs: 82, 83, and 84, and a light chain variable region (VL) comprising a CDRL1, a CDRL2, and a CDRL3 comprising the amino acid sequences of SEQ ID NOs: 86, 87, and 88; a CD8 α hinge and transmembrane domain; a CD137 costimulatory domain; and a CD3 ζ primary signaling domain.
2. The particle of claim 1, wherein the anti-CD3 ϵ scFv is isolated from TR66.
3. The particle of claim 1, wherein the anti-CD3 ϵ scFv comprises the amino acid sequence of any one of SEQ ID NOs: 153, 154, 163, 164, 173, 174, 183, 184, 193, 194, 203, 204, 213, 214, 223, and 224.
4. An isolated immune effector cell transduced with the particle of claim 1.
5. A composition comprising the immune effector cell of claim 4.
6. A composition comprising the particle of claim 1.
7. A pharmaceutical composition comprising a pharmaceutically acceptable carrier and the particle of claim 1.
8. A kit comprising the particle of claim 1, a pharmaceutically acceptable carrier, and instructions for use.
9. A recombinant lentiviral particle comprising:
 - (a) a viral envelope comprising (i) a mutated viral envelope glycoprotein comprising the amino acid sequence of any one of SEQ ID NOs: 332, 333, 334, and 335 and (ii) a non-viral membrane-bound tropism polypeptide comprising the amino acid sequence of any one of SEQ ID NOs: 324, 325, 326, 327, 328, 329, 330, and 331; and
 - (b) a recombinant lentiviral vector comprising a 5' long terminal repeat (LTR) comprising R and U5 regions; a Psi (Ψ) packaging signal; a cPPT/FLAP; a rev response element (RRE); a polynucleotide encoding an MND promoter or an EF1 α promoter operably linked to a polynucleotide encoding a signal peptide and an anti-BCMA chimeric antigen receptor (CAR) comprising the amino acid sequence of SEQ ID NO: 270; optionally a WPRE; a 3' LTR comprising U3 and R regions; a polyadenylation signal; and a poly(A) tail.
10. The particle of claim 9, wherein the mutated viral envelope glycoprotein comprises the amino acid sequence of SEQ ID NO: 335.
11. The particle of claim 9, wherein the non-viral membrane-bound tropism polypeptide comprises the amino acid sequence of SEQ ID NO: 324.
12. An isolated immune effector cell transduced with the particle of claim 9.
13. A composition comprising the particle of claim 9.
14. A pharmaceutical composition comprising a pharmaceutically acceptable carrier and the particle of claim 9.
15. A kit comprising the particle of claim 9, a pharmaceutically acceptable carrier, and instructions for use.
16. A recombinant lentiviral particle comprising:
 - (a) a viral envelope comprising (i) a mutated viral envelope glycoprotein comprising the amino acid sequence of SEQ ID NO: 335 and (ii) a non-viral membrane-bound tropism polypeptide comprising the amino acid sequence of SEQ ID NO: 324; and
 - (b) a recombinant lentiviral vector comprising a 5' LTR comprising R and U5 regions; a Ψ packaging signal; a cPPT/FLAP; a rev response element (RRE); a polynucleotide encoding an EF1 α promoter operably linked to a polynucleotide encoding a signal peptide and an anti-BCMA CAR comprising the amino acid sequence

of SEQ ID NO: 270; optionally a WPRE; a 3' LTR comprising U3 and R regions; a polyadenylation signal; and a poly(A) tail.

17. The particle of claim 16, wherein the EF1 α promoter comprises the polynucleotide sequence of SEQ ID NO: 319. 5

18. The particle of claim 16, wherein the recombinant lentiviral vector is a recombinant HIV-1 lentiviral vector or a recombinant HIV-2 lentiviral vector.

19. The particle of claim 16, wherein the polynucleotide encoding the anti-BCMA CAR comprises the polynucleotide sequence of SEQ ID NO: 302. 10

20. The particle of claim 16, wherein the signal peptide is isolated from a polypeptide selected from the group consisting of: CD8 α , mIgGk, hIgGk, CD33, tPA, SEAP, hGM-CSF, CSF2R, and B2M. 15

21. The particle of claim 16, wherein the lentiviral vector further comprises a WPRE operably linked to the 3' end of the polynucleotide encoding the anti-BCMA CAR.

22. An isolated immune effector cell transduced with the particle of claim 16. 20

23. The immune effector cell of claim 22, wherein the immune effector cell is a T cell or natural killer (NK) cell.

24. A composition comprising the particle of claim 16.

25. A pharmaceutical composition comprising a pharmaceutically acceptable carrier and the particle of claim 16. 25

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